Mathematical Foundations of Revised Spherical Cap Harmonic Analysis

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Abstract

This handbook presents the analytical construction of revised spherical cap harmonic analysis (R-SCHA) on a source-free spherical-cap shell. Starting from two complementary mixed boundary-value problems, it derives the Legendre, Mehler, constant-radial, and radial Sturm–Liouville factors under specified branch and normalisation conventions. It develops the associated Dirichlet-energy orthogonality, field and tensor formulas, coordinate rotations, flux identities, and spherical-harmonic coefficient projections. Volume energy, fixed-surface full-gradient energy, and a Legendre-only surface approximation are treated separately. Weak solutions, classical boundary values, and convergence of infinite expansions are distinguished; surface-energy statements require a square-integrable gradient trace on the chosen surface. The classical one-surface R-SCHA2D construction is described with its different boundary assumptions. The scope and limitations of the results are stated alongside the relevant formulas.

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Keywords. Revised spherical cap harmonic analysis; Laplace equation; mixed boundary-value problems; Sturm–Liouville theory; associated Legendre functions; Mehler functions; Dirichlet energy; regional potential-field theory.

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Part I
Foundations

1 Introduction

1.1 Historical background

Global spherical-harmonic analysis represents a harmonic potential by basis functions that are regular on the complete sphere. Regional observations, however, occupy only a proper subset of the sphere and may be distributed on more than one radius. Haines introduced spherical cap harmonic

analysis (SCHA) to retain the potential-field constraint while replacing the integer degrees of global spherical harmonics by cap-dependent non-integer degrees [3]. Later work clarified the relation between ordinary spherical harmonics and spherical-cap harmonics and supplied coefficient rotations for relocating the cap axis [1, 2]. For a review of the formulations and their limitations, see [11].

Revised spherical cap harmonic analysis (R-SCHA) was developed to represent a harmonic potential in a three-dimensional regional shell bounded by two spherical surfaces and a conical lateral surface. For the boundary assignment used here, the potential is decomposed into solutions of complementary mixed boundary-value problems. The separated solutions have two non-integer Legendre radial branches, positive radial eigenmodes paired with Mehler angular factors, and a constant-radial branch paired with zero-eigenvalue angular factors [9, 8, 6, 7].

A distinct one-surface formulation, called R-SCHA2D, was introduced for regional data lying on one sphere or within a narrow altitude interval [10]. It is not the collapsed-shell limit of the three-dimensional spectral problem: it solves a different, deliberately incomplete boundary-value problem and selects an exceptional degree-$-1/2$ radial branch. The exact Dirichlet energy in the full three-dimensional shell is diagonal in the boundary-adapted energy decomposition. The full-gradient energy on one fixed-radius cap is not generally diagonal, because the radial integrations responsible for the three-dimensional orthogonality have disappeared. A third object, the simplified Legendre-dominant surface spectrum, is a useful approximation but is not an exact Parseval identity for the complete four-family field. The three constructions must therefore be derived and named separately.

1.2 Purpose of the handbook

The purpose of this handbook is to make the mathematical chain explicit:

$$\begin{aligned} \text{partial differential equation} &\longrightarrow \text{boundary conditions} \longrightarrow \text{separation constants} \\ &\longrightarrow \text{eigenfunctions} \longrightarrow \text{series coefficients} \\ &\longrightarrow \text{energy and tensor identities.} \end{aligned} \tag{1.1}$$

The derivations and their assumptions are given in Sections 4 to 6, 13, 17 and 18. The notation register provides the conventions used across these sections.

1.3 Analytical scope

The handbook is intentionally confined to mathematical structure: the defining differential equations, boundary operators, special-function systems, Hilbert-space identities, and exact coefficient projections.

Coefficient determination is treated only as an exact projection or as an abstract finite-dimensional variational problem. Statements about orthogonality, completeness, convergence, or energy are made in explicitly identified Hilbert spaces.

1.4 Provenance and status of the derivations

The boundary-value architecture and the four-family expansion follow the foundational R-SCHA literature [9, 8, 7]. The separate one-surface branch follows the classical R-SCHA2D construction [10]. The special-function, orthogonality, tensor, energy, flux, and projection sections are derived here directly from the stated differential equations, boundary operators, and normalisation conventions, with the cited literature used to identify the historical construction and established formulas.

The presentation distinguishes:

1. analytical identities under stated regularity, normalisation, and boundary assumptions;
2. mathematical reconstructions that derive, in a unified notation, formulas in the chosen notation;
3. modelling approximations and questions not resolved by the arguments presented here. The latter are not claims that the corresponding questions are open in the entire literature.

2 Geometric setting, boundary decomposition, and notation

Throughout this handbook, the physical domain of three-dimensional revised spherical cap harmonic analysis is a truncated spherical cone, or equivalently a spherical-cap shell. The construction follows the geometry underlying the R-SCHA boundary-value problem [9, 8, 7].

2.1 Spherical-cap coordinates

Let (r, θ, λ) denote spherical coordinates centred at the origin, where $r > 0$ is the radial distance, $\theta \in [0, \pi]$ is the co-latitude measured from the positive polar axis, and $\lambda \in \mathbb{R}/(2\pi\mathbb{Z})$ is longitude. Fix a cap half-angle $\theta_0 \in (0, \pi)$ and two radii

$$0 < r_{\mathrm{a}} < r_{\mathrm{b}}. \tag{2.1}$$

Definition 2.1 (Three-dimensional spherical-cap shell)**.** The open R-SCHA domain is

$$\Omega_{\theta_0}^{r_{\mathrm{a}}, r_{\mathrm{b}}} = \left\{ \boldsymbol{x} \in \mathbb{R}^3 : r_{\mathrm{a}} < |\boldsymbol{x}| < r_{\mathrm{b}}, \quad x_3 > |\boldsymbol{x}| \cos\theta_0 \right\}. \tag{2.2}$$

In spherical coordinates this is $r_{\mathrm{a}} < r < r_{\mathrm{b}}$ and $0 \le \theta < \theta_0$, with all longitudes at $\theta = 0$ identified as the same physical point. The Cartesian definition makes explicit that Ω is open and contains its polar axis. The axis is not a physical boundary of the domain. It is a coordinate singularity at which admissible scalar fields must satisfy the appropriate regularity and single-valuedness conditions. Similarly, $\lambda = 0$ and $\lambda = 2\pi$ represent the same meridian and are identified periodically.

2.2 Boundary decomposition

The boundary of $\Omega_{\theta_0}^{r_{\mathrm{a}}, r_{\mathrm{b}}}$ consists of the inner spherical cap, the outer spherical cap, and the lateral conical surface:

$$\Gamma_{\mathrm{a}} = \{(r, \theta, \lambda) : r = r_{\mathrm{a}},\ 0 \le \theta \le \theta_0,\ \lambda \in \mathbb{R}/(2\pi\mathbb{Z})\}, \tag{2.3a}$$

$$\Gamma_{\mathrm{b}} = \{(r, \theta, \lambda) : r = r_{\mathrm{b}},\ 0 \le \theta \le \theta_0,\ \lambda \in \mathbb{R}/(2\pi\mathbb{Z})\}, \tag{2.3b}$$

$$\Gamma_{\theta_0} = \{(r, \theta, \lambda) : r_{\mathrm{a}} \le r \le r_{\mathrm{b}},\ \theta = \theta_0,\ \lambda \in \mathbb{R}/(2\pi\mathbb{Z})\}. \tag{2.3c}$$

Thus,

$$\partial\Omega_{\theta_0}^{r_{\mathrm{a}}, r_{\mathrm{b}}} = \Gamma_{\mathrm{a}} \cup \Gamma_{\mathrm{b}} \cup \Gamma_{\theta_0}, \tag{2.4}$$

where intersections along the two circular edges have surface measure zero.

Let $\{\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_\lambda\}$ be the standard orthonormal spherical frame. The following normals are defined on the relative interiors of the faces, hence almost everywhere on the boundary; the circular edges do not have a single face-independent normal. The outward unit normal is

$$\boldsymbol{n} = \begin{cases} -\boldsymbol{e}_r, & \text{on } \Gamma_{\mathrm{a}}, \\ \boldsymbol{e}_r, & \text{on } \Gamma_{\mathrm{b}}, \\ \boldsymbol{e}_\theta, & \text{on } \Gamma_{\theta_0}. \end{cases} \tag{2.5}$$

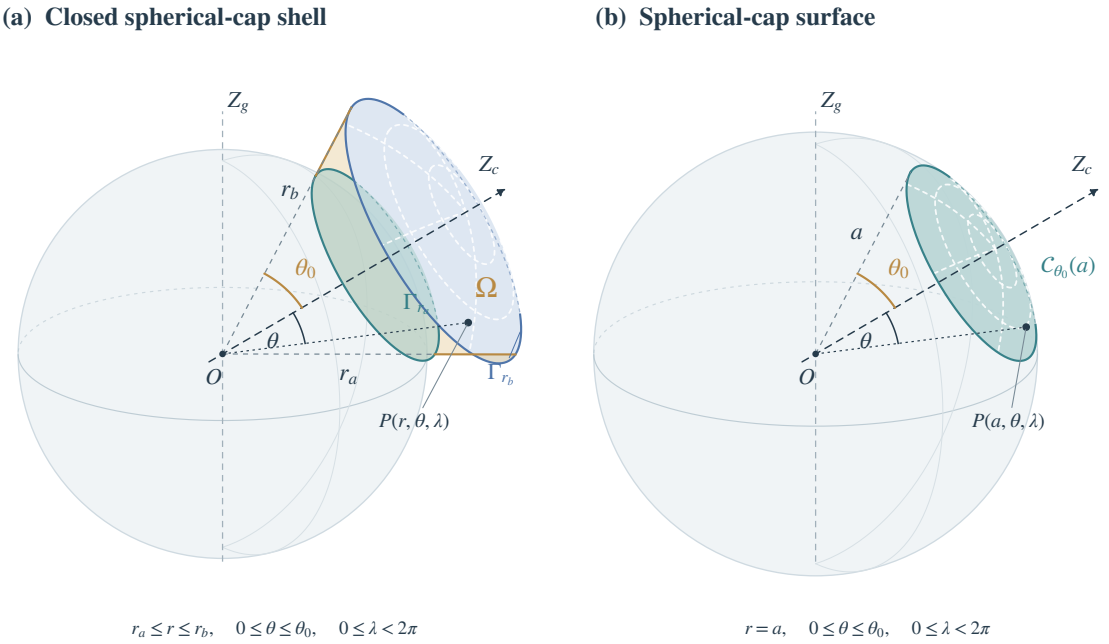


Figure 1: Three-dimensional (a) and two-dimensional (b) modelling domains of revised spherical cap harmonic analysis. The three-dimensional domain is bounded by two concentric spherical caps and a conical lateral surface.

Consequently, the outward normal derivative of a differentiable scalar field u is

$$\frac{\partial u}{\partial n} = \nabla u \cdot \boldsymbol{n} = \begin{cases} -\dfrac{\partial u}{\partial r}, & \text{on } \Gamma_{\mathrm{a}}, \\ \dfrac{\partial u}{\partial r}, & \text{on } \Gamma_{\mathrm{b}}, \\ \dfrac{1}{r}\dfrac{\partial u}{\partial \theta}, & \text{on } \Gamma_{\theta_0}. \end{cases} \tag{2.6}$$

2.3 Coordinate bases and metric

The position vector is

$$\boldsymbol{x}(r,\theta,\lambda) = r \begin{bmatrix} \sin\theta\cos\lambda \\ \sin\theta\sin\lambda \\ \cos\theta \end{bmatrix}. \tag{2.7}$$

The covariant coordinate basis is

$$\boldsymbol{g}_r = \frac{\partial \boldsymbol{x}}{\partial r} = \boldsymbol{e}_r, \qquad \boldsymbol{g}_\theta = \frac{\partial \boldsymbol{x}}{\partial \theta} = r\boldsymbol{e}_\theta, \qquad \boldsymbol{g}_\lambda = \frac{\partial \boldsymbol{x}}{\partial \lambda} = r\sin\theta\,\boldsymbol{e}_\lambda. \tag{2.8}$$

The corresponding scale factors are

$$h_r = 1, \qquad h_\theta = r, \qquad h_\lambda = r\sin\theta. \tag{2.9}$$

The metric tensor and its inverse are therefore

$$[g_{ij}] = \operatorname{diag}\left(1, r^2, r^2\sin^2\theta\right), \qquad [g^{ij}] = \operatorname{diag}\left(1, r^{-2}, r^{-2}\sin^{-2}\theta\right). \tag{2.10}$$

For a scalar field u, the physical gradient components in the orthonormal spherical frame are

$$\nabla u = \boldsymbol{e}_r \frac{\partial u}{\partial r} + \boldsymbol{e}_\theta \frac{1}{r} \frac{\partial u}{\partial \theta} + \boldsymbol{e}_\lambda \frac{1}{r \sin\theta} \frac{\partial u}{\partial \lambda}. \tag{2.11}$$

The spherical-coordinate Laplacian is

$$\Delta u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 u}{\partial \lambda^2}. \tag{2.12}$$

2.4 Volume and surface measures

The volume element in the cap shell is

$$\mathrm{d}\Omega = r^2 \sin\theta \ \mathrm{d}r \ \mathrm{d}\theta \ \mathrm{d}\lambda. \tag{2.13}$$

The induced surface elements on the three boundary components are

$$\mathrm{d}S_\mathrm{a} = r_\mathrm{a}^2 \sin\theta \ \mathrm{d}\theta \ \mathrm{d}\lambda, \tag{2.14a}$$

$$\mathrm{d}S_\mathrm{b} = r_\mathrm{b}^2 \sin\theta \ \mathrm{d}\theta \ \mathrm{d}\lambda, \tag{2.14b}$$

$$\mathrm{d}S_{\theta_0} = r \sin\theta_0 \ \mathrm{d}r \ \mathrm{d}\lambda. \tag{2.14c}$$

Hence, for an integrable function f,

$$\int_{\partial\Omega_{\theta_0}^{r_\mathrm{a},r_\mathrm{b}}} f \ \mathrm{d}S = \int_{\Gamma_\mathrm{a}} f \ \mathrm{d}S_\mathrm{a} + \int_{\Gamma_\mathrm{b}} f \ \mathrm{d}S_\mathrm{b} + \int_{\Gamma_{\theta_0}} f \ \mathrm{d}S_{\theta_0}. \tag{2.15}$$

2.5 Function spaces and regularity

Let $L^2(\Omega)$ and $H^k(\Omega)$ denote the usual Lebesgue and Sobolev spaces on $\Omega = \Omega_{\theta_0}^{r_\mathrm{a},r_\mathrm{b}}$. The Dirichlet bilinear form on the real space $H^1(\Omega)$ is

$$(u, v)_\mathcal{E} = \int_\Omega \nabla u \cdot \nabla v \ \mathrm{d}\Omega. \tag{2.16}$$

Because constants have zero Dirichlet energy, this expression is an inner product only after an appropriate boundary condition is imposed or after quotienting by the constant functions.

We denote the harmonic subspace by

$$\mathcal{H}(\Omega) = \left\{ u \in H^1(\Omega) : \Delta u = 0 \text{ in the weak sense} \right\}. \tag{2.17}$$

Weakly harmonic H^1 functions are smooth in the interior. Thus interior pointwise field components require no additional boundary regularity. Classical values up to all boundary faces and edges may instead require $u \in C^1(\overline{\Omega})$ or $u \in C^2(\overline{\Omega})$, respectively; such regularity is not asserted for arbitrary weak boundary data. At the polar axis, admissible solutions must be bounded, single-valued, and compatible with the azimuthal Fourier order.

2.6 Green identities

For $u, v \in H^2(\Omega)$, Green's first identity reads

$$\int_\Omega \nabla u \cdot \nabla v \ \mathrm{d}\Omega = \int_{\partial\Omega} u \frac{\partial v}{\partial n} \ \mathrm{d}S - \int_\Omega u \, \Delta v \ \mathrm{d}\Omega. \tag{2.18}$$

If both fields are harmonic, then

$$\int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}\Omega = \int_{\partial\Omega} u \frac{\partial v}{\partial n} \, \mathrm{d}S. \tag{2.19}$$

This identity will later provide the shortest proof of the gradient orthogonality of the two complementary R-SCHA potential subspaces.

3 Laplace equation and separation of variables

The mathematical starting point of spherical harmonic analysis, spherical cap harmonic analysis, and revised spherical cap harmonic analysis is the same: the scalar potential is harmonic in a source-free region. This section derives the separated solutions of the Laplace equation in spherical coordinates without imposing the R-SCHA boundary conditions. The boundary conditions introduced in Section 4 will subsequently select the admissible angular degrees and radial combinations.

3.1 Harmonic scalar potentials

Let

$$\Omega = \Omega_{\theta_0}^{r_\mathrm{a}, r_\mathrm{b}}$$

be the spherical-cap shell defined in Section 2. A scalar potential V is harmonic in Ω if

$$\Delta V = 0 \qquad \text{in } \Omega. \tag{3.1}$$

Using Equation (2.12), this equation becomes

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \lambda^2} = 0. \tag{3.2}$$

The derivation below is local in the radial coordinate and therefore applies to any spherical shell that does not contain the origin. Conditions at the polar axis and at the cap boundary will be imposed separately.

3.2 Separated ansatz

Assume first that V is a non-trivial separated solution,

$$V(r, \theta, \lambda) = R(r)\,\Theta(\theta)\,\Lambda(\lambda), \tag{3.3}$$

where none of the three factors is identically zero. Substituting Equation (3.3) into Equation (3.2) and dividing by $R\Theta\Lambda$ gives

$$\frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 \frac{\mathrm{d}R}{\mathrm{d}r}\right) + \frac{1}{\Theta \sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta \frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + \frac{1}{\Lambda \sin^2\theta}\frac{\mathrm{d}^2\Lambda}{\mathrm{d}\lambda^2} = 0. \tag{3.4}$$

The first term depends only on r, whereas the remaining terms depend only on (θ, λ). Therefore both sides must equal a separation constant. It is convenient to parameterise that constant as $\nu(\nu+1)$:

$$\frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 \frac{\mathrm{d}R}{\mathrm{d}r}\right) = \nu(\nu+1), \tag{3.5a}$$

$$\frac{1}{\Theta \sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta \frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + \frac{1}{\Lambda \sin^2\theta}\frac{\mathrm{d}^2\Lambda}{\mathrm{d}\lambda^2} = -\nu(\nu+1). \tag{3.5b}$$

At this stage, ν is a spectral parameter and is not assumed to be an integer. The use of $\nu(\nu+1)$ rather than an arbitrary constant anticipates the standard form of the associated Legendre equation.

Multiplying Equation (3.5b) by $\sin^2\theta$ yields

$$\frac{\sin\theta}{\Theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)+\nu(\nu+1)\sin^2\theta+\frac{1}{\Lambda}\frac{\mathrm{d}^2\Lambda}{\mathrm{d}\lambda^2}=0. \tag{3.6}$$

The first two terms depend only on θ, while the last depends only on λ. They must therefore equal a second separation constant. We write this constant as $-m^2$, obtaining

$$r^2\frac{\mathrm{d}^2R}{\mathrm{d}r^2}+2r\frac{\mathrm{d}R}{\mathrm{d}r}-\nu(\nu+1)R=0, \tag{3.7a}$$

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)+\left[\nu(\nu+1)-\frac{m^2}{\sin^2\theta}\right]\Theta=0, \tag{3.7b}$$

$$\frac{\mathrm{d}^2\Lambda}{\mathrm{d}\lambda^2}+m^2\Lambda=0. \tag{3.7c}$$

Thus, every separated harmonic mode is obtained by combining a solution of the radial Euler equation, a solution of the associated Legendre equation, and a periodic azimuthal mode.

3.3 Azimuthal equation and periodicity

For $m\neq 0$, the real-valued general solution of Equation (3.7c) is

$$\Lambda_m(\lambda)=A_m\cos(m\lambda)+B_m\sin(m\lambda). \tag{3.8}$$

For $m=0$, the differential equation has the algebraic solution

$$\Lambda_0(\lambda)=A_0+B_0\lambda. \tag{3.9}$$

Single-valuedness and 2π-periodicity require

$$V(r,\theta,\lambda+2\pi)=V(r,\theta,\lambda). \tag{3.10}$$

Consequently,

$$m\in\mathbb{Z}, \tag{3.11}$$

and the non-periodic term $B_0\lambda$ in Equation (3.9) must vanish. For a real expansion, negative orders carry no independent information because

$$\cos(-m\lambda)=\cos(m\lambda),\qquad \sin(-m\lambda)=-\sin(m\lambda).$$

It is therefore sufficient to use

$$m=0,1,2,\ldots, \tag{3.12}$$

with the convention that the sine coefficient is zero when $m=0$.

Equivalently, the complex azimuthal basis is

$$\Lambda_m(\lambda)=\mathrm{e}^{\mathrm{i}m\lambda},\qquad m\in\mathbb{Z}. \tag{3.13}$$

The real and complex forms span the same periodic function space.

3.4 Radial Euler equation

Seeking a power-law solution

$$R(r) = r^q \tag{3.14}$$

in Equation (3.7a) gives the characteristic equation

$$q(q-1) + 2q - \nu(\nu+1) = 0, \tag{3.15}$$

or

$$q^2 + q - \nu(\nu+1) = 0. \tag{3.16}$$

Its roots are

$$q_1 = \nu, \qquad q_2 = -(\nu+1). \tag{3.17}$$

For $\nu \neq -1/2$, the general radial solution is

$$R_\nu(r) = C_\nu^+ r^\nu + C_\nu^- r^{-(\nu+1)}. \tag{3.18}$$

At the repeated degree $\nu = -1/2$, the two powers coalesce and the independent solutions are $r^{-1/2}$ and $r^{-1/2}\ln(r/a)$, for any fixed reference radius $a > 0$. The repeated-root solution is developed explicitly in Equation (12.13).

Introducing a positive reference radius a, the distinct-root solution may be written in dimensionless form as

$$R_\nu(r) = A_\nu \left(\frac{a}{r}\right)^{\nu+1} + B_\nu \left(\frac{r}{a}\right)^{\nu}. \tag{3.19}$$

The two branches are commonly called the internal-type and external-type radial solutions, respectively:

$$R_\nu^{\mathrm{int}}(r) = \left(\frac{a}{r}\right)^{\nu+1}, \tag{3.20a}$$

$$R_\nu^{\mathrm{ext}}(r) = \left(\frac{r}{a}\right)^{\nu}. \tag{3.20b}$$

For complete-sphere integer-degree harmonics, these labels describe sources interior or exterior to the observation shell under the corresponding global continuation assumptions. For a non-integer cap degree they specify radial behaviour; the radial power alone does not locate physical sources. In a finite shell both branches are admissible until boundary conditions are imposed.

3.5 Associated Legendre equation

Set

$$x = \cos\theta, \qquad -1 \le x \le 1. \tag{3.21}$$

Using

$$\frac{\mathrm{d}}{\mathrm{d}\theta} = -\sin\theta \frac{\mathrm{d}}{\mathrm{d}x},$$

Equation (3.7b) becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[(1-x^2)\frac{\mathrm{d}\Theta}{\mathrm{d}x}\right] + \left[\nu(\nu+1) - \frac{m^2}{1-x^2}\right]\Theta = 0. \tag{3.22}$$

Equivalently,

$$(1-x^2)\frac{\mathrm{d}^2\Theta}{\mathrm{d}x^2} - 2x\frac{\mathrm{d}\Theta}{\mathrm{d}x} + \left[\nu(\nu+1) - \frac{m^2}{1-x^2}\right]\Theta = 0. \tag{3.23}$$

For non-degenerate degree ν and integer order m, a fundamental pair of solutions is formed by the associated Legendre functions of the first and second kinds:

$$\Theta_{\nu m}(\theta) = C_{\nu m} P_\nu^m(\cos\theta) + D_{\nu m} Q_\nu^m(\cos\theta). \tag{3.24}$$

The exact branch and phase conventions for P_ν^m and Q_ν^m will be fixed in Section 8. At this stage, it is important only that the cap problem does not in general require ν to be an integer. The admissible non-integer degrees are selected later by the boundary conditions at $\theta = \theta_0$.

3.6 Elementary separated harmonic modes

Combining the radial, angular, and azimuthal factors yields the elementary separated modes for $\nu \neq -1/2$,

$$\begin{aligned} V_{\nu m}(r,\theta,\lambda) = &\left[A_\nu \left(\frac{a}{r}\right)^{\nu+1} + B_\nu \left(\frac{r}{a}\right)^{\nu}\right] \\ &\times \left[C_{\nu m} P_\nu^m(\cos\theta) + D_{\nu m} Q_\nu^m(\cos\theta)\right] \\ &\times \left[E_m \cos(m\lambda) + F_m \sin(m\lambda)\right]. \end{aligned} \tag{3.25}$$

Complex degrees may require conjugate combinations to obtain a real potential. Finite sums are harmonic by linearity. An infinite sum is harmonic when, for example, it converges locally uniformly; pointwise convergence alone does not justify interchanging the sum and the Laplace operator.

Proposition 3.1 (Separated harmonic modes)**.** *Let R, Θ, and Λ solve Equation* (3.7) *for the same pair (ν, m). Then*

$$V(r,\theta,\lambda) = R(r)\Theta(\theta)\Lambda(\lambda)$$

satisfies $\Delta V = 0$ wherever the factors are twice differentiable.

Proof. Substituting the three ordinary differential equations into Equation (3.4) gives

$$\nu(\nu+1) - \nu(\nu+1) = 0.$$

Therefore Equation (3.2) is satisfied. □

3.7 The complete-sphere limit

The classical spherical harmonic expansion is recovered when the angular domain is the complete sphere and the solution is required to be bounded and regular at both poles. Longitude periodicity already requires $m \in \mathbb{Z}$. Regularity at $\theta = 0$ and $\theta = \pi$ turns the angular equation into a discrete Sturm–Liouville eigenproblem, for which

$$\nu = \ell, \qquad \ell = 0, 1, 2, \ldots, \qquad 0 \le m \le \ell. \tag{3.26}$$

The second-kind solution $Q_\ell^m(\cos\theta)$ is singular at one or both poles and is excluded. The regular angular basis therefore consists of $P_\ell^m(\cos\theta)$.

With an unnormalised associated Legendre convention, the general real-valued harmonic expansion in a spherical shell may be written as

$$\begin{aligned} V(r,\theta,\lambda) = &\, a \sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} \left(\frac{a}{r}\right)^{\ell+1} \left[g_{\ell m}^{\mathrm{int}} \cos(m\lambda) + h_{\ell m}^{\mathrm{int}} \sin(m\lambda)\right] P_\ell^m(\cos\theta) \\ &+ a \sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} \left(\frac{r}{a}\right)^{\ell} \left[g_{\ell m}^{\mathrm{ext}} \cos(m\lambda) + h_{\ell m}^{\mathrm{ext}} \sin(m\lambda)\right] P_\ell^m(\cos\theta). \end{aligned} \tag{3.27}$$

For a truncated representation, the two sums may have different maximum degrees:

$$V(r,\theta,\lambda) = a\sum_{\ell=0}^{L_{\mathrm{int}}}\sum_{m=0}^{\ell}\left(\frac{a}{r}\right)^{\ell+1}\left[g_{\ell m}^{\mathrm{int}}\cos(m\lambda)+h_{\ell m}^{\mathrm{int}}\sin(m\lambda)\right]P_\ell^m(\cos\theta) + a\sum_{\ell=0}^{L_{\mathrm{ext}}}\sum_{m=0}^{\ell}\left(\frac{r}{a}\right)^{\ell}\left[g_{\ell m}^{\mathrm{ext}}\cos(m\lambda)+h_{\ell m}^{\mathrm{ext}}\sin(m\lambda)\right]P_\ell^m(\cos\theta). \tag{3.28}$$

Here, $g_{\ell m}^{\mathrm{int}}$, $h_{\ell m}^{\mathrm{int}}$, $g_{\ell m}^{\mathrm{ext}}$, and $h_{\ell m}^{\mathrm{ext}}$ are real expansion coefficients. The normalisation adopted for P_ℓ^m must be stated together with the coefficients, since a change of normalisation rescales the coefficient set.

Remark 3.2 (The degree-zero magnetic term). The degree-zero internal radial term is proportional to $1/r$ and represents a monopole-type harmonic potential. It is mathematically admissible in a shell excluding the origin, but it is removed in a physically solenoidal magnetic model when zero net magnetic flux is imposed. The degree-zero external term is spatially constant and therefore contributes no magnetic field.

3.8 Transition from global harmonics to R-SCHA

The full-sphere conditions in Equation (3.26) are not appropriate for a regional cap. R-SCHA replaces regularity at the southern pole by boundary conditions on the lateral surface $\theta = \theta_0$ and on the two radial surfaces $r = r_{\mathrm{a}}$ and $r = r_{\mathrm{b}}$. As a result:

1. the azimuthal order m remains an integer because longitude is still periodic;
2. the angular degree is generally non-integer and is determined by a cap-boundary eigenvalue equation;
3. both radial power-law branches are retained and recombined to satisfy the radial boundary conditions;
4. additional Mehler and constant angular modes are required to obtain the revised, boundary-adapted representation.

These modifications do not change the separated differential equations derived above. They change the admissible spectra, the normalisations, and the combinations of separated solutions. The complete R-SCHA boundary-value formulation is developed next in Section 4.

4 Boundary-value formulation of R-SCHA

The separated differential equations derived in Section 3 determine the local form of harmonic functions, but they do not by themselves determine which angular degrees, radial combinations, or coefficient families are admissible. Revised spherical cap harmonic analysis is obtained by imposing a mixed boundary-value problem on the spherical-cap shell

$$\Omega = \Omega_{\theta_0}^{r_{\mathrm{a}},r_{\mathrm{b}}},$$

whose boundary decomposition was defined in Section 2. The essential idea is to replace the single regional expansion by two complementary harmonic subproblems whose boundary conditions are adapted to different parts of $\partial\Omega$.

4.1 The mixed Laplace boundary-value problem

Let

$$F = F(r, \lambda)$$

denote the prescribed potential on the lateral boundary Γ_{θ_0}, and let

$$G_{\mathrm{a}} = G_{\mathrm{a}}(\theta, \lambda), \qquad G_{\mathrm{b}} = G_{\mathrm{b}}(\theta, \lambda)$$

denote the prescribed outward normal derivatives on the inner and outer spherical boundaries, respectively. The complete R-SCHA boundary-value problem is

$$\Delta V = 0 \qquad \text{in } \Omega, \tag{4.1a}$$

$$V = F(r, \lambda) \qquad \text{on } \Gamma_{\theta_0}, \tag{4.1b}$$

$$\frac{\partial V}{\partial n} = G_{\mathrm{a}}(\theta, \lambda) \qquad \text{on } \Gamma_{\mathrm{a}}, \tag{4.1c}$$

$$\frac{\partial V}{\partial n} = G_{\mathrm{b}}(\theta, \lambda) \qquad \text{on } \Gamma_{\mathrm{b}}. \tag{4.1d}$$

Using the outward-normal convention in Equation (2.6), the radial forms of the Neumann conditions are

$$-\left.\frac{\partial V}{\partial r}\right|_{r=r_{\mathrm{a}}} = G_{\mathrm{a}}(\theta, \lambda), \tag{4.2a}$$

$$\left.\frac{\partial V}{\partial r}\right|_{r=r_{\mathrm{b}}} = G_{\mathrm{b}}(\theta, \lambda). \tag{4.2b}$$

The minus sign in Equation (4.2a) is geometric rather than conventional: the outward normal on the inner spherical boundary is $-\boldsymbol{e}_r$, whereas the outward normal on the outer spherical boundary is $+\boldsymbol{e}_r$.

4.2 Boundary data induced by a global spherical-harmonic potential

A useful theoretical construction is obtained when the boundary functions in Equation (4.1) are inherited from a known global spherical-harmonic potential. Let

$$\begin{aligned} V^{\mathrm{SH}}(r, \theta, \lambda) = a \sum_{\ell=0}^{L_{\mathrm{int}}} \sum_{m=0}^{\ell} \left(\frac{a}{r}\right)^{\ell+1} C_{\ell m}^{\mathrm{int}}(\lambda) P_\ell^m(\cos\theta) \\ + a \sum_{\ell=0}^{L_{\mathrm{ext}}} \sum_{m=0}^{\ell} \left(\frac{r}{a}\right)^{\ell} C_{\ell m}^{\mathrm{ext}}(\lambda) P_\ell^m(\cos\theta), \end{aligned} \tag{4.3}$$

where

$$C_{\ell m}^{\mathrm{int}}(\lambda) = g_{\ell m}^{\mathrm{int}} \cos(m\lambda) + h_{\ell m}^{\mathrm{int}} \sin(m\lambda), \tag{4.4a}$$

$$C_{\ell m}^{\mathrm{ext}}(\lambda) = g_{\ell m}^{\mathrm{ext}} \cos(m\lambda) + h_{\ell m}^{\mathrm{ext}} \sin(m\lambda). \tag{4.4b}$$

Here, $a > 0$ is the reference radius used in the spherical-harmonic representation; it is distinct from the inner boundary radius r_{a}.

The lateral Dirichlet data are the trace of V^{SH} at $\theta = \theta_0$:

$$\begin{aligned} F(r,\lambda) &= V^{\mathrm{SH}}(r,\theta_0,\lambda) \\ &= a\sum_{\ell=0}^{L_{\mathrm{int}}}\sum_{m=0}^{\ell}\left(\frac{a}{r}\right)^{\ell+1} C_{\ell m}^{\mathrm{int}}(\lambda)P_\ell^m(\cos\theta_0) \\ &\quad + a\sum_{\ell=0}^{L_{\mathrm{ext}}}\sum_{m=0}^{\ell}\left(\frac{r}{a}\right)^{\ell} C_{\ell m}^{\mathrm{ext}}(\lambda)P_\ell^m(\cos\theta_0). \end{aligned} \tag{4.5}$$

Differentiating Equation (4.3) with respect to r gives

$$\begin{aligned} \frac{\partial V^{\mathrm{SH}}}{\partial r} &= -\sum_{\ell=0}^{L_{\mathrm{int}}}\sum_{m=0}^{\ell}(\ell+1)\left(\frac{a}{r}\right)^{\ell+2} C_{\ell m}^{\mathrm{int}}(\lambda)P_\ell^m(\cos\theta) \\ &\quad + \sum_{\ell=0}^{L_{\mathrm{ext}}}\sum_{m=0}^{\ell}\ell\left(\frac{r}{a}\right)^{\ell-1} C_{\ell m}^{\mathrm{ext}}(\lambda)P_\ell^m(\cos\theta). \end{aligned} \tag{4.6}$$

Therefore, the inner-boundary outward normal data are

$$\begin{aligned} G_{\mathrm{a}}(\theta,\lambda) &= -\left.\frac{\partial V^{\mathrm{SH}}}{\partial r}\right|_{r=r_{\mathrm{a}}} \\ &= \sum_{\ell=0}^{L_{\mathrm{int}}}\sum_{m=0}^{\ell}(\ell+1)\left(\frac{a}{r_{\mathrm{a}}}\right)^{\ell+2} C_{\ell m}^{\mathrm{int}}(\lambda)P_\ell^m(\cos\theta) \\ &\quad - \sum_{\ell=0}^{L_{\mathrm{ext}}}\sum_{m=0}^{\ell}\ell\left(\frac{r_{\mathrm{a}}}{a}\right)^{\ell-1} C_{\ell m}^{\mathrm{ext}}(\lambda)P_\ell^m(\cos\theta), \end{aligned} \tag{4.7}$$

whereas the outer-boundary outward normal data are

$$\begin{aligned} G_{\mathrm{b}}(\theta,\lambda) &= \left.\frac{\partial V^{\mathrm{SH}}}{\partial r}\right|_{r=r_{\mathrm{b}}} \\ &= -\sum_{\ell=0}^{L_{\mathrm{int}}}\sum_{m=0}^{\ell}(\ell+1)\left(\frac{a}{r_{\mathrm{b}}}\right)^{\ell+2} C_{\ell m}^{\mathrm{int}}(\lambda)P_\ell^m(\cos\theta) \\ &\quad + \sum_{\ell=0}^{L_{\mathrm{ext}}}\sum_{m=0}^{\ell}\ell\left(\frac{r_{\mathrm{b}}}{a}\right)^{\ell-1} C_{\ell m}^{\mathrm{ext}}(\lambda)P_\ell^m(\cos\theta). \end{aligned} \tag{4.8}$$

Equations (4.5), (4.7) and (4.8) should be understood as a boundary-data construction rather than as a restriction on R-SCHA itself. In an inverse problem, the three boundary functions may instead be inferred from regional observations or prescribed by another physically admissible potential-field model.

4.3 Decomposition into two complementary harmonic subproblems

The boundary conditions in Equation (4.1) are non-homogeneous on all three boundary components. R-SCHA separates them by writing

$$V = V_1 + V_2, \tag{4.9}$$

where V_1 carries the normal-derivative data on the two spherical boundaries, while V_2 carries the potential data on the lateral boundary.

The first subproblem is

$$\Delta V_1 = 0 \qquad \text{in } \Omega, \tag{4.10a}$$
$$V_1 = 0 \qquad \text{on } \Gamma_{\theta_0}, \tag{4.10b}$$
$$\frac{\partial V_1}{\partial n} = G_{\mathrm{a}}(\theta, \lambda) \qquad \text{on } \Gamma_{\mathrm{a}}, \tag{4.10c}$$
$$\frac{\partial V_1}{\partial n} = G_{\mathrm{b}}(\theta, \lambda) \qquad \text{on } \Gamma_{\mathrm{b}}. \tag{4.10d}$$

Equivalently,

$$-\left.\frac{\partial V_1}{\partial r}\right|_{r=r_{\mathrm{a}}} = G_{\mathrm{a}}(\theta, \lambda), \tag{4.11a}$$
$$\left.\frac{\partial V_1}{\partial r}\right|_{r=r_{\mathrm{b}}} = G_{\mathrm{b}}(\theta, \lambda). \tag{4.11b}$$

The second subproblem is

$$\Delta V_2 = 0 \qquad \text{in } \Omega, \tag{4.12a}$$
$$V_2 = F(r, \lambda) \qquad \text{on } \Gamma_{\theta_0}, \tag{4.12b}$$
$$\frac{\partial V_2}{\partial n} = 0 \qquad \text{on } \Gamma_{\mathrm{a}}, \tag{4.12c}$$
$$\frac{\partial V_2}{\partial n} = 0 \qquad \text{on } \Gamma_{\mathrm{b}}. \tag{4.12d}$$

Equivalently,

$$\left.\frac{\partial V_2}{\partial r}\right|_{r=r_{\mathrm{a}}} = \left.\frac{\partial V_2}{\partial r}\right|_{r=r_{\mathrm{b}}} = 0. \tag{4.13}$$

Proposition 4.1 (Superposition of the complementary subproblems)**.** *If V_1 and V_2 satisfy Equations* (4.10) *and* (4.12)*, then*

$$V = V_1 + V_2$$

satisfies the complete mixed boundary-value problem Equation (4.1)*.*

Proof. Linearity of the Laplace operator gives

$$\Delta V = \Delta V_1 + \Delta V_2 = 0.$$

On the lateral boundary,

$$V = V_1 + V_2 = 0 + F = F.$$

On the inner and outer spherical boundaries,

$$\frac{\partial V}{\partial n} = \frac{\partial V_1}{\partial n} + \frac{\partial V_2}{\partial n} = G_{\mathrm{a}} + 0$$

and

$$\frac{\partial V}{\partial n} = G_{\mathrm{b}} + 0,$$

respectively. □

4.4 Mathematical roles of the two subspaces

The lateral homogeneous Dirichlet condition Equation (4.10b) requires the angular factor of each separated V_1 mode to vanish at the cap boundary. For associated Legendre modes this leads to the eigenvalue condition

$$P_\nu^m(\cos\theta_0) = 0. \tag{4.14}$$

For each fixed integer order m, the admissible degrees form a discrete sequence of generally non-integer roots, denoted later by

$$\nu = \nu_{km}, \qquad k = m, m+1, m+2, \dots.$$

The resulting V_1 expansion consists of internal- and external-type Legendre modes.

By contrast, V_2 has homogeneous Neumann conditions at $r = r_\mathrm{a}$ and $r = r_\mathrm{b}$ but non-homogeneous Dirichlet data at $\theta = \theta_0$. Its separated radial functions must therefore be adapted to the two radial boundaries. The corresponding construction gives rise to the Mehler and constant-angular families developed in Sections 9 to 11.

Accordingly, the complete R-SCHA potential has the structural decomposition

$$V = \underbrace{V_\mathrm{Int} + V_\mathrm{Ext}}_{V_1} + \underbrace{V_\mathrm{Meh} + V_\mathrm{Cons}}_{V_2}, \tag{4.15}$$

where the explicit series representations will be derived in Section 7.

4.5 Weak formulation

Let

$$\Gamma_D = \Gamma_{\theta_0}, \qquad \Gamma_N = \Gamma_\mathrm{a} \cup \Gamma_\mathrm{b}. \tag{4.16}$$

For the piecewise smooth shell, the trace space on the relatively open Dirichlet part is understood as the quotient space

$$H^{1/2}(\Gamma_D) := \mathrm{Tr}_{\Gamma_D} H^1(\Omega), \qquad \|f\|_{H^{1/2}(\Gamma_D)} := \inf_{\mathrm{Tr}_{\Gamma_D} u = f} \|u\|_{H^1(\Omega)}. \tag{4.17}$$

The natural Neumann datum is a continuous functional on the traces of test functions on Γ_N. Equivalently, it belongs to the dual of the Lions–Magenes trace space

$$H_{00}^{1/2}(\Gamma_N) := \overline{C_c^\infty(\Gamma_N)}^{H^{1/2}(\partial\Omega)}. \tag{4.18}$$

This formulation avoids treating the three boundary faces as if they were closed smooth manifolds independent of their circular edges.

The mixed boundary-value problem can be stated naturally in Sobolev spaces. Define

$$\mathcal{V}_0 = \left\{ w \in H^1(\Omega) : \mathrm{Tr}\, w = 0 \text{ on } \Gamma_{\theta_0} \right\}. \tag{4.19}$$

The weak form of the V_1 problem is: find $V_1 \in \mathcal{V}_0$ such that

$$\int_\Omega \nabla V_1 \cdot \nabla w \, \mathrm{d}\Omega = \langle G_N, \mathrm{Tr}_{\Gamma_N} w \rangle \qquad \text{for every } w \in \mathcal{V}_0. \tag{4.20}$$

When G_a and G_b are L^2 densities, the duality pairing on the right is the sum of the two surface integrals over Γ_a and Γ_b.

For the V_2 problem, let $\widetilde{F} \in H^1(\Omega)$ be any lifting whose trace on Γ_{θ_0} equals F. Write

$$V_2 = \widetilde{F} + u_2, \qquad u_2 \in \mathcal{V}_0. \tag{4.21}$$

Then u_2 satisfies

$$\int_\Omega \nabla u_2 \cdot \nabla w \, \mathrm{d}\Omega = -\int_\Omega \nabla \widetilde{F} \cdot \nabla w \, \mathrm{d}\Omega \qquad \text{for every } w \in \mathcal{V}_0. \tag{4.22}$$

The zero-Neumann conditions on $\Gamma_\mathrm{a} \cup \Gamma_\mathrm{b}$ are natural boundary conditions in this variational formulation.

4.6 Uniqueness

Theorem 4.2 (Uniqueness of the mixed R-SCHA boundary-value problem)**.** *Assume that $u \in H^1(\Omega)$ is weakly harmonic, has zero trace on Γ_{θ_0}, and satisfies*

$$\frac{\partial u}{\partial n} = 0 \quad \text{on } \Gamma_\mathrm{a} \cup \Gamma_\mathrm{b}.$$

Then $u = 0$ in Ω.

Proof. Testing the homogeneous weak equation with $w = u$ gives zero Dirichlet energy. For smooth fields the same calculation is Green's first identity,

$$\int_\Omega |\nabla u|^2 \, \mathrm{d}\Omega = \int_{\partial\Omega} u \frac{\partial u}{\partial n} \, \mathrm{d}S - \int_\Omega u \Delta u \, \mathrm{d}\Omega. \tag{4.23}$$

The volume term vanishes because u is harmonic. The boundary term vanishes on Γ_{θ_0} because $u = 0$ there, and it vanishes on $\Gamma_\mathrm{a} \cup \Gamma_\mathrm{b}$ because the normal derivative is zero. Hence

$$\int_\Omega |\nabla u|^2 \, \mathrm{d}\Omega = 0,$$

so u is constant almost everywhere. Its zero trace on the non-empty lateral boundary forces that constant to be zero. □

Corollary 4.3. *Each of the two complementary subproblems Equations* (4.10) *and* (4.12)*, and hence the complete problem Equation* (4.1)*, has at most one weak solution for prescribed boundary data.*

Because a non-empty Dirichlet boundary is present, the mixed problem does not have the additive-constant indeterminacy of a pure Neumann problem. Likewise, no global zero-flux compatibility condition is required merely for solvability of the mixed problem. Magnetic zero-flux constraints arise instead from the physical solenoidal condition and will be analysed in Section 16.

4.7 Existence and regularity

Assume

$$F \in H^{1/2}(\Gamma_D), \qquad G_N \in \left(H_{00}^{1/2}(\Gamma_N)\right)', \tag{4.24}$$

where G_N restricts to G_a and G_b on the two spherical faces. Choose any lifting of F in $H^1(\Omega)$. Poincaré's inequality on $\mathcal{V}_0$ makes the Dirichlet form coercive, while the Neumann functional is bounded by the trace theorem. The Lax–Milgram theorem therefore gives a unique weak solution of

each variational problem Equations (4.20) and (4.22). Classical solutions require stronger smoothness of the boundary data and compatibility at the circular edges

$$\Gamma_{\mathrm{a}} \cap \Gamma_{\theta_0}, \qquad \Gamma_{\mathrm{b}} \cap \Gamma_{\theta_0}.$$

Possible loss of regularity at these edges is a property of the mixed boundary geometry and should be distinguished from truncation-induced oscillations in finite R-SCHA expansions.

4.8 Preview of the gradient-orthogonal structure

The complementary boundary conditions also imply

$$\int_{\Omega} \nabla V_1 \cdot \nabla V_2 \, \mathrm{d}\Omega = 0. \tag{4.25}$$

Indeed, Green's identity converts the volume integral into

$$\int_{\partial\Omega} V_1 \frac{\partial V_2}{\partial n} \, \mathrm{d}S.$$

On the lateral boundary, $V_1 = 0$; on the two spherical boundaries, $\partial V_2/\partial n = 0$. A complete discussion of the resulting orthogonal decomposition in the Dirichlet-energy inner product is given in Section 13.

Part II
Three-Dimensional Eigenfunction Construction

5 Derivation of the first subpotential V_1

The first subpotential is the solution of the mixed problem Equation (4.10). Its lateral trace is homogeneous, while its normal derivatives on the two spherical boundaries carry the prescribed radial boundary data. This section derives its complete separated series and the exact coefficient equations generated by those two boundaries.

5.1 Separated equations

Let

$$V_1(r, \theta, \lambda) = R(r)\Theta(\theta)\Lambda(\lambda). \tag{5.1}$$

Substitution into the spherical Laplace equation gives

$$\Lambda'' + m^2 \Lambda = 0, \tag{5.2a}$$

$$\frac{1}{\sin\theta} \frac{\mathrm{d}}{\mathrm{d}\theta} \left(\sin\theta \Theta' \right) + \left[\nu(\nu+1) - \frac{m^2}{\sin^2\theta} \right] \Theta = 0, \tag{5.2b}$$

$$r^2 R'' + 2rR' - \nu(\nu+1)R = 0. \tag{5.2c}$$

The periodic longitude functions are

$$L_m^{\mathrm{c}}(\lambda) = \cos(m\lambda), \qquad L_m^{\mathrm{s}}(\lambda) = \sin(m\lambda), \tag{5.3}$$

with $m \in \mathbb{N}_0$ and no sine mode at $m = 0$.

5.2 Lateral Dirichlet spectrum

Regularity at the cap pole selects

$$\Theta(\theta) = P_\nu^m(\cos\theta). \tag{5.4}$$

The homogeneous lateral boundary condition

$$V_1(r, \theta_0, \lambda) = 0 \tag{5.5}$$

requires

$$P_\nu^m(\cos\theta_0) = 0. \tag{5.6}$$

For every fixed m, denote the ordered real roots on the selected branch by

$$\nu_{mm} < \nu_{m+1,m} < \nu_{m+2,m} < \cdots, \tag{5.7}$$

using the same index convention $k = m, m+1, \ldots$ as the complete expansion and the notation register. The associated eigenvalue is

$$\lambda_{km} = \nu_{km}(\nu_{km} + 1) > 0. \tag{5.8}$$

The fixed-m eigenfunctions are orthogonal in $L^2((0, \theta_0), \sin\theta\, \mathrm{d}\theta)$; the detailed spectral theory is given in Section 8.

5.3 Internal and external radial branches

The Euler equation Equation (5.2c) has the two independent solutions

$$U_{km}^{\mathrm{Int}}(r) = \left(\frac{r_\mathrm{a}}{r}\right)^{\nu_{km}+1}, \qquad U_{km}^{\mathrm{Ext}}(r) = \left(\frac{r}{r_\mathrm{a}}\right)^{\nu_{km}}. \tag{5.9}$$

The names "internal" and "external" refer to their radial continuation: U^{Int} decays outward and is singular only at the excluded origin, whereas U^{Ext} grows outward and tends to zero as $r \downarrow 0$ for $\nu_{km} > 0$. This radial limit alone does not establish a smooth harmonic extension through the origin for non-integer degree. Within the finite conical shell both are admissible radial factors.

Their derivatives are

$$\frac{\mathrm{d}U_{km}^{\mathrm{Int}}}{\mathrm{d}r} = -\frac{\nu_{km}+1}{r} U_{km}^{\mathrm{Int}}, \tag{5.10a}$$

$$\frac{\mathrm{d}U_{km}^{\mathrm{Ext}}}{\mathrm{d}r} = \frac{\nu_{km}}{r} U_{km}^{\mathrm{Ext}}. \tag{5.10b}$$

5.4 General separated expansion of V_1

For each longitude parity $\sigma \in \{\mathrm{c}, \mathrm{s}\}$, write

$$V_1 = r_\mathrm{a} \sum_{m,\sigma} \sum_k \left[a_{km}^{\mathrm{Int},\sigma} U_{km}^{\mathrm{Int}}(r) + a_{km}^{\mathrm{Ext},\sigma} U_{km}^{\mathrm{Ext}}(r) \right] P_{\nu_{km}}^m(\cos\theta) L_m^\sigma(\lambda). \tag{5.11}$$

Every term is harmonic and vanishes on the lateral boundary. The remaining coefficients are fixed by the two radial normal-derivative traces.

5.5 Projection of the radial boundary data

The following surface-integral formulae are stated for $G_{\mathrm{a}}, G_{\mathrm{b}} \in L^2$ on their respective caps. For the more general weak data of Equation (4.24), the projections must instead be written as trace-space duality pairings.

Let

$$\mathcal{N}_{km}^{P} = \int_0^{\theta_0} \left[P_{\nu_{km}}^{m}(\cos\theta)\right]^2 \sin\theta \, \mathrm{d}\theta \tag{5.12}$$

and

$$\ell_m^{\mathrm{c}} = \begin{cases} 2\pi, & m = 0, \\ \pi, & m \geq 1, \end{cases} \qquad \ell_m^{\mathrm{s}} = \pi \quad (m \geq 1). \tag{5.13}$$

For $\Gamma \in \{\mathrm{a}, \mathrm{b}\}$, define the boundary-data projection

$$b_{km}^{\Gamma,\sigma} = \frac{1}{\mathcal{N}_{km}^{P} \ell_m^{\sigma}} \int_0^{2\pi} \int_0^{\theta_0} G_{\Gamma}(\theta, \lambda) P_{\nu_{km}}^{m}(\cos\theta) L_m^{\sigma}(\lambda) \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda. \tag{5.14}$$

These are the exact Fourier–Legendre coefficients of the two normal traces in the boundary eigensystem.

5.6 The lower-boundary coefficient equation

At $r = r_{\mathrm{a}}$,

$$U_{km}^{\mathrm{Int}}(r_{\mathrm{a}}) = U_{km}^{\mathrm{Ext}}(r_{\mathrm{a}}) = 1. \tag{5.15}$$

The outward normal is $-\boldsymbol{e}_r$, and hence

$$\frac{\partial V_1}{\partial n} = -\frac{\partial V_1}{\partial r} \quad \text{on } \Gamma_{\mathrm{a}}. \tag{5.16}$$

Projection onto one angular-longitude mode gives

$$(\nu_{km} + 1) a_{km}^{\mathrm{Int},\sigma} - \nu_{km} a_{km}^{\mathrm{Ext},\sigma} = b_{km}^{\mathrm{a},\sigma}. \tag{5.17}$$

5.7 The upper-boundary coefficient equation

Introduce

$$\alpha_{km} = \left(\frac{r_{\mathrm{a}}}{r_{\mathrm{b}}}\right)^{\nu_{km}+2}, \qquad \beta_{km} = \left(\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}\right)^{\nu_{km}-1}. \tag{5.18}$$

At the outer boundary the outward normal is $\boldsymbol{e}_r$. Using Equation (5.10), projection gives

$$-(\nu_{km} + 1)\alpha_{km} a_{km}^{\mathrm{Int},\sigma} + \nu_{km}\beta_{km} a_{km}^{\mathrm{Ext},\sigma} = b_{km}^{\mathrm{b},\sigma}. \tag{5.19}$$

5.8 Solution of the two-boundary system

The coefficient pair satisfies

$$\begin{bmatrix} \nu_{km} + 1 & -\nu_{km} \\ -(\nu_{km} + 1)\alpha_{km} & \nu_{km}\beta_{km} \end{bmatrix} \begin{bmatrix} a_{km}^{\mathrm{Int},\sigma} \\ a_{km}^{\mathrm{Ext},\sigma} \end{bmatrix} = \begin{bmatrix} b_{km}^{\mathrm{a},\sigma} \\ b_{km}^{\mathrm{b},\sigma} \end{bmatrix}. \tag{5.20}$$

Its determinant is

$$\nu_{km}(\nu_{km} + 1)\,(\beta_{km} - \alpha_{km})\,. \tag{5.21}$$

For $r_{\mathrm{b}} > r_{\mathrm{a}}$ and the positive Dirichlet spectrum, this is non-zero. Therefore,

$$a_{km}^{\mathrm{Int},\sigma} = \frac{\beta_{km} b_{km}^{\mathrm{a},\sigma} + b_{km}^{\mathrm{b},\sigma}}{(\nu_{km}+1)(\beta_{km}-\alpha_{km})}, \tag{5.22a}$$

$$a_{km}^{\mathrm{Ext},\sigma} = \frac{\alpha_{km} b_{km}^{\mathrm{a},\sigma} + b_{km}^{\mathrm{b},\sigma}}{\nu_{km}(\beta_{km}-\alpha_{km})}. \tag{5.22b}$$

Thus the two radial branches are not arbitrary duplicate functions. They are the unique pair required to satisfy two independent radial normal traces for each lateral Dirichlet eigenmode.

5.9 Final series for V_1

In the conventional real notation,

$$\begin{aligned} V_1 = {} & r_{\mathrm{a}} \sum_{k,m} \left(\frac{r_{\mathrm{a}}}{r}\right)^{\nu_{km}+1} \left[g_{km}^{\mathrm{Int}} \cos(m\lambda) + h_{km}^{\mathrm{Int}} \sin(m\lambda)\right] P_{\nu_{km}}^{m}(\cos\theta) \\ & + r_{\mathrm{a}} \sum_{k,m} \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}} \left[g_{km}^{\mathrm{Ext}} \cos(m\lambda) + h_{km}^{\mathrm{Ext}} \sin(m\lambda)\right] P_{\nu_{km}}^{m}(\cos\theta). \end{aligned} \tag{5.23}$$

The sums may be infinite in the exact function-space representation or finite in a chosen approximation subspace. The coefficient formulae Equations (5.14) and (5.22) make the boundary origin of every term explicit.

6 Derivation of the second subpotential V_2 from the partial differential equation

The second subpotential satisfies homogeneous Neumann conditions on the two spherical boundaries and carries the non-homogeneous potential trace on the lateral conical boundary. The Mehler and constant families arise from a single radial Sturm–Liouville problem. Positive radial eigenvalues generate complex Legendre degrees and hence Mehler functions; the zero radial eigenvalue generates the constant radial and angular branch.

This section derives the complete series directly from Equation (4.12).

6.1 Boundary-value problem

The equations are

$$\Delta V_2 = 0 \qquad \text{in } \Omega, \tag{6.1a}$$

$$V_2(r,\theta_0,\lambda) = F(r,\lambda) \qquad \text{on } \Gamma_{\theta_0}, \tag{6.1b}$$

$$\left.\frac{\partial V_2}{\partial r}\right|_{r=r_{\mathrm{a}}} = 0, \tag{6.1c}$$

$$\left.\frac{\partial V_2}{\partial r}\right|_{r=r_{\mathrm{b}}} = 0. \tag{6.1d}$$

No homogeneous condition is imposed on the angular function at $\theta = \theta_0$. The lateral value is instead reproduced by the series coefficients.

6.2 Separation of variables

Let

$$V_2(r,\theta,\lambda) = R(r)A(\theta)L(\lambda). \tag{6.2}$$

Substitution into

$$\Delta V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 V}{\partial\lambda^2} \tag{6.3}$$

gives

$$\frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2R'\right) + \frac{1}{A\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta A'\right) + \frac{1}{L\sin^2\theta}L'' = 0. \tag{6.4}$$

Periodicity in longitude gives

$$L'' + m^2L = 0, \qquad m \in \mathbb{N}_0, \tag{6.5}$$

with

$$L_m^{\mathrm{c}}(\lambda) = \cos(m\lambda), \qquad L_m^{\mathrm{s}}(\lambda) = \sin(m\lambda). \tag{6.6}$$

The remaining radial expression depends only on r, whereas the angular expression depends only on θ. Introduce a real separation constant μ by

$$-\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2R'\right) = \mu R, \tag{6.7a}$$

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta A'\right) - \frac{m^2}{\sin^2\theta}A - \mu A = 0. \tag{6.7b}$$

The radial boundary conditions are

$$R'(r_{\mathrm{a}}) = R'(r_{\mathrm{b}}) = 0. \tag{6.8}$$

6.3 Non-negativity of the radial spectrum

Multiply Equation (6.7a) by R and integrate:

$$\begin{aligned}\mu\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R^2\,\mathrm{d}r &= -\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R(r^2R')'\,\mathrm{d}r \\ &= -\left[r^2RR'\right]_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} + \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} r^2(R')^2\,\mathrm{d}r \\ &= \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} r^2(R')^2\,\mathrm{d}r \geq 0.\end{aligned} \tag{6.9}$$

Every radial eigenvalue is therefore non-negative. Negative values of μ are excluded before the characteristic equation is solved.

6.4 Logarithmic radial reduction

Define

$$t = \ln\frac{r}{r_{\mathrm{a}}}, \qquad s = \ln\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}, \qquad 0 \leq t \leq s. \tag{6.10}$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}r} = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}t}, \tag{6.11}$$

the radial equation becomes

$$\frac{\mathrm{d}^2R}{\mathrm{d}t^2} + \frac{\mathrm{d}R}{\mathrm{d}t} + \mu R = 0. \tag{6.12}$$

The Neumann conditions are equivalent to

$$R_t(0) = R_t(s) = 0. \tag{6.13}$$

With

$$R(t) = \mathrm{e}^{-t/2}Y(t), \tag{6.14}$$

one obtains

$$Y'' + \left(\mu - \frac{1}{4}\right) Y = 0, \tag{6.15}$$

and

$$Y'(0) - \frac{1}{2}Y(0) = 0, \qquad Y'(s) - \frac{1}{2}Y(s) = 0. \tag{6.16}$$

This elementary constant-coefficient problem contains the entire radial spectrum.

6.5 Zero radial eigenvalue

When $\mu = 0$, Equation (6.12) has

$$R(t) = A + B\mathrm{e}^{-t}, \tag{6.17}$$

or equivalently

$$R(r) = A + B\frac{r_\mathrm{a}}{r}. \tag{6.18}$$

The condition $R_t(0) = 0$ gives $B = 0$, and the second condition is then automatic. Thus

$$R_0(r) = \text{constant}. \tag{6.19}$$

With the normalisation

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_0^2 \, \mathrm{d}r = r_\mathrm{a}, \tag{6.20}$$

one may take

$$R_0 = \left(\frac{r_\mathrm{a}}{r_\mathrm{b} - r_\mathrm{a}}\right)^{1/2}. \tag{6.21}$$

6.6 The interval $0 < \mu < 1/4$

Let

$$\eta = \sqrt{\frac{1}{4} - \mu}, \qquad 0 < \eta < \frac{1}{2}. \tag{6.22}$$

Then

$$R(t) = A\mathrm{e}^{(-1/2+\eta)t} + B\mathrm{e}^{(-1/2-\eta)t}. \tag{6.23}$$

The two Neumann equations have determinant

$$\begin{aligned} \mathcal{D}_\eta &= \left(-\frac{1}{2} + \eta\right)\left(-\frac{1}{2} - \eta\right)\left[\mathrm{e}^{(-1/2-\eta)s} - \mathrm{e}^{(-1/2+\eta)s}\right] \\ &= \mu\mathrm{e}^{-s/2}\left(\mathrm{e}^{-\eta s} - \mathrm{e}^{\eta s}\right). \end{aligned} \tag{6.24}$$

For $\mu > 0$, $\eta > 0$, and $s > 0$, this determinant is non-zero. Hence no non-trivial radial eigenfunction exists in this interval.

6.7 The repeated value $\mu = 1/4$

When $\mu = 1/4$,

$$R(t) = \mathrm{e}^{-t/2}(A + Bt). \tag{6.25}$$

The first Neumann condition gives

$$B = \frac{A}{2}. \tag{6.26}$$

The second gives

$$-\frac{As}{4}\mathrm{e}^{-s/2} = 0. \tag{6.27}$$

Since $s > 0$ for a non-degenerate shell, $A = B = 0$. Thus $\mu = 1/4$ is not an eigenvalue of the three-dimensional radial problem. The same repeated Euler root is used under the distinct one-surface conditions of R-SCHA2D; it is not a limit eigenmode obtained by setting $s = 0$ in this closed-shell problem. See Section 12.

6.8 Positive oscillatory spectrum

For $\mu > 1/4$, define

$$\tau = \sqrt{\mu - \frac{1}{4}} > 0. \tag{6.28}$$

Then

$$R(t) = \mathrm{e}^{-t/2}\left[A\cos(\tau t) + B\sin(\tau t)\right]. \tag{6.29}$$

Differentiation gives

$$R_t(t) = \mathrm{e}^{-t/2}\left[\left(-\frac{A}{2} + B\tau\right)\cos(\tau t) + \left(-A\tau - \frac{B}{2}\right)\sin(\tau t)\right]. \tag{6.30}$$

The condition at $t = 0$ yields

$$A = 2\tau B. \tag{6.31}$$

Substituting this into $R_t(s) = 0$ gives

$$\left(\tau^2 + \frac{1}{4}\right)\sin(\tau s) = 0. \tag{6.32}$$

Since $\tau > 0$,

$$\tau_p = \frac{p\pi}{s}, \qquad p = 1, 2, \ldots, \tag{6.33}$$

and

$$\mu_p = \frac{1}{4} + \tau_p^2 = \frac{1}{4} + \left(\frac{p\pi}{s}\right)^2. \tag{6.34}$$

An unnormalised eigenfunction is

$$\mathcal{R}_p(r) = \left(\frac{r_\mathrm{a}}{r}\right)^{1/2}\left[2p\pi\cos\left(\frac{p\pi}{s}\ln\frac{r}{r_\mathrm{a}}\right) + s\sin\left(\frac{p\pi}{s}\ln\frac{r}{r_\mathrm{a}}\right)\right]. \tag{6.35}$$

With

$$\mathcal{A}_p = \left\{\frac{2}{s[(2p\pi)^2 + s^2]}\right\}^{1/2}, \tag{6.36}$$

define

$$R_p(r) = \mathcal{A}_p\mathcal{R}_p(r). \tag{6.37}$$

Then

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p R_q \,\mathrm{d}r = r_\mathrm{a}\delta_{pq}. \tag{6.38}$$

6.9 From the radial eigenvalue to the Mehler degree

The standard associated Legendre equation is

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta A'\right)+\left[\nu(\nu+1)-\frac{m^2}{\sin^2\theta}\right]A=0. \tag{6.39}$$

Comparison with Equation (6.7b) gives

$$\nu(\nu+1)=-\mu. \tag{6.40}$$

For the positive radial eigenvalue Equation (6.34),

$$\nu_p=-\frac{1}{2}\pm\mathrm{i}\tau_p. \tag{6.41}$$

The two values are related by the degree symmetry $P^m_{-\nu-1}=P^m_\nu$ on the adopted branch, so one may use

$$\nu_p=-\frac{1}{2}+\mathrm{i}\tau_p. \tag{6.42}$$

The regular angular solution is therefore

$$K^m_p(\theta)=P^m_{-1/2+\mathrm{i}\tau_p}(\cos\theta). \tag{6.43}$$

It satisfies

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta(K^m_p)'\right)-\left[\mu_p+\frac{m^2}{\sin^2\theta}\right]K^m_p=0. \tag{6.44}$$

Thus the complex degree is not introduced independently: it is the algebraic image of the positive radial Neumann spectrum under Equation (6.40).

6.10 Angular equation of the zero radial mode

For $\mu_0=0$, the angular equation becomes

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta C'_m\right)-\frac{m^2}{\sin^2\theta}C_m=0. \tag{6.45}$$

Introduce

$$u=\ln\tan\frac{\theta}{2}. \tag{6.46}$$

Since

$$\frac{\mathrm{d}u}{\mathrm{d}\theta}=\frac{1}{\sin\theta}, \tag{6.47}$$

one has

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}C_m}{\mathrm{d}\theta}\right)=\frac{1}{\sin^2\theta}\frac{\mathrm{d}^2C_m}{\mathrm{d}u^2}. \tag{6.48}$$

Equation Equation (6.45) therefore reduces to

$$\frac{\mathrm{d}^2C_m}{\mathrm{d}u^2}-m^2C_m=0. \tag{6.49}$$

For $m\geq 1$,

$$C_m=A_m\mathrm{e}^{mu}+B_m\mathrm{e}^{-mu}=A_m\tan^m\frac{\theta}{2}+B_m\cot^m\frac{\theta}{2}. \tag{6.50}$$

Regularity at $\theta = 0$ removes the cotangent branch, leaving

$$C_m(\theta) = A_m \tan^m \frac{\theta}{2}, \qquad m \geq 1. \tag{6.51}$$

For $m = 0$,

$$C_0(\theta) = A_0 + B_0 u. \tag{6.52}$$

Regularity at the pole removes the logarithmic branch, so

$$C_0(\theta) = A_0. \tag{6.53}$$

After normalisation, these are the constant angular functions developed in Section 10.

6.11 Separated modes of V_2

The positive-eigenvalue modes are

$$\Phi_{pm}^{\mathrm{Meh},\sigma} = r_{\mathrm{a}} R_p(r) K_p^m(\theta) L_m^\sigma(\lambda), \qquad p \geq 1. \tag{6.54}$$

The zero-eigenvalue modes are

$$\Phi_{0m}^{\mathrm{Cons},\sigma} = r_{\mathrm{a}} R_0 C_m(\theta) L_m^\sigma(\lambda). \tag{6.55}$$

Every mode is harmonic because the radial and angular separation constants cancel:

$$\frac{1}{R}(r^2 R')' + \frac{1}{A}\left[\frac{1}{\sin\theta}(\sin\theta A')' - \frac{m^2}{\sin^2\theta} A\right] = -\mu + \mu = 0. \tag{6.56}$$

Lemma 6.1 (Non-vanishing lateral trace of a positive radial mode)**.** *For every admissible pair (p, m) with $p \geq 1$,*

$$K_p^m(\theta_0) \neq 0. \tag{6.57}$$

Proof. If $K_p^m(\theta_0) = 0$, the separated mode $R_p(r) K_p^m(\theta) L_m^\sigma(\lambda)$ would be harmonic, would have zero Dirichlet trace on Γ_{θ_0}, and would have zero normal derivative on both spherical boundaries. Uniqueness of the homogeneous mixed problem would force the mode to vanish identically, contradicting the non-triviality of the radial and angular eigenfunctions. □

The constant angular trace $C_m(\theta_0)$ is also non-zero by its explicit construction. Hence every denominator in the boundary projection below is well defined.

6.12 Fourier decomposition of the lateral trace

For $\sigma \in \{\mathrm{c}, \mathrm{s}\}$, define

$$\ell_m^{\mathrm{c}} = \begin{cases} 2\pi, & m = 0, \\ \pi, & m \geq 1, \end{cases} \qquad \ell_m^{\mathrm{s}} = \pi \quad (m \geq 1), \tag{6.58}$$

and the Fourier coefficient of the boundary trace by

$$F_m^\sigma(r) = \frac{1}{\ell_m^\sigma} \int_0^{2\pi} F(r, \lambda) L_m^\sigma(\lambda)\, \mathrm{d}\lambda. \tag{6.59}$$

Then

$$F(r, \lambda) = \sum_{m,\sigma} F_m^\sigma(r) L_m^\sigma(\lambda) \tag{6.60}$$

in the corresponding Fourier sense.

The projection formulae below assume $F \in L^2$ on the lateral face. They give an L^2 boundary expansion; convergence to an $H^1(\Omega)$ harmonic solution requires the trace and energy conditions stated below.

6.13 Radial expansion of each boundary Fourier coefficient

The radial Sturm–Liouville system

$$\{R_0, R_1, R_2, \ldots\} \tag{6.61}$$

is complete in $L^2([r_\mathrm{a}, r_\mathrm{b}], \mathrm{d}r)$. Therefore, for every fixed (m, σ),

$$F_m^\sigma(r) = r_\mathrm{a} \left[d_{0m}^\sigma R_0 C_m(\theta_0) + \sum_{p=1}^{\infty} d_{pm}^\sigma R_p(r) K_p^m(\theta_0) \right]. \tag{6.62}$$

Using

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p R_q \, \mathrm{d}r = r_\mathrm{a} \delta_{pq}, \tag{6.63}$$

the coefficients are

$$d_{pm}^\sigma = \frac{1}{r_\mathrm{a}^2 K_p^m(\theta_0)} \int_{r_\mathrm{a}}^{r_\mathrm{b}} F_m^\sigma(r) R_p(r) \, \mathrm{d}r, \qquad p \geq 1, \tag{6.64a}$$

$$d_{0m}^\sigma = \frac{1}{r_\mathrm{a}^2 C_m(\theta_0)} \int_{r_\mathrm{a}}^{r_\mathrm{b}} F_m^\sigma(r) R_0 \, \mathrm{d}r. \tag{6.64b}$$

Equivalently, define

$$A_{qm}(\theta) = \begin{cases} C_m(\theta), & q = 0, \\ K_q^m(\theta), & q \geq 1. \end{cases} \tag{6.65}$$

Then, before taking the longitude projection, all coefficients satisfy

$$d_{qm}^\sigma = \frac{1}{r_\mathrm{a}^2 A_{qm}(\theta_0) \ell_m^\sigma} \int_{r_\mathrm{a}}^{r_\mathrm{b}} \int_0^{2\pi} F(r, \lambda) R_q(r) L_m^\sigma(\lambda) \, \mathrm{d}\lambda \, \mathrm{d}r, \qquad q = 0, 1, 2, \ldots. \tag{6.66}$$

6.14 Complete series solution

Substitution of the projected coefficients gives

$$\begin{aligned} V_2(r, \theta, \lambda) = {} & r_\mathrm{a} \sum_{m,\sigma} \sum_{p=1}^{\infty} d_{pm}^\sigma R_p(r) K_p^m(\theta) L_m^\sigma(\lambda) \\ & + r_\mathrm{a} \sum_{m,\sigma} d_{0m}^\sigma R_0 C_m(\theta) L_m^\sigma(\lambda). \end{aligned} \tag{6.67}$$

In the conventional cosine–sine notation,

$$\begin{aligned} V_2 = {} & r_\mathrm{a} \sum_{p=1}^{\infty} \sum_{m=0}^{\infty} R_p(r) \left[G_p^m \cos(m\lambda) + H_p^m \sin(m\lambda) \right] K_p^m(\theta) \\ & + r_\mathrm{a} R_0 \sum_{m=0}^{\infty} \left[G_0^m \cos(m\lambda) + H_0^m \sin(m\lambda) \right] C_m(\theta). \end{aligned} \tag{6.68}$$

The sine coefficients at $m = 0$ are zero.

6.15 Why the constant branch must be separated

The zero mode cannot be obtained by substituting $p = 0$ into the normalised positive-spectrum formula. The positive radial functions contain the factor

$$\mathcal{A}_p \left[2p\pi \cos\left(\frac{p\pi t}{s}\right) + s \sin\left(\frac{p\pi t}{s}\right) \right], \tag{6.69}$$

whose $p \to 0$ limit does not reproduce the independently normalised constant function without a separate limiting analysis. More fundamentally, the angular eigenvalue changes from $-\mu_p < 0$ to zero and the regular angular solution changes from a conical function to $\tan^m(\theta/2)$. The constant family is therefore an independent zero-eigenvalue branch of the self-adjoint radial operator.

6.16 Passage from boundary expansion to a weak harmonic solution

The completeness of $\{R_q\}_{q\geq 0}$ in radial L^2 and Fourier completeness in longitude determine the coefficients of the lateral trace. Those boundary statements alone do not automatically justify termwise interior differentiation of the infinite series. Let $V_{2,N}$ denote any increasing sequence of finite partial sums. If

$$\sum_j \int_\Omega |\nabla(c_j \phi_j)|^2 \, \mathrm{d}\Omega < \infty \tag{6.70}$$

and the lateral traces of $V_{2,N}$ converge to F in $H^{1/2}(\Gamma_{\theta_0})$, then $V_{2,N}$ is Cauchy in the Dirichlet-energy norm. Its limit is the unique weak solution of the mixed problem Equation (4.12). Thus the series is an H^1 solution under a finite-energy and trace-convergence hypothesis; mere radial L^2 convergence of each Fourier coefficient is not, by itself, sufficient.

6.17 Verification of the boundary conditions

For every positive mode,

$$R_p'(r_\mathrm{a}) = R_p'(r_\mathrm{b}) = 0, \tag{6.71}$$

and for the zero mode,

$$R_0' = 0. \tag{6.72}$$

Hence the complete series satisfies the homogeneous radial Neumann conditions term by term. At $\theta = \theta_0$, the series reduces to Equation (6.62), which equals the prescribed lateral trace by construction. Thus the coefficient projections are not post hoc fitting formulae: they are the Fourier–Sturm–Liouville solution of the mixed boundary-value problem.

6.18 Uniqueness of the V_2 solution

Suppose that W satisfies the homogeneous version of the V_2 problem:

$$\Delta W = 0, \qquad W|_{\Gamma_{\theta_0}} = 0, \qquad \partial_r W|_{r_\mathrm{a}, r_\mathrm{b}} = 0. \tag{6.73}$$

Green's first identity gives

$$\int_\Omega |\nabla W|^2 \, \mathrm{d}\Omega = \int_{\partial\Omega} W \frac{\partial W}{\partial n} \, \mathrm{d}S = 0. \tag{6.74}$$

Therefore W is constant; the zero lateral trace forces that constant to be zero. The mixed problem and its series solution are unique.

7 Complete revised spherical-cap harmonic expansion

The boundary-value decomposition established in Section 4 leads to four families of harmonic basis functions. The first subpotential,

$$V_1 = V^{\mathrm{Int}} + V^{\mathrm{Ext}},$$

is represented by internal- and external-type associated Legendre modes that vanish on the lateral boundary. The second subpotential,

$$V_2 = V^{\mathrm{Meh}} + V^{\mathrm{Cons}},$$

is represented by Mehler modes and constant angular modes whose radial functions satisfy homogeneous Neumann conditions at the inner and outer spherical boundaries. Consequently,

$$V = V_1 + V_2 = V^{\mathrm{Int}} + V^{\mathrm{Ext}} + V^{\mathrm{Meh}} + V^{\mathrm{Cons}}. \tag{7.1}$$

This section fixes the notation and gives the complete truncated expansion. The detailed construction, normalisation, and orthogonality of the individual angular and radial functions are developed in Sections 8 to 11.

7.1 Non-integer Legendre spectrum

For each integer azimuthal order $m \geq 0$, let

$$\nu_{km}, \qquad k = m, m+1, m+2, \ldots, \tag{7.2}$$

denote the positive roots of the lateral Dirichlet eigenvalue equation

$$P^m_{\nu_{km}}(\cos\theta_0) = 0. \tag{7.3}$$

For fixed m, the roots are ordered as

$$0 < \nu_{mm} < \nu_{m+1,m} < \nu_{m+2,m} < \cdots . \tag{7.4}$$

The first index k is therefore a degree-like triangular index satisfying $0 \leq m \leq k$, whereas the second index records the azimuthal order on which the non-integer root depends. This explicit notation avoids suppressing the m-dependence of the roots.

Define the real longitudinal combinations

$$C^{\mathrm{Int}}_{km}(\lambda) = g^{\mathrm{Int}}_{km}\cos(m\lambda) + h^{\mathrm{Int}}_{km}\sin(m\lambda), \tag{7.5a}$$

$$C^{\mathrm{Ext}}_{km}(\lambda) = g^{\mathrm{Ext}}_{km}\cos(m\lambda) + h^{\mathrm{Ext}}_{km}\sin(m\lambda). \tag{7.5b}$$

For $m = 0$, the sine terms vanish and we set

$$h^{\mathrm{Int}}_{k0} = h^{\mathrm{Ext}}_{k0} = 0. \tag{7.6}$$

7.2 Internal Legendre contribution

The internal-type Legendre contribution is

$$\begin{aligned} V^{\mathrm{Int}}(r,\theta,\lambda) &= r_{\mathrm{a}} \sum_{k=0}^{N_{\mathrm{Int}}} \sum_{m=0}^{k} \left(\frac{r_{\mathrm{a}}}{r}\right)^{\nu_{km}+1} C^{\mathrm{Int}}_{km}(\lambda) P^m_{\nu_{km}}(\cos\theta) \\ &= r_{\mathrm{a}} \sum_{k=0}^{N_{\mathrm{Int}}} \sum_{m=0}^{k} \left(\frac{r_{\mathrm{a}}}{r}\right)^{\nu_{km}+1} \left[g^{\mathrm{Int}}_{km}\cos(m\lambda) + h^{\mathrm{Int}}_{km}\sin(m\lambda)\right] P^m_{\nu_{km}}(\cos\theta). \end{aligned} \tag{7.7}$$

Here, N_{Int} is the maximum triangular index retained in the internal Legendre sequence. Every term satisfies the Laplace equation, decays with increasing radius, and vanishes at $\theta = \theta_0$ by Equation (7.3).

7.3 External Legendre contribution

The external-type Legendre contribution is

$$V^{\mathrm{Ext}}(r,\theta,\lambda) = r_{\mathrm{a}} \sum_{k=0}^{N_{\mathrm{Ext}}} \sum_{m=0}^{k} \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}} C_{km}^{\mathrm{Ext}}(\lambda) P_{\nu_{km}}^{m}(\cos\theta)$$
$$= r_{\mathrm{a}} \sum_{k=0}^{N_{\mathrm{Ext}}} \sum_{m=0}^{k} \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}} \left[g_{km}^{\mathrm{Ext}} \cos(m\lambda) + h_{km}^{\mathrm{Ext}} \sin(m\lambda)\right] P_{\nu_{km}}^{m}(\cos\theta). \tag{7.8}$$

The truncation index N_{Ext} need not equal N_{Int}. Each mode grows algebraically with radius inside the finite modelling shell and satisfies the same homogeneous lateral Dirichlet condition as the internal contribution.

Combining Equations (7.7) and (7.8) gives

$$V_1 = V^{\mathrm{Int}} + V^{\mathrm{Ext}}. \tag{7.9}$$

7.4 Logarithmic radial coordinate

The radial functions associated with V_2 are most naturally expressed in the logarithmic coordinate

$$t = \ln\left(\frac{r}{r_{\mathrm{a}}}\right), \qquad 0 \le t \le s, \qquad s = \ln\left(\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}\right). \tag{7.10}$$

For every positive integer p, define

$$\tau_p = \frac{p\pi}{s}. \tag{7.11}$$

The corresponding complex angular degree is

$$\nu_p^{\mathrm{Meh}} = -\frac{1}{2} + \mathrm{i}\tau_p. \tag{7.12}$$

Because

$$\nu_p^{\mathrm{Meh}}\left(\nu_p^{\mathrm{Meh}} + 1\right) = -\left(\frac{1}{4} + \tau_p^2\right),$$

the angular equation is coupled to an oscillatory logarithmic radial equation.

7.5 Boundary-adapted Mehler radial functions

For $p = 1, 2, \ldots$, define

$$R_p(r) = \left[\frac{2}{s\left[(2p\pi)^2 + s^2\right]}\right]^{1/2} \left(\frac{r_{\mathrm{a}}}{r}\right)^{1/2}$$
$$\times \left[2p\pi \cos\left(\frac{p\pi}{s}\ln\frac{r}{r_{\mathrm{a}}}\right) + s\sin\left(\frac{p\pi}{s}\ln\frac{r}{r_{\mathrm{a}}}\right)\right]. \tag{7.13}$$

These functions satisfy

$$\left.\frac{\mathrm{d}R_p}{\mathrm{d}r}\right|_{r=r_{\mathrm{a}}} = \left.\frac{\mathrm{d}R_p}{\mathrm{d}r}\right|_{r=r_{\mathrm{b}}} = 0, \tag{7.14}$$

and the normalisation

$$\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R_p^2(r)\ \mathrm{d}r = r_{\mathrm{a}}. \tag{7.15}$$

The derivation of Equations (7.13) and (7.15) is presented in Section 11.

Let

$$K_p^m(\theta) = P^m_{-\frac{1}{2}+\mathrm{i}\tau_p}(\cos\theta) \tag{7.16}$$

denote the Mehler, or conical, angular function under the branch and normalisation convention specified later in Section 9. Define

$$C^{\mathrm{Meh}}_{pm}(\lambda) = G_p^m \cos(m\lambda) + H_p^m \sin(m\lambda), \tag{7.17}$$

with

$$H_p^0 = 0. \tag{7.18}$$

7.6 Mehler contribution

The separated partial differential equation gives the series

$$V^{\mathrm{Meh}}(r,\theta,\lambda) = r_{\mathrm{a}} \sum_{p=1}^{\infty} \sum_{m=0}^{\infty} R_p(r) \left[G_p^m \cos(m\lambda) + H_p^m \sin(m\lambda) \right] K_p^m(\theta). \tag{7.19}$$

For finite approximation, choose independent bounds $P_{\max} \geq 1$ and $M_{\mathrm{Meh}} \geq 0$. The truncated Mehler contribution is

$$V^{\mathrm{Meh}}_{P,M}(r,\theta,\lambda) = r_{\mathrm{a}} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} R_p(r) \left[G_p^m \cos(m\lambda) + H_p^m \sin(m\lambda) \right] K_p^m(\theta). \tag{7.20}$$

7.7 Constant radial and angular modes

The exceptional real roots of the radial characteristic equation are 0 and -1. Enforcing homogeneous Neumann conditions at both radial boundaries removes the r^{-1} branch and leaves a constant radial function. With the same radial norm used in Equation (7.15), define

$$R_0 = \left(\frac{r_{\mathrm{a}}}{r_{\mathrm{b}} - r_{\mathrm{a}}} \right)^{1/2}. \tag{7.21}$$

Then

$$\frac{\mathrm{d}R_0}{\mathrm{d}r} = 0, \qquad \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R_0^2 \, \mathrm{d}r = r_{\mathrm{a}}. \tag{7.22}$$

Let

$$\mathcal{C}_m(\theta) \equiv P_0^m(\theta) \tag{7.23}$$

denote the constant angular function in the conventional R-SCHA notation. For $m > 0$, this symbol should not be confused with the classical associated Legendre polynomial obtained by differentiating the degree-zero Legendre polynomial. Its precise definition, normalisation, and limiting interpretation are given in Section 10.

Define

$$C^{\mathrm{Cons}}_{0m}(\lambda) = G_0^m \cos(m\lambda) + H_0^m \sin(m\lambda), \tag{7.24}$$

where

$$H_0^0 = 0. \tag{7.25}$$

7.8 Constant-function contribution

The constant-function contribution is

$$V^{\mathrm{Cons}}(r,\theta,\lambda) = r_{\mathrm{a}} \sum_{m=0}^{M_{\max}} R_0 C_{0m}^{\mathrm{Cons}}(\lambda)\mathcal{C}_m(\theta)$$
$$= r_{\mathrm{a}} \sum_{m=0}^{M_{\max}} R_0 \left[G_0^m \cos(m\lambda) + H_0^m \sin(m\lambda)\right] \mathcal{C}_m(\theta). \tag{7.26}$$

The integer $M_{\max}$ is the maximum azimuthal order retained in the constant-function sequence.

Combining Equations (7.20) and (7.26) gives

$$V_2 = V^{\mathrm{Meh}} + V^{\mathrm{Cons}}. \tag{7.27}$$

7.9 Complete truncated R-SCHA potential

Substitution of Equations (7.7), (7.8), (7.20) and (7.26) into Equation (7.1) gives the complete real-valued truncated R-SCHA representation:

$$\begin{aligned} V(r,\theta,\lambda) = {} & r_{\mathrm{a}} \sum_{k=0}^{N_{\mathrm{Int}}} \sum_{m=0}^{k} \left(\frac{r_{\mathrm{a}}}{r}\right)^{\nu_{km}+1} \left[g_{km}^{\mathrm{Int}} \cos(m\lambda) + h_{km}^{\mathrm{Int}} \sin(m\lambda)\right] P_{\nu_{km}}^m(\cos\theta) \\ & + r_{\mathrm{a}} \sum_{k=0}^{N_{\mathrm{Ext}}} \sum_{m=0}^{k} \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}} \left[g_{km}^{\mathrm{Ext}} \cos(m\lambda) + h_{km}^{\mathrm{Ext}} \sin(m\lambda)\right] P_{\nu_{km}}^m(\cos\theta) \\ & + r_{\mathrm{a}} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} R_p(r) \left[G_p^m \cos(m\lambda) + H_p^m \sin(m\lambda)\right] K_p^m(\theta) \\ & + r_{\mathrm{a}} \sum_{m=0}^{M_{\max}} R_0 \left[G_0^m \cos(m\lambda) + H_0^m \sin(m\lambda)\right] \mathcal{C}_m(\theta). \end{aligned} \tag{7.28}$$

The four lines of Equation (7.28) are, respectively, the internal Legendre, external Legendre, Mehler, and constant angular sequences.

7.10 Complex-valued coefficient form

For analytical purposes, the longitude dependence may alternatively be written with complex exponentials. Introduce complex coefficients $\widehat{g}_{km}^{\mathrm{Int}}$, $\widehat{g}_{km}^{\mathrm{Ext}}$, $\widehat{G}_{pm}$, and $\widehat{G}_{0m}$ satisfying the conjugate-symmetry conditions required for a real potential. Then

$$V^{\mathrm{Int}} = r_{\mathrm{a}} \sum_{k=0}^{N_{\mathrm{Int}}} \sum_{m=-k}^{k} \widehat{g}_{km}^{\mathrm{Int}} \left(\frac{r_{\mathrm{a}}}{r}\right)^{\nu_{k|m|}+1} P_{\nu_{k|m|}}^{|m|}(\cos\theta) \mathrm{e}^{\mathrm{i}m\lambda}, \tag{7.29}$$

$$V^{\mathrm{Ext}} = r_{\mathrm{a}} \sum_{k=0}^{N_{\mathrm{Ext}}} \sum_{m=-k}^{k} \widehat{g}_{km}^{\mathrm{Ext}} \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{k|m|}} P_{\nu_{k|m|}}^{|m|}(\cos\theta) \mathrm{e}^{\mathrm{i}m\lambda}, \tag{7.30}$$

$$V^{\mathrm{Meh}} = r_{\mathrm{a}} \sum_{p=1}^{P_{\max}} \sum_{m=-M_{\mathrm{Meh}}}^{M_{\mathrm{Meh}}} R_p(r) \widehat{G}_{pm} K_p^{|m|}(\theta) \mathrm{e}^{\mathrm{i}m\lambda}, \tag{7.31}$$

$$V^{\mathrm{Cons}} = r_{\mathrm{a}} \sum_{m=-M_{\max}}^{M_{\max}} R_0 \widehat{G}_{0m} \mathcal{C}_{|m|}(\theta) \mathrm{e}^{\mathrm{i}m\lambda}. \tag{7.32}$$

For $m > 0$, the corresponding complex coefficients are $(g^m - \mathrm{i}h^m)/2$ at order $+m$ and $(g^m + \mathrm{i}h^m)/2$ at order $-m$; the order-zero complex coefficient equals the real cosine coefficient. The same conversion applies to each family. The real and complex formulations are algebraically equivalent; the complex form is often more compact in proofs.

7.11 Coefficient counts

The numbers of real cosine–sine coefficients in the four truncated sequences are

$$d_{\mathrm{Int}} = (N_{\mathrm{Int}} + 1)^2, \tag{7.33a}$$

$$d_{\mathrm{Ext}} = (N_{\mathrm{Ext}} + 1)^2, \tag{7.33b}$$

$$d_{\mathrm{Meh}} = P_{\max}(2M_{\mathrm{Meh}} + 1), \tag{7.33c}$$

$$d_{\mathrm{Cons}} = 2M_{\max} + 1. \tag{7.33d}$$

Indeed, for a fixed triangular index k, the orders $m = 0, \ldots, k$ contribute one coefficient for $m = 0$ and two coefficients for each $m \geq 1$, giving $2k + 1$ coefficients. Thus,

$$\sum_{k=0}^{N} (2k + 1) = (N + 1)^2.$$

For each of the $P_{\max}$ positive radial indices, the Mehler family contains one order-zero cosine coefficient and two coefficients for each order $m = 1, \ldots, M_{\mathrm{Meh}}$, hence $P_{\max}(2M_{\mathrm{Meh}} + 1)$ coefficients. The following count includes the additive-potential constant in the constant family. Fixing that gauge removes one degree of freedom when the model is interpreted as a magnetic field. The total number of real potential coefficients is

$$d_{\text{R-SCHA}} = (N_{\mathrm{Int}} + 1)^2 + (N_{\mathrm{Ext}} + 1)^2 + P_{\max}(2M_{\mathrm{Meh}} + 1) + 2M_{\max} + 1. \tag{7.34}$$

7.12 Normalisation dependence

The representation Equation (7.28) is unchanged if a basis function is multiplied by a non-zero constant and its coefficient is divided by the same constant. Hence, the coefficient values have meaning only after the normalisation conventions for

$$P^m_{\nu_{km}}, \qquad K^m_p, \qquad \mathcal{C}_m, \qquad R_p, \qquad R_0$$

have been fixed. The radial normalisations have already been stated in Equations (7.15) and (7.22); the angular normalisations are derived in the following sections.

7.13 Boundary properties of the complete expansion

The four basis families are adapted to the complementary boundary conditions as follows:

$$V^{\mathrm{Int}} + V^{\mathrm{Ext}} = 0 \qquad \text{on } \Gamma_{\theta_0}, \tag{7.35}$$

$$\frac{\partial}{\partial n}\left(V^{\mathrm{Meh}} + V^{\mathrm{Cons}}\right) = 0 \qquad \text{on } \Gamma_{\mathrm{a}} \cup \Gamma_{\mathrm{b}}. \tag{7.36}$$

Therefore, the Legendre coefficient families are determined by the normal derivative data on the spherical boundaries, whereas the Mehler and constant coefficient families supply the remaining lateral-boundary trace. This boundary-adapted division is the structural feature that distinguishes the revised formulation from a direct restriction of a global spherical-harmonic series.

8 Associated Legendre functions of non-integer degree

The internal and external components of R-SCHA are constructed from associated Legendre functions whose degrees are generally non-integer. For each fixed integer azimuthal order m, the admissible degrees are selected by the homogeneous Dirichlet condition on the lateral boundary of the spherical cap. This section develops the required function definitions, derivative identities, normalisation conventions, orthogonality relations, norm formulae, and completeness theorem.

8.1 Associated Legendre equation

Let

$$x = \cos\theta, \qquad x_0 = \cos\theta_0, \qquad x \in [x_0, 1]. \tag{8.1}$$

For a separated angular factor

$$f(x) = \Theta(\theta),$$

the angular Equation (3.7b) becomes

$$(1-x^2)\frac{\mathrm{d}^2 f}{\mathrm{d}x^2} - 2x\frac{\mathrm{d}f}{\mathrm{d}x} + \left[\nu(\nu+1) - \frac{m^2}{1-x^2}\right] f = 0. \tag{8.2}$$

Here,

$$m = 0, 1, 2, \ldots,$$

whereas the degree ν is not assumed to be an integer.

For non-degenerate degree and order parameters, a fundamental pair of solutions of Equation (8.2) is denoted by

$$P_\nu^m(x), \qquad Q_\nu^m(x),$$

the associated Legendre functions of the first and second kinds. On a cap containing the polar axis, the R-SCHA Legendre modes use the first-kind solution because it is the regular branch at $x = 1$; the second-kind solution contains the singular local branch and is excluded from this part of the basis.

8.2 Hypergeometric representations

For integer order $m \geq 0$, the first-kind associated Legendre function is represented by

$$\begin{aligned} P_\nu^m(\cos\theta) = {} & \frac{(-1)^m \Gamma(\nu+m+1)}{2^m m!\,\Gamma(\nu-m+1)} \sin^m\theta \\ & \times {}_2F_1\left(m-\nu,\, \nu+m+1;\, m+1;\, \sin^2\frac{\theta}{2}\right). \end{aligned} \tag{8.3}$$

An equivalent representation is

$$\begin{aligned} P_\nu^m(\cos\theta) = {} & \frac{(-1)^m \Gamma(\nu+m+1)}{m!\,\Gamma(\nu-m+1)} \tan^m\frac{\theta}{2} \\ & \times {}_2F_1\left(-\nu,\, \nu+1;\, m+1;\, \sin^2\frac{\theta}{2}\right). \end{aligned} \tag{8.4}$$

These formulae use the Condon–Shortley phase $(-1)^m$ and the Ferrers branch on $-1 < x < 1$ [5].

The Gaussian hypergeometric function is

$$ {}_2F_1(a,b;c;z)=\sum_{j=0}^{\infty}\frac{(a)_j(b)_j}{(c)_j\,j!}z^j,\qquad |z|<1, \tag{8.5}$$

where

$$ (a)_j=\prod_{q=0}^{j-1}(a+q)=\frac{\Gamma(a+j)}{\Gamma(a)} \tag{8.6}$$

is the Pochhammer symbol, with $(a)_0=1$. The finite product is the definition for every a; the Gamma quotient is used where it is defined and otherwise by continuation. This distinction includes the terminating series at non-positive integer a. In Equation (8.3), write

$$ {}_2F_1\left(m-\nu,\nu+m+1;m+1;\sin^2\frac{\theta}{2}\right)=\sum_{j=0}^{\infty}\delta_j\sin^{2j}\frac{\theta}{2}. \tag{8.7}$$

The coefficients satisfy

$$ \delta_0=1, \tag{8.8a}$$

$$ \delta_j=\frac{(m-\nu+j-1)(\nu+m+j)}{j(m+j)}\delta_{j-1},\qquad j\geq 1. \tag{8.8b}$$

Since the hypergeometric factor tends to unity as $\theta\to 0$, the regular endpoint behaviour is

$$ P_\nu^m(\cos\theta)=\frac{(-1)^m\Gamma(\nu+m+1)}{2^m m!\Gamma(\nu-m+1)}\theta^m\left[1+\mathcal{O}(\theta^2)\right]. \tag{8.9}$$

Thus, the first-kind branch is bounded at the polar axis and vanishes there for every $m\geq 1$.

8.3 Cap-boundary eigenvalue equation

The Legendre part of the R-SCHA potential belongs to V_1 and therefore satisfies the homogeneous lateral Dirichlet condition

$$ V_1(r,\theta_0,\lambda)=0. $$

For a non-trivial separated mode, this requires

$$ P_\nu^m(x_0)=0,\qquad x_0=\cos\theta_0. \tag{8.10}$$

For every fixed m, let

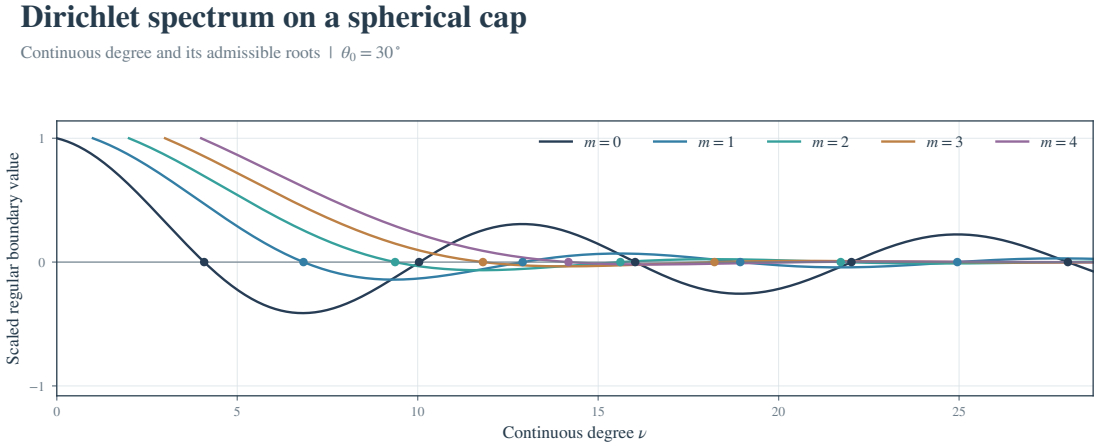
$$ \{\nu_{km}\}_{k=m}^{\infty} \tag{8.11}$$

denote the selected real roots of Equation (8.10), ordered so that

$$ \nu_{mm}<\nu_{m+1,m}<\nu_{m+2,m}<\cdots. \tag{8.12}$$

Because the differential equation depends on ν only through $\nu(\nu+1)$, the degrees ν and $-\nu-1$ represent the same eigenvalue. The R-SCHA convention selects the branch

$$ \nu_{km}>-\frac{1}{2}. \tag{8.13}$$



Admissible non-integer degrees ν_{km}

	m=0	m=1	m=2	m=3	m=4
$k=0$	4.083687	—	—	—	—
$k=1$	10.038551	6.835398	—	—	—
$k=2$	16.024836	12.908284	9.373283	—	—
$k=3$	22.018287	18.936446	15.615372	11.807398	—
$k=4$	28.014464	24.951381	21.720958	18.221949	14.177771

Each boundary curve is divided by its maximum absolute value. Dots show the roots used in the complete atlas.

Figure 2: Regular associated Legendre boundary values as functions of the continuous degree ν for $\theta_0 = 30°$. Each curve is divided by its maximum absolute value over the displayed interval. Zero crossings determine the admissible non-integer degrees; the table lists all ν_{km} with $0 \le m \le k \le 4$.

The selected roots correspond to non-trivial angular functions. In particular, for $m \ge 1$ the values $\nu = 0, \dots, m-1$, at which the classical Ferrers function vanishes identically, are not cap eigenmodes. For a proper Dirichlet cap $0 < \theta_0 < \pi$, the selected degrees satisfy $\nu_{km} > m$: extending a zero-trace cap function by zero to the full sphere and applying the fixed-order Rayleigh bound gives $\lambda_{km} > m(m+1)$. The inequality is strict because the lowest full-sphere fixed-order eigenfunction cannot vanish on an open complement.

The corresponding eigenvalues are

$$\lambda_{km} = \nu_{km}(\nu_{km} + 1). \tag{8.14}$$

8.4 First-order derivative identities

The hypergeometric derivative identity

$$\frac{\mathrm{d}}{\mathrm{d}z}{}_2F_1(a, b; c; z) = \frac{ab}{c}{}_2F_1(a+1, b+1; c+1; z) \tag{8.15}$$

and the chain rule applied to Equation (8.3) give

$$\frac{\partial}{\partial\theta}P_\nu^m(\cos\theta) = m\cot\theta\, P_\nu^m(\cos\theta) + P_\nu^{m+1}(\cos\theta). \tag{8.16}$$

A second form follows from the order recurrence:

$$\frac{\partial}{\partial\theta}P_\nu^m(\cos\theta) = -m\cot\theta\, P_\nu^m(\cos\theta) - (\nu+m)(\nu-m+1)P_\nu^{m-1}(\cos\theta). \tag{8.17}$$

The raising and lowering forms are algebraically complementary. The first involves order $m+1$, whereas the second involves order $m-1$; together they cover the endpoint orders without extending the order outside the stated family.

The compatible degree and order recurrences are

$$(\nu-m+1)P_{\nu+1}^m(x) - (2\nu+1)xP_\nu^m(x) + (\nu+m)P_{\nu-1}^m(x) = 0, \tag{8.18}$$

$$P_\nu^{m+1}(x) + \frac{2mx}{\sqrt{1-x^2}}P_\nu^m(x) + (\nu+m)(\nu-m+1)P_\nu^{m-1}(x) = 0. \tag{8.19}$$

8.5 Second-order derivative identities

Differentiating Equation (8.16) and eliminating the order-$(m+2)$ function gives

$$\begin{aligned}\frac{\partial^2}{\partial\theta^2}P_\nu^m(\cos\theta) = &\left[-\frac{m}{\sin^2\theta} + m^2\cot^2\theta - (\nu-m)(\nu+m+1)\right]P_\nu^m(\cos\theta)\\ &- \cot\theta\, P_\nu^{m+1}(\cos\theta).\end{aligned} \tag{8.20}$$

Likewise, differentiating the lowering form yields

$$\begin{aligned}\frac{\partial^2}{\partial\theta^2}P_\nu^m(\cos\theta) = &\left[\frac{m}{\sin^2\theta} + m^2\cot^2\theta - (\nu+m)(\nu-m+1)\right]P_\nu^m(\cos\theta)\\ &+ (\nu+m)(\nu-m+1)\cot\theta\, P_\nu^{m-1}(\cos\theta).\end{aligned} \tag{8.21}$$

The angular differential equation itself provides the independent check

$$\frac{\partial^2 P_\nu^m}{\partial\theta^2} + \cot\theta\frac{\partial P_\nu^m}{\partial\theta} + \left[\nu(\nu+1) - \frac{m^2}{\sin^2\theta}\right]P_\nu^m = 0, \tag{8.22}$$

where all functions in Equation (8.22) are evaluated at $\cos\theta$. This relation is useful for verifying the independently evaluated first- and second-order derivatives.

8.6 Geophysical normalisation conventions

Two normalisation conventions are common in potential-field studies. For non-integer degree, factorials are interpreted through the Gamma function:

$$(\nu\pm m)! := \Gamma(\nu\pm m+1). \tag{8.23}$$

The full-normalisation factor is

$$\mathcal{N}_{\nu m}^{\text{full}} = (-1)^m\left[\left(\nu+\frac{1}{2}\right)\frac{\Gamma(\nu-m+1)}{\Gamma(\nu+m+1)}\right]^{1/2}. \tag{8.24}$$

The Schmidt semi-normalisation factor used in this review is

$$\mathcal{N}_{\nu m}^{\text{Sch}} = \begin{cases}1, & m=0,\\ (-1)^m\left[2\dfrac{\Gamma(\nu-m+1)}{\Gamma(\nu+m+1)}\right]^{1/2}, & m>0.\end{cases} \tag{8.25}$$

The Schmidt semi-normalised function is denoted by

$$\widetilde{P}_\nu^m(x) = \mathcal{N}_{\nu m}^{\mathrm{Sch}} P_\nu^m(x). \tag{8.26}$$

These real normalisation factors are used here on the non-trivial Dirichlet cap spectrum $\nu_{km} > m$, so their Gamma ratios are finite and positive. Expressions at other degree/order values require their own branch or limiting convention; a Gamma pole is not a non-zero cap eigenfunction.

Remark 8.1 (Global scaling versus cap normalisation). The factors in Equations (8.24) and (8.25) are conventional geophysical scalings inherited from spherical harmonic analysis and extended to non-integer degree through Gamma functions. They should not be confused with unit normalisation on the cap interval $[x_0, 1]$. The exact cap norm is derived in Equations (8.44) and (8.45).

For compact notation, define

$$A_{\nu m} = (\nu - m)(\nu + m + 1), \tag{8.27a}$$
$$B_{\nu m} = (\nu + m)(\nu - m + 1). \tag{8.27b}$$

The first derivatives of the Schmidt semi-normalised functions are

$$\frac{\partial}{\partial\theta}\widetilde{P}_\nu^m(\cos\theta) = m\cot\theta\,\widetilde{P}_\nu^m(\cos\theta) - 2^{-\delta_{0m}/2}\sqrt{A_{\nu m}}\,\widetilde{P}_\nu^{m+1}(\cos\theta), \tag{8.28}$$
$$\frac{\partial}{\partial\theta}\widetilde{P}_\nu^m(\cos\theta) = -m\cot\theta\,\widetilde{P}_\nu^m(\cos\theta) + 2^{\delta_{1m}/2}\sqrt{B_{\nu m}}\,\widetilde{P}_\nu^{m-1}(\cos\theta). \tag{8.29}$$

Here, δ_{ij} is the Kronecker delta.

The corresponding second derivatives are

$$\begin{aligned}\frac{\partial^2}{\partial\theta^2}\widetilde{P}_\nu^m(\cos\theta) = &\Big[-\frac{m}{\sin^2\theta} + m^2\cot^2\theta - A_{\nu m}\Big]\,\widetilde{P}_\nu^m(\cos\theta)\\ &+ 2^{-\delta_{0m}/2}\sqrt{A_{\nu m}}\,\cot\theta\,\widetilde{P}_\nu^{m+1}(\cos\theta),\end{aligned} \tag{8.30}$$
$$\begin{aligned}\frac{\partial^2}{\partial\theta^2}\widetilde{P}_\nu^m(\cos\theta) = &\Big[\frac{m}{\sin^2\theta} + m^2\cot^2\theta - B_{\nu m}\Big]\,\widetilde{P}_\nu^m(\cos\theta)\\ &- 2^{\delta_{1m}/2}\sqrt{B_{\nu m}}\,\cot\theta\,\widetilde{P}_\nu^{m-1}(\cos\theta).\end{aligned} \tag{8.31}$$

The degree and order recurrences for the normalised functions may be written without ambiguity in terms of normalisation ratios:

$$\begin{aligned}(\nu - m + 1)\frac{\mathcal{N}_{\nu m}^{\mathrm{Sch}}}{\mathcal{N}_{\nu+1,m}^{\mathrm{Sch}}}\widetilde{P}_{\nu+1}^m(x) - (2\nu+1)x\widetilde{P}_\nu^m(x)&\\ + (\nu+m)\frac{\mathcal{N}_{\nu m}^{\mathrm{Sch}}}{\mathcal{N}_{\nu-1,m}^{\mathrm{Sch}}}\widetilde{P}_{\nu-1}^m(x) = 0,&\end{aligned} \tag{8.32}$$
$$\begin{aligned}\frac{\mathcal{N}_{\nu m}^{\mathrm{Sch}}}{\mathcal{N}_{\nu,m+1}^{\mathrm{Sch}}}\widetilde{P}_\nu^{m+1}(x) + \frac{2mx}{\sqrt{1-x^2}}\widetilde{P}_\nu^m(x)&\\ + (\nu+m)(\nu-m+1)\frac{\mathcal{N}_{\nu m}^{\mathrm{Sch}}}{\mathcal{N}_{\nu,m-1}^{\mathrm{Sch}}}\widetilde{P}_\nu^{m-1}(x) = 0.&\end{aligned} \tag{8.33}$$

The lowering derivative formulas and the order recurrence Equations (8.19) and (8.33) are used for $m \geq 1$. At $m = 0$, the raising formulas give the derivatives without introducing a negative order. The degree recurrence keeps the azimuthal order fixed.

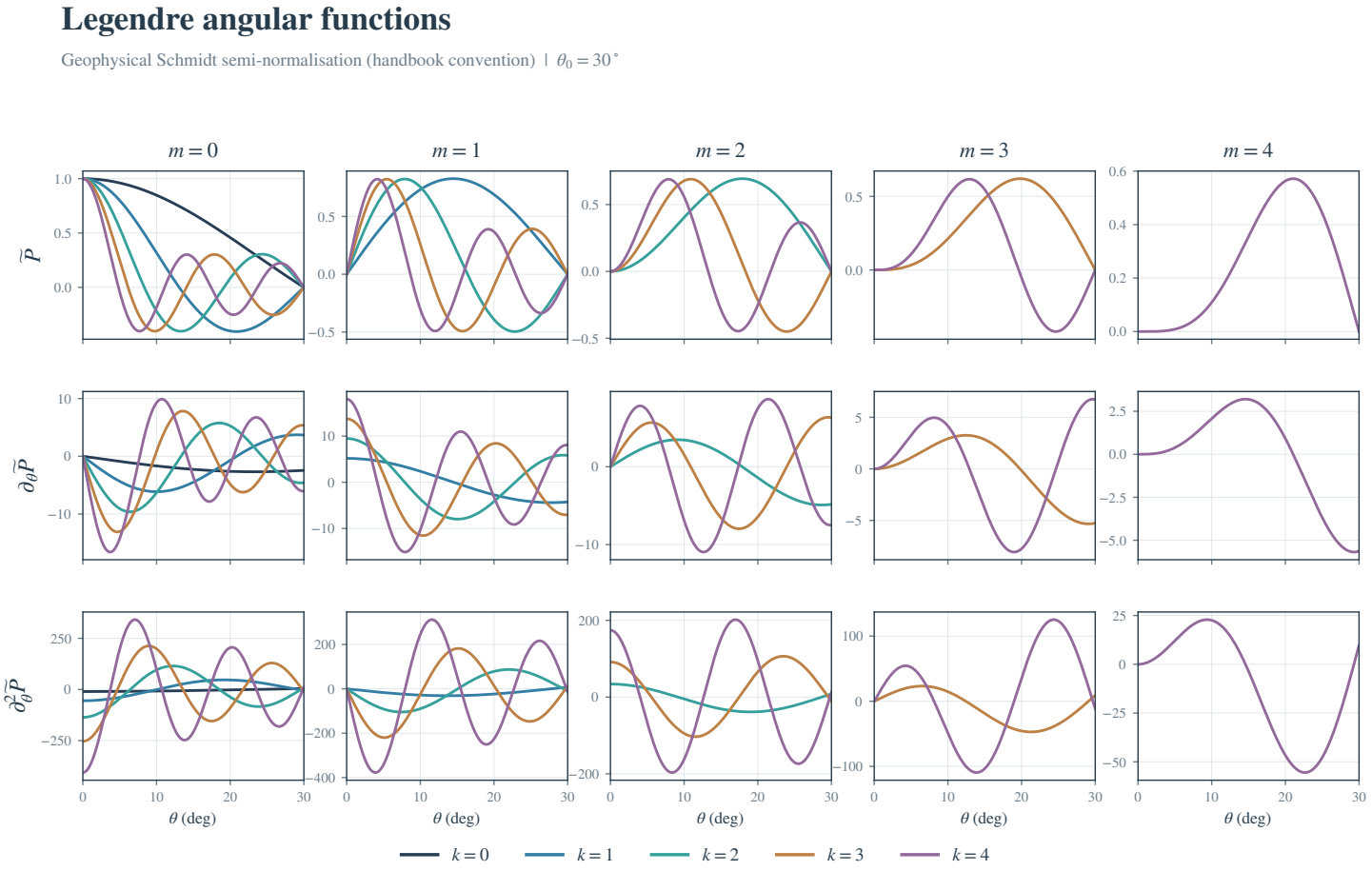


Figure 3: Schmidt semi-normalised associated Legendre functions and their first- and second-order colatitudinal derivatives for every pair $0 \le m \le k \le 4$, with $\theta_0 = 30°$. Columns fix the azimuthal order m and colours identify k. Derivatives are taken with respect to radians; all coordinate axes are linear.

8.7 Sturm–Liouville formulation

For fixed m, define the differential expression

$$\mathscr{L}_m y = -\frac{\mathrm{d}}{\mathrm{d}x}\left[(1-x^2)\frac{\mathrm{d}y}{\mathrm{d}x}\right] + \frac{m^2}{1-x^2}y. \tag{8.34}$$

Then

$$\mathscr{L}_m P_\nu^m(x) = \nu(\nu+1)P_\nu^m(x). \tag{8.35}$$

Equivalently, in the co-latitude coordinate,

$$\mathcal{L}_m u = -\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}u}{\mathrm{d}\theta}\right) + \frac{m^2}{\sin^2\theta}u, \qquad 0<\theta<\theta_0. \tag{8.36}$$

The natural Hilbert space is

$$\mathcal{H}_m = L^2\left((0,\theta_0), \sin\theta\ \mathrm{d}\theta\right). \tag{8.37}$$

For sufficiently regular u and v, the Green–Lagrange identity is

$$\begin{aligned}\int_0^{\theta_0} &[v\,\mathcal{L}_m u - u\,\mathcal{L}_m v]\sin\theta\ \mathrm{d}\theta \\ &= \sin\theta\left(u\frac{\mathrm{d}v}{\mathrm{d}\theta} - v\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)\Big|_{\theta=0}^{\theta=\theta_0}.\end{aligned} \tag{8.38}$$

In the variable x, the same identity is

$$\begin{aligned}\int_{x_0}^{1} &[v\,\mathscr{L}_m u - u\,\mathscr{L}_m v]\,\mathrm{d}x \\ &= (1-x^2)\left(u\frac{\mathrm{d}v}{\mathrm{d}x} - v\frac{\mathrm{d}u}{\mathrm{d}x}\right)\Big|_{x=x_0}^{x=1}.\end{aligned} \tag{8.39}$$

8.8 Orthogonality of distinct cap eigenfunctions

Let

$$y_i(x) = P_{\nu_{im}}^m(x), \qquad y_j(x) = P_{\nu_{jm}}^m(x), \tag{8.40}$$

with distinct eigenvalues

$$\lambda_{im} = \nu_{im}(\nu_{im}+1), \qquad \lambda_{jm} = \nu_{jm}(\nu_{jm}+1).$$

Multiplying their eigenvalue equations by y_j and y_i, respectively, subtracting, and integrating gives

$$\begin{aligned}[\lambda_{im} - \lambda_{jm}]&\int_{x_0}^{1} y_i(x)y_j(x)\ \mathrm{d}x \\ &= (1-x_0^2)\left[y_j(x_0)y_i'(x_0) - y_i(x_0)y_j'(x_0)\right].\end{aligned} \tag{8.41}$$

The contribution at $x=1$ vanishes because the first-kind solutions satisfy the regular endpoint condition.

Both eigenfunctions satisfy the cap-boundary condition

$$y_i(x_0) = y_j(x_0) = 0.$$

Therefore,

$$\int_{x_0}^{1} P^m_{\nu_{im}}(x) P^m_{\nu_{jm}}(x)\ \mathrm{d}x = 0, \qquad i \neq j. \tag{8.42}$$

Equivalently,

$$\int_{0}^{\theta_0} P^m_{\nu_{im}}(\cos\theta) P^m_{\nu_{jm}}(\cos\theta) \sin\theta\ \mathrm{d}\theta = 0, \qquad i \neq j. \tag{8.43}$$

Theorem 8.2 (Orthogonality of the non-integer Legendre modes)**.** *For fixed integer order m, the first-kind associated Legendre functions selected by*

$$P^m_{\nu_{km}}(\cos\theta_0) = 0$$

are mutually orthogonal in $L^2((0,\theta_0), \sin\theta\ \mathrm{d}\theta)$.

8.9 Cap norm

Let

$$y_\nu(x) = P^m_\nu(x).$$

To evaluate the diagonal limit, regard one of the degree parameters in Equation (8.41) as continuous before imposing the Dirichlet root condition. Differentiating both the boundary numerator and $\nu(\nu+1)$ and applying l'Hôpital's rule gives

$$\begin{aligned} \left\| P^m_{\nu_{km}} \right\|^2_{\mathrm{cap}} &:= \int_{x_0}^{1} \left[P^m_{\nu_{km}}(x) \right]^2 \mathrm{d}x \\ &= \frac{1 - x_0^2}{2\nu_{km}+1} \left[P^m_\nu(x_0) \frac{\partial^2 P^m_\nu(x_0)}{\partial\nu\,\partial x} \right. \\ &\qquad \left. - \frac{\partial P^m_\nu(x_0)}{\partial\nu} \frac{\partial P^m_\nu(x_0)}{\partial x} \right]_{\nu=\nu_{km}}. \end{aligned} \tag{8.44}$$

At a Dirichlet root, $P^m_{\nu_{km}}(x_0) = 0$, so this simplifies to

$$\left\| P^m_{\nu_{km}} \right\|^2_{\mathrm{cap}} = -\frac{1 - x_0^2}{2\nu_{km}+1} \left. \frac{\partial P^m_\nu(x_0)}{\partial\nu} \frac{\partial P^m_\nu(x_0)}{\partial x} \right|_{\nu=\nu_{km}}. \tag{8.45}$$

The right-hand side is positive because it equals the integral of the square; accordingly, the two boundary derivatives have opposite signs when $2\nu_{km}+1 > 0$. As a direct sign check, for $m = 0$, $x_0 = 0$, and $\nu = 1$, one obtains

$$\int_0^1 P_1^2(x)\ \mathrm{d}x = \frac{1}{3},$$

which is reproduced by the negative sign in Equation (8.45). The equivalent θ-domain norm is

$$\left\| P^m_{\nu_{km}} \right\|^2_{\mathrm{cap}} = \int_0^{\theta_0} \left[P^m_{\nu_{km}}(\cos\theta) \right]^2 \sin\theta\ \mathrm{d}\theta. \tag{8.46}$$

The cap-unit-normalised eigenfunction is therefore

$$\widehat{P}^m_{\nu_{km}}(\cos\theta) = \frac{P^m_{\nu_{km}}(\cos\theta)}{\left\| P^m_{\nu_{km}} \right\|_{\mathrm{cap}}}. \tag{8.47}$$

This normalisation is mathematically distinct from Schmidt semi-normalisation.

8.10 Self-adjoint realisation

The endpoint $\theta = \theta_0$ is regular, whereas $\theta = 0$ is singular. The local leading equation at the polar endpoint is

$$u'' + \frac{1}{\theta}u' - \frac{m^2}{\theta^2}u = 0. \tag{8.48}$$

For $m \geq 1$, its independent local branches satisfy

$$u_1(\theta) = \mathcal{O}(\theta^m), \qquad u_2(\theta) = \mathcal{O}(\theta^{-m}), \qquad \theta \to 0^+. \tag{8.49}$$

Only the first branch belongs to $\mathcal{H}_m$ near the polar endpoint. For $m = 0$, the branches satisfy

$$u_1(\theta) = \mathcal{O}(1), \qquad u_2(\theta) = \mathcal{O}(\log\theta), \qquad \theta \to 0^+. \tag{8.50}$$

Both are locally square-integrable with the weight $\sin\theta$, and the bounded branch is selected as the regular endpoint condition.

A self-adjoint operator associated with Equation (8.36) is obtained on the domain

$$\mathcal{D}(\mathcal{L}_m) = \left\{ u \in \mathcal{H}_m : \begin{array}{l} u \text{ and } \sin\theta\, u' \text{ are locally absolutely continuous,} \\ \mathcal{L}_m u \in \mathcal{H}_m, \quad u(\theta_0) = 0, \quad u \text{ satisfies the regular condition at } 0 \end{array} \right\}. \tag{8.51}$$

For $m = 0$, the regular endpoint condition may be represented by

$$\lim_{\theta\to 0^+} \sin\theta\, u'(\theta) = 0 \tag{8.52}$$

together with boundedness. On this domain, the boundary form in Equation (8.38) vanishes.

8.11 Compactness of the resolvent

Let

$$z \in \rho(\mathcal{L}_m),$$

where $\rho(\mathcal{L}_m)$ is the resolvent set. Let ϕ_z be a solution of

$$(\mathcal{L}_m - zI)u = 0$$

satisfying the regular condition at $\theta = 0$, and let ψ_z be an independent solution satisfying

$$\psi_z(\theta_0) = 0. \tag{8.53}$$

Define the weighted Wronskian

$$\Omega_z = \sin\theta \left[\phi_z(\theta)\psi_z'(\theta) - \phi_z'(\theta)\psi_z(\theta)\right]. \tag{8.54}$$

Abel's identity implies that Ω_z is independent of θ, and $\Omega_z \neq 0$ because z is not an eigenvalue.

The Green kernel is

$$G_z(\theta,\eta) = -\frac{1}{\Omega_z}\begin{cases} \phi_z(\theta)\psi_z(\eta), & 0 < \theta \leq \eta < \theta_0, \\ \phi_z(\eta)\psi_z(\theta), & 0 < \eta < \theta \leq \theta_0. \end{cases} \tag{8.55}$$

Thus,

$$\left[(\mathcal{L}_m - zI)^{-1} f\right](\theta) = \int_0^{\theta_0} G_z(\theta,\eta) f(\eta) \sin\eta \, \mathrm{d}\eta. \tag{8.56}$$

Near the polar endpoint,

$$\phi_z(\theta) = \begin{cases} \mathcal{O}(\theta^m), & m \geq 1, \\ \mathcal{O}(1), & m = 0, \end{cases} \tag{8.57}$$

whereas the independent solution has

$$\psi_z(\theta) = \begin{cases} \mathcal{O}(\theta^{-m}), & m \geq 1, \\ \mathcal{O}(1 + |\log\theta|), & m = 0. \end{cases} \tag{8.58}$$

Therefore, for $0 < \theta \leq \eta < \varepsilon$,

$$|G_z(\theta,\eta)| \leq \begin{cases} C\theta^m \eta^{-m}, & m \geq 1, \\ C(1 + |\log\eta|), & m = 0. \end{cases} \tag{8.59}$$

For $m \geq 1$,

$$\begin{aligned} &\int_0^\varepsilon \int_0^\eta |G_z(\theta,\eta)|^2 \sin\theta \sin\eta \, \mathrm{d}\theta \, \mathrm{d}\eta \\ &\qquad \leq C \int_0^\varepsilon \int_0^\eta \theta^{2m+1} \eta^{1-2m} \, \mathrm{d}\theta \, \mathrm{d}\eta < \infty. \end{aligned} \tag{8.60}$$

For $m = 0$,

$$\begin{aligned} &\int_0^\varepsilon \int_0^\eta |G_z(\theta,\eta)|^2 \sin\theta \sin\eta \, \mathrm{d}\theta \, \mathrm{d}\eta \\ &\qquad \leq C \int_0^\varepsilon \eta^3 \left[1 + |\log\eta|^2\right] \mathrm{d}\eta < \infty. \end{aligned} \tag{8.61}$$

By symmetry, the same estimate holds in the complementary triangular region $\eta \leq \theta$. Hence,

$$G_z \in L^2\left((0,\theta_0)^2, \sin\theta \sin\eta \, \mathrm{d}\theta \, \mathrm{d}\eta\right). \tag{8.62}$$

The resolvent is therefore a Hilbert–Schmidt operator and, in particular, a compact operator.

8.12 Completeness

Theorem 8.3 (Completeness of the cap Legendre eigenfunctions)**.** *For every fixed integer order $m \geq 0$, the self-adjoint operator $\mathcal{L}_m$ defined by Equations* (8.36) *and* (8.51) *has compact resolvent. Its spectrum consists of a discrete sequence of real eigenvalues*

$$\lambda_{km} = \nu_{km}(\nu_{km} + 1), \qquad k = m, m+1, \ldots, \tag{8.63}$$

and the corresponding eigenfunctions

$$u_{km}(\theta) = P^m_{\nu_{km}}(\cos\theta) \tag{8.64}$$

form an orthogonal complete system in

$$L^2\left((0,\theta_0), \sin\theta \, \mathrm{d}\theta\right).$$

Consequently, for

$$f \in L^2\left((0,\theta_0), \sin\theta \, \mathrm{d}\theta\right),$$

the expansion

$$f(\theta) = \sum_{k=m}^{\infty} c_{km} P^m_{\nu_{km}}(\cos\theta) \tag{8.65}$$

holds in the weighted L^2 sense, with coefficients

$$c_{km} = \frac{\displaystyle\int_0^{\theta_0} f(\theta) P^m_{\nu_{km}}(\cos\theta) \sin\theta \, \mathrm{d}\theta}{\displaystyle\int_0^{\theta_0} \left[P^m_{\nu_{km}}(\cos\theta)\right]^2 \sin\theta \, \mathrm{d}\theta}. \tag{8.66}$$

Equivalently, for $f \in L^2([x_0, 1])$,

$$f(x) = \sum_{k=m}^{\infty} c_{km} P^m_{\nu_{km}}(x), \tag{8.67}$$

where

$$c_{km} = \frac{\displaystyle\int_{x_0}^{1} f(x) P^m_{\nu_{km}}(x) \, \mathrm{d}x}{\left\| P^m_{\nu_{km}} \right\|^2_{\mathrm{cap}}}. \tag{8.68}$$

Using Equation (8.44), the denominator can be evaluated from boundary values and derivatives with respect to the continuous degree.

8.13 Real spherical-cap harmonics

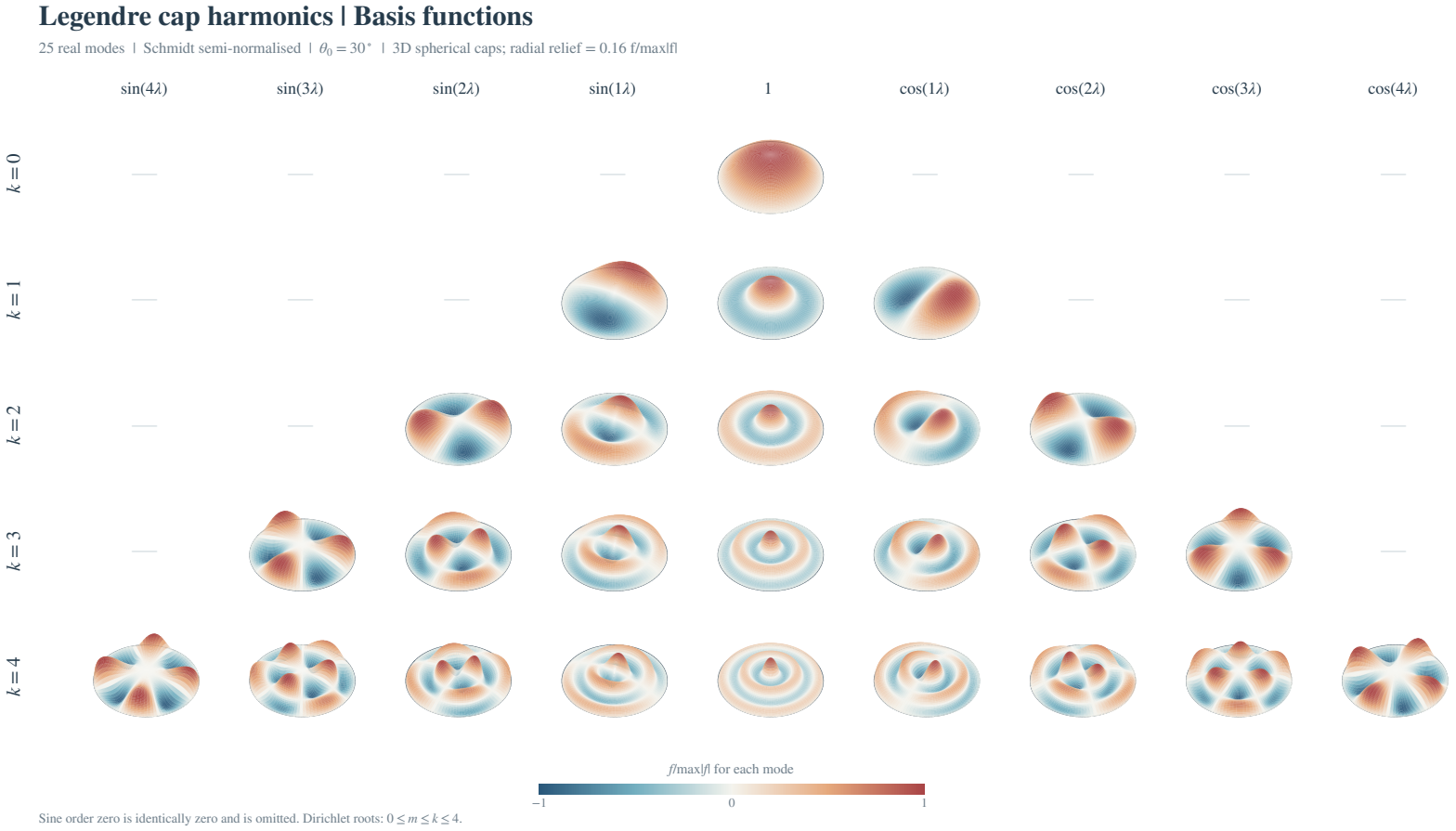
For each admissible pair (k, m), define the real Legendre-type spherical-cap harmonics

$$Y^{\mathrm{c}}_{km}(\theta,\lambda) = \widetilde{P}^m_{\nu_{km}}(\cos\theta)\cos(m\lambda), \tag{8.69a}$$

$$Y^{\mathrm{s}}_{km}(\theta,\lambda) = \widetilde{P}^m_{\nu_{km}}(\cos\theta)\sin(m\lambda), \qquad m \geq 1. \tag{8.69b}$$

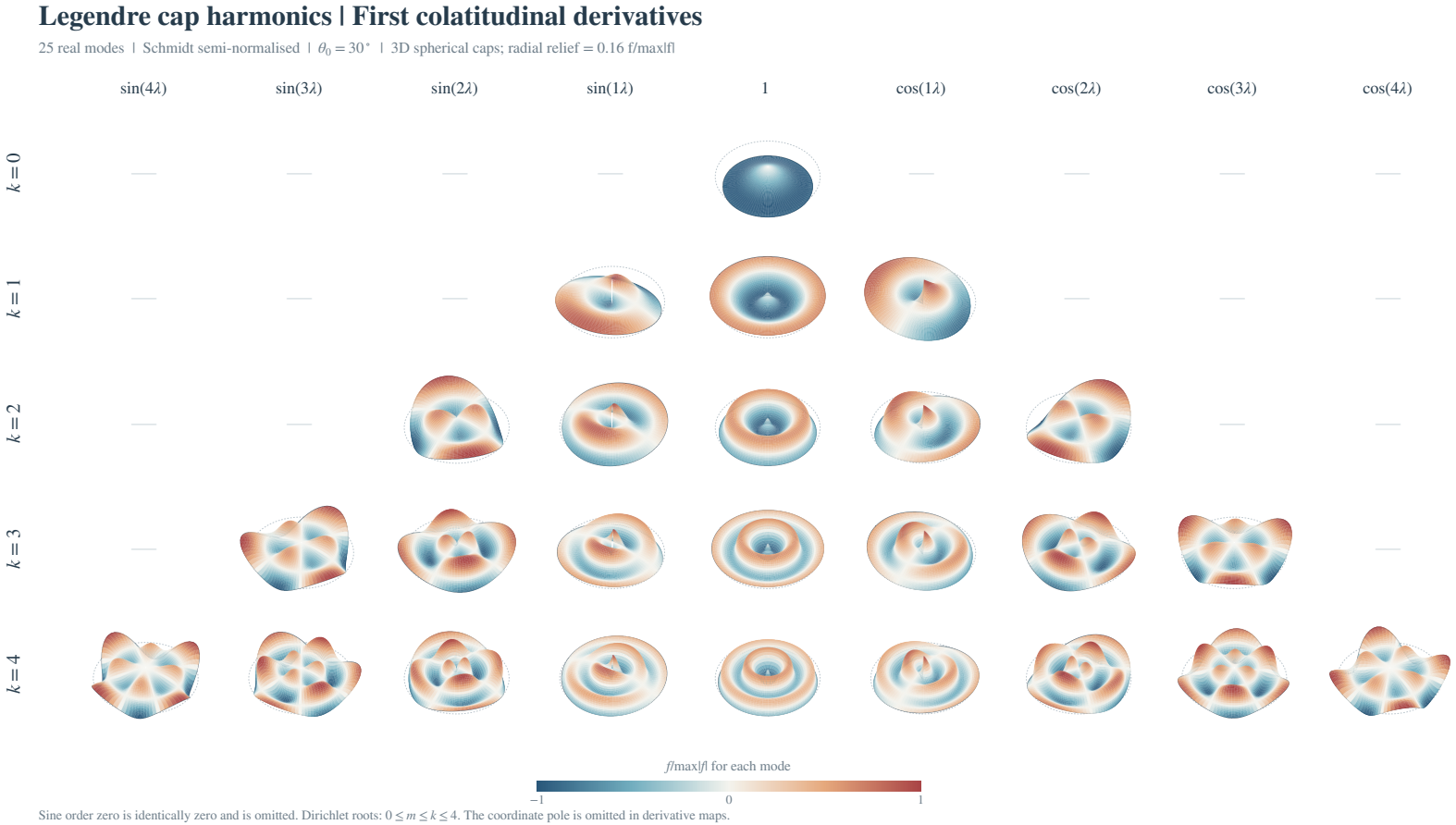
Their first and second colatitudinal derivatives are obtained by replacing $\widetilde{P}^m_{\nu_{km}}$ with the derivatives in Equations (8.28) and (8.30). Longitudinal derivatives follow directly from

$$\frac{\partial}{\partial\lambda}\begin{bmatrix}\cos(m\lambda)\\ \sin(m\lambda)\end{bmatrix} = m\begin{bmatrix}-\sin(m\lambda)\\ \cos(m\lambda)\end{bmatrix}. \tag{8.70}$$



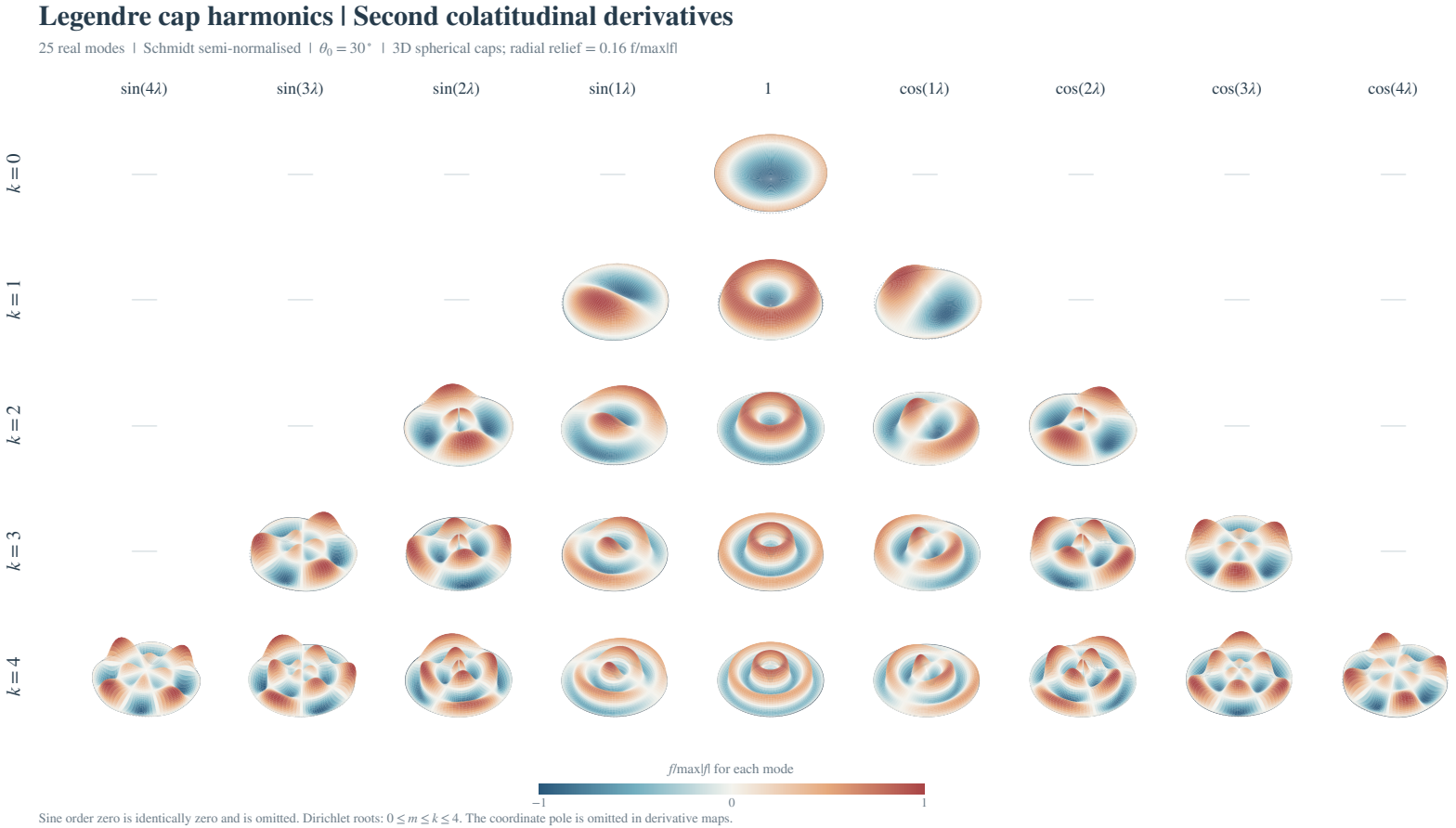
(a) Basis functions.

Figure 4: Complete real Legendre cap harmonics through $k = 4$: 25 modes for $0 \le m \le k \le 4$ and $\theta_0 = 30°$. Columns show sine modes in decreasing order, followed by cosine modes in increasing order; the identically zero $m = 0$ sine term is omitted. Colour shows each mode divided by its maximum absolute value. All panels use three-dimensional reference caps with display radius $1 + 0.16 f / \max|f|$; the coordinate pole is omitted in derivative panels.



(b) First-order colatitudinal derivatives.

Figure 4: Spherical-cap harmonics constructed from Schmidt semi-normalised associated Legendre functions (continued).



(c) Second-order colatitudinal derivatives.

Figure 4: Spherical-cap harmonics constructed from Schmidt semi-normalised associated Legendre functions (continued).

9 Mehler functions

The second R-SCHA subpotential V_2 contains angular functions associated with the complex Legendre degree

$$\nu = -\frac{1}{2} + \mathrm{i}\tau, \qquad \tau > 0. \tag{9.1}$$

For the three-dimensional spherical-cap shell,

$$\tau = \tau_p = \frac{p\pi}{s}, \qquad s = \ln\left(\frac{r_\mathrm{b}}{r_\mathrm{a}}\right), \qquad p = 1, 2, \ldots. \tag{9.2}$$

The resulting angular functions are conventionally called Mehler functions in R-SCHA and belong to the broader class of conical functions [9, 8, 7]. Their radial partners are oscillatory functions of $\ln r$ and satisfy the homogeneous Neumann conditions on the two spherical boundaries.

9.1 Angular differential equation

Substituting Equation (9.1) into

$$\nu(\nu+1)$$

gives

$$\nu(\nu+1) = -\left(\tau^2 + \frac{1}{4}\right). \tag{9.3}$$

Accordingly, the associated Legendre Equation (3.7b) becomes

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}K_\tau^m}{\mathrm{d}\theta}\right) - \left[\tau^2 + \frac{1}{4} + \frac{m^2}{\sin^2\theta}\right]K_\tau^m = 0. \tag{9.4}$$

Equivalently,

$$\frac{\mathrm{d}^2K_\tau^m}{\mathrm{d}\theta^2} + \cot\theta\frac{\mathrm{d}K_\tau^m}{\mathrm{d}\theta} - \left[\tau^2 + \frac{1}{4} + \frac{m^2}{\sin^2\theta}\right]K_\tau^m = 0. \tag{9.5}$$

The function used in R-SCHA is the regular first-kind solution

$$K_\tau^m(\theta) := P^m_{-\frac{1}{2}+\mathrm{i}\tau}(\cos\theta). \tag{9.6}$$

For the discrete R-SCHA spectrum, write

$$K_p^m(\theta) := K_{\tau_p}^m(\theta). \tag{9.7}$$

The symmetry

$$P^m_{-\frac{1}{2}+\mathrm{i}\tau}(x) = P^m_{-\frac{1}{2}-\mathrm{i}\tau}(x), \qquad -1 < x < 1, \tag{9.8}$$

shows that $K_\tau^m(\theta)$ is real-valued for real τ and $\theta \in (0, \pi)$ under the branch convention adopted here.

9.2 Hypergeometric representation

Substituting

$$\nu = -\frac{1}{2} + \mathrm{i}\tau$$

into the hypergeometric representation Equation (8.3) gives

$$K_\tau^m(\theta) = \frac{(-1)^m}{2^m m!}\frac{\Gamma\left(m+\frac{1}{2}+\mathrm{i}\tau\right)}{\Gamma\left(\frac{1}{2}-m+\mathrm{i}\tau\right)}\sin^m\theta \\ \times {}_2F_1\left(m+\frac{1}{2}-\mathrm{i}\tau,\, m+\frac{1}{2}+\mathrm{i}\tau;\, m+1;\, \sin^2\frac{\theta}{2}\right). \tag{9.9}$$

For integer $m \geq 1$, the Gamma-function ratio may be reduced to the finite product

$$\frac{\Gamma\left(m+\frac{1}{2}+\mathrm{i}\tau\right)}{\Gamma\left(\frac{1}{2}-m+\mathrm{i}\tau\right)} = (-1)^m \prod_{j=1}^{m}\left[\tau^2+\left(j-\frac{1}{2}\right)^2\right]. \tag{9.10}$$

Define

$$\chi_m(\tau) = \prod_{j=1}^{m}\left[\tau^2+\left(j-\frac{1}{2}\right)^2\right], \qquad \chi_0(\tau)=1. \tag{9.11}$$

Then

$$K_\tau^m(\theta) = \frac{\chi_m(\tau)}{2^m m!}\sin^m\theta\, F_\tau^m(\theta), \tag{9.12}$$

$$F_\tau^m(\theta) := {}_2F_1\left(m+\frac{1}{2}-\mathrm{i}\tau,\, m+\frac{1}{2}+\mathrm{i}\tau;\, m+1;\, \sin^2\frac{\theta}{2}\right). \tag{9.13}$$

This form makes the reality and positivity of the multiplicative prefactor explicit.

9.3 Power-series coefficients

Write

$$F_\tau^m(\theta) = \sum_{k=0}^{\infty}\delta_k^m(\tau)\sin^{2k}\frac{\theta}{2}. \tag{9.14}$$

The coefficients satisfy

$$\delta_0^m(\tau) = 1, \tag{9.15a}$$

$$\delta_k^m(\tau) = \frac{\left(m-\frac{1}{2}+k\right)^2+\tau^2}{k(m+k)}\delta_{k-1}^m(\tau), \qquad k\geq 1. \tag{9.15b}$$

For real τ, every coefficient in Equation (9.14) is positive. Hence,

$$F_\tau^m(\theta) > 0 \qquad \text{for } 0\leq\theta<\pi, \tag{9.16}$$

and therefore

$$K_\tau^m(\theta) \geq 0 \qquad \text{for } 0\leq\theta<\pi. \tag{9.17}$$

The regular polar behaviour is

$$K_\tau^m(\theta) = \frac{\chi_m(\tau)}{2^m m!}\theta^m\left[1+\mathcal{O}(\theta^2)\right], \qquad \theta\to 0^+. \tag{9.18}$$

In particular,

$$K_\tau^0(0)=1, \qquad K_\tau^m(0)=0 \quad (m\geq 1). \tag{9.19}$$

9.4 First- and second-order derivatives

The associated Legendre derivative identity Equation (8.16) gives

$$\frac{\partial K_\tau^m}{\partial \theta} = m \cot\theta \, K_\tau^m(\theta) + K_\tau^{m+1}(\theta). \tag{9.20}$$

Differentiating once more yields

$$\begin{aligned}\frac{\partial^2 K_\tau^m}{\partial \theta^2} = &\left[m^2 \cot^2\theta - \frac{m}{\sin^2\theta}\right] K_\tau^m(\theta) \\ &+ (2m+1)\cot\theta \, K_\tau^{m+1}(\theta) + K_\tau^{m+2}(\theta).\end{aligned} \tag{9.21}$$

Equations (9.20) and (9.21) are exact but contain factors $\cot\theta$ and $\sin^{-2}\theta$. At the polar axis, their finite limiting values should be obtained from Equation (9.18) rather than by direct evaluation of the singular-looking expressions.

For a standard regional cap satisfying

$$0 < \theta_0 \le \frac{\pi}{2}, \tag{9.22}$$

Equations (9.17) and (9.20) imply

$$\frac{\partial K_\tau^m}{\partial \theta} > 0 \qquad \text{for } 0 < \theta \le \theta_0. \tag{9.23}$$

Thus, the unnormalised Mehler functions are positive and monotone increasing on such caps. Their magnitude, however, may grow rapidly with both m and τ, which motivates the complete cap normalisation derived below.

9.5 A bilinear integral identity

Let $\tau, \alpha > 0$ with $\tau \neq \alpha$. The functions K_τ^m and K_α^m satisfy

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta \frac{\mathrm{d}K_\tau^m}{\mathrm{d}\theta}\right) - \left[\tau^2 + \frac{1}{4} + \frac{m^2}{\sin^2\theta}\right] K_\tau^m = 0, \tag{9.24a}$$

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta \frac{\mathrm{d}K_\alpha^m}{\mathrm{d}\theta}\right) - \left[\alpha^2 + \frac{1}{4} + \frac{m^2}{\sin^2\theta}\right] K_\alpha^m = 0. \tag{9.24b}$$

Multiplying the first equation by K_α^m, the second by K_τ^m, subtracting, and integrating gives

$$\begin{aligned}&\int_0^{\theta_0} K_\tau^m(\theta) K_\alpha^m(\theta) \sin\theta \, \mathrm{d}\theta \\ &\quad = \frac{\sin\theta_0}{\tau^2 - \alpha^2}\left[K_\alpha^m(\theta_0)\frac{\partial K_\tau^m(\theta_0)}{\partial\theta} - K_\tau^m(\theta_0)\frac{\partial K_\alpha^m(\theta_0)}{\partial\theta}\right].\end{aligned} \tag{9.25}$$

The boundary contribution at $\theta = 0$ vanishes by the regular endpoint behaviour.

Unlike the Legendre eigenfunctions in Section 8, the Mehler functions do not satisfy a common homogeneous self-adjoint boundary condition at $\theta = \theta_0$. The right-hand side of Equation (9.25) therefore does not generally vanish. Consequently, the angular family

$$\left\{K_{\tau_p}^m\right\}_{p\ge 1}$$

is not, in general, an orthogonal system on the spherical-cap interval. The orthogonality of the complete V_2 modes arises instead from their radial functions and is treated in Sections 11 and 13.

9.6 Norm obtained by a parameter limit

Letting $\alpha \to \tau$ in Equation (9.25) and applying l'Hospital's rule gives

$$\begin{aligned}\|K_\tau^m\|_{\text{cap}}^2 &:= \int_0^{\theta_0} [K_\tau^m(\theta)]^2 \sin\theta \, \mathrm{d}\theta \\ &= \frac{\sin\theta_0}{2\tau}\left[K_\tau^m \frac{\partial}{\partial\tau}\left(\frac{\partial K_\tau^m}{\partial\theta}\right) - \frac{\partial K_\tau^m}{\partial\tau}\frac{\partial K_\tau^m}{\partial\theta}\right]_{\theta=\theta_0}.\end{aligned} \tag{9.26}$$

Using Equation (9.20), the terms proportional to $m\cot\theta_0$ cancel, yielding the simpler expression

$$\|K_\tau^m\|_{\text{cap}}^2 = \frac{\sin\theta_0}{2\tau}\left[K_\tau^m \frac{\partial K_\tau^{m+1}}{\partial\tau} - K_\tau^{m+1}\frac{\partial K_\tau^m}{\partial\tau}\right]_{\theta=\theta_0}. \tag{9.27}$$

9.7 Factorisation used in the normalisation proof

Introduce

$$C_m(\theta) = \frac{(-1)^m \sin^m\theta}{2^m m!}, \tag{9.28a}$$

$$\mathcal{R}_\tau^m = \frac{\Gamma\left(m + \frac{1}{2} + \mathrm{i}\tau\right)}{\Gamma\left(\frac{1}{2} - m + \mathrm{i}\tau\right)}, \tag{9.28b}$$

$$F_\tau^m(\theta) = {}_2F_1\left(m + \frac{1}{2} - \mathrm{i}\tau,\, m + \frac{1}{2} + \mathrm{i}\tau;\, m+1;\, \sin^2\frac{\theta}{2}\right). \tag{9.28c}$$

Then

$$K_\tau^m(\theta) = C_m(\theta)\mathcal{R}_\tau^m F_\tau^m(\theta). \tag{9.29}$$

The adjacent-order prefactors satisfy

$$C_m(\theta)C_{m+1}(\theta) = -C_m^2(\theta)\frac{\sin\theta}{2(m+1)}, \tag{9.30}$$

$$\mathcal{R}_\tau^m\mathcal{R}_\tau^{m+1} = -\left(\mathcal{R}_\tau^m\right)^2\left[\tau^2 + \left(m + \frac{1}{2}\right)^2\right]. \tag{9.31}$$

Define

$$d_m(\tau) = \frac{2\tau}{\tau^2 + \left(m + \frac{1}{2}\right)^2}. \tag{9.32}$$

Further define

$$\begin{aligned}H_\tau^m(\theta) &= F_\tau^m(\theta)F_\tau^{m+1}(\theta)d_m(\tau) \\ &\quad + F_\tau^m(\theta)\frac{\partial F_\tau^{m+1}(\theta)}{\partial\tau} - F_\tau^{m+1}(\theta)\frac{\partial F_\tau^m(\theta)}{\partial\tau}.\end{aligned} \tag{9.33}$$

Substitution into Equation (9.27) gives

$$\begin{aligned}\|K_\tau^m\|_{\text{cap}}^2 &= \left[\sin\theta_0\, C_m(\theta_0)\mathcal{R}_\tau^m\right]^2 \\ &\quad \times \frac{\left[\tau^2 + \left(m + \frac{1}{2}\right)^2\right]H_\tau^m(\theta_0)}{4\tau(m+1)}.\end{aligned} \tag{9.34}$$

9.8 Parameter derivative of the hypergeometric factor

Let

$$B_k^m(\tau,\theta) = \frac{\left(m+\frac{1}{2}+\mathrm{i}\tau\right)_k \left(m+\frac{1}{2}-\mathrm{i}\tau\right)_k}{(m+1)_k\, k!} \sin^{2k}\frac{\theta}{2}. \tag{9.35}$$

Then

$$F_\tau^m(\theta) = \sum_{k=0}^{\infty} B_k^m(\tau,\theta). \tag{9.36}$$

Differentiation with respect to τ gives the real series

$$\frac{\partial F_\tau^m(\theta)}{\partial\tau} = \sum_{k=0}^{\infty} B_k^m(\tau,\theta) \left[\sum_{j=0}^{k-1} \frac{2\tau}{\left(m+\frac{1}{2}+j\right)^2+\tau^2}\right]. \tag{9.37}$$

The inner sum is defined to be zero when $k=0$. This representation follows from

$$\frac{\partial}{\partial\tau}\log\left[\left(m+\frac{1}{2}+\mathrm{i}\tau\right)_k \left(m+\frac{1}{2}-\mathrm{i}\tau\right)_k\right] = \sum_{j=0}^{k-1} \frac{2\tau}{\left(m+\frac{1}{2}+j\right)^2+\tau^2}. \tag{9.38}$$

9.9 Complete cap normalisation

Define the unit-normalised Mehler function by

$$\widetilde{K}_\tau^m(\theta) = \frac{K_\tau^m(\theta)}{\|K_\tau^m\|_{\mathrm{cap}}}, \tag{9.39}$$

so that

$$\int_0^{\theta_0} \left[\widetilde{K}_\tau^m(\theta)\right]^2 \sin\theta\, \mathrm{d}\theta = 1. \tag{9.40}$$

Using Equation (9.34), the Gamma-function and factorial prefactors cancel, and the normalised function may be evaluated as

$$\widetilde{K}_\tau^m(\theta) = \widetilde{\mathcal{N}}_\tau^m(\theta) F_\tau^m(\theta), \tag{9.41}$$

where the normalising prefunction is

$$\widetilde{\mathcal{N}}_\tau^m(\theta) = \left[\frac{4\tau(m+1)}{\left[\tau^2+\left(m+\frac{1}{2}\right)^2\right] H_\tau^m(\theta_0)}\right]^{1/2} \frac{\sin^m\theta}{\sin^{m+1}\theta_0}. \tag{9.42}$$

The notation "prefunction" is deliberate: $\widetilde{\mathcal{N}}_\tau^m(\theta)$ depends on θ through $\sin^m\theta$ and is not merely a scalar constant.

For the discrete three-dimensional R-SCHA spectrum, define

$$\widetilde{K}_p^m(\theta) = \widetilde{K}_{\tau_p}^m(\theta), \qquad p = 1, 2, \ldots. \tag{9.43}$$

9.10 Derivatives of the normalised Mehler function

Differentiating Equation (9.41) gives

$$\frac{\partial \widetilde{K}_\tau^m}{\partial \theta} = \widetilde{\mathcal{N}}_\tau^m(\theta) \{ m \cot\theta \, F_\tau^m(\theta) + \frac{\sin\theta}{2(m+1)} \left[\tau^2 + \left(m + \frac{1}{2} \right)^2 \right] F_\tau^{m+1}(\theta) \Bigg\}. \tag{9.44}$$

The second derivative is

$$\begin{aligned} \frac{\partial^2 \widetilde{K}_\tau^m}{\partial \theta^2} = \widetilde{\mathcal{N}}_\tau^m(\theta) \Bigg\{ & \left[-\frac{m}{\sin^2\theta} + m^2 \cot^2\theta \right] F_\tau^m(\theta) \\ & + \frac{(2m+1)\cos\theta}{2(m+1)} \left[\tau^2 + \left(m + \frac{1}{2} \right)^2 \right] F_\tau^{m+1}(\theta) \\ & + \frac{\sin^2\theta}{4(m+1)(m+2)} \left[\tau^2 + \left(m + \frac{1}{2} \right)^2 \right] \\ & \qquad \times \left[\tau^2 + \left(m + \frac{3}{2} \right)^2 \right] F_\tau^{m+2}(\theta) \Bigg\}. \end{aligned} \tag{9.45}$$

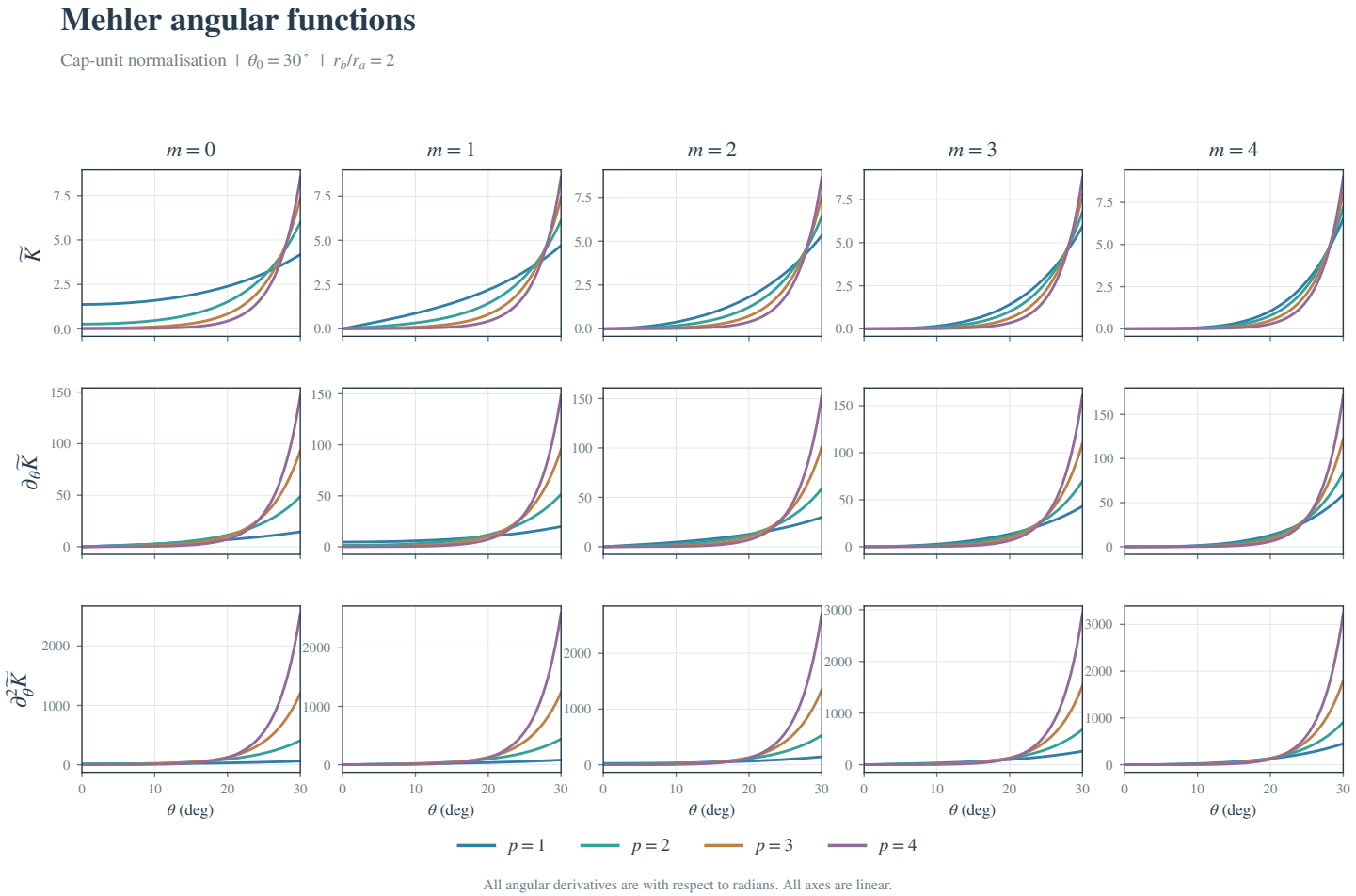


Figure 5: Completely cap-normalised Mehler functions and their first- and second-order colatitudinal derivatives for $1 \le p \le 4$, $0 \le m \le 4$. The reference geometry is $\theta_0 = 30^\circ$ and $r_\mathrm{b}/r_\mathrm{a} = 2$, so $\tau_p = p\pi/\ln 2$. Columns fix m and colours identify p. Derivatives use radians; all coordinate axes are linear.

9.11 Real Mehler spherical-cap harmonics

For the discrete parameter τ_p, define

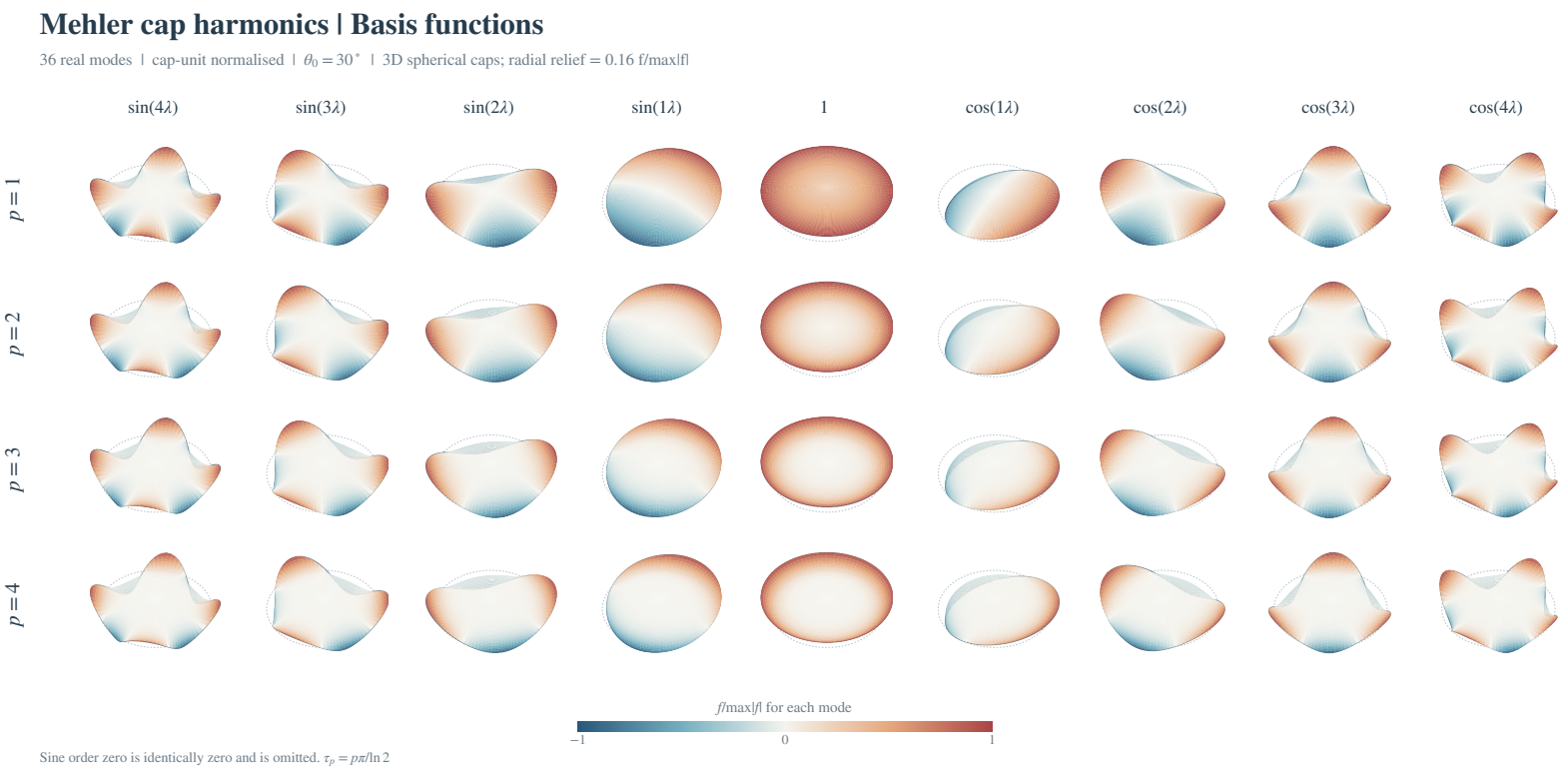
$$\mathcal{K}^{\mathrm{c}}_{pm}(\theta,\lambda) = \widetilde{K}^m_p(\theta)\cos(m\lambda), \tag{9.46a}$$

$$\mathcal{K}^{\mathrm{s}}_{pm}(\theta,\lambda) = \widetilde{K}^m_p(\theta)\sin(m\lambda), \qquad m \geq 1. \tag{9.46b}$$

Their colatitudinal derivatives are obtained from Equations (9.44) and (9.45), while their longitudinal derivatives are

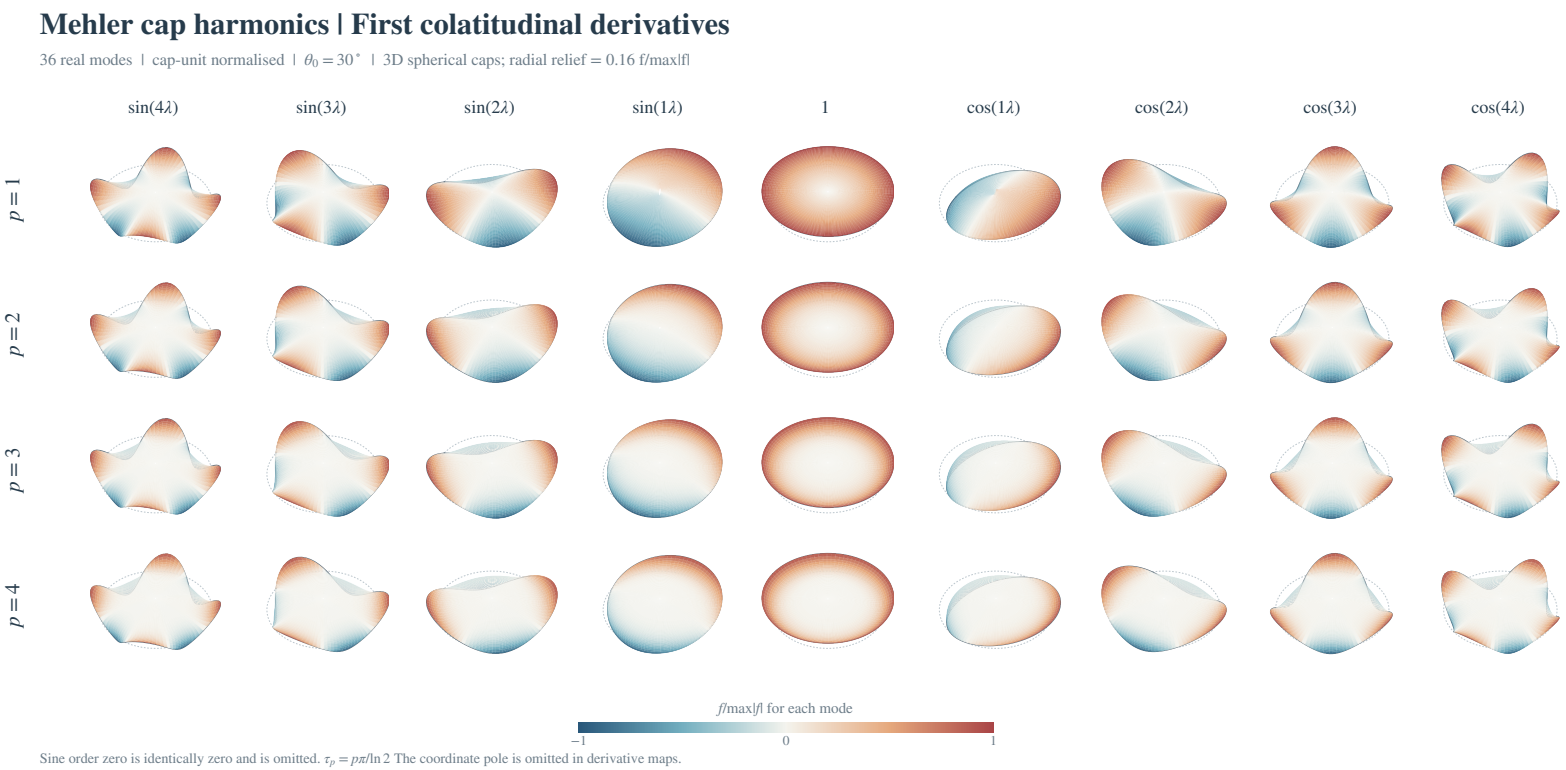
$$\frac{\partial \mathcal{K}^{\mathrm{c}}_{pm}}{\partial \lambda} = -m\widetilde{K}^m_p(\theta)\sin(m\lambda), \tag{9.47}$$

$$\frac{\partial \mathcal{K}^{\mathrm{s}}_{pm}}{\partial \lambda} = m\widetilde{K}^m_p(\theta)\cos(m\lambda). \tag{9.48}$$



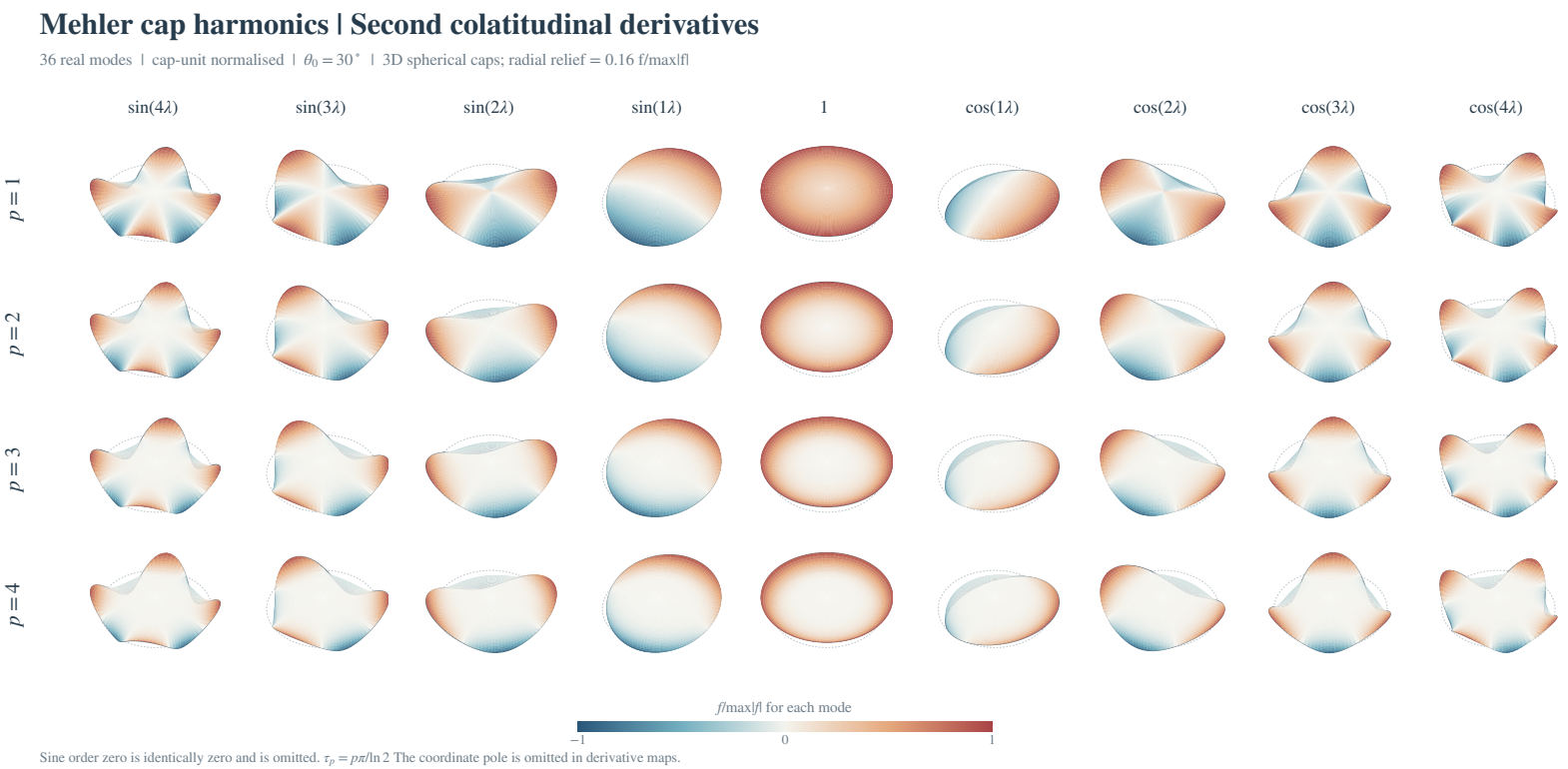
(a) Basis functions.

Figure 6: Complete real Mehler cap harmonics for $1 \le p \le 4$ and $0 \le m \le 4$: 36 modes at $\theta_0 = 30^\circ$ and $r_\mathrm{b}/r_\mathrm{a} = 2$. The zero-order sine term is omitted. Colour is scaled separately for each mode. All panels use three-dimensional reference caps with display radius $1 + 0.16 f / \max|f|$; the coordinate pole is omitted in derivative panels.



(b) First-order colatitudinal derivatives.

Figure 6: Spherical-cap harmonics constructed from completely normalised Mehler functions (continued).



(c) Second-order colatitudinal derivatives.

Figure 6: Spherical-cap harmonics constructed from completely normalised Mehler functions (continued).

9.12 The exceptional case $\tau = 0$

The normalisation formula Equation (9.42) was derived for $\tau > 0$ and has a 0/0 parameter quotient at $\tau = 0$. The normalised conical function itself has a finite limit there, as follows from its integral norm and continuous dependence on τ; the quotient must be evaluated by that limit. For a non-degenerate shell $r_{\mathrm{a}} < r_{\mathrm{b}}$, $\tau = 0$ would correspond to $\mu = 1/4$, which is not an eigenvalue of the radial Neumann problem. The positive Mehler sequence therefore begins at

$$p = 1. \tag{9.49}$$

The zero radial eigenvalue is represented separately by the constant radial and angular modes developed in Sections 10 and 11; it is not the value $p = 0$ of the positive-spectrum formula. Classical R-SCHA2D uses the degree $-1/2$ conical function in a distinct one-surface boundary problem. It is therefore not obtained by substituting $p = 0$ into the normalised three-dimensional Mehler spectrum; see Section 12.

10 Constant angular modes

The Mehler sequence described in Section 9 contains the oscillatory radial modes indexed by $p \geq 1$. The homogeneous radial Neumann problem also possesses a zero-eigenvalue mode whose radial factor is constant. The angular equation paired with this radial mode is the associated Legendre equation with

$$\nu(\nu + 1) = 0, \qquad \nu = 0 \quad \text{or} \quad \nu = -1. \tag{10.1}$$

The corresponding regular angular solutions are called *constant functions* in the R-SCHA literature [9, 8, 7].

The terminology requires clarification. Except for the axisymmetric mode $m = 0$, these functions are not constant with respect to co-latitude. The name refers to the zero separation eigenvalue and to the constant radial factor with which the angular functions are paired.

10.1 Degenerate associated Legendre equation

Setting Equation (10.1) in the angular Equation (3.7b) gives

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) - \frac{m^2}{\sin^2\theta}\Theta = 0, \qquad 0 < \theta < \theta_0. \tag{10.2}$$

Equivalently,

$$\frac{\mathrm{d}^2\Theta}{\mathrm{d}\theta^2} + \cot\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta} - \frac{m^2}{\sin^2\theta}\Theta = 0. \tag{10.3}$$

Introduce the logarithmic angular coordinate

$$t = \ln\tan\frac{\theta}{2}. \tag{10.4}$$

Since

$$\frac{\mathrm{d}t}{\mathrm{d}\theta} = \frac{1}{\sin\theta}, \qquad \frac{\mathrm{d}\theta}{\mathrm{d}t} = \sin\theta, \tag{10.5}$$

and $\Theta(\theta) = Y(t)$, one has

$$\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta} = \frac{\mathrm{d}Y}{\mathrm{d}t}. \tag{10.6}$$

Differentiating once more gives

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)=\frac{1}{\sin\theta}\frac{\mathrm{d}^2Y}{\mathrm{d}t^2}. \tag{10.7}$$

Substitution into Equation (10.2) reduces the singular angular equation to the constant-coefficient equation

$$\frac{\mathrm{d}^2Y}{\mathrm{d}t^2}-m^2Y=0. \tag{10.8}$$

For $m\geq 1$, the general solution is

$$Y(t)=A_m\mathrm{e}^{mt}+B_m\mathrm{e}^{-mt}. \tag{10.9}$$

Using

$$\mathrm{e}^t=\tan\frac{\theta}{2},$$

this becomes

$$\Theta_m(\theta)=A_m\tan^m\frac{\theta}{2}+B_m\cot^m\frac{\theta}{2}. \tag{10.10}$$

The second branch diverges as $\theta\to 0^+$ and is excluded by regularity at the polar axis. Hence, up to a non-zero multiplicative constant, the regular solution is

$$\Theta_m^{\mathrm{reg}}(\theta)=\tan^m\frac{\theta}{2},\qquad m\geq 1. \tag{10.11}$$

For $m=0$, Equation (10.8) reduces to

$$\frac{\mathrm{d}^2Y}{\mathrm{d}t^2}=0, \tag{10.12}$$

with solution

$$\Theta_0(\theta)=A_0+B_0\ln\tan\frac{\theta}{2}. \tag{10.13}$$

The logarithmic branch is singular at the polar axis, so the regular axisymmetric solution is the constant

$$\Theta_0^{\mathrm{reg}}(\theta)=1. \tag{10.14}$$

10.2 Regularised Legendre construction

The same functions can be obtained as a regularised degree-zero limit of the first-kind associated Legendre function. For $m\geq 1$, start from

$$\begin{aligned}P_\nu^m(\cos\theta)=&\frac{(-1)^m\Gamma(\nu+m+1)}{m!\,\Gamma(\nu-m+1)}\tan^m\frac{\theta}{2}\\&\times{}_2F_1\left(-\nu,\,\nu+1;\,m+1;\,\sin^2\frac{\theta}{2}\right).\end{aligned} \tag{10.15}$$

At $\nu=0$, the denominator $\Gamma(1-m)$ has a pole, so direct substitution gives the classical degree-zero Ferrers value $P_0^m=0$ for $m\geq 1$. R-SCHA instead defines the regularised branch

$$P_0^m(\theta):=\lim_{\varepsilon\to 0^+}\frac{P_{-\varepsilon}^m(\cos\theta)}{\varepsilon},\qquad m=1,2,\ldots. \tag{10.16}$$

The hypergeometric factor satisfies

$$ {}_2F_1\left(\varepsilon,\, 1-\varepsilon;\, m+1;\, \sin^2\frac{\theta}{2}\right) = 1 + \mathcal{O}(\varepsilon). \tag{10.17} $$

Near the Gamma-function pole,

$$ \Gamma(1-m-\varepsilon) \sim \frac{(-1)^m}{(m-1)!\,\varepsilon}, \qquad \varepsilon \to 0^+, \tag{10.18} $$

and

$$ \Gamma(m+1-\varepsilon) = m! + \mathcal{O}(\varepsilon). \tag{10.19} $$

Substitution into Equation (10.16) yields

$$ P_0^m(\theta) = \Gamma(m)\tan^m\frac{\theta}{2}, \qquad m \geq 1. \tag{10.20} $$

For $m = 0$, define

$$ P_0^0(\theta) = 1. \tag{10.21} $$

The direct construction Equation (10.11) and the regularised construction Equation (10.20) differ only by the constant factor $\Gamma(m)$. After unit normalisation on the cap, they produce the same basis function.

Remark 10.1 (Difference from the classical Ferrers polynomial). For $m \geq 1$, the R-SCHA function $P_0^m(\theta)$ defined by Equation (10.16) is not the classical integer-degree Ferrers polynomial $P_0^m(x)$, which vanishes identically. The R-SCHA function is a regularised zero-eigenvalue branch.

The degree symmetry

$$ P_\nu^m(x) = P_{-\nu-1}^m(x) \tag{10.22} $$

shows that the two roots $\nu = 0$ and $\nu = -1$ of Equation (10.1) correspond to the same regular solution space.

10.3 First- and second-order derivatives

For $m \geq 1$,

$$ \frac{\mathrm{d}}{\mathrm{d}\theta}\ln\tan\frac{\theta}{2} = \frac{1}{\sin\theta}. \tag{10.23} $$

Therefore,

$$ \frac{\partial P_0^m}{\partial\theta} = \frac{m}{\sin\theta}P_0^m(\theta). \tag{10.24} $$

Differentiating again gives

$$ \begin{aligned} \frac{\partial^2 P_0^m}{\partial\theta^2} &= \left[\frac{m^2}{\sin^2\theta} - \frac{m\cos\theta}{\sin^2\theta}\right] P_0^m(\theta) \\ &= \frac{m(m-\cos\theta)}{\sin^2\theta}P_0^m(\theta). \end{aligned} \tag{10.25} $$

For $m = 0$, both derivatives vanish.

Although Equations (10.24) and (10.25) contain apparently singular trigonometric factors, the products have finite limits at the polar axis because

$$ P_0^m(\theta) = \Gamma(m)\left(\frac{\theta}{2}\right)^m \left[1 + \mathcal{O}(\theta^2)\right]. \tag{10.26} $$

The asymptotic form proves that the apparent singular factors are removable at the polar axis.

10.4 Angular norm

Define

$$I_m = \int_0^{\theta_0} [P_0^m(\theta)]^2 \sin\theta \, \mathrm{d}\theta. \tag{10.27}$$

For $m = 0$,

$$I_0 = \int_0^{\theta_0} \sin\theta \, \mathrm{d}\theta = 1 - \cos\theta_0. \tag{10.28}$$

For $m \geq 1$, let

$$u = \tan\frac{\theta}{2}, \qquad u_0 = \tan\frac{\theta_0}{2}. \tag{10.29}$$

Then

$$\sin\theta = \frac{2u}{1+u^2}, \qquad \mathrm{d}\theta = \frac{2\,\mathrm{d}u}{1+u^2}, \qquad \sin\theta \, \mathrm{d}\theta = \frac{4u}{(1+u^2)^2} \, \mathrm{d}u. \tag{10.30}$$

Using Equation (10.20),

$$I_m = 4\Gamma^2(m) \int_0^{u_0} \frac{u^{2m+1}}{(1+u^2)^2} \, \mathrm{d}u. \tag{10.31}$$

Integration by parts gives

$$I_m = -2\Gamma^2(m)\frac{u_0^{2m}}{1+u_0^2} + 4m\Gamma^2(m) \int_0^{u_0} \frac{u^{2m-1}}{1+u^2} \, \mathrm{d}u. \tag{10.32}$$

With $q = u^2$,

$$2\int_0^{u_0} \frac{u^{2m-1}}{1+u^2} \, \mathrm{d}u = \int_0^{u_0^2} \frac{q^{m-1}}{1+q} \, \mathrm{d}q. \tag{10.33}$$

For $u_0^2 < 1$, equivalently $0 < \theta_0 < \pi/2$, expand

$$(1+q)^{-1} = \sum_{j=0}^{\infty} (-1)^j q^j.$$

Then

$$\begin{aligned} \int_0^{u_0^2} \frac{q^{m-1}}{1+q} \, \mathrm{d}q &= u_0^{2m} \sum_{j=0}^{\infty} \frac{(-u_0^2)^j}{j+m} \\ &= u_0^{2m} \Phi(-u_0^2, 1, m), \end{aligned} \tag{10.34}$$

where the Lerch transcendent is defined by

$$\Phi(z, \alpha, v) = \sum_{j=0}^{\infty} \frac{z^j}{(j+v)^\alpha}. \tag{10.35}$$

Consequently,

$$\begin{aligned} I_m &= 2u_0^{2m}\Gamma^2(m) \left[m\Phi(-u_0^2, 1, m) - \frac{1}{1+u_0^2} \right] \\ &= 2\tan^{2m}\frac{\theta_0}{2}\Gamma^2(m) \left[m\Phi\left(-\tan^2\frac{\theta_0}{2}, 1, m\right) - \cos^2\frac{\theta_0}{2} \right]. \end{aligned} \tag{10.36}$$

The integral expression Equation (10.31) remains valid for every $0 < \theta_0 < \pi$. The power-series definition Equation (10.35) directly yields Equation (10.36) when $\theta_0 < \pi/2$; for larger caps, the Lerch function must be interpreted by analytic continuation or the norm should be evaluated from the integral.

10.5 Completely normalised constant functions

Define

$$\widetilde{P}_0^m(\theta) = \frac{P_0^m(\theta)}{\sqrt{I_m}}. \tag{10.37}$$

By construction,

$$\int_0^{\theta_0} \left[\widetilde{P}_0^m(\theta)\right]^2 \sin\theta \, \mathrm{d}\theta = 1. \tag{10.38}$$

For $m = 0$,

$$\widetilde{P}_0^0(\theta) = \frac{1}{\sqrt{1-\cos\theta_0}}. \tag{10.39}$$

For $m \geq 1$,

$$\widetilde{P}_0^m(\theta) = \left[\frac{\tan(\theta/2)}{\tan(\theta_0/2)}\right]^m \times \left\{2\left[m\Phi\left(-\tan^2\frac{\theta_0}{2}, 1, m\right) - \cos^2\frac{\theta_0}{2}\right]\right\}^{-1/2}. \tag{10.40}$$

The Gamma factor in Equation (10.20) cancels completely in the normalised expression.

An equivalent representation that avoids explicit use of the Lerch transcendent is

$$\widetilde{P}_0^m(\theta) = \frac{\tan^m(\theta/2)}{\left[4\displaystyle\int_0^{u_0} \frac{u^{2m+1}}{(1+u^2)^2} \, \mathrm{d}u\right]^{1/2}}, \qquad m \geq 1. \tag{10.41}$$

This integral form is equivalent to the closed normalisation formula in Equation (10.40).

The derivatives of the normalised functions retain the same multiplicative relations:

$$\frac{\partial \widetilde{P}_0^m}{\partial\theta} = \frac{m}{\sin\theta}\widetilde{P}_0^m(\theta), \tag{10.42}$$

$$\frac{\partial^2 \widetilde{P}_0^m}{\partial\theta^2} = \frac{m(m-\cos\theta)}{\sin^2\theta}\widetilde{P}_0^m(\theta). \tag{10.43}$$

From this point onward, the angular factor $\mathcal{C}_m$ introduced in Equation (7.23) is fixed as

$$\mathcal{C}_m(\theta) := \widetilde{P}_0^m(\theta). \tag{10.44}$$

10.6 Qualitative properties

For $m \geq 1$ and $0 < \theta < \theta_0$,

$$\widetilde{P}_0^m(\theta) > 0, \tag{10.45}$$

and

$$\frac{\partial \widetilde{P}_0^m}{\partial\theta} = \frac{m}{\sin\theta}\widetilde{P}_0^m(\theta) > 0. \tag{10.46}$$

Thus, every positive-order constant function is strictly increasing with co-latitude and exhibits no angular oscillation. Its endpoint behaviour is

$$\widetilde{P}_0^m(\theta) = \mathcal{O}(\theta^m) \qquad \text{as } \theta \to 0^+. \tag{10.47}$$

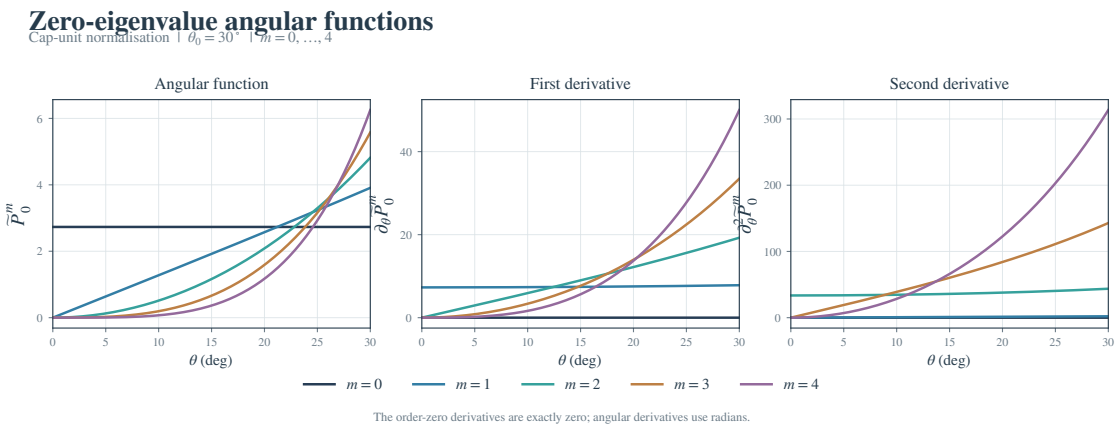


Figure 7: Completely cap-normalised zero-eigenvalue angular functions and their first- and second-order colatitudinal derivatives for every order $m = 0, \ldots, 4$, with $\theta_0 = 30^\circ$. Angular derivatives use radians. Both derivatives of the order-zero function vanish identically.

For distinct orders $m \neq n$, the angular functions alone do not generally satisfy

$$\int_0^{\theta_0} \widetilde{P}_0^m(\theta) \widetilde{P}_0^n(\theta) \sin\theta \, \mathrm{d}\theta = 0.$$

They are unit normalised individually but do not form an orthogonal family in the one-dimensional angular space.

10.7 Real constant spherical-cap harmonics

Define the real constant-function spherical-cap harmonics by

$$\mathcal{C}_m^{\mathrm{c}}(\theta, \lambda) = \widetilde{P}_0^m(\theta) \cos(m\lambda), \qquad m = 0, 1, 2, \ldots, \tag{10.48a}$$

$$\mathcal{C}_m^{\mathrm{s}}(\theta, \lambda) = \widetilde{P}_0^m(\theta) \sin(m\lambda), \qquad m = 1, 2, \ldots. \tag{10.48b}$$

Although the angular factors are not mutually orthogonal for distinct orders, the complete two-dimensional harmonics are orthogonal through their longitude dependence. With the cap-surface angular measure

$$\mathrm{d}\omega = \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda, \tag{10.49}$$

one obtains

$$\int_0^{2\pi} \int_0^{\theta_0} \mathcal{C}_m^{\mathrm{c}} \mathcal{C}_n^{\mathrm{c}} \, \mathrm{d}\omega = \begin{cases} 2\pi, & m = n = 0, \\ \pi\delta_{mn}, & m, n \geq 1, \\ 0, & \text{otherwise}, \end{cases} \tag{10.50}$$

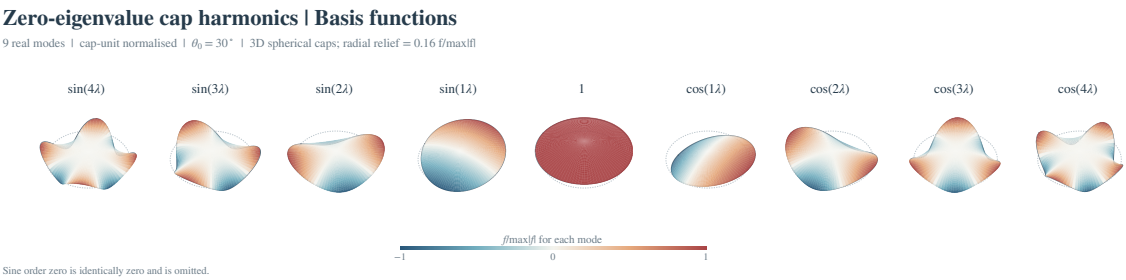
$$\int_0^{2\pi} \int_0^{\theta_0} \mathcal{C}_m^{\mathrm{s}} \mathcal{C}_n^{\mathrm{s}} \, \mathrm{d}\omega = \pi\delta_{mn}, \qquad m, n \geq 1, \tag{10.51}$$

$$\int_0^{2\pi} \int_0^{\theta_0} \mathcal{C}_m^{\mathrm{c}} \mathcal{C}_n^{\mathrm{s}} \, \mathrm{d}\omega = 0. \tag{10.52}$$

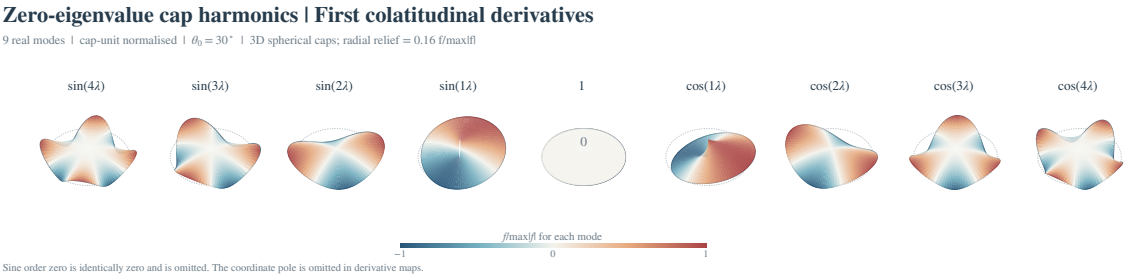
Their longitudinal derivatives are

$$\frac{\partial \mathcal{C}_m^{\mathrm{c}}}{\partial \lambda} = -m\widetilde{P}_0^m(\theta)\sin(m\lambda), \tag{10.53}$$

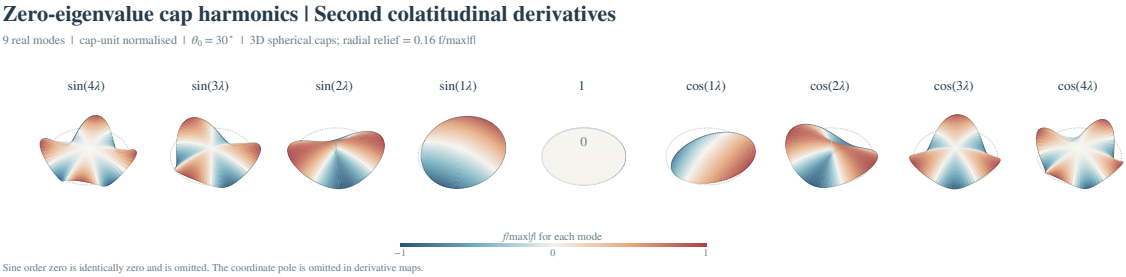
$$\frac{\partial \mathcal{C}_m^{\mathrm{s}}}{\partial \lambda} = m\widetilde{P}_0^m(\theta)\cos(m\lambda). \tag{10.54}$$



(a) Basis functions.



(b) First-order colatitudinal derivatives.



(c) Second-order colatitudinal derivatives.

Figure 8: Complete real zero-eigenvalue cap harmonics through $m = 4$: nine modes at $\theta_0 = 30^\circ$, with the identically zero order-zero sine term omitted. Colour shows the signed function divided by its maximum absolute value. All panels use three-dimensional reference caps with display radius $1 + 0.16 f / \max |f|$; the coordinate pole is omitted in derivative panels.

10.8 Harmonicity of the constant modes

Let R_0 be the constant radial function defined in Equation (7.21). For $m \geq 0$, consider

$$\Psi_{0m}(r, \theta, \lambda) = R_0 \widetilde{P}_0^m(\theta) \begin{cases} \cos(m\lambda), \\ \sin(m\lambda), \end{cases} \tag{10.55}$$

where the sine mode is omitted for $m = 0$.

Proposition 10.2 (Harmonicity and radial boundary conditions)**.** *Every mode in Equation* (10.55) *satisfies*

$$\Delta \Psi_{0m} = 0 \qquad \text{in } \Omega, \tag{10.56}$$

and

$$\left.\frac{\partial \Psi_{0m}}{\partial r}\right|_{r=r_\mathrm{a}} = \left.\frac{\partial \Psi_{0m}}{\partial r}\right|_{r=r_\mathrm{b}} = 0. \tag{10.57}$$

Proof. The radial derivative vanishes because R_0 is constant. The radial part of the spherical Laplacian is therefore zero. The remaining angular and longitudinal terms cancel by Equation (10.2) and

$$\frac{\partial^2}{\partial \lambda^2} \begin{cases} \cos(m\lambda), \\ \sin(m\lambda) \end{cases} = -m^2 \begin{cases} \cos(m\lambda), \\ \sin(m\lambda). \end{cases}$$

□

These modes supply the zero-radial-eigenvalue branch that is not contained in the $p \geq 1$ Mehler sequence. Together with the Mehler modes, they form the V_2 part of the complete R-SCHA expansion.

11 Radial functions

The Mehler and constant angular modes of V_2 are coupled to radial functions satisfying homogeneous Neumann conditions on the inner and outer spherical boundaries. This section derives the complete radial spectrum, normalises the resulting functions, establishes orthogonality and completeness, gives compact first- and second-order derivative formulae, and collects the radial inner products required in the mapping between spherical harmonics and R-SCHA. The development follows the radial construction in R-SCHA and the calculations given in the original radial-inner-product derivation [9, 8, 7].

11.1 Radial Sturm–Liouville problem

The separated radial equation derived in Equation (3.7a) is

$$r^2 \frac{\mathrm{d}^2 R}{\mathrm{d}r^2} + 2r \frac{\mathrm{d}R}{\mathrm{d}r} - \nu(\nu+1) R = 0. \tag{11.1}$$

For the V_2 subproblem, it is convenient to define

$$\mu = -\nu(\nu+1). \tag{11.2}$$

Then

$$-\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 \frac{\mathrm{d}R}{\mathrm{d}r}\right) = \mu R, \qquad r_\mathrm{a} < r < r_\mathrm{b}, \tag{11.3}$$

with homogeneous Neumann boundary conditions

$$R'(r_{\mathrm{a}}) = R'(r_{\mathrm{b}}) = 0. \tag{11.4}$$

Define the radial operator

$$\mathscr{R}R = -\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 \frac{\mathrm{d}R}{\mathrm{d}r}\right) \tag{11.5}$$

on

$$\mathcal{D}(\mathscr{R}) = \left\{R \in H^2\left(r_{\mathrm{a}}, r_{\mathrm{b}}\right) : R'(r_{\mathrm{a}}) = R'(r_{\mathrm{b}}) = 0\right\}. \tag{11.6}$$

Because r^2 is strictly positive on the closed interval $[r_{\mathrm{a}}, r_{\mathrm{b}}]$, this is a regular self-adjoint Sturm–Liouville problem in

$$L^2\left([r_{\mathrm{a}}, r_{\mathrm{b}}], \mathrm{d}r\right). \tag{11.7}$$

For $R \in \mathcal{D}(\mathscr{R})$,

$$\begin{aligned}\langle \mathscr{R}R, R\rangle &= -\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R \frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 R'\right)\mathrm{d}r \\ &= \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} r^2 \left(R'\right)^2 \mathrm{d}r \geq 0.\end{aligned} \tag{11.8}$$

Hence, every radial eigenvalue satisfies

$$\mu \geq 0. \tag{11.9}$$

11.2 Logarithmic radial coordinate

Introduce

$$t = \ln\left(\frac{r}{r_{\mathrm{a}}}\right), \qquad s = \ln\left(\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}\right), \tag{11.10}$$

so that

$$r = r_{\mathrm{a}}\mathrm{e}^t, \qquad 0 \leq t \leq s. \tag{11.11}$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}r} = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}t}, \tag{11.12}$$

Equation (11.3) becomes

$$\frac{\mathrm{d}^2 R}{\mathrm{d}t^2} + \frac{\mathrm{d}R}{\mathrm{d}t} + \mu R = 0. \tag{11.13}$$

The Neumann conditions are equivalently

$$\left.\frac{\mathrm{d}R}{\mathrm{d}t}\right|_{t=0} = \left.\frac{\mathrm{d}R}{\mathrm{d}t}\right|_{t=s} = 0. \tag{11.14}$$

Set

$$R(t) = \mathrm{e}^{-t/2} Y(t). \tag{11.15}$$

Then Equation (11.13) reduces to

$$-Y''(t) + \frac{1}{4}Y(t) = \mu Y(t), \qquad 0 < t < s, \tag{11.16}$$

and the boundary conditions become

$$Y'(0) - \frac{1}{2}Y(0) = 0, \qquad Y'(s) - \frac{1}{2}Y(s) = 0. \tag{11.17}$$

The radial norm transforms according to

$$\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R_1(r) R_2(r)\, \mathrm{d}r = r_{\mathrm{a}} \int_0^s Y_1(t) Y_2(t)\, \mathrm{d}t. \tag{11.18}$$

11.3 Classification of the radial spectrum

The characteristic equation associated with Equation (11.13) is

$$q^2 + q + \mu = 0, \tag{11.19}$$

with roots

$$q_\pm = \frac{-1 \pm \sqrt{1 - 4\mu}}{2}. \tag{11.20}$$

11.3.1 The zero eigenvalue

When $\mu = 0$, the two power-law solutions are 1 and r^{-1}. The Neumann conditions eliminate the r^{-1} branch, leaving

$$R(r) = A_0. \tag{11.21}$$

This is the constant radial mode.

11.3.2 The interval $0 < \mu < 1/4$

In this case, the two characteristic roots are distinct and real. Writing

$$R(r) = A \left(\frac{r}{r_\mathrm{a}}\right)^{q_+} + B \left(\frac{r}{r_\mathrm{a}}\right)^{q_-},$$

the two Neumann conditions have a non-trivial solution only if

$$q_+ q_- \left[\left(\frac{r_\mathrm{b}}{r_\mathrm{a}}\right)^{q_+} - \left(\frac{r_\mathrm{b}}{r_\mathrm{a}}\right)^{q_-}\right] = 0. \tag{11.22}$$

For $0 < \mu < 1/4$, one has

$$q_+ q_- = \mu > 0$$

and $q_+ \neq q_-$, so Equation (11.22) cannot hold. Therefore, there are no radial eigenvalues in this interval.

11.3.3 The repeated-root case $\mu = 1/4$

When $\mu = 1/4$, the repeated characteristic root is $q = -1/2$ and

$$R(r) = \left[A_1 \ln\left(\frac{r}{r_\mathrm{a}}\right) + A_2\right] \left(\frac{r_\mathrm{a}}{r}\right)^{1/2}. \tag{11.23}$$

For every non-degenerate interval $s > 0$, the two Neumann conditions force $A_1 = A_2 = 0$. Thus $\mu = 1/4$ is not an admissible radial eigenvalue. Collapsing the two endpoints removes this interval eigenproblem; it does not create an additional mode of the closed-shell spectrum.

11.3.4 The oscillatory spectrum

For $\mu > 1/4$, define

$$\tau = \sqrt{\mu - \frac{1}{4}}. \tag{11.24}$$

The characteristic roots are

$$q_\pm = -\frac{1}{2} \pm \mathrm{i}\tau, \tag{11.25}$$

and the real solution is

$$R(t) = \mathrm{e}^{-t/2}\left[A_1\cos(\tau t) + A_2\sin(\tau t)\right]. \tag{11.26}$$

The boundary condition at $t = 0$ gives

$$A_1 = 2\tau A_2. \tag{11.27}$$

The boundary condition at $t = s$ then reduces to

$$\left(\tau^2 + \frac{1}{4}\right)\sin(\tau s) = 0. \tag{11.28}$$

Since $\tau > 0$,

$$\tau_p = \frac{p\pi}{s}, \qquad p = 1, 2, \ldots. \tag{11.29}$$

The corresponding eigenvalues are

$$\mu_p = \frac{1}{4} + \left(\frac{p\pi}{s}\right)^2, \qquad p = 1, 2, \ldots. \tag{11.30}$$

11.4 Normalised radial eigenfunctions

Define

$$\mathcal{A}_p = \left[\frac{2}{s\left[(2p\pi)^2 + s^2\right]}\right]^{1/2}, \qquad p \geq 1. \tag{11.31}$$

The normalised oscillatory radial function is

$$\begin{aligned} R_p(r) = \mathcal{A}_p &\left(\frac{r_\mathrm{a}}{r}\right)^{1/2} \\ &\times \left[2p\pi\cos\left(\frac{p\pi}{s}\ln\frac{r}{r_\mathrm{a}}\right) + s\sin\left(\frac{p\pi}{s}\ln\frac{r}{r_\mathrm{a}}\right)\right], \qquad p \geq 1. \end{aligned} \tag{11.32}$$

It satisfies

$$R_p'(r_\mathrm{a}) = R_p'(r_\mathrm{b}) = 0 \tag{11.33}$$

and

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p^2(r)\,\mathrm{d}r = r_\mathrm{a}. \tag{11.34}$$

The constant radial mode is

$$R_0 = \left(\frac{r_\mathrm{a}}{r_\mathrm{b} - r_\mathrm{a}}\right)^{1/2} = \frac{1}{\sqrt{\mathrm{e}^s - 1}}. \tag{11.35}$$

It satisfies

$$R_0' = 0, \qquad \int_{r_\mathrm{a}}^{r_\mathrm{b}} R_0^2\,\mathrm{d}r = r_\mathrm{a}. \tag{11.36}$$

The definitions in Equations (11.32) and (11.35) agree with the functions introduced structurally in Equations (7.13) and (7.21).

11.5 First- and second-order derivatives

Let

$$t(r) = \ln\left(\frac{r}{r_\mathrm{a}}\right), \qquad \tau_p = \frac{p\pi}{s}. \tag{11.37}$$

Direct differentiation of Equation (11.32), followed by algebraic simplification, gives

$$\frac{\mathrm{d}R_p}{\mathrm{d}r} = -\left[\frac{r_\mathrm{a}\left[(2p\pi)^2 + s^2\right]}{2s^3r^3}\right]^{1/2} \sin\left(\tau_p t(r)\right). \tag{11.38}$$

The boundary conditions in Equation (11.33) follow immediately because

$$\tau_p t(r_\mathrm{a}) = 0, \qquad \tau_p t(r_\mathrm{b}) = p\pi.$$

The second derivative is

$$\begin{aligned}\frac{\mathrm{d}^2R_p}{\mathrm{d}r^2} = &\left[\frac{r_\mathrm{a}\left[(2p\pi)^2 + s^2\right]}{8s^5r^5}\right]^{1/2} \\ &\times \left[3s\sin\left(\tau_p t(r)\right) - 2p\pi\cos\left(\tau_p t(r)\right)\right].\end{aligned} \tag{11.39}$$

An equivalent second-derivative identity follows directly from the radial differential equation:

$$\frac{\mathrm{d}^2R_p}{\mathrm{d}r^2} = -\frac{2}{r}\frac{\mathrm{d}R_p}{\mathrm{d}r} - \frac{\tau_p^2 + \frac{1}{4}}{r^2}R_p(r). \tag{11.40}$$

For the constant mode,

$$R_0'(r) = 0, \qquad R_0''(r) = 0. \tag{11.41}$$

11.6 Orthogonality

Let R_p and R_q be two eigenfunctions of Equation (11.3) with eigenvalues μ_p and μ_q. Multiplying their equations by R_q and R_p, respectively, subtracting, and integrating gives

$$\begin{aligned}(\mu_p - \mu_q)&\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p(r)R_q(r)\,\mathrm{d}r \\ &= r^2\left[R_p(r)R_q'(r) - R_q(r)R_p'(r)\right]\big|_{r=r_\mathrm{a}}^{r=r_\mathrm{b}}.\end{aligned} \tag{11.42}$$

The boundary term vanishes by Equation (11.4). Hence,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p(r)R_q(r)\,\mathrm{d}r = 0, \qquad p \neq q. \tag{11.43}$$

With the normalisations already chosen,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p(r)R_q(r)\,\mathrm{d}r = r_\mathrm{a}\delta_{pq}, \qquad p, q = 0, 1, 2, \ldots. \tag{11.44}$$

In particular,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_0R_p(r)\,\mathrm{d}r = 0, \qquad p \geq 1. \tag{11.45}$$

The last identity can also be verified explicitly. Using $t = \ln(r/r_\mathrm{a})$,

$$\begin{aligned}\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_0R_p(r)\,\mathrm{d}r &= R_0r_\mathrm{a}\mathcal{A}_p\int_0^s \mathrm{e}^{t/2}\left[2p\pi\cos(\tau_p t) + s\sin(\tau_p t)\right]\mathrm{d}t \\ &= 0.\end{aligned} \tag{11.46}$$

11.7 Completeness

Theorem 11.1 (Completeness of the radial R-SCHA functions)**.** *The operator $\mathscr{R}$ defined by Equations* (11.5) *and* (11.6) *is self-adjoint, non-negative, and has compact resolvent. Its spectrum is*

$$\mu_0 = 0, \qquad \mu_p = \frac{1}{4} + \left(\frac{p\pi}{s}\right)^2, \quad p = 1, 2, \ldots, \tag{11.47}$$

and the corresponding normalised eigenfunctions

$$\{R_p\}_{p=0}^{\infty} \tag{11.48}$$

form a complete orthogonal system in $L^2([r_\mathrm{a}, r_\mathrm{b}], \mathrm{d}r)$.

The compactness statement follows from standard regular Sturm–Liouville theory on the finite interval $[r_\mathrm{a}, r_\mathrm{b}]$.

Consequently, for

$$f \in L^2\left([r_\mathrm{a}, r_\mathrm{b}], \mathrm{d}r\right), \tag{11.49}$$

one has

$$f(r) = \sum_{p=0}^{\infty} c_p R_p(r) \tag{11.50}$$

in the L^2 sense, where

$$c_p = \frac{1}{r_\mathrm{a}} \int_{r_\mathrm{a}}^{r_\mathrm{b}} f(r) R_p(r)\ \mathrm{d}r. \tag{11.51}$$

For smoother functions satisfying compatible endpoint conditions, stronger forms of convergence follow from Sturm–Liouville expansion theory.

11.8 Inner products with spherical-harmonic radial factors

The mapping from a global spherical-harmonic potential to the V_2 component of R-SCHA requires inner products between the R-SCHA radial functions and the classical internal- and external-type radial powers. Let $r_\mathrm{ref} > 0$ be a spherical-harmonic reference radius, and define

$$I_n(r) = \left(\frac{r_\mathrm{ref}}{r}\right)^{n+1}, \tag{11.52a}$$

$$E_n(r) = \left(\frac{r}{r_\mathrm{ref}}\right)^{n}, \qquad n = 0, 1, 2, \ldots. \tag{11.52b}$$

Here, r_ref is conceptually distinct from the inner shell radius r_a.

For compactness, define

$$D_{pn} = \left(n + \frac{1}{2}\right)^2 + \left(\frac{p\pi}{s}\right)^2. \tag{11.53}$$

For $p \geq 1$,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} I_n(r) R_p(r)\ \mathrm{d}r = 2p\pi\mathcal{A}_p \frac{(n+1)\left[1 - (-1)^p \mathrm{e}^{-(n+\frac{1}{2})s}\right]}{D_{pn}} \frac{r_\mathrm{ref}^{n+1}}{r_\mathrm{a}^{n}}. \tag{11.54}$$

Likewise,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} E_n(r) R_p(r)\ \mathrm{d}r = 2p\pi\mathcal{A}_p \frac{n\left[(-1)^p \mathrm{e}^{(n+\frac{1}{2})s} - 1\right]}{D_{pn}} \frac{r_\mathrm{a}^{n+1}}{r_\mathrm{ref}^{n}}. \tag{11.55}$$

For $n = 0$, the right-hand side of Equation (11.55) vanishes, consistently with the orthogonality of R_p to the constant radial function.

For the constant radial mode and $n \neq 0$,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_0 I_n(r)\ \mathrm{d}r = R_0 r_\mathrm{ref} \left(\frac{r_\mathrm{ref}}{r_\mathrm{a}}\right)^n \frac{1 - \mathrm{e}^{-ns}}{n}. \tag{11.56}$$

At $n = 0$, the limiting value is

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_0 \frac{r_\mathrm{ref}}{r}\ \mathrm{d}r = R_0 r_\mathrm{ref} s. \tag{11.57}$$

For $n \geq 0$,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_0 E_n(r)\ \mathrm{d}r = R_0 r_\mathrm{a} \left(\frac{r_\mathrm{a}}{r_\mathrm{ref}}\right)^n \frac{\mathrm{e}^{(n+1)s} - 1}{n+1}. \tag{11.58}$$

11.9 Derivation of the mapping integrals

The preceding expressions follow from the elementary antiderivatives

$$\int \mathrm{e}^{Lx} \cos(Mx)\ \mathrm{d}x = \frac{\mathrm{e}^{Lx}}{L^2 + M^2} \left[L\cos(Mx) + M\sin(Mx)\right], \tag{11.59a}$$

$$\int \mathrm{e}^{Lx} \sin(Mx)\ \mathrm{d}x = \frac{\mathrm{e}^{Lx}}{L^2 + M^2} \left[L\sin(Mx) - M\cos(Mx)\right], \tag{11.59b}$$

where $L^2 + M^2 \neq 0$.

For example, substituting $x = \ln(r/r_\mathrm{a})$ into the internal-type product gives

$$\begin{aligned} &\int_{r_\mathrm{a}}^{r_\mathrm{b}} I_n(r) R_p(r)\ \mathrm{d}r \\ &= \mathcal{A}_p \frac{r_\mathrm{ref}^{n+1}}{r_\mathrm{a}^n} \int_0^s \mathrm{e}^{-(n+\frac{1}{2})x} \left[2p\pi \cos\left(\frac{p\pi x}{s}\right) + s \sin\left(\frac{p\pi x}{s}\right)\right] \mathrm{d}x, \end{aligned} \tag{11.60}$$

which evaluates to Equation (11.54). The external-type formula is obtained by replacing the exponential factor with $\mathrm{e}^{(n+\frac{1}{2})x}$.

11.10 Supplementary power-law inner products

The remaining inner products among the classical radial powers are elementary. For non-negative integers n and k,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} I_n(r) I_k(r)\ \mathrm{d}r = \frac{r_\mathrm{ref}^{n+k+2}}{n+k+1} \left[r_\mathrm{a}^{-(n+k+1)} - r_\mathrm{b}^{-(n+k+1)}\right], \tag{11.61}$$

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} E_n(r) E_k(r)\ \mathrm{d}r = \frac{r_\mathrm{b}^{n+k+1} - r_\mathrm{a}^{n+k+1}}{(n+k+1) r_\mathrm{ref}^{n+k}}. \tag{11.62}$$

For the mixed product,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} I_n(r) E_k(r)\ \mathrm{d}r = r_\mathrm{ref}^{n+1-k} \begin{cases} \dfrac{r_\mathrm{b}^{k-n} - r_\mathrm{a}^{k-n}}{k-n}, & k \neq n, \\[2ex] s, & k = n. \end{cases} \tag{11.63}$$

These supplementary identities are not needed to establish the radial eigenbasis, but they are useful when normal matrices or cross-basis mappings are formed explicitly.

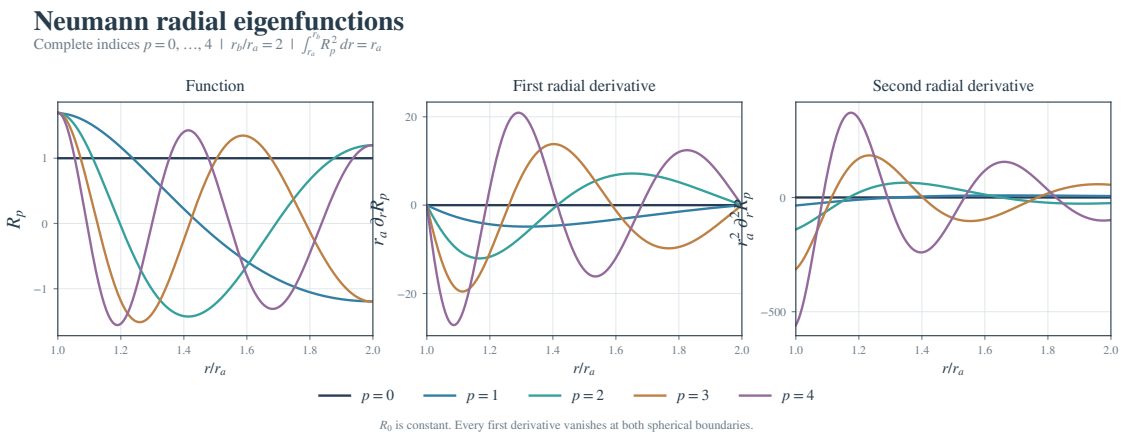


Figure 9: Normalised radial functions $R_p(r)$ for every index $p = 0, \ldots, 4$ over the reference interval $r_\mathrm{a} \le r \le r_\mathrm{b}$ with $r_\mathrm{b}/r_\mathrm{a} = 2$. The derivative panels show $r_\mathrm{a} R_p'$ and $r_\mathrm{a}^2 R_p''$ against r/r_a. The constant mode and homogeneous Neumann endpoint conditions are included.

12 A one-surface branch: classical R-SCHA2D

The three-dimensional theory developed in the preceding sections is adapted to data distributed between two distinct radial boundaries. A different construction, called R-SCHA2D, was introduced by Thébault for regional magnetic data lying on one sphere, or on surfaces whose altitude variation is small compared with the horizontal scale of the model [10]. R-SCHA2D is not obtained by setting $r_\mathrm{a} = r_\mathrm{b}$ in the three-dimensional Sturm–Liouville problem. It is the solution of a distinct, deliberately incomplete one-surface boundary-value problem. This section records that classical branch because its repeated degree $-1/2$ is often confused with either the zero radial mode or the $p \to 0$ limit of the three-dimensional Mehler spectrum.

12.1 Geometry and component convention

Let

$$\mathcal{C}_{\theta_0}(a) = \{(r, \theta, \lambda) : r = a,\ 0 \le \theta \le \theta_0,\ 0 \le \lambda < 2\pi\} \tag{12.1}$$

be the observation cap. In the cap-centred local north–east–down frame,

$$X = \frac{1}{r}\frac{\partial V}{\partial \theta}, \qquad Y = -\frac{1}{r \sin\theta}\frac{\partial V}{\partial \lambda}, \qquad Z = \frac{\partial V}{\partial r}. \tag{12.2}$$

The prescribed one-surface data will be denoted by $\widetilde{X}(\lambda)$ on the circular boundary and $\widetilde{Z}(\theta, \lambda)$ on the cap surface.

12.2 The first potential and the Neumann cap spectrum

The first potential is chosen to solve

$$\Delta V_1 = 0, \tag{12.3a}$$

$$\left.\frac{\partial V_1}{\partial \theta}\right|_{\theta=\theta_0} = 0, \tag{12.3b}$$

$$\left.\frac{\partial V_1}{\partial r}\right|_{r=a} = \widetilde{Z}(\theta, \lambda). \tag{12.3c}$$

This is an incomplete boundary-value problem because no condition is imposed at infinity or on a second radial surface.

With

$$V_1(r,\theta,\lambda) = R(r)\Theta(\theta)\Lambda(\lambda), \tag{12.4}$$

separation of Laplace's equation gives

$$\Lambda'' + m^2\Lambda = 0, \tag{12.5a}$$

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + \left[n(n+1) - \frac{m^2}{\sin^2\theta}\right]\Theta = 0, \tag{12.5b}$$

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right) - n(n+1)R = 0. \tag{12.5c}$$

Periodicity gives the real longitude factors $\cos(m\lambda)$ and $\sin(m\lambda)$. Regularity at the cap pole selects $P_n^m(\cos\theta)$, and the homogeneous lateral condition becomes

$$\frac{\mathrm{d}}{\mathrm{d}\theta}P_n^m(\cos\theta)\bigg|_{\theta=\theta_0} = 0. \tag{12.6}$$

For each fixed order m, let

$$\mathscr{N}_m = \left\{ n_k^m : \frac{\mathrm{d}}{\mathrm{d}\theta}P_{n_k^m}^m(\cos\theta)\bigg|_{\theta=\theta_0} = 0 \right\} \tag{12.7}$$

be the admissible real degree set, ordered increasingly on the regular branch. Only $n \geq 0$ with P_n^m not identically zero are retained; degrees related by $n \mapsto -n-1$ are counted once. The degree $n = 0$ at $m = 0$ belongs to this set and is essential when the mean vertical field on the cap is non-zero.

For every $n = n_k^m$, the radial equation has the two Euler branches

$$R_n^{\mathrm{i}}(r) = \left(\frac{a}{r}\right)^{n+1}, \qquad R_n^{\mathrm{e}}(r) = \left(\frac{r}{a}\right)^{n}. \tag{12.8}$$

Accordingly,

$$V_1(r,\theta,\lambda) = a\sum_{m=0}^{\infty}\sum_{n_k^m\in\mathscr{N}_m}\Bigg\{\left(\frac{a}{r}\right)^{n_k^m+1}\left[G_{n_k^m}^{\mathrm{i},m}\cos(m\lambda) + H_{n_k^m}^{\mathrm{i},m}\sin(m\lambda)\right] + \left(\frac{r}{a}\right)^{n_k^m}\left[G_{n_k^m}^{\mathrm{e},m}\cos(m\lambda) + H_{n_k^m}^{\mathrm{e},m}\sin(m\lambda)\right]\Bigg\}P_{n_k^m}^m(\cos\theta). \tag{12.9}$$

The superscripts i and e describe the radial powers. Because only one observation radius is prescribed, they do not by themselves provide a unique physical separation of internal and external sources.

12.3 Why a complementary potential is required

The homogeneous condition Equation (12.3b) implies

$$X_1(a,\theta_0,\lambda) = 0. \tag{12.10}$$

A general regional magnetic field need not satisfy this restriction. A second potential is therefore introduced so that

$$\Delta V_2 = 0, \tag{12.11a}$$

$$\frac{1}{a}\frac{\partial V_2}{\partial \theta}\bigg|_{r=a,\,\theta=\theta_0} = \widetilde{X}(\lambda), \tag{12.11b}$$

$$\frac{\partial V_2}{\partial r}\bigg|_{r=a} = 0. \tag{12.11c}$$

Here the lateral datum is specified on the observation rim, not on the entire conical surface. These data alone leave the continuation incomplete. A statement of uniqueness requires a specified continuation domain and additional boundary or admissibility conditions; decay at infinity alone does not replace unspecified data on the lateral surface.

12.4 The repeated degree $-1/2$

For a separated radial factor, the Euler equation Equation (12.5c) has distinct-power solutions when $n(n+1) \neq -1/4$. At the repeated value

$$n = -\frac{1}{2}, \qquad n(n+1) = -\frac{1}{4}, \tag{12.12}$$

the radial solution is

$$R(r) = \left(\frac{a}{r}\right)^{1/2}\left[A + B\ln\left(\frac{r}{a}\right)\right]. \tag{12.13}$$

The condition $R'(a) = 0$ gives $B = A/2$ if the constant in Equation (12.13) is named as written. Equivalently, one may absorb the common factor and write

$$R_2(r) = A_2\left(\frac{a}{r}\right)^{1/2}\left[\ln\left(\frac{r}{a}\right) + 2\right]. \tag{12.14}$$

The convenient surface normalisation $R_2(a) = 1$ corresponds to $A_2 = 1/2$.

The angular partner of the repeated radial degree is

$$P^m_{-1/2}(\cos\theta), \tag{12.15}$$

a degree-$-1/2$ conical function. The complementary potential may therefore be chosen as

$$V_2(r,\theta,\lambda) = aR_2(r)\sum_{m=0}^{\infty}\left[G^m_{-1/2}\cos(m\lambda) + H^m_{-1/2}\sin(m\lambda)\right]P^m_{-1/2}(\cos\theta). \tag{12.16}$$

The sine coefficient at $m = 0$ is omitted. Other continuations are possible because the second problem is incomplete; the repeated-root branch is the classical R-SCHA2D choice.

12.5 Complete one-surface representation

The classical R-SCHA2D potential is

$$V = V_1 + V_2, \tag{12.17}$$

with V_1 and V_2 given by Equations (12.9) and (12.16). On the observation sphere, it removes the implicit restriction $X(a,\theta_0,\lambda) = 0$ while preserving the prescribed radial derivative. The construction is therefore adapted to regional vector data on one quasi-constant-radius surface.

Remark 12.1 (Relation to three-dimensional R-SCHA). The degree-$-1/2$ radial form also appears when the Euler equation has a repeated characteristic root. This algebraic resemblance does not make R-SCHA2D the limit $r_{\mathrm{a}} \uparrow r_{\mathrm{b}}$ of the closed-shell R-SCHA spectrum. In the three-dimensional Neumann eigenproblem, the positive frequencies $p\pi/\ln(r_{\mathrm{b}}/r_{\mathrm{a}})$ diverge as the shell collapses, while the zero radial eigenvalue produces the separate constant family. The one-surface repeated-root mode is selected by the distinct incomplete boundary problem Equation (12.11).

12.6 Scope and mathematical status

The derivative-root Legendre system in V_1 is an ordinary Sturm–Liouville system for each fixed order m and has the corresponding weighted orthogonality and completeness properties. The complete union of that system with the single degree-$-1/2$ family is not orthogonal in the field-energy pairing because ∇V_1 and ∇V_2 are not generally orthogonal. The original work includes numerical tests of the representation [10]; the construction given here does not establish a Riesz-basis or frame theorem for the combined family.

The normal trace at one radius alone does not separate the two radial coefficients: for one mode it determines only $z = -(n+1)g^{\mathrm{i}} + ng^{\mathrm{e}}$. If the surface-potential coefficient $c = g^{\mathrm{i}} + g^{\mathrm{e}}$ is also known, the two coefficients are determined because the corresponding determinant is $2n+1 > 0$ on this real Neumann spectrum. In particular, complete exact tangential and normal field data on an open cap provide harmonic Cauchy data, up to a potential constant; their uniqueness must not be confused with stability of continuation from finite or noisy observations. The incomplete prescribed data above do not, by themselves, supply those Cauchy data or a global source separation.

Part III
Structural Identities

13 Orthogonality, completeness, and direct-sum structure

The preceding sections established three distinct orthogonality mechanisms: the non-integer Legendre functions are orthogonal in co-latitude for fixed azimuthal order, the trigonometric functions are orthogonal in longitude, and the radial functions $\{R_p\}_{p\geq 0}$ are orthogonal on $[r_{\mathrm{a}}, r_{\mathrm{b}}]$. These properties do not imply that every pair of angular factors occurring in R-SCHA is orthogonal. In particular, Mehler functions with different radial indices are generally not orthogonal on the cap interval. The orthogonality of the complete three-dimensional modes must therefore be derived from the combined radial–angular structure.

A further structural result is that the two complementary harmonic subpotentials V_1 and V_2 are orthogonal in the Dirichlet energy pairing. This follows directly from their boundary conditions and does not require the explicit special-function expansions. The result was introduced in the original R-SCHA theory and is proved below both at the function-space level and at the separated-mode level [9, 7].

13.1 Dirichlet-energy pairing and gauge freedom

For $u, v \in H^1(\Omega)$, define the Dirichlet-energy bilinear form

$$\mathcal{E}(u,v) = \int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}\Omega. \tag{13.1}$$

The associated energy seminorm is

$$|u|_{\mathcal{E}} = [\mathcal{E}(u,u)]^{1/2} = \left[\int_\Omega |\nabla u|^2 \, \mathrm{d}\Omega\right]^{1/2}. \tag{13.2}$$

Because

$$\mathcal{E}(u+c,v) = \mathcal{E}(u,v)$$

for every constant c, this is a seminorm on $H^1(\Omega)$. It becomes a norm either after imposing a non-empty Dirichlet boundary condition, after fixing a gauge such as

$$\int_\Omega u \, \mathrm{d}\Omega = 0, \tag{13.3}$$

or after passing to the quotient space

$$\dot{H}^1(\Omega) = H^1(\Omega)/\mathbb{R}. \tag{13.4}$$

The axisymmetric constant potential contained in the V_2 sequence is therefore a zero-energy gauge mode. It changes the scalar potential but not the magnetic field.

Define the harmonic energy space

$$\mathscr{H}(\Omega) = \left\{u \in H^1(\Omega) : \Delta u = 0 \text{ weakly in } \Omega\right\}. \tag{13.5}$$

The two boundary-adapted subspaces are

$$\mathscr{H}_1 = \left\{u \in \mathscr{H}(\Omega) : \operatorname{Tr} u = 0 \text{ on } \Gamma_{\theta_0}\right\}, \tag{13.6}$$

$$\mathscr{H}_2 = \left\{v \in \mathscr{H}(\Omega) : \frac{\partial v}{\partial n} = 0 \text{ on } \Gamma_{\mathrm{a}} \cup \Gamma_{\mathrm{b}}\right\}. \tag{13.7}$$

Normal derivatives in this definition are weak conormal traces, with the zero condition understood variationally on the two spherical faces. Thus $v \in \mathscr{H}_2$ satisfies $\mathcal{E}(v,w) = 0$ for every $w \in \mathcal{V}_0$ of Equation (4.19). For smooth functions this agrees with the facewise classical condition. The subpotentials satisfy $V_1 \in \mathscr{H}_1$ and $V_2 \in \mathscr{H}_2$.

13.2 Longitude orthogonality

For non-negative integers m, n, define

$$c_m = \pi\,(1 + \delta_{0m}). \tag{13.8}$$

Then

$$\int_0^{2\pi} \cos(m\lambda)\cos(n\lambda) \, \mathrm{d}\lambda = c_m \delta_{mn}, \tag{13.9a}$$

$$\int_0^{2\pi} \sin(m\lambda)\sin(n\lambda) \, \mathrm{d}\lambda = \pi\delta_{mn}, \qquad m,n \geq 1, \tag{13.9b}$$

$$\int_0^{2\pi} \cos(m\lambda)\sin(n\lambda) \, \mathrm{d}\lambda = 0. \tag{13.9c}$$

The derivative products satisfy

$$\int_0^{2\pi} \frac{\partial \cos(m\lambda)}{\partial\lambda}\frac{\partial \cos(n\lambda)}{\partial\lambda} \, \mathrm{d}\lambda = m^2 c_m \delta_{mn}, \tag{13.10a}$$

$$\int_0^{2\pi} \frac{\partial \sin(m\lambda)}{\partial\lambda}\frac{\partial \sin(n\lambda)}{\partial\lambda} \, \mathrm{d}\lambda = m^2 \pi\delta_{mn}, \qquad m,n \geq 1. \tag{13.10b}$$

Consequently, modes with different azimuthal orders, or with cosine and sine longitude factors, are orthogonal in both the L^2 and energy pairings.

13.3 Angular energy orthogonality of Legendre modes

Let

$$\widehat{P}_{km}(\theta) = \widehat{P}^m_{\nu_{km}}(\cos\theta) \tag{13.11}$$

denote the cap-unit-normalised eigenfunction defined in Equation (8.47). Then

$$\int_0^{\theta_0} \widehat{P}_{km}(\theta)\widehat{P}_{jm}(\theta)\sin\theta \ \mathrm{d}\theta = \delta_{kj}. \tag{13.12}$$

The Sturm–Liouville equation gives the stronger angular-energy identity

$$\begin{aligned}\int_0^{\theta_0} \left[\frac{\mathrm{d}\widehat{P}_{km}}{\mathrm{d}\theta}\frac{\mathrm{d}\widehat{P}_{jm}}{\mathrm{d}\theta} + \frac{m^2}{\sin^2\theta}\widehat{P}_{km}\widehat{P}_{jm}\right]\sin\theta \ \mathrm{d}\theta \\ = \lambda_{km}\delta_{kj}, \qquad \lambda_{km} = \nu_{km}(\nu_{km}+1).\end{aligned} \tag{13.13}$$

Indeed, multiply

$$\mathcal{L}_m\widehat{P}_{km} = \lambda_{km}\widehat{P}_{km}$$

by $\widehat{P}_{jm}\sin\theta$, integrate, and use the regular polar condition together with

$$\widehat{P}_{km}(\theta_0) = 0.$$

13.4 Energy factorisation for separated modes

Let

$$\Phi(r,\theta,\lambda) = r_{\mathrm{a}}U(r)A(\theta)L(\lambda), \qquad \Psi(r,\theta,\lambda) = r_{\mathrm{a}}W(r)B(\theta)M(\lambda). \tag{13.14}$$

Using the spherical gradient and volume element,

$$\begin{aligned}\mathcal{E}(\Phi,\Psi) = r_{\mathrm{a}}^2\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} r^2U'(r)W'(r)\ \mathrm{d}r \int_0^{\theta_0} A(\theta)B(\theta)\sin\theta \ \mathrm{d}\theta \int_0^{2\pi} L(\lambda)M(\lambda)\ \mathrm{d}\lambda \\ + r_{\mathrm{a}}^2\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} U(r)W(r)\ \mathrm{d}r \\ \times \int_0^{2\pi}\int_0^{\theta_0} \big[A'(\theta)B'(\theta)L(\lambda)M(\lambda) \\ + \frac{A(\theta)B(\theta)L'(\lambda)M'(\lambda)}{\sin^2\theta}\Big]\sin\theta \ \mathrm{d}\theta \ \mathrm{d}\lambda.\end{aligned} \tag{13.15}$$

This factorisation explains how angular, longitudinal, and radial orthogonality combine in three dimensions.

13.5 Energy orthogonality within the Legendre subspace

Define the internal and external radial factors

$$U^{\mathrm{Int}}_{km}(r) = \left(\frac{r_{\mathrm{a}}}{r}\right)^{\nu_{km}+1}, \tag{13.16a}$$

$$U^{\mathrm{Ext}}_{km}(r) = \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}}. \tag{13.16b}$$

For $\sigma \in \{\mathrm{c}, \mathrm{s}\}$, let

$$L_m^{\mathrm{c}}(\lambda) = \cos(m\lambda), \qquad L_m^{\mathrm{s}}(\lambda) = \sin(m\lambda), \tag{13.17}$$

where the sine mode is omitted for $m = 0$. The unit-angular-norm Legendre basis modes are

$$\Phi_{km}^{\mathrm{Int},\sigma} = r_{\mathrm{a}} U_{km}^{\mathrm{Int}}(r) \widehat{P}_{km}(\theta) L_m^{\sigma}(\lambda), \tag{13.18}$$

$$\Phi_{km}^{\mathrm{Ext},\sigma} = r_{\mathrm{a}} U_{km}^{\mathrm{Ext}}(r) \widehat{P}_{km}(\theta) L_m^{\sigma}(\lambda). \tag{13.19}$$

By Equations (13.9) and (13.13), two such modes are energy-orthogonal whenever their azimuthal orders, angular-root indices, or longitude parities differ. The only remaining possible coupling is between the internal and external modes carrying the same (k, m, σ). Their radial factors satisfy

$$\frac{\mathrm{d}U_{km}^{\mathrm{Int}}}{\mathrm{d}r} = -\frac{\nu_{km}+1}{r} U_{km}^{\mathrm{Int}}, \qquad \frac{\mathrm{d}U_{km}^{\mathrm{Ext}}}{\mathrm{d}r} = \frac{\nu_{km}}{r} U_{km}^{\mathrm{Ext}}. \tag{13.20}$$

Therefore,

$$\begin{aligned} r^2 \frac{\mathrm{d}U_{km}^{\mathrm{Int}}}{\mathrm{d}r} \frac{\mathrm{d}U_{km}^{\mathrm{Ext}}}{\mathrm{d}r} + \lambda_{km} U_{km}^{\mathrm{Int}} U_{km}^{\mathrm{Ext}} \\ = -\nu_{km}(\nu_{km}+1) U_{km}^{\mathrm{Int}} U_{km}^{\mathrm{Ext}} + \lambda_{km} U_{km}^{\mathrm{Int}} U_{km}^{\mathrm{Ext}} = 0. \end{aligned} \tag{13.21}$$

Hence,

$$\mathcal{E}\left(\Phi_{km}^{\mathrm{Int},\sigma}, \Phi_{km}^{\mathrm{Ext},\sigma}\right) = 0. \tag{13.22}$$

Theorem 13.1 (Energy orthogonality of the Legendre modes)**.** *The complete internal and external Legendre basis*

$$\left\{\Phi_{km}^{\mathrm{Int},\sigma}, \Phi_{km}^{\mathrm{Ext},\sigma}\right\}$$

is mutually orthogonal in the Dirichlet-energy pairing, provided the angular functions are selected by the same homogeneous lateral Dirichlet eigenproblem.

For later energy calculations, their squared energy norms are

$$\mathcal{E}\left(\Phi_{km}^{\mathrm{Int},\sigma}, \Phi_{km}^{\mathrm{Int},\sigma}\right) = \ell_m^{\sigma} r_{\mathrm{a}}^3 \left(\nu_{km}+1\right)\left[1 - \mathrm{e}^{-(2\nu_{km}+1)s}\right], \tag{13.23}$$

$$\mathcal{E}\left(\Phi_{km}^{\mathrm{Ext},\sigma}, \Phi_{km}^{\mathrm{Ext},\sigma}\right) = \ell_m^{\sigma} r_{\mathrm{a}}^3 \nu_{km}\left[\mathrm{e}^{(2\nu_{km}+1)s} - 1\right], \tag{13.24}$$

where

$$\ell_m^{\mathrm{c}} = c_m, \qquad \ell_m^{\mathrm{s}} = \pi \quad (m \geq 1). \tag{13.25}$$

These expressions assume cap-unit-normalised angular functions. A different angular normalisation multiplies the right-hand sides by the corresponding squared cap norm.

13.6 Energy orthogonality within the V_2 subspace

Let

$$A_{pm}(\theta) = \widetilde{K}_p^m(\theta), \qquad p \geq 1, \tag{13.26}$$

and let

$$A_{0m}(\theta) = \widetilde{P}_0^m(\theta). \tag{13.27}$$

The corresponding three-dimensional modes are

$$\Psi_{pm}^{\sigma} = r_{\mathrm{a}} R_p(r) A_{pm}(\theta) L_m^{\sigma}(\lambda), \qquad p = 0, 1, 2, \ldots. \tag{13.28}$$

For $p, q \geq 0$, radial Sturm–Liouville theory gives

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p R_q \, \mathrm{d}r = r_\mathrm{a} \delta_{pq}, \tag{13.29}$$

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} r^2 R_p' R_q' \, \mathrm{d}r = \mu_p r_\mathrm{a} \delta_{pq}, \tag{13.30}$$

where

$$\mu_0 = 0, \qquad \mu_p = \frac{1}{4} + \tau_p^2 \quad (p \geq 1). \tag{13.31}$$

The second identity follows by integration by parts:

$$\begin{aligned} \int_{r_\mathrm{a}}^{r_\mathrm{b}} r^2 R_p' R_q' \, \mathrm{d}r &= -\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_q \frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 R_p'\right) \mathrm{d}r \\ &= \mu_p \int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p R_q \, \mathrm{d}r. \end{aligned} \tag{13.32}$$

It follows from Equations (13.15), (13.29) and (13.30) that

$$\mathcal{E}\left(\Psi_{pm}^{\sigma}, \Psi_{qn}^{\rho}\right) = 0 \tag{13.33}$$

whenever

$$p \neq q, \qquad m \neq n, \qquad \text{or} \qquad \sigma \neq \rho.$$

This remains true even though

$$\int_0^{\theta_0} \widetilde{K}_p^m \widetilde{K}_q^m \sin\theta \, \mathrm{d}\theta$$

does not generally vanish for $p \neq q$: both radial factors in Equation (13.15) vanish separately by Equations (13.29) and (13.30).

For a single V_2 mode, the angular equation implies

$$\begin{aligned} &\int_0^{\theta_0} \left[\left(A_{pm}'(\theta)\right)^2 + \frac{m^2}{\sin^2\theta} A_{pm}^2(\theta)\right] \sin\theta \, \mathrm{d}\theta \\ &\qquad = \sin\theta_0 A_{pm}(\theta_0) A_{pm}'(\theta_0) - \mu_p \int_0^{\theta_0} A_{pm}^2(\theta) \sin\theta \, \mathrm{d}\theta. \end{aligned} \tag{13.34}$$

Since the angular functions are unit normalised and

$$\int R_p^2 \, \mathrm{d}r = r_\mathrm{a}, \qquad \int r^2 (R_p')^2 \, \mathrm{d}r = \mu_p r_\mathrm{a},$$

the two terms proportional to μ_p cancel in the three-dimensional energy. Therefore,

$$\mathcal{E}\left(\Psi_{pm}^{\sigma}, \Psi_{pm}^{\sigma}\right) = \ell_m^{\sigma} r_\mathrm{a}^3 \sin\theta_0 A_{pm}(\theta_0) A_{pm}'(\theta_0). \tag{13.35}$$

For $p = m = 0$, the angular and radial factors are both constant, so

$$\mathcal{E}\left(\Psi_{00}^{\mathrm{c}}, \Psi_{00}^{\mathrm{c}}\right) = 0. \tag{13.36}$$

13.7 Gradient orthogonality of V_1 and V_2

Theorem 13.2 (Gradient orthogonality)**.** *Let*

$$V_1 \in \mathscr{H}_1, \qquad V_2 \in \mathscr{H}_2.$$

Then

$$\mathcal{E}(V_1, V_2) = \int_\Omega \nabla V_1 \cdot \nabla V_2 \, \mathrm{d}\Omega = 0. \tag{13.37}$$

Proof. Since $V_1 \in \mathcal{V}_0$, the weak zero-Neumann formulation of V_2 gives $\mathcal{E}(V_1, V_2) = 0$ directly. For functions smooth enough to have classical boundary integrals, the same argument is Green's identity:

$$\int_\Omega \nabla V_1 \cdot \nabla V_2 \, \mathrm{d}\Omega = \int_{\partial\Omega} V_1 \frac{\partial V_2}{\partial n} \, \mathrm{d}S. \tag{13.38}$$

Using the boundary decomposition,

$$\begin{aligned}\mathcal{E}(V_1, V_2) = &\int_{\Gamma_{\theta_0}} V_1 \frac{1}{r}\frac{\partial V_2}{\partial \theta} \, \mathrm{d}S - \int_{\Gamma_\mathrm{a}} V_1 \frac{\partial V_2}{\partial r} \, \mathrm{d}S \\ &+ \int_{\Gamma_\mathrm{b}} V_1 \frac{\partial V_2}{\partial r} \, \mathrm{d}S.\end{aligned} \tag{13.39}$$

The three surface measures are

$$\mathrm{d}S_{\theta_0} = r \sin\theta_0 \, \mathrm{d}r \, \mathrm{d}\lambda, \tag{13.40a}$$

$$\mathrm{d}S_\mathrm{a} = r_\mathrm{a}^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda, \tag{13.40b}$$

$$\mathrm{d}S_\mathrm{b} = r_\mathrm{b}^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda. \tag{13.40c}$$

On Γ_{θ_0},

$$V_1 = 0,$$

and on $\Gamma_\mathrm{a} \cup \Gamma_\mathrm{b}$,

$$\frac{\partial V_2}{\partial r} = 0.$$

Every term in Equation (13.39) therefore vanishes. □

This proof is independent of truncation, special-function normalisation, and coefficient values. It applies to the exact harmonic subspaces as soon as the traces and normal derivatives are defined.

The same argument applies to every pair of separated modes:

$$\mathcal{E}\big(\Phi_{km}^{\alpha,\sigma}, \Psi_{pn}^{\rho}\big) = 0, \qquad \alpha \in \{\mathrm{Int}, \mathrm{Ext}\}. \tag{13.41}$$

The Legendre factor has zero lateral trace and the V_2 radial factor has zero normal derivative on both spherical faces.

13.8 Orthogonal direct-sum theorem

Theorem 13.3 (Boundary-adapted direct-sum decomposition)**.** *Every $V \in \mathscr{H}(\Omega)$ has the unique decomposition*

$$V = V_1 + V_2, \qquad V_1 \in \mathscr{H}_1, \qquad V_2 \in \mathscr{H}_2, \tag{13.42}$$

and

$$\mathscr{H}_1 \cap \mathscr{H}_2 = \{0\}. \tag{13.43}$$

In the quotient by constant potentials,

$$\mathscr{H}(\Omega)/\mathbb{R} = \mathscr{H}_1 \overset{\mathcal{E}}{\oplus} (\mathscr{H}_2/\mathbb{R})\,. \tag{13.44}$$

Proof. The Dirichlet form is coercive on $\mathcal{V}_0$ by the Poincaré inequality. Lax–Milgram therefore gives a unique $V_1 \in \mathcal{V}_0$ such that

$$\mathcal{E}(V_1, w) = \mathcal{E}(V, w) \qquad (w \in \mathcal{V}_0).$$

Testing with $w \in H_0^1(\Omega)$ shows that V_1 is harmonic, because V is harmonic. For $V_2 = V - V_1$, the defining equality gives $\mathcal{E}(V_2, w) = 0$ on $\mathcal{V}_0$, precisely the weak zero-Neumann condition on the spherical faces. Hence $V_1 \in \mathscr{H}_1$ and $V_2 \in \mathscr{H}_2$, and they are energy-orthogonal. An element of their intersection has zero energy by the same identity and is a constant; its zero lateral trace makes it zero. Only the constant contained in $\mathscr{H}_2$ is removed in the quotient. □

A direct consequence is the Pythagorean identity

$$\int_\Omega |\nabla V|^2 \,\mathrm{d}\Omega = \int_\Omega |\nabla V_1|^2 \,\mathrm{d}\Omega + \int_\Omega |\nabla V_2|^2 \,\mathrm{d}\Omega. \tag{13.45}$$

This identity is the basis of the three-dimensional R-SCHA energy spectrum derived in Section 17.

13.9 Completeness of the component systems

The completeness results established previously may be summarised as follows.

1. For every fixed integer order m, the cap Legendre eigenfunctions
$$\left\{P^m_{\nu_{km}}(\cos\theta)\right\}_{k=m}^{\infty}$$
form a complete orthogonal system in $L^2((0,\theta_0), \sin\theta \,\mathrm{d}\theta)$.
2. The Fourier functions
$$1, \quad \cos(m\lambda), \quad \sin(m\lambda), \qquad m \geq 1,$$
form a complete orthogonal system in $L^2([0, 2\pi])$.
3. The radial functions
$$\{R_p\}_{p=0}^{\infty}$$
form a complete orthogonal system in $L^2([r_\mathrm{a}, r_\mathrm{b}], \mathrm{d}r)$.

To obtain completeness in the harmonic energy spaces, componentwise L^2 completeness must also be connected to the energy quadratic forms. Write $L^\mathrm{c}_m(\lambda) = \cos(m\lambda)$ and $L^\mathrm{s}_m(\lambda) = \sin(m\lambda)$, retaining only the cosine parity for $m = 0$. In the sums below, σ denotes the retained parity and $\ell^\sigma_m = \int_0^{2\pi} (L^\sigma_m)^2 \,\mathrm{d}\lambda$ is its longitude norm: 2π for the constant mode and π otherwise. For $u \in \mathscr{H}_1$, expand in the cap-unit Dirichlet angular basis:

$$u = \sum_{k,m,\sigma} f^\sigma_{km}(r) \widehat{P}_{km}(\theta) L^\sigma_m(\lambda).$$

Parseval's identity for the angular Dirichlet form and for radial derivatives gives

$$\mathcal{E}(u,u) = \sum_{k,m,\sigma} \ell^\sigma_m \int_{r_\mathrm{a}}^{r_\mathrm{b}} \left(r^2 |(f^\sigma_{km})'|^2 + \lambda_{km} |f^\sigma_{km}|^2\right) \mathrm{d}r. \tag{13.46}$$

Projecting the weak Laplace equation onto an angular mode yields $(r^2 f')' - \lambda_{km} f = 0$. Thus every coefficient is a linear combination of $(r_\mathrm{a}/r)^{\nu_{km}+1}$ and $(r/r_\mathrm{a})^{\nu_{km}}$. The non-negative summands in Equation (13.46) have vanishing tails, proving energy convergence of the internal/external expansion.

For $u \in \mathscr{H}_2$, instead expand in the radial Neumann and longitude bases:

$$u = \sum_{p,m,\sigma} a^\sigma_{pm}(\theta) R_p(r) L^\sigma_m(\lambda).$$

The spectral theorem for the radial Neumann quadratic form, whose form domain is $H^1((r_\mathrm{a}, r_\mathrm{b}))$, gives

$$\mathcal{E}(u,u) = r_\mathrm{a} \sum_{p,m,\sigma} \ell^\sigma_m \int_0^{\theta_0} \left[|(a^\sigma_{pm})'|^2 + \left(\mu_p + \frac{m^2}{\sin^2\theta} \right) |a^\sigma_{pm}|^2 \right] \sin\theta \, \mathrm{d}\theta. \tag{13.47}$$

The weak Laplace equation and radial zero-Neumann condition imply $(\sin\theta\, a')' - (\mu_p \sin\theta + m^2/\sin\theta) a = 0$. Finite energy excludes the singular polar branch, so each coefficient is proportional to the regular function A_{pm}. Again the non-negative energy tail tends to zero. For $p = m = 0$ the regular solution is constant; its coefficient is fixed separately by a gauge. These arguments establish energy completeness of the separated modes, rather than inferring it solely from L^2 completeness of their factors.

13.10 Finite truncation and exact orthogonality

The orthogonality identities above are properties of the analytical basis functions. Any finite subset inherits the same exact pairwise orthogonality. Thus truncation changes the approximation space but does not alter the Dirichlet inner products among the retained modes.

Let $\{\phi_j\}_{j \in J_N}$ be a finite energy-orthogonal set with

$$\mathcal{E}(\phi_i, \phi_j) = e_j \delta_{ij}. \tag{13.48}$$

For

$$V_N = \sum_{j \in J_N} c_j \phi_j, \tag{13.49}$$

one has the exact finite Parseval identity

$$\mathcal{E}(V_N, V_N) = \sum_{j \in J_N} e_j |c_j|^2. \tag{13.50}$$

The only zero-energy mode is the additive constant, which is removed by the gauge quotient or by fixing one potential value.

14 Differential and tensor calculus for R-SCHA

This section develops the first- and second-order differential tensors of the R-SCHA scalar potential. The construction uses covariant tensor notation together with the local north–east–down Cartesian convention adopted throughout this handbook. Coordinate rotations between the geocentric Cartesian, spherical, local Cartesian, and spherical-cap frames are postponed to Section 15; here the emphasis is the covariant Hessian in spherical coordinates and its explicit action on the four R-SCHA basis families.

14.1 Covariant and contravariant bases

Let

$$\boldsymbol{x} = \boldsymbol{x}(u^1, u^2, u^3)$$

parameterise a three-dimensional Euclidean domain. The covariant coordinate basis is

$$\boldsymbol{g}_p = \frac{\partial \boldsymbol{x}}{\partial u^p}, \qquad p = 1, 2, 3, \tag{14.1}$$

and the contravariant basis $\{\boldsymbol{g}^p\}$ is defined by

$$\boldsymbol{g}^p \cdot \boldsymbol{g}_q = \delta^p_q. \tag{14.2}$$

The covariant and contravariant metric components are

$$g_{pq} = \boldsymbol{g}_p \cdot \boldsymbol{g}_q, \tag{14.3a}$$
$$g^{pq} = \boldsymbol{g}^p \cdot \boldsymbol{g}^q, \tag{14.3b}$$

with

$$g^{ps} g_{sq} = \delta^p_q. \tag{14.4}$$

Indices are raised and lowered by

$$a_p = g_{pq} a^q, \qquad a^p = g^{pq} a_q. \tag{14.5}$$

In an orthogonal coordinate system, define the scale factors

$$h_p = \sqrt{g_{pp}} = \|\boldsymbol{g}_p\|, \tag{14.6}$$

where no summation is intended in Equation (14.6). The associated orthonormal physical basis is

$$\boldsymbol{e}_{\hat{p}} = \frac{\boldsymbol{g}_p}{h_p}. \tag{14.7}$$

For a vector

$$\boldsymbol{a} = a^p \boldsymbol{g}_p,$$

its physical components are

$$a_{\hat{p}} = h_p a^p = \frac{a_p}{h_p}. \tag{14.8}$$

An nth-order tensor can be written with covariant, contravariant, or mixed components, provided that each component index is paired with the corresponding dual basis vector. For example,

$$\boldsymbol{T} = T^{ijk} \boldsymbol{g}_i \boldsymbol{g}_j \boldsymbol{g}_k = T_{ijk} \boldsymbol{g}^i \boldsymbol{g}^j \boldsymbol{g}^k = T^i{}_j{}^k \boldsymbol{g}_i \boldsymbol{g}^j \boldsymbol{g}_k. \tag{14.9}$$

Einstein summation is used whenever an index is repeated once in an upper position and once in a lower position.

14.2 Tensor transformation law

Let $u^{p'} = u^{p'}(u^1, u^2, u^3)$ be a smooth change of coordinates with a non-singular Jacobian. A tensor of type (r, s) transforms by one Jacobian factor for each contravariant index and one inverse-Jacobian factor for each covariant index. For a mixed third-order tensor,

$$T^{i'}{}_{j'}{}^{k'} = \frac{\partial u^{i'}}{\partial u^i} \frac{\partial u^j}{\partial u^{j'}} \frac{\partial u^{k'}}{\partial u^k} T^i{}_j{}^k. \tag{14.10}$$

In orthogonal coordinate systems, the transformation of physical components also contains the appropriate ratios of scale factors. The resulting orthogonal-frame transformation matrices are developed explicitly in Section 15.

14.3 Gradient and covariant Hessian

The gradient operator is

$$\nabla = \boldsymbol{g}^p \frac{\partial}{\partial u^p}. \tag{14.11}$$

For a scalar potential V,

$$\nabla V = V_{,p} \boldsymbol{g}^p, \qquad V_{,p} := \frac{\partial V}{\partial u^p}. \tag{14.12}$$

In an orthogonal physical basis,

$$\nabla V = \sum_{p=1}^{3} \boldsymbol{e}_{\hat{p}} \frac{1}{h_p} \frac{\partial V}{\partial u^p}. \tag{14.13}$$

The Christoffel symbols of the second kind are

$$\Gamma^s_{pq} = \frac{1}{2} g^{st} \left(\frac{\partial g_{qt}}{\partial u^p} + \frac{\partial g_{pt}}{\partial u^q} - \frac{\partial g_{pq}}{\partial u^t} \right). \tag{14.14}$$

The covariant Hessian of a scalar field is

$$V_{;pq} = \frac{\partial^2 V}{\partial u^p \partial u^q} - \Gamma^s_{pq} \frac{\partial V}{\partial u^s}. \tag{14.15}$$

Its physical components in an orthogonal coordinate system are

$$V_{;\hat{p}\hat{q}} = \frac{1}{h_p h_q} V_{;pq}. \tag{14.16}$$

Because

$$\Gamma^s_{pq} = \Gamma^s_{qp}$$

for the Levi–Civita connection and mixed partial derivatives commute for a twice continuously differentiable scalar field,

$$V_{;\hat{p}\hat{q}} = V_{;\hat{q}\hat{p}}. \tag{14.17}$$

14.4 Direct derivation in the spherical orthonormal frame

The Christoffel-symbol calculation below can also be obtained directly from the derivatives of the spherical unit vectors. The non-zero derivatives are

$$\frac{\partial \boldsymbol{e}_r}{\partial \theta} = \boldsymbol{e}_\theta, \qquad \frac{\partial \boldsymbol{e}_\theta}{\partial \theta} = -\boldsymbol{e}_r, \tag{14.18a}$$

$$\frac{\partial \boldsymbol{e}_r}{\partial \lambda} = \sin\theta\, \boldsymbol{e}_\lambda, \qquad \frac{\partial \boldsymbol{e}_\theta}{\partial \lambda} = \cos\theta\, \boldsymbol{e}_\lambda, \tag{14.18b}$$

$$\frac{\partial \boldsymbol{e}_\lambda}{\partial \lambda} = -\sin\theta\, \boldsymbol{e}_r - \cos\theta\, \boldsymbol{e}_\theta. \tag{14.18c}$$

Write

$$\nabla V = V_{,r} \boldsymbol{e}_r + \frac{V_{,\theta}}{r} \boldsymbol{e}_\theta + \frac{V_{,\lambda}}{r \sin\theta} \boldsymbol{e}_\lambda. \tag{14.19}$$

The Hessian component in directions $\boldsymbol{e}_p$ and $\boldsymbol{e}_q$ is

$$V^*_{;pq} = \boldsymbol{e}_p \cdot \left[(\boldsymbol{e}_q \cdot \nabla) \nabla V \right]. \tag{14.20}$$

For example,

$$\begin{aligned} V^*_{;r\theta} &= \boldsymbol{e}_r \cdot \left[\frac{1}{r} \frac{\partial}{\partial \theta} \left(V_{,r} \boldsymbol{e}_r + \frac{V_{,\theta}}{r} \boldsymbol{e}_\theta + \frac{V_{,\lambda}}{r \sin\theta} \boldsymbol{e}_\lambda \right) \right] \\ &= \frac{1}{r} V_{,r\theta} - \frac{1}{r^2} V_{,\theta}. \end{aligned} \tag{14.21}$$

The second term is produced by $\partial_\theta \boldsymbol{e}_\theta = -\boldsymbol{e}_r$; it would be missed by naively differentiating only the scalar physical components. Similarly,

$$\begin{aligned} V^*_{;\theta\lambda} &= \boldsymbol{e}_\theta \cdot \left[\frac{1}{r \sin\theta} \frac{\partial}{\partial \lambda} \nabla V \right] \\ &= \frac{1}{r^2 \sin\theta} V_{,\theta\lambda} - \frac{\cos\theta}{r^2 \sin^2\theta} V_{,\lambda}. \end{aligned} \tag{14.22}$$

Applying Equation (14.18) to all ordered pairs produces the six physical components listed in Equation (14.24). This derivation makes clear why ordinary second partial derivatives are not tensor components in a curvilinear basis.

14.5 Christoffel symbols in spherical coordinates

For

$$(u^1, u^2, u^3) = (r, \theta, \lambda),$$

the scale factors are

$$h_r = 1, \qquad h_\theta = r, \qquad h_\lambda = r \sin\theta.$$

The non-zero Christoffel symbols are

$$\Gamma^r_{\theta\theta} = -r, \qquad \Gamma^r_{\lambda\lambda} = -r \sin^2\theta, \tag{14.23a}$$

$$\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{1}{r}, \qquad \Gamma^\theta_{\lambda\lambda} = -\sin\theta\cos\theta, \tag{14.23b}$$

$$\Gamma^\lambda_{r\lambda} = \Gamma^\lambda_{\lambda r} = \frac{1}{r}, \qquad \Gamma^\lambda_{\theta\lambda} = \Gamma^\lambda_{\lambda\theta} = \cot\theta. \tag{14.23c}$$

Substitution into Equations (14.15) and (14.16) gives the six independent physical Hessian components:

$$V^*_{;rr} = V_{,rr}, \tag{14.24a}$$

$$V^*_{;r\theta} = V^*_{;\theta r} = \frac{1}{r} V_{,r\theta} - \frac{1}{r^2} V_{,\theta}, \tag{14.24b}$$

$$V^*_{;r\lambda} = V^*_{;\lambda r} = \frac{1}{r \sin\theta} V_{,r\lambda} - \frac{1}{r^2 \sin\theta} V_{,\lambda}, \tag{14.24c}$$

$$V^*_{;\theta\theta} = \frac{1}{r^2} V_{,\theta\theta} + \frac{1}{r} V_{,r}, \tag{14.24d}$$

$$V^*_{;\theta\lambda} = V^*_{;\lambda\theta} = \frac{1}{r^2 \sin\theta} V_{,\theta\lambda} - \frac{\cos\theta}{r^2 \sin^2\theta} V_{,\lambda}, \tag{14.24e}$$

$$V^*_{;\lambda\lambda} = \frac{1}{r^2 \sin^2\theta} V_{,\lambda\lambda} + \frac{1}{r} V_{,r} + \frac{\cot\theta}{r^2} V_{,\theta}. \tag{14.24f}$$

The trace of the physical Hessian is the spherical Laplacian:

$$V^*_{;rr} + V^*_{;\theta\theta} + V^*_{;\lambda\lambda} = \Delta V. \tag{14.25}$$

14.6 Local north–east–down convention

Let

$$(\boldsymbol{e}_x, \boldsymbol{e}_y, \boldsymbol{e}_z)$$

denote the local north–east–down orthonormal frame. Relative to the spherical orthonormal frame,

$$\boldsymbol{e}_x = -\boldsymbol{e}_\theta, \qquad \boldsymbol{e}_y = \boldsymbol{e}_\lambda, \qquad \boldsymbol{e}_z = -\boldsymbol{e}_r. \tag{14.26}$$

The magnetic field is defined by

$$\boldsymbol{B} = -\nabla V. \tag{14.27}$$

Its local physical components are therefore

$$B_x = \frac{1}{r} V_{,\theta}, \tag{14.28a}$$

$$B_y = -\frac{1}{r \sin\theta} V_{,\lambda}, \tag{14.28b}$$

$$B_z = V_{,r}. \tag{14.28c}$$

The local second-order magnetic gradient tensor is defined as

$$\boldsymbol{T} = -\operatorname{Hess} V \tag{14.29}$$

and expressed in the symmetric matrix

$$\boldsymbol{T} = \begin{bmatrix} T_{xx} & T_{xy} & T_{zx} \\ T_{xy} & T_{yy} & T_{zy} \\ T_{zx} & T_{zy} & T_{zz} \end{bmatrix}. \tag{14.30}$$

The ordering in Equation (14.30) uses T_{zx} and T_{zy} for the vertical cross-components, consistently with the local north–east–down convention of this handbook.

Using Equations (14.26) and (14.24), the six independent local components are

$$T_{xx} = -\frac{1}{r^2} V_{,\theta\theta} - \frac{1}{r} V_{,r}, \tag{14.31a}$$

$$T_{xy} = \frac{1}{r^2 \sin\theta} V_{,\theta\lambda} - \frac{\cos\theta}{r^2 \sin^2\theta} V_{,\lambda}, \tag{14.31b}$$

$$T_{yy} = -\frac{1}{r^2 \sin^2\theta} V_{,\lambda\lambda} - \frac{1}{r} V_{,r} - \frac{\cot\theta}{r^2} V_{,\theta}, \tag{14.31c}$$

$$T_{zx} = -\frac{1}{r} V_{,r\theta} + \frac{1}{r^2} V_{,\theta}, \tag{14.31d}$$

$$T_{zy} = \frac{1}{r \sin\theta} V_{,r\lambda} - \frac{1}{r^2 \sin\theta} V_{,\lambda}, \tag{14.31e}$$

$$T_{zz} = -V_{,rr}. \tag{14.31f}$$

If V is harmonic, then

$$T_{xx} + T_{yy} + T_{zz} = -\Delta V = 0. \tag{14.32}$$

Thus, symmetry reduces the nine entries to six independent entries, and the trace-free condition reduces the number of independent scalar components to five.

14.7 Generic separated-mode differentiation

The four R-SCHA families can be differentiated with one common template. Consider the real separated mode

$$\mathcal{V}_m(r,\theta,\lambda) = r_{\mathrm{a}} U(r) A(\theta) C_m(\lambda), \tag{14.33}$$

where

$$C_m(\lambda) = c_m \cos(m\lambda) + s_m \sin(m\lambda), \tag{14.34a}$$
$$S_m(\lambda) = c_m \sin(m\lambda) - s_m \cos(m\lambda). \tag{14.34b}$$

Then

$$\frac{\partial C_m}{\partial \lambda} = -m S_m, \qquad \frac{\partial S_m}{\partial \lambda} = m C_m. \tag{14.35}$$

A prime on U denotes differentiation with respect to r, whereas a prime on A denotes differentiation with respect to θ.

Substitution into Equation (14.28) gives

$$B_x[\mathcal{V}_m] = \frac{r_{\mathrm{a}}}{r} U A' C_m, \tag{14.36a}$$
$$B_y[\mathcal{V}_m] = \frac{r_{\mathrm{a}}}{r \sin\theta} m U A S_m, \tag{14.36b}$$
$$B_z[\mathcal{V}_m] = r_{\mathrm{a}} U' A C_m. \tag{14.36c}$$

The second-order components are

$$T_{xx}[\mathcal{V}_m] = -r_{\mathrm{a}} C_m \left[\frac{U}{r^2} A'' + \frac{U'}{r} A \right], \tag{14.37a}$$
$$T_{xy}[\mathcal{V}_m] = \frac{r_{\mathrm{a}} m U S_m}{r^2} \left[\frac{\cos\theta}{\sin^2\theta} A - \frac{1}{\sin\theta} A' \right], \tag{14.37b}$$
$$T_{yy}[\mathcal{V}_m] = r_{\mathrm{a}} C_m \left[\frac{m^2 U}{r^2 \sin^2\theta} A - \frac{U'}{r} A - \frac{U \cot\theta}{r^2} A' \right], \tag{14.37c}$$
$$T_{zx}[\mathcal{V}_m] = r_{\mathrm{a}} C_m A' \left[\frac{U}{r^2} - \frac{U'}{r} \right], \tag{14.37d}$$
$$T_{zy}[\mathcal{V}_m] = \frac{r_{\mathrm{a}} m S_m A}{\sin\theta} \left[\frac{U}{r^2} - \frac{U'}{r} \right], \tag{14.37e}$$
$$T_{zz}[\mathcal{V}_m] = -r_{\mathrm{a}} U'' A C_m. \tag{14.37f}$$

Equations (14.36) and (14.37) are the differentiation kernel for all four R-SCHA series.

14.8 Coefficient combinations for the four families

For the Legendre families, retain the cosine combinations defined in Equation (7.5) and introduce

$$S^{\mathrm{Int}}_{km}(\lambda) = g^{\mathrm{Int}}_{km} \sin(m\lambda) - h^{\mathrm{Int}}_{km} \cos(m\lambda), \tag{14.38a}$$
$$S^{\mathrm{Ext}}_{km}(\lambda) = g^{\mathrm{Ext}}_{km} \sin(m\lambda) - h^{\mathrm{Ext}}_{km} \cos(m\lambda). \tag{14.38b}$$

For the Mehler and constant families, define

$$C^{\mathrm{Meh}}_{pm}(\lambda) = G^m_p \cos(m\lambda) + H^m_p \sin(m\lambda), \tag{14.39a}$$
$$S^{\mathrm{Meh}}_{pm}(\lambda) = G^m_p \sin(m\lambda) - H^m_p \cos(m\lambda), \tag{14.39b}$$
$$C^{\mathrm{Cons}}_{0m}(\lambda) = G^m_0 \cos(m\lambda) + H^m_0 \sin(m\lambda), \tag{14.39c}$$
$$S^{\mathrm{Cons}}_{0m}(\lambda) = G^m_0 \sin(m\lambda) - H^m_0 \cos(m\lambda). \tag{14.39d}$$

Every S combination multiplied by m vanishes at $m = 0$.

For compactness, write

$$P_{km}(\theta) := P^m_{\nu_{km}}(\cos\theta), \qquad K_{pm}(\theta) := K^m_p(\theta), \qquad C_m(\theta) := \mathcal{C}_m(\theta). \tag{14.40}$$

14.9 Internal Legendre field and tensor

Define

$$\rho^{\mathrm{Int}}_{km}(r) = \left(\frac{r_\mathrm{a}}{r}\right)^{\nu_{km}+2}, \tag{14.41a}$$

$$\eta^{\mathrm{Int}}_{km}(r) = \frac{1}{r_\mathrm{a}}\left(\frac{r_\mathrm{a}}{r}\right)^{\nu_{km}+3}. \tag{14.41b}$$

Substitution of

$$U(r) = \left(\frac{r_\mathrm{a}}{r}\right)^{\nu_{km}+1}$$

into the generic kernel gives

$$B^{\mathrm{Int}}_x = \sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} \rho^{\mathrm{Int}}_{km} C^{\mathrm{Int}}_{km} P'_{km}, \tag{14.42a}$$

$$B^{\mathrm{Int}}_y = \sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} \rho^{\mathrm{Int}}_{km} \frac{m}{\sin\theta} S^{\mathrm{Int}}_{km} P_{km}, \tag{14.42b}$$

$$B^{\mathrm{Int}}_z = -\sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} (\nu_{km}+1)\, \rho^{\mathrm{Int}}_{km} C^{\mathrm{Int}}_{km} P_{km}. \tag{14.42c}$$

The corresponding second-order components are

$$T^{\mathrm{Int}}_{xx} = \sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} \eta^{\mathrm{Int}}_{km} C^{\mathrm{Int}}_{km} \left[(\nu_{km}+1)\, P_{km} - P''_{km}\right], \tag{14.43a}$$

$$T^{\mathrm{Int}}_{xy} = \sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} \eta^{\mathrm{Int}}_{km} m S^{\mathrm{Int}}_{km} \left[\frac{\cos\theta}{\sin^2\theta} P_{km} - \frac{1}{\sin\theta} P'_{km}\right], \tag{14.43b}$$

$$T^{\mathrm{Int}}_{yy} = \sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} \eta^{\mathrm{Int}}_{km} C^{\mathrm{Int}}_{km} \left[\frac{m^2}{\sin^2\theta} P_{km} + (\nu_{km}+1)\, P_{km} - \cot\theta P'_{km}\right], \tag{14.43c}$$

$$T^{\mathrm{Int}}_{zx} = \sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} (\nu_{km}+2)\, \eta^{\mathrm{Int}}_{km} C^{\mathrm{Int}}_{km} P'_{km}, \tag{14.43d}$$

$$T^{\mathrm{Int}}_{zy} = \sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} (\nu_{km}+2)\, \eta^{\mathrm{Int}}_{km} \frac{m}{\sin\theta} S^{\mathrm{Int}}_{km} P_{km}, \tag{14.43e}$$

$$T^{\mathrm{Int}}_{zz} = -\sum_{k=0}^{N_{\mathrm{Int}}}\sum_{m=0}^{k} (\nu_{km}+1)\,(\nu_{km}+2)\, \eta^{\mathrm{Int}}_{km} C^{\mathrm{Int}}_{km} P_{km}. \tag{14.43f}$$

14.10 External Legendre field and tensor

Define

$$\rho_{km}^{\mathrm{Ext}}(r) = \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}-1}, \tag{14.44a}$$

$$\eta_{km}^{\mathrm{Ext}}(r) = \frac{1}{r_{\mathrm{a}}}\left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}-2}. \tag{14.44b}$$

For

$$U(r) = \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}},$$

the field components are

$$B_x^{\mathrm{Ext}} = \sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} \rho_{km}^{\mathrm{Ext}} C_{km}^{\mathrm{Ext}} P'_{km}, \tag{14.45a}$$

$$B_y^{\mathrm{Ext}} = \sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} \rho_{km}^{\mathrm{Ext}} \frac{m}{\sin\theta} S_{km}^{\mathrm{Ext}} P_{km}, \tag{14.45b}$$

$$B_z^{\mathrm{Ext}} = \sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} \nu_{km}\rho_{km}^{\mathrm{Ext}} C_{km}^{\mathrm{Ext}} P_{km}. \tag{14.45c}$$

The tensor components are

$$T_{xx}^{\mathrm{Ext}} = -\sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} \eta_{km}^{\mathrm{Ext}} C_{km}^{\mathrm{Ext}} \left[P''_{km} + \nu_{km} P_{km}\right], \tag{14.46a}$$

$$T_{xy}^{\mathrm{Ext}} = \sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} \eta_{km}^{\mathrm{Ext}} m S_{km}^{\mathrm{Ext}} \left[\frac{\cos\theta}{\sin^2\theta} P_{km} - \frac{1}{\sin\theta} P'_{km}\right], \tag{14.46b}$$

$$T_{yy}^{\mathrm{Ext}} = \sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} \eta_{km}^{\mathrm{Ext}} C_{km}^{\mathrm{Ext}} \left[\frac{m^2}{\sin^2\theta} P_{km} - \nu_{km} P_{km} - \cot\theta P'_{km}\right], \tag{14.46c}$$

$$T_{zx}^{\mathrm{Ext}} = \sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} (1-\nu_{km})\, \eta_{km}^{\mathrm{Ext}} C_{km}^{\mathrm{Ext}} P'_{km}, \tag{14.46d}$$

$$T_{zy}^{\mathrm{Ext}} = \sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} (1-\nu_{km})\, \eta_{km}^{\mathrm{Ext}} \frac{m}{\sin\theta} S_{km}^{\mathrm{Ext}} P_{km}, \tag{14.46e}$$

$$T_{zz}^{\mathrm{Ext}} = -\sum_{k=0}^{N_{\mathrm{Ext}}}\sum_{m=0}^{k} \nu_{km}\,(\nu_{km}-1)\, \eta_{km}^{\mathrm{Ext}} C_{km}^{\mathrm{Ext}} P_{km}. \tag{14.46f}$$

14.11 Mehler field and tensor

For the Mehler contribution, set

$$U(r) = R_p(r), \qquad A(\theta) = K_{pm}(\theta).$$

The first-order components are

$$B_x^{\mathrm{Meh}} = \frac{r_\mathrm{a}}{r} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} R_p C_{pm}^{\mathrm{Meh}} K'_{pm}, \tag{14.47a}$$

$$B_y^{\mathrm{Meh}} = \frac{r_\mathrm{a}}{r} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} R_p \frac{m}{\sin\theta} S_{pm}^{\mathrm{Meh}} K_{pm}, \tag{14.47b}$$

$$B_z^{\mathrm{Meh}} = r_\mathrm{a} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} R'_p C_{pm}^{\mathrm{Meh}} K_{pm}. \tag{14.47c}$$

The tensor components are

$$T_{xx}^{\mathrm{Meh}} = -r_\mathrm{a} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} C_{pm}^{\mathrm{Meh}} \left[\frac{R_p}{r^2} K''_{pm} + \frac{R'_p}{r} K_{pm} \right], \tag{14.48a}$$

$$T_{xy}^{\mathrm{Meh}} = r_\mathrm{a} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} \frac{m R_p}{r^2} S_{pm}^{\mathrm{Meh}} \left[\frac{\cos\theta}{\sin^2\theta} K_{pm} - \frac{1}{\sin\theta} K'_{pm} \right], \tag{14.48b}$$

$$T_{yy}^{\mathrm{Meh}} = r_\mathrm{a} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} C_{pm}^{\mathrm{Meh}} \left[\frac{m^2 R_p}{r^2 \sin^2\theta} K_{pm} - \frac{R'_p}{r} K_{pm} - \frac{R_p \cot\theta}{r^2} K'_{pm} \right], \tag{14.48c}$$

$$T_{zx}^{\mathrm{Meh}} = r_\mathrm{a} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} C_{pm}^{\mathrm{Meh}} K'_{pm} \left[\frac{R_p}{r^2} - \frac{R'_p}{r} \right], \tag{14.48d}$$

$$T_{zy}^{\mathrm{Meh}} = r_\mathrm{a} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} \frac{m}{\sin\theta} S_{pm}^{\mathrm{Meh}} K_{pm} \left[\frac{R_p}{r^2} - \frac{R'_p}{r} \right], \tag{14.48e}$$

$$T_{zz}^{\mathrm{Meh}} = -r_\mathrm{a} \sum_{p=1}^{P_{\max}} \sum_{m=0}^{M_{\mathrm{Meh}}} R''_p C_{pm}^{\mathrm{Meh}} K_{pm}. \tag{14.48f}$$

14.12 Constant-function field and tensor

For the constant contribution,

$$U(r) = R_0, \qquad U'(r) = U''(r) = 0, \qquad A(\theta) = C_m(\theta).$$

The first-order components are

$$B_x^{\mathrm{Cons}} = \frac{r_\mathrm{a} R_0}{r} \sum_{m=0}^{M_{\max}} C_{0m}^{\mathrm{Cons}} C'_m, \tag{14.49a}$$

$$B_y^{\mathrm{Cons}} = \frac{r_\mathrm{a} R_0}{r} \sum_{m=0}^{M_{\max}} \frac{m}{\sin\theta} S_{0m}^{\mathrm{Cons}} C_m, \tag{14.49b}$$

$$B_z^{\mathrm{Cons}} = 0. \tag{14.49c}$$

The tensor components are

$$T_{xx}^{\mathrm{Cons}} = -\frac{r_{\mathrm{a}} R_0}{r^2} \sum_{m=0}^{M_{\max}} C_{0m}^{\mathrm{Cons}} C_m'', \tag{14.50a}$$

$$T_{xy}^{\mathrm{Cons}} = \frac{r_{\mathrm{a}} R_0}{r^2} \sum_{m=0}^{M_{\max}} m S_{0m}^{\mathrm{Cons}} \left[\frac{\cos\theta}{\sin^2\theta} C_m - \frac{1}{\sin\theta} C_m' \right], \tag{14.50b}$$

$$T_{yy}^{\mathrm{Cons}} = \frac{r_{\mathrm{a}} R_0}{r^2} \sum_{m=0}^{M_{\max}} C_{0m}^{\mathrm{Cons}} \left[\frac{m^2}{\sin^2\theta} C_m - \cot\theta C_m' \right], \tag{14.50c}$$

$$T_{zx}^{\mathrm{Cons}} = \frac{r_{\mathrm{a}} R_0}{r^2} \sum_{m=0}^{M_{\max}} C_{0m}^{\mathrm{Cons}} C_m', \tag{14.50d}$$

$$T_{zy}^{\mathrm{Cons}} = \frac{r_{\mathrm{a}} R_0}{r^2} \sum_{m=0}^{M_{\max}} \frac{m}{\sin\theta} S_{0m}^{\mathrm{Cons}} C_m, \tag{14.50e}$$

$$T_{zz}^{\mathrm{Cons}} = 0. \tag{14.50f}$$

14.13 Assembly of the complete field and tensor

By linearity,

$$\boldsymbol{B} = \boldsymbol{B}^{\mathrm{Int}} + \boldsymbol{B}^{\mathrm{Ext}} + \boldsymbol{B}^{\mathrm{Meh}} + \boldsymbol{B}^{\mathrm{Cons}}, \tag{14.51}$$

and

$$\boldsymbol{T} = \boldsymbol{T}^{\mathrm{Int}} + \boldsymbol{T}^{\mathrm{Ext}} + \boldsymbol{T}^{\mathrm{Meh}} + \boldsymbol{T}^{\mathrm{Cons}}. \tag{14.52}$$

Equations (14.42), (14.45), (14.47) and (14.49) give all three local magnetic components, while Equations (14.43), (14.46), (14.48) and (14.50) give the six independent second-order tensor components.

14.14 Modewise trace-free verification

For the generic separated mode Equation (14.33), the trace of the local magnetic tensor is

$$T_{xx} + T_{yy} + T_{zz} = -r_{\mathrm{a}} C_m \Bigg\{ \left[U'' + \frac{2}{r} U' \right] A + \frac{U}{r^2} \left[A'' + \cot\theta A' - \frac{m^2}{\sin^2\theta} A \right] \Bigg\}. \tag{14.53}$$

The radial and angular separated equations imply

$$U'' + \frac{2}{r} U' = \frac{\nu(\nu+1)}{r^2} U \tag{14.54}$$

and

$$A'' + \cot\theta A' - \frac{m^2}{\sin^2\theta} A = -\nu(\nu+1) A. \tag{14.55}$$

Therefore,

$$T_{xx} + T_{yy} + T_{zz} = 0 \tag{14.56}$$

for every R-SCHA basis mode separately.

14.15 Polar-axis regularity

Several expressions above contain

$$\frac{mA}{\sin\theta}, \qquad \frac{m^2A}{\sin^2\theta}, \qquad \cot\theta A', \qquad \frac{mA'}{\sin\theta}.$$

These factors appear singular when written separately, but the regular angular functions satisfy

$$A(\theta) = \mathcal{O}(\theta^m) \qquad \text{as } \theta \to 0^+. \tag{14.57}$$

Here regularity means the regular solution of the angular differential equation, with a local expansion $A(\theta) = \theta^m(a_0 + a_2\theta^2 + \cdots)$; the bound $A = \mathcal{O}(\theta^m)$ alone would not control its derivatives. For these regular modes, the combined vector and tensor remain bounded as the axis is approached along a fixed meridian. The spherical and NED basis vectors themselves depend on longitude at the pole, so their component limits need not be independent of that approach. A unique value on the axis is obtained by transforming the complete field or tensor to a fixed Cartesian frame before taking the limit; smoothness of the regular mode then gives a direction-independent Cartesian limit. Direct evaluation at or extremely close to the polar axis should use the series and limiting formulae developed in Sections 8 to 10, rather than evaluating the individual singular-looking factors independently.

15 Coordinate systems and tensor transformations

The analytical R-SCHA expressions are evaluated naturally in a spherical coordinate system whose polar axis passes through the centre of the modelling cap. Magnetic observations and regional products, however, are commonly expressed in a global geocentric frame or in a local north–east–down frame. A complete R-SCHA formulation therefore requires transformations among four frames:

1. the global geocentric Cartesian frame;
2. the global spherical frame;
3. the local north–east–down frame;
4. the cap-centred Cartesian and spherical frames.

The transformations may be expressed through Jacobian matrices and metric scale factors. In this section, the relations are written directly for physical components in orthonormal frames. This removes scale-factor ambiguity and reduces every vector and second-rank tensor transformation to multiplication by an orthogonal matrix.

15.1 Column-vector convention

All vector components in this section are column vectors. Suppose that $\{\boldsymbol{e}_1^{\mathrm{old}}, \boldsymbol{e}_2^{\mathrm{old}}, \boldsymbol{e}_3^{\mathrm{old}}\}$ and $\{\boldsymbol{e}_1^{\mathrm{new}}, \boldsymbol{e}_2^{\mathrm{new}}, \boldsymbol{e}_3^{\mathrm{new}}\}$ are two right-handed orthonormal frames. Define the direction-cosine matrix

$$\boldsymbol{R}_{\mathrm{new}\leftarrow\mathrm{old}} = \left[\boldsymbol{e}_i^{\mathrm{new}} \cdot \boldsymbol{e}_j^{\mathrm{old}}\right]_{i,j=1}^{3}. \tag{15.1}$$

Then

$$\boldsymbol{v}_{\mathrm{new}} = \boldsymbol{R}_{\mathrm{new}\leftarrow\mathrm{old}}\boldsymbol{v}_{\mathrm{old}}, \tag{15.2}$$

and a symmetric second-rank tensor transforms as

$$\boldsymbol{T}_{\text{new}} = \boldsymbol{R}_{\text{new}\leftarrow\text{old}}\boldsymbol{T}_{\text{old}}\boldsymbol{R}_{\text{new}\leftarrow\text{old}}^{\mathsf{T}}. \tag{15.3}$$

Because the frames are orthonormal,

$$\boldsymbol{R}_{\text{new}\leftarrow\text{old}}^{-1} = \boldsymbol{R}_{\text{new}\leftarrow\text{old}}^{\mathsf{T}}. \tag{15.4}$$

Remark 15.1 (Active and passive rotations). Equation (15.2) is a passive component transformation: the geometrical vector is unchanged while its components are reported in a different frame. The active rotation carrying the old basis onto the new basis is represented in the old frame by $\boldsymbol{R}_{\text{new}\leftarrow\text{old}}^{\mathsf{T}}$. This specifies which active rotation is meant; an arbitrary active rotation must instead be defined by its own action on the vectors.

15.2 General six-component transformation

Let

$$\boldsymbol{R} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}, \tag{15.5}$$

and define the six-component representation of a symmetric tensor by

$$\operatorname{vec}_6(\boldsymbol{T}) = \begin{bmatrix} T_{11} & T_{22} & T_{33} & T_{12} & T_{13} & T_{23} \end{bmatrix}^{\mathsf{T}}. \tag{15.6}$$

Then

$$\operatorname{vec}_6(\boldsymbol{T}_{\text{new}}) = \boldsymbol{M}(\boldsymbol{R})\operatorname{vec}_6(\boldsymbol{T}_{\text{old}}), \tag{15.7}$$

where

$$\boldsymbol{M}(\boldsymbol{R}) = \begin{bmatrix} r_{11}^2 & r_{12}^2 & r_{13}^2 & 2r_{11}r_{12} & 2r_{11}r_{13} & 2r_{12}r_{13} \\ r_{21}^2 & r_{22}^2 & r_{23}^2 & 2r_{21}r_{22} & 2r_{21}r_{23} & 2r_{22}r_{23} \\ r_{31}^2 & r_{32}^2 & r_{33}^2 & 2r_{31}r_{32} & 2r_{31}r_{33} & 2r_{32}r_{33} \\ r_{11}r_{21} & r_{12}r_{22} & r_{13}r_{23} & r_{11}r_{22}+r_{12}r_{21} & r_{11}r_{23}+r_{13}r_{21} & r_{12}r_{23}+r_{13}r_{22} \\ r_{11}r_{31} & r_{12}r_{32} & r_{13}r_{33} & r_{11}r_{32}+r_{12}r_{31} & r_{11}r_{33}+r_{13}r_{31} & r_{12}r_{33}+r_{13}r_{32} \\ r_{21}r_{31} & r_{22}r_{32} & r_{23}r_{33} & r_{21}r_{32}+r_{22}r_{31} & r_{21}r_{33}+r_{23}r_{31} & r_{22}r_{33}+r_{23}r_{32} \end{bmatrix} \tag{15.8}$$

Equation (15.8) is the compact matrix equivalent of writing all six second-order component relations separately.

15.3 Global geocentric Cartesian frame

Let

$$(X, Y, Z)$$

denote the global geocentric Cartesian coordinates. The origin is the centre of the reference sphere, the positive X axis intersects the equator at the prime meridian, the positive Z axis points towards the north pole, and the positive Y axis completes a right-handed system. The associated constant orthonormal basis is

$$\{\boldsymbol{e}_X, \boldsymbol{e}_Y, \boldsymbol{e}_Z\}. \tag{15.9}$$

For spherical coordinates (r, θ, λ),

$$X = r \sin\theta \cos\lambda, \tag{15.10a}$$
$$Y = r \sin\theta \sin\lambda, \tag{15.10b}$$
$$Z = r \cos\theta. \tag{15.10c}$$

The inverse transformation is

$$r = \sqrt{X^2 + Y^2 + Z^2}, \tag{15.11a}$$
$$\theta = \operatorname{atan2}\left(\sqrt{X^2 + Y^2}, Z\right), \tag{15.11b}$$
$$\lambda = \operatorname{atan2}(Y, X). \tag{15.11c}$$

The equivalent expression

$$\theta = \arccos(Z/r)$$

is valid away from round-off violations of $|Z/r| \leq 1$.

Remark 15.2 (Quadrant-safe longitude). The two-argument function $\operatorname{atan2}(Y, X)$ must be used instead of $\arctan(Y/X)$. The latter loses quadrant information and is undefined on the plane $X = 0$.

15.4 Global spherical orthonormal frame

The global spherical orthonormal basis is

$$\boldsymbol{e}_r = \sin\theta\cos\lambda\, \boldsymbol{e}_X + \sin\theta\sin\lambda\, \boldsymbol{e}_Y + \cos\theta\, \boldsymbol{e}_Z, \tag{15.12a}$$
$$\boldsymbol{e}_\theta = \cos\theta\cos\lambda\, \boldsymbol{e}_X + \cos\theta\sin\lambda\, \boldsymbol{e}_Y - \sin\theta\, \boldsymbol{e}_Z, \tag{15.12b}$$
$$\boldsymbol{e}_\lambda = -\sin\lambda\, \boldsymbol{e}_X + \cos\lambda\, \boldsymbol{e}_Y. \tag{15.12c}$$

Define

$$\boldsymbol{Q}(\theta, \lambda) = \begin{bmatrix} \sin\theta\cos\lambda & \cos\theta\cos\lambda & -\sin\lambda \\ \sin\theta\sin\lambda & \cos\theta\sin\lambda & \cos\lambda \\ \cos\theta & -\sin\theta & 0 \end{bmatrix}. \tag{15.13}$$

Its columns are the spherical unit vectors expressed in the geocentric Cartesian frame:

$$\begin{bmatrix} \boldsymbol{e}_r & \boldsymbol{e}_\theta & \boldsymbol{e}_\lambda \end{bmatrix} = \begin{bmatrix} \boldsymbol{e}_X & \boldsymbol{e}_Y & \boldsymbol{e}_Z \end{bmatrix} \boldsymbol{Q}(\theta, \lambda). \tag{15.14}$$

Therefore,

$$\boldsymbol{Q}^\mathsf{T}\boldsymbol{Q} = \boldsymbol{Q}\boldsymbol{Q}^\mathsf{T} = \boldsymbol{I}. \tag{15.15}$$

Remark 15.3 (Inverse basis relation). The inverse of Equation (15.12) is obtained by transposition of Equation (15.13). In particular, the coefficient of $\boldsymbol{e}_\theta$ in $\boldsymbol{e}_X$ is $\cos\theta\cos\lambda$, not $\cos\theta\sin\lambda$.

15.5 Vectors and tensors between spherical and geocentric frames

Let

$$\boldsymbol{v}_{\text{sph}} = \begin{bmatrix} v_r & v_\theta & v_\lambda \end{bmatrix}^\mathsf{T}, \qquad \boldsymbol{v}_{\text{ECEF}} = \begin{bmatrix} v_X & v_Y & v_Z \end{bmatrix}^\mathsf{T}. \tag{15.16}$$

Then

$$\boldsymbol{v}_{\text{ECEF}} = \boldsymbol{Q}\boldsymbol{v}_{\text{sph}}, \tag{15.17a}$$
$$\boldsymbol{v}_{\text{sph}} = \boldsymbol{Q}^\mathsf{T}\boldsymbol{v}_{\text{ECEF}}. \tag{15.17b}$$

Explicitly,

$$v_X = v_r \sin\theta \cos\lambda + v_\theta \cos\theta \cos\lambda - v_\lambda \sin\lambda, \tag{15.18a}$$
$$v_Y = v_r \sin\theta \sin\lambda + v_\theta \cos\theta \sin\lambda + v_\lambda \cos\lambda, \tag{15.18b}$$
$$v_Z = v_r \cos\theta - v_\theta \sin\theta. \tag{15.18c}$$

The inverse relations are

$$v_r = v_X \sin\theta \cos\lambda + v_Y \sin\theta \sin\lambda + v_Z \cos\theta, \tag{15.19a}$$
$$v_\theta = v_X \cos\theta \cos\lambda + v_Y \cos\theta \sin\lambda - v_Z \sin\theta, \tag{15.19b}$$
$$v_\lambda = -v_X \sin\lambda + v_Y \cos\lambda. \tag{15.19c}$$

For a symmetric second-rank tensor,

$$\boldsymbol{T}_{\mathrm{ECEF}} = \boldsymbol{Q}\boldsymbol{T}_{\mathrm{sph}}\boldsymbol{Q}^{\mathsf{T}}, \tag{15.20a}$$
$$\boldsymbol{T}_{\mathrm{sph}} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{T}_{\mathrm{ECEF}}\boldsymbol{Q}. \tag{15.20b}$$

All six explicit component formulae follow by substituting $\boldsymbol{R} = \boldsymbol{Q}^{\mathsf{T}}$ or $\boldsymbol{R} = \boldsymbol{Q}$ into Equation (15.8).

15.6 Local north–east–down frame

At the point (r, θ, λ), define

$$\boldsymbol{e}_x = -\boldsymbol{e}_\theta, \qquad \boldsymbol{e}_y = \boldsymbol{e}_\lambda, \qquad \boldsymbol{e}_z = -\boldsymbol{e}_r, \tag{15.21}$$

where x, y, and z point north, east, and down, respectively. The basis matrix whose columns are these unit vectors in the ECEF frame is

$$\boldsymbol{C}_{\mathrm{NED}}(\theta, \lambda) = \begin{bmatrix} -\cos\theta\cos\lambda & -\sin\lambda & -\sin\theta\cos\lambda \\ -\cos\theta\sin\lambda & \cos\lambda & -\sin\theta\sin\lambda \\ \sin\theta & 0 & -\cos\theta \end{bmatrix}. \tag{15.22}$$

Thus,

$$\boldsymbol{v}_{\mathrm{ECEF}} = \boldsymbol{C}_{\mathrm{NED}}\boldsymbol{v}_{\mathrm{NED}}, \tag{15.23a}$$
$$\boldsymbol{v}_{\mathrm{NED}} = \boldsymbol{C}_{\mathrm{NED}}^{\mathsf{T}}\boldsymbol{v}_{\mathrm{ECEF}}, \tag{15.23b}$$
$$\boldsymbol{T}_{\mathrm{ECEF}} = \boldsymbol{C}_{\mathrm{NED}}\boldsymbol{T}_{\mathrm{NED}}\boldsymbol{C}_{\mathrm{NED}}^{\mathsf{T}}, \tag{15.23c}$$
$$\boldsymbol{T}_{\mathrm{NED}} = \boldsymbol{C}_{\mathrm{NED}}^{\mathsf{T}}\boldsymbol{T}_{\mathrm{ECEF}}\boldsymbol{C}_{\mathrm{NED}}. \tag{15.23d}$$

The direct spherical-to-NED transformation is

$$\boldsymbol{v}_{\mathrm{NED}} = \boldsymbol{S}\boldsymbol{v}_{\mathrm{sph}}, \qquad \boldsymbol{S} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}. \tag{15.24}$$

Hence,

$$v_x = -v_\theta, \qquad v_y = v_\lambda, \qquad v_z = -v_r. \tag{15.25}$$

For second-rank tensors,

$$\boldsymbol{T}_{\mathrm{NED}} = \boldsymbol{S}\boldsymbol{T}_{\mathrm{sph}}\boldsymbol{S}^{\mathsf{T}}. \tag{15.26}$$

The independent component relations are

$$T_{xx} = T_{\theta\theta}, \qquad T_{xy} = -T_{\theta\lambda}, \qquad T_{yy} = T_{\lambda\lambda}, \tag{15.27a}$$
$$T_{zx} = T_{r\theta}, \qquad T_{zy} = -T_{r\lambda}, \qquad T_{zz} = T_{rr}. \tag{15.27b}$$

These are the component relations used in Section 14.

15.7 Definition of the cap-centred Cartesian frame

Let the centre of the spherical cap have global spherical coordinates

$$(r, \theta, \lambda) = (r, \theta_{\mathrm{c}}, \lambda_{\mathrm{c}}), \tag{15.28}$$

where θ_{c} is the global co-latitude and λ_{c} is the global longitude of the cap centre. The cap-centred Cartesian frame is obtained by two passive rotations:

1. rotate the global Cartesian coordinate axes about the Z axis by λ_{c};
2. rotate the resulting axes about the intermediate Y' axis by θ_{c}.

The first rotation is

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \boldsymbol{R}_z(\lambda_{\mathrm{c}}) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}, \qquad \boldsymbol{R}_z(\lambda_{\mathrm{c}}) = \begin{bmatrix} \cos\lambda_{\mathrm{c}} & \sin\lambda_{\mathrm{c}} & 0 \\ -\sin\lambda_{\mathrm{c}} & \cos\lambda_{\mathrm{c}} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{15.29}$$

The second rotation is

$$\begin{bmatrix} X_{\mathrm{c}} \\ Y_{\mathrm{c}} \\ Z_{\mathrm{c}} \end{bmatrix} = \boldsymbol{R}_y(\theta_{\mathrm{c}}) \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix}, \qquad \boldsymbol{R}_y(\theta_{\mathrm{c}}) = \begin{bmatrix} \cos\theta_{\mathrm{c}} & 0 & -\sin\theta_{\mathrm{c}} \\ 0 & 1 & 0 \\ \sin\theta_{\mathrm{c}} & 0 & \cos\theta_{\mathrm{c}} \end{bmatrix}. \tag{15.30}$$

The combined global-to-cap Cartesian rotation is

$$\boldsymbol{A}_{\mathrm{c}} = \boldsymbol{R}_y(\theta_{\mathrm{c}}) \boldsymbol{R}_z(\lambda_{\mathrm{c}}), \tag{15.31}$$

or explicitly,

$$\boldsymbol{A}_{\mathrm{c}} = \begin{bmatrix} \cos\theta_{\mathrm{c}}\cos\lambda_{\mathrm{c}} & \cos\theta_{\mathrm{c}}\sin\lambda_{\mathrm{c}} & -\sin\theta_{\mathrm{c}} \\ -\sin\lambda_{\mathrm{c}} & \cos\lambda_{\mathrm{c}} & 0 \\ \sin\theta_{\mathrm{c}}\cos\lambda_{\mathrm{c}} & \sin\theta_{\mathrm{c}}\sin\lambda_{\mathrm{c}} & \cos\theta_{\mathrm{c}} \end{bmatrix}. \tag{15.32}$$

Therefore,

$$\boldsymbol{x}_{\mathrm{c}} = \boldsymbol{A}_{\mathrm{c}} \boldsymbol{x}_{\mathrm{g}}, \tag{15.33a}$$

$$\boldsymbol{x}_{\mathrm{g}} = \boldsymbol{A}_{\mathrm{c}}^{\mathsf{T}} \boldsymbol{x}_{\mathrm{c}}. \tag{15.33b}$$

The matrix is orthogonal and proper:

$$\boldsymbol{A}_{\mathrm{c}} \boldsymbol{A}_{\mathrm{c}}^{\mathsf{T}} = \boldsymbol{I}, \qquad \det \boldsymbol{A}_{\mathrm{c}} = 1. \tag{15.34}$$

The global unit vector pointing towards the cap centre is mapped to the positive cap Z_{c} axis:

$$\boldsymbol{A}_{\mathrm{c}} \begin{bmatrix} \sin\theta_{\mathrm{c}}\cos\lambda_{\mathrm{c}} \\ \sin\theta_{\mathrm{c}}\sin\lambda_{\mathrm{c}} \\ \cos\theta_{\mathrm{c}} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \tag{15.35}$$

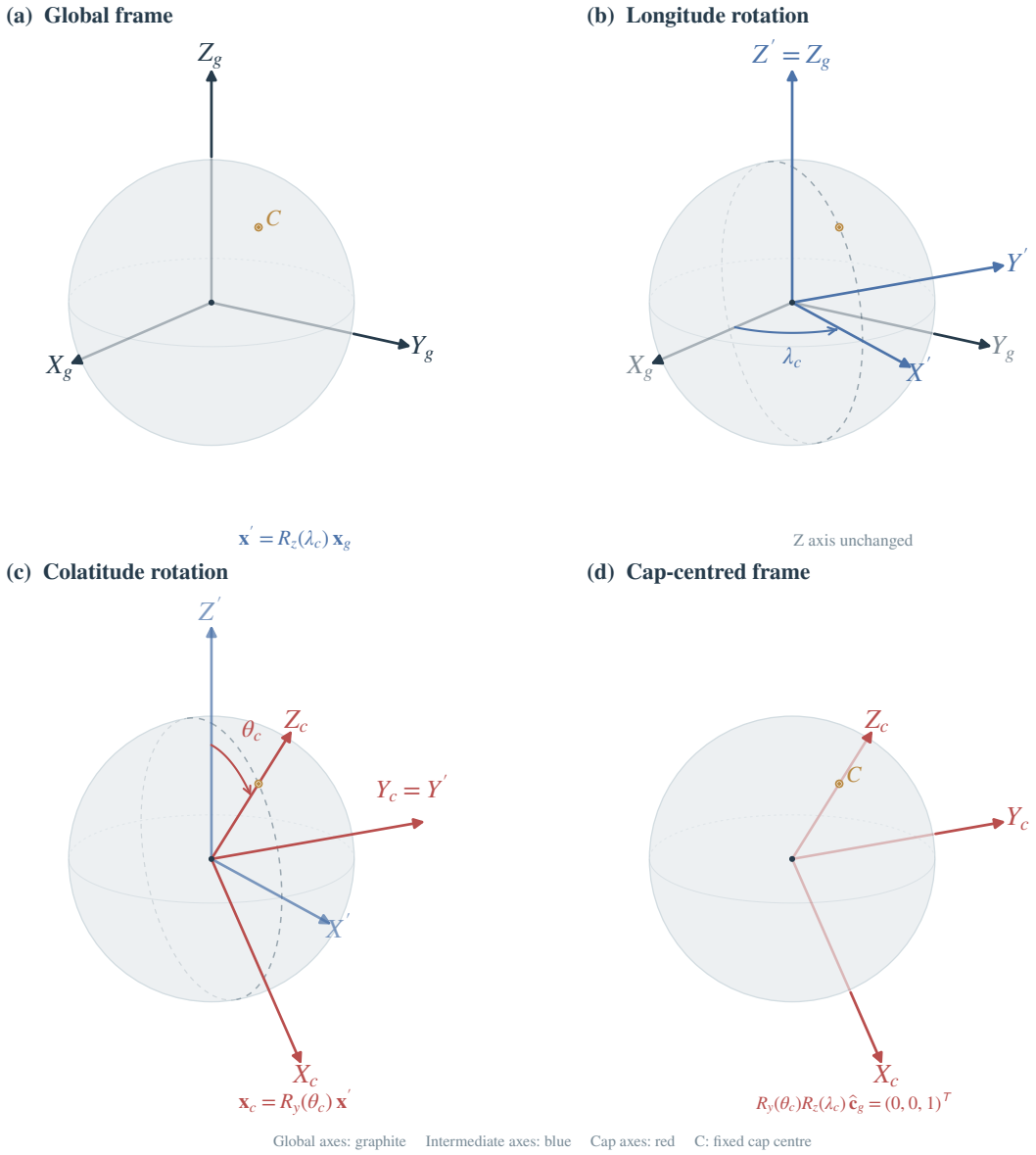


Figure 10: Two passive rotations displayed in four stages: (a) the global frame; (b) longitude rotation about Z_g by λ_c; (c) colatitude rotation about Y' by θ_c; and (d) the final cap-centred frame. Global, intermediate, and final axes are shown in graphite, blue, and red, respectively. The fixed cap centre C is mapped to the positive Z_c axis, with $Y_c = Y'$.

15.8 Global spherical coordinates to cap spherical coordinates

Let

$$(r, \theta', \lambda')$$

denote the spherical coordinates associated with $(X_\mathrm{c}, Y_\mathrm{c}, Z_\mathrm{c})$. The radial coordinate is unchanged:

$$r' = r. \tag{15.36}$$

Using Equations (15.10) and (15.32), one obtains

$$\sin\theta' \cos\lambda' = \cos\theta_\mathrm{c} \sin\theta \cos(\lambda - \lambda_\mathrm{c}) - \sin\theta_\mathrm{c} \cos\theta, \tag{15.37a}$$

$$\sin\theta' \sin\lambda' = \sin\theta \sin(\lambda - \lambda_\mathrm{c}), \tag{15.37b}$$

$$\cos\theta' = \sin\theta_\mathrm{c} \sin\theta \cos(\lambda - \lambda_\mathrm{c}) + \cos\theta_\mathrm{c} \cos\theta. \tag{15.37c}$$

A quadrant-complete definition is

$$\theta' = \mathrm{atan2}\left(\sqrt{X_\mathrm{c}^2 + Y_\mathrm{c}^2}, Z_\mathrm{c}\right), \tag{15.38a}$$

$$\lambda' = \mathrm{atan2}\left(Y_\mathrm{c}, X_\mathrm{c}\right). \tag{15.38b}$$

At the exact cap centre,

$$\theta' = 0,$$

and the cap longitude is geometrically indeterminate. Any fixed longitude convention represents the same point; for definiteness one may set $\lambda' = 0$ there.

15.9 Vector and tensor rotation between global and cap Cartesian frames

The physical vector and tensor components transform as

$$\boldsymbol{v}_\mathrm{c} = \boldsymbol{A}_\mathrm{c} \boldsymbol{v}_\mathrm{g}, \tag{15.39a}$$

$$\boldsymbol{v}_\mathrm{g} = \boldsymbol{A}_\mathrm{c}^\mathsf{T} \boldsymbol{v}_\mathrm{c}, \tag{15.39b}$$

$$\boldsymbol{T}_\mathrm{c} = \boldsymbol{A}_\mathrm{c} \boldsymbol{T}_\mathrm{g} \boldsymbol{A}_\mathrm{c}^\mathsf{T}, \tag{15.39c}$$

$$\boldsymbol{T}_\mathrm{g} = \boldsymbol{A}_\mathrm{c}^\mathsf{T} \boldsymbol{T}_\mathrm{c} \boldsymbol{A}_\mathrm{c}. \tag{15.39d}$$

The individual longitude and co-latitude component relations are obtained from Equation (15.8) using $\boldsymbol{R}_z(\lambda_\mathrm{c})$ and $\boldsymbol{R}_y(\theta_\mathrm{c})$, respectively.

15.10 Direct transformation between global and cap spherical frames

Let

$$\boldsymbol{Q}_\mathrm{g} = \boldsymbol{Q}(\theta, \lambda), \qquad \boldsymbol{Q}_\mathrm{c} = \boldsymbol{Q}(\theta', \lambda'). \tag{15.40}$$

The direct rotation between global and cap spherical physical components is

$$\boldsymbol{R}_{\text{cap-sph}\leftarrow\text{global-sph}} = \boldsymbol{Q}_\mathrm{c}^\mathsf{T} \boldsymbol{A}_\mathrm{c} \boldsymbol{Q}_\mathrm{g}. \tag{15.41}$$

Thus,

$$\boldsymbol{v}_\text{cap-sph} = \boldsymbol{R}_{\text{cap-sph}\leftarrow\text{global-sph}} \boldsymbol{v}_\text{global-sph}, \tag{15.42a}$$

$$\boldsymbol{T}_\text{cap-sph} = \boldsymbol{R}_{\text{cap-sph}\leftarrow\text{global-sph}} \boldsymbol{T}_\text{global-sph} \boldsymbol{R}_{\text{cap-sph}\leftarrow\text{global-sph}}^\mathsf{T}. \tag{15.42b}$$

The inverse transformation uses the transpose of Equation (15.41).

15.11 Direct transformation between global and cap NED frames

Let

$$\boldsymbol{C}_{\mathrm{g}} = \boldsymbol{C}_{\mathrm{NED}}(\theta, \lambda), \qquad \boldsymbol{C}_{\mathrm{c}} = \boldsymbol{C}_{\mathrm{NED}}(\theta', \lambda'). \tag{15.43}$$

Then

$$\boldsymbol{R}_{\text{cap-NED}\leftarrow\text{global-NED}} = \boldsymbol{C}_{\mathrm{c}}^{\mathsf{T}} \boldsymbol{A}_{\mathrm{c}} \boldsymbol{C}_{\mathrm{g}}. \tag{15.44}$$

Accordingly,

$$\boldsymbol{v}_{\text{cap-NED}} = \boldsymbol{R}_{\text{cap-NED}\leftarrow\text{global-NED}} \boldsymbol{v}_{\text{global-NED}}, \tag{15.45a}$$

$$\boldsymbol{T}_{\text{cap-NED}} = \boldsymbol{R}_{\text{cap-NED}\leftarrow\text{global-NED}} \boldsymbol{T}_{\text{global-NED}} \boldsymbol{R}_{\text{cap-NED}\leftarrow\text{global-NED}}^{\mathsf{T}}. \tag{15.45b}$$

This composite rotation implements the conversion chain

$$\begin{aligned} \text{global NED} &\longrightarrow \text{global spherical} \longrightarrow \text{global ECEF} \\ &\longrightarrow \text{cap ECEF} \longrightarrow \text{cap spherical} \longrightarrow \text{cap NED}. \end{aligned} \tag{15.46}$$

15.12 Transformation invariants

For every orthogonal matrix $\boldsymbol{R}$,

$$\operatorname{tr}\left(\boldsymbol{R}\boldsymbol{T}\boldsymbol{R}^{\mathsf{T}}\right) = \operatorname{tr}\boldsymbol{T}, \tag{15.47a}$$

$$\det\left(\boldsymbol{R}\boldsymbol{T}\boldsymbol{R}^{\mathsf{T}}\right) = \det\boldsymbol{T}, \tag{15.47b}$$

$$\left\|\boldsymbol{R}\boldsymbol{T}\boldsymbol{R}^{\mathsf{T}}\right\|_{\mathrm{F}} = \|\boldsymbol{T}\|_{\mathrm{F}}, \tag{15.47c}$$

$$\operatorname{spec}\left(\boldsymbol{R}\boldsymbol{T}\boldsymbol{R}^{\mathsf{T}}\right) = \operatorname{spec}(\boldsymbol{T}). \tag{15.47d}$$

Therefore, symmetry, the harmonic trace-free condition, determinant, Frobenius norm, and tensor eigenvalues are unchanged by any of the frame rotations above.

16 Solenoidal property and magnetic-flux identities

In a source-free region, the magnetic field must be solenoidal. For an R-SCHA model this property follows immediately from the harmonicity of the scalar potential, but its boundary interpretation is more informative: the flux through the inner spherical boundary, the outer spherical boundary, and the lateral conical boundary must balance exactly. The balance can be verified by expanding six boundary integrals for V_1 and V_2. Here the result is first established by the divergence theorem and is then resolved into explicit boundary-component and mode-by-mode identities.

Throughout this section,

$$\boldsymbol{B} = -\nabla V, \qquad \Delta V = 0 \quad \text{in } \Omega. \tag{16.1}$$

16.1 Pointwise and weak divergence-free conditions

For a twice continuously differentiable harmonic potential,

$$\nabla \cdot \boldsymbol{B} = -\nabla \cdot \nabla V = -\Delta V = 0 \qquad \text{pointwise in } \Omega. \tag{16.2}$$

Thus, every exact R-SCHA basis function and every finite linear combination of basis functions is divergence-free in the interior of the spherical-cap shell.

The same statement holds weakly. If

$$V \in H^1(\Omega)$$

is weakly harmonic, then for every $\varphi \in C_c^\infty(\Omega)$,

$$\begin{aligned}\langle \nabla \cdot \boldsymbol{B}, \varphi \rangle &= -\int_\Omega \boldsymbol{B} \cdot \nabla\varphi \, \mathrm{d}\Omega \\ &= \int_\Omega \nabla V \cdot \nabla\varphi \, \mathrm{d}\Omega = 0. \end{aligned} \tag{16.3}$$

Hence,

$$\nabla \cdot \boldsymbol{B} = 0 \tag{16.4}$$

in the sense of distributions.

The local magnetic-gradient tensor defined in Equation (14.29) satisfies

$$\operatorname{tr} \boldsymbol{T} = -\Delta V = \nabla \cdot \boldsymbol{B}. \tag{16.5}$$

Therefore, the solenoidal condition and the trace-free condition Equation (14.32) are equivalent descriptions of the same differential constraint.

16.2 Integral flux theorem

Assume first that $V \in C^2(\Omega) \cap C^1(\overline{\Omega})$, so that the classical divergence theorem applies. Then

$$\int_{\partial\Omega} \boldsymbol{B} \cdot \boldsymbol{n} \, \mathrm{d}S = \int_\Omega \nabla \cdot \boldsymbol{B} \, \mathrm{d}\Omega = 0. \tag{16.6}$$

Theorem 16.1 (Zero net flux through the closed spherical-cap shell)**.** *Let V be harmonic in Ω and sufficiently regular for its normal derivative to be defined on $\partial\Omega$. Then the magnetic field $\boldsymbol{B} = -\nabla V$ satisfies*

$$\Phi_{\partial\Omega}[V] := \int_{\partial\Omega} \boldsymbol{B} \cdot \boldsymbol{n} \, \mathrm{d}S = 0. \tag{16.7}$$

For weak solutions, the same result follows from the normal-trace theorem for $H(\mathrm{div}; \Omega)$ fields. Since

$$\boldsymbol{B} = -\nabla V \in L^2(\Omega)^3, \qquad \nabla \cdot \boldsymbol{B} = 0,$$

the total weak normal trace satisfies

$$\langle \boldsymbol{B} \cdot \boldsymbol{n}, 1 \rangle_{\partial\Omega} = 0. \tag{16.8}$$

16.3 Flux through the three boundary components

Using the outward unit normals from Equation (2.5),

$$\boldsymbol{n}_{\mathrm{a}} = -\boldsymbol{e}_r, \qquad \boldsymbol{n}_{\mathrm{b}} = \boldsymbol{e}_r, \qquad \boldsymbol{n}_{\theta_0} = \boldsymbol{e}_\theta, \tag{16.9}$$

one obtains

$$\boldsymbol{B} \cdot \boldsymbol{n}|_{\Gamma_{\mathrm{a}}} = \left.\frac{\partial V}{\partial r}\right|_{r=r_{\mathrm{a}}}, \tag{16.10a}$$

$$\boldsymbol{B} \cdot \boldsymbol{n}|_{\Gamma_{\mathrm{b}}} = -\left.\frac{\partial V}{\partial r}\right|_{r=r_{\mathrm{b}}}, \tag{16.10b}$$

$$\boldsymbol{B} \cdot \boldsymbol{n}|_{\Gamma_{\theta_0}} = -\left.\frac{1}{r}\frac{\partial V}{\partial \theta}\right|_{\theta=\theta_0}. \tag{16.10c}$$

The signs differ from the outward normal derivative of V because $\boldsymbol{B} = -\nabla V$.

Define

$$\Phi_{\mathrm{a}}[V] = \int_{\Gamma_{\mathrm{a}}} \boldsymbol{B} \cdot \boldsymbol{n} \, \mathrm{d}S, \tag{16.11a}$$

$$\Phi_{\mathrm{b}}[V] = \int_{\Gamma_{\mathrm{b}}} \boldsymbol{B} \cdot \boldsymbol{n} \, \mathrm{d}S, \tag{16.11b}$$

$$\Phi_{\theta_0}[V] = \int_{\Gamma_{\theta_0}} \boldsymbol{B} \cdot \boldsymbol{n} \, \mathrm{d}S. \tag{16.11c}$$

Using the surface elements in Equation (2.14),

$$\Phi_{\mathrm{a}}[V] = r_{\mathrm{a}}^2 \int_0^{2\pi} \int_0^{\theta_0} \left.\frac{\partial V}{\partial r}\right|_{r=r_{\mathrm{a}}} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda, \tag{16.12a}$$

$$\Phi_{\mathrm{b}}[V] = -r_{\mathrm{b}}^2 \int_0^{2\pi} \int_0^{\theta_0} \left.\frac{\partial V}{\partial r}\right|_{r=r_{\mathrm{b}}} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda, \tag{16.12b}$$

$$\Phi_{\theta_0}[V] = -\sin\theta_0 \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} \int_0^{2\pi} \left.\frac{\partial V}{\partial \theta}\right|_{\theta=\theta_0} \mathrm{d}\lambda \, \mathrm{d}r. \tag{16.12c}$$

The total-flux identity is therefore

$$\Phi_{\mathrm{a}}[V] + \Phi_{\mathrm{b}}[V] + \Phi_{\theta_0}[V] = 0. \tag{16.13}$$

16.4 Flux identities for the two complementary subpotentials

Because

$$V = V_1 + V_2,$$

linearity gives

$$\Phi_\Gamma[V] = \Phi_\Gamma[V_1] + \Phi_\Gamma[V_2] \tag{16.14}$$

for every boundary component Γ.

For V_1, the radial normal derivatives are the prescribed data G_{a} and G_{b}. Since

$$\boldsymbol{B}_1 \cdot \boldsymbol{n} = -\frac{\partial V_1}{\partial n},$$

one has

$$\Phi_{\mathrm{a}}[V_1] = -\int_{\Gamma_{\mathrm{a}}} G_{\mathrm{a}} \, \mathrm{d}S, \tag{16.15a}$$

$$\Phi_{\mathrm{b}}[V_1] = -\int_{\Gamma_{\mathrm{b}}} G_{\mathrm{b}} \, \mathrm{d}S. \tag{16.15b}$$

Consequently,

$$\Phi_{\theta_0}[V_1] = \int_{\Gamma_{\mathrm{a}}} G_{\mathrm{a}} \, \mathrm{d}S + \int_{\Gamma_{\mathrm{b}}} G_{\mathrm{b}} \, \mathrm{d}S. \tag{16.16}$$

Thus, the lateral flux of V_1 generally does not vanish; it balances the normal fluxes through the two spherical boundaries.

For V_2,

$$\frac{\partial V_2}{\partial r} = 0 \quad \text{at } r = r_{\mathrm{a}}, r_{\mathrm{b}}.$$

Therefore,

$$\Phi_{\mathrm{a}}[V_2] = \Phi_{\mathrm{b}}[V_2] = 0, \tag{16.17}$$

and the total-flux theorem forces

$$\Phi_{\theta_0}[V_2] = 0. \tag{16.18}$$

This is the compact explanation of the cancellation that also appears when the six boundary integrals are expanded separately.

16.5 Generic separated-mode flux identity

Consider a single real separated harmonic mode

$$\mathcal{V}_m^\sigma = r_{\mathrm{a}}\gamma U(r)A(\theta)L_m^\sigma(\lambda), \tag{16.19}$$

where

$$L_m^{\mathrm{c}}(\lambda) = \cos(m\lambda), \qquad L_m^{\mathrm{s}}(\lambda) = \sin(m\lambda),$$

and the sine mode is omitted for $m = 0$. Define

$$J[A] = \int_0^{\theta_0} A(\theta)\sin\theta \ \mathrm{d}\theta, \tag{16.20a}$$

$$L_m^\sigma = \int_0^{2\pi} L_m^\sigma(\lambda) \ \mathrm{d}\lambda. \tag{16.20b}$$

The longitude means are

$$L_m^{\mathrm{c}} = 2\pi\delta_{0m}, \qquad L_m^{\mathrm{s}} = 0. \tag{16.21}$$

The three boundary fluxes of Equation (16.19) are

$$\Phi_{\mathrm{a}}\left[\mathcal{V}_m^\sigma\right] = r_{\mathrm{a}}^3\gamma U'(r_{\mathrm{a}})J[A]L_m^\sigma, \tag{16.22a}$$

$$\Phi_{\mathrm{b}}\left[\mathcal{V}_m^\sigma\right] = -r_{\mathrm{a}}r_{\mathrm{b}}^2\gamma U'(r_{\mathrm{b}})J[A]L_m^\sigma, \tag{16.22b}$$

$$\Phi_{\theta_0}\left[\mathcal{V}_m^\sigma\right] = -r_{\mathrm{a}}\gamma\sin\theta_0 A'(\theta_0)\left[\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} U(r) \ \mathrm{d}r\right] L_m^\sigma. \tag{16.22c}$$

Let Λ denote the separation constant. Suppose that U and A satisfy

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2U'\right) - \Lambda U = 0, \tag{16.23a}$$

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta A'\right) + \left[\Lambda\sin\theta - \frac{m^2}{\sin\theta}\right]A = 0. \tag{16.23b}$$

When $m > 0$, or when the longitude factor is a sine function, Equation (16.21) shows that every integrated boundary flux vanishes separately.

For the axisymmetric cosine mode $m = 0$, integration of the radial and angular equations gives

$$r_{\mathrm{b}}^2U'(r_{\mathrm{b}}) - r_{\mathrm{a}}^2U'(r_{\mathrm{a}}) = \Lambda\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} U(r) \ \mathrm{d}r, \tag{16.24a}$$

$$\sin\theta_0 A'(\theta_0) + \Lambda J[A] = 0. \tag{16.24b}$$

Substitution into Equation (16.22) gives

$$\Phi_{\mathrm{a}}\left[\mathcal{V}_0^{\mathrm{c}}\right] + \Phi_{\mathrm{b}}\left[\mathcal{V}_0^{\mathrm{c}}\right] + \Phi_{\theta_0}\left[\mathcal{V}_0^{\mathrm{c}}\right] = 0. \tag{16.25}$$

Theorem 16.2 (Modewise flux conservation)**.** *Every regular separated harmonic R-SCHA mode has zero net magnetic flux through the closed spherical-cap shell. Non-axisymmetric and sine modes have zero integrated flux through each boundary component separately. An axisymmetric Legendre mode may have non-zero flux through the individual boundary components, but the three contributions cancel exactly.*

16.6 Axisymmetric internal Legendre mode

Let

$$\mathcal{V}_\nu^{\mathrm{Int}} = r_{\mathrm{a}}\gamma\left(\frac{r_{\mathrm{a}}}{r}\right)^{\nu+1} P_\nu^0(\cos\theta), \qquad P_\nu^0(\cos\theta_0) = 0. \tag{16.26}$$

Define

$$J_\nu = \int_0^{\theta_0} P_\nu^0(\cos\theta)\sin\theta \ \mathrm{d}\theta. \tag{16.27}$$

For $\nu > 0$, the angular equation gives

$$J_\nu = -\frac{\sin\theta_0}{\nu(\nu+1)} \left.\frac{\partial P_\nu^0(\cos\theta)}{\partial\theta}\right|_{\theta=\theta_0}. \tag{16.28}$$

The three fluxes are

$$\Phi_{\mathrm{a}}^{\mathrm{Int}} = -2\pi\gamma r_{\mathrm{a}}^2(\nu+1)J_\nu, \tag{16.29a}$$

$$\Phi_{\mathrm{b}}^{\mathrm{Int}} = 2\pi\gamma r_{\mathrm{a}}^2(\nu+1)\left(\frac{r_{\mathrm{a}}}{r_{\mathrm{b}}}\right)^{\nu} J_\nu, \tag{16.29b}$$

$$\Phi_{\theta_0}^{\mathrm{Int}} = -\frac{2\pi\gamma r_{\mathrm{a}}^2\sin\theta_0}{\nu}\left[1-\left(\frac{r_{\mathrm{a}}}{r_{\mathrm{b}}}\right)^{\nu}\right] \left.\frac{\partial P_\nu^0(\cos\theta)}{\partial\theta}\right|_{\theta=\theta_0}. \tag{16.29c}$$

Using Equation (16.28),

$$\Phi_{\mathrm{a}}^{\mathrm{Int}} + \Phi_{\mathrm{b}}^{\mathrm{Int}} = -\Phi_{\theta_0}^{\mathrm{Int}}. \tag{16.30}$$

16.7 Axisymmetric external Legendre mode

For

$$\mathcal{V}_\nu^{\mathrm{Ext}} = r_{\mathrm{a}}\gamma\left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu} P_\nu^0(\cos\theta), \qquad P_\nu^0(\cos\theta_0) = 0, \tag{16.31}$$

the corresponding fluxes are

$$\Phi_{\mathrm{a}}^{\mathrm{Ext}} = 2\pi\gamma r_{\mathrm{a}}^2\nu J_\nu, \tag{16.32a}$$

$$\Phi_{\mathrm{b}}^{\mathrm{Ext}} = -2\pi\gamma r_{\mathrm{a}}^2\nu\left(\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}\right)^{\nu+1} J_\nu, \tag{16.32b}$$

$$\Phi_{\theta_0}^{\mathrm{Ext}} = -\frac{2\pi\gamma r_{\mathrm{a}}^2\sin\theta_0}{\nu+1}\left[\left(\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}\right)^{\nu+1}-1\right] \left.\frac{\partial P_\nu^0(\cos\theta)}{\partial\theta}\right|_{\theta=\theta_0}. \tag{16.32c}$$

Again,

$$\Phi_{\mathrm{a}}^{\mathrm{Ext}} + \Phi_{\mathrm{b}}^{\mathrm{Ext}} = -\Phi_{\theta_0}^{\mathrm{Ext}}. \tag{16.33}$$

16.8 Mehler-mode flux identities

Consider the Mehler mode

$$\mathcal{V}_{pm}^{\mathrm{Meh},\sigma} = r_{\mathrm{a}}\gamma R_p(r)\widetilde{K}_p^m(\theta)L_m^\sigma(\lambda), \qquad p \geq 1. \tag{16.34}$$

The radial Neumann conditions imply

$$\Phi_{\mathrm{a}}\left[\mathcal{V}_{pm}^{\mathrm{Meh},\sigma}\right] = \Phi_{\mathrm{b}}\left[\mathcal{V}_{pm}^{\mathrm{Meh},\sigma}\right] = 0. \tag{16.35}$$

For $m > 0$ and for every sine mode, the lateral integrated flux is zero by longitude orthogonality.

For the axisymmetric cosine mode,

$$\Phi_{\theta_0}\left[\mathcal{V}_{p0}^{\mathrm{Meh,c}}\right] = -2\pi r_{\mathrm{a}}\gamma \sin\theta_0 \left.\frac{\partial \widetilde{K}_p^0}{\partial\theta}\right|_{\theta=\theta_0} \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R_p(r)\ \mathrm{d}r. \tag{16.36}$$

The radial eigenvalue equation

$$-\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 R_p'\right) = \mu_p R_p, \qquad \mu_p > 0,$$

and the two Neumann conditions give

$$\mu_p \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R_p(r)\ \mathrm{d}r = -\left. r^2 R_p'(r)\right|_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} = 0. \tag{16.37}$$

Hence,

$$\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R_p(r)\ \mathrm{d}r = 0 \tag{16.38}$$

and

$$\Phi_{\theta_0}\left[\mathcal{V}_{p0}^{\mathrm{Meh,c}}\right] = 0. \tag{16.39}$$

This is also a consequence of the radial orthogonality

$$\int R_0 R_p\ \mathrm{d}r = 0,$$

because R_0 is a non-zero constant.

16.9 Constant-mode flux identities

For a constant-family mode,

$$\mathcal{V}_{0m}^{\mathrm{Cons},\sigma} = r_{\mathrm{a}}\gamma R_0 \widetilde{P}_0^m(\theta)L_m^\sigma(\lambda), \tag{16.40}$$

the radial derivative vanishes identically, so

$$\Phi_{\mathrm{a}}\left[\mathcal{V}_{0m}^{\mathrm{Cons},\sigma}\right] = \Phi_{\mathrm{b}}\left[\mathcal{V}_{0m}^{\mathrm{Cons},\sigma}\right] = 0. \tag{16.41}$$

For $m > 0$ and for every sine mode, the longitude integral vanishes. For $m = 0$, the normalised angular function is constant and

$$\frac{\partial \widetilde{P}_0^0}{\partial\theta} = 0.$$

Thus,

$$\Phi_{\theta_0}\left[\mathcal{V}_{0m}^{\mathrm{Cons},\sigma}\right] = 0 \tag{16.42}$$

for every constant-family mode.

16.10 Finite and infinite R-SCHA expansions

A finite R-SCHA expansion is a finite sum of exact harmonic modes. Therefore, its divergence and total flux satisfy

$$\nabla \cdot \boldsymbol{B}_N = 0, \qquad \int_{\partial\Omega} \boldsymbol{B}_N \cdot \boldsymbol{n} \, \mathrm{d}S = 0 \tag{16.43}$$

analytically, independently of the truncation degrees.

For an infinite series, termwise differentiation and boundary integration require convergence assumptions. A sufficient classical condition is convergence in $C^1(\overline{\Omega})$ together with locally uniform convergence of the second derivatives in Ω. More generally, if the partial sums converge in $H^1(\Omega)$ to a weakly harmonic limit, then the solenoidal condition holds distributionally and the flux identity holds in the weak normal-trace sense.

16.11 Local flux conservation and the magnetic-monopole constraint

The shell identity

$$\Phi_{\mathrm{a}} + \Phi_{\mathrm{b}} + \Phi_{\theta_0} = 0$$

is a local consequence of $\Delta V = 0$ in the modelled shell. It does not imply that the flux through each individual boundary component is zero.

It also does not, by itself, exclude an effective source in the region removed from the shell. For example, the potential

$$V_{\mathrm{mono}}(r) = \frac{c}{r} \tag{16.44}$$

is harmonic for $r > 0$ and has zero total flux through a shell boundary because the inner and outer spherical fluxes cancel. Nevertheless, its flux through a single complete sphere enclosing the origin is non-zero. Consequently, the physical absence of magnetic monopoles requires the stronger global condition

$$\int_{S_r} \boldsymbol{B} \cdot \boldsymbol{n} \, \mathrm{d}S = 0 \tag{16.45}$$

for a complete closed surface enclosing the internal source region. In a global spherical-harmonic representation, this removes the degree-zero internal magnetic coefficient.

Therefore:

1. zero total flux through the complete R-SCHA shell is an automatic local consistency property of the harmonic expansion;
2. zero flux through each separate cap boundary is not generally required;
3. exclusion of a global magnetic monopole is an additional physical constraint that cannot be inferred solely from the shell balance.

Part IV
Spectral Energy Theory

17 Exact Dirichlet energy in the three-dimensional spherical-cap shell

The first spectral object is the Dirichlet energy in the complete three-dimensional shell. It is the natural quadratic form of the Laplace operator and, because $\boldsymbol{B} = -\nabla V$, is also the volume

integral of the squared magnetic-field magnitude. The energy is diagonal in the boundary-adapted R-SCHA decomposition. This section derives that statement from both the volume integral and Green's boundary identity, retaining all cross terms until the reason for their disappearance has been identified. The mode calculations and ordinary face integrals below are first understood for finite expansions, or for harmonic potentials with sufficient boundary regularity. For an $H^1(\Omega)$ weak solution, Green's identity uses the duality between its conormal derivative and its boundary trace; facewise products and termwise boundary integration require the corresponding trace regularity. The final coefficient identity extends by convergence in the Dirichlet-energy norm, as stated below.

17.1 Dirichlet pairing and energy

For real-valued $u, v \in H^1(\Omega)$, define

$$\mathcal{E}_\Omega(u,v) = \int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}\Omega, \tag{17.1}$$

and

$$E_\Omega[V] = \mathcal{E}_\Omega(V,V) = \int_\Omega |\nabla V|^2 \, \mathrm{d}\Omega. \tag{17.2}$$

The additive constant in V lies in the null space of this energy. The magnetic energy in vacuum is $E_\Omega[V]/(2\mu_0)$. This physical interpretation assumes that V is a potential for $\boldsymbol{B}$ in vacuum; the Dirichlet quadratic form itself does not depend on that interpretation.

In spherical coordinates,

$$E_\Omega[V] = \int_{r_\mathrm{a}}^{r_\mathrm{b}} \int_0^{\theta_0} \int_0^{2\pi} \left[r^2 \left(\frac{\partial V}{\partial r}\right)^2 + \left(\frac{\partial V}{\partial \theta}\right)^2 + \frac{1}{\sin^2\theta} \left(\frac{\partial V}{\partial \lambda}\right)^2 \right] \sin\theta \, \mathrm{d}\lambda \, \mathrm{d}\theta \, \mathrm{d}r. \tag{17.3}$$

17.2 Generic separated-mode factorisation

Let

$$\phi_i = r_\mathrm{a} c_i U_i(r) A_i(\theta) L_{m_i}^{\sigma_i}(\lambda), \tag{17.4}$$

and define

$$J_{ij}^{(r)} = \int_{r_\mathrm{a}}^{r_\mathrm{b}} r^2 U_i'(r) U_j'(r) \, \mathrm{d}r, \tag{17.5a}$$

$$J_{ij}^{(0)} = \int_{r_\mathrm{a}}^{r_\mathrm{b}} U_i(r) U_j(r) \, \mathrm{d}r, \tag{17.5b}$$

$$I_{ij}^{(0)} = \int_0^{\theta_0} A_i(\theta) A_j(\theta) \sin\theta \, \mathrm{d}\theta, \tag{17.5c}$$

$$I_{ij}^{(1,m)} = \int_0^{\theta_0} \left[A_i' A_j' + \frac{m^2}{\sin^2\theta} A_i A_j \right] \sin\theta \, \mathrm{d}\theta. \tag{17.5d}$$

The longitude norms are

$$\ell_m^\mathrm{c} = \begin{cases} 2\pi, & m = 0, \\ \pi, & m \geq 1, \end{cases} \qquad \ell_m^\mathrm{s} = \pi \quad (m \geq 1). \tag{17.6}$$

Fourier orthogonality gives

$$\mathcal{E}_\Omega(\phi_i, \phi_j) = r_\mathrm{a}^2 c_i c_j \delta_{m_i m_j} \delta_{\sigma_i \sigma_j} \ell_{m_i}^{\sigma_i} \left[J_{ij}^{(r)} I_{ij}^{(0)} + J_{ij}^{(0)} I_{ij}^{(1,m_i)} \right]. \tag{17.7}$$

Equation Equation (17.7) contains every three-dimensional cross term. All subsequent orthogonality statements are consequences of one of its radial, angular, or longitude factors.

17.3 Green representation and the six boundary terms

For a harmonic potential,

$$E_\Omega[V] = \int_{\partial\Omega} V \frac{\partial V}{\partial n}\, \mathrm{d}S. \tag{17.8}$$

Write $V = V_1 + V_2$ and use the three outward normals. Expanding before applying the boundary conditions gives six self terms and, in principle, cross terms. The gradient orthogonality

$$\mathcal{E}_\Omega(V_1, V_2) = 0 \tag{17.9}$$

removes the cross contribution. The remaining self terms are

$$\begin{aligned} E_\Omega[V] = {} & \underbrace{\int_{\Gamma_{\theta_0}} V_1 \frac{1}{r}\frac{\partial V_1}{\partial \theta}\, \mathrm{d}S}_{I_1} + \underbrace{\int_{\Gamma_{\theta_0}} V_2 \frac{1}{r}\frac{\partial V_2}{\partial \theta}\, \mathrm{d}S}_{I_2} \\ & - \underbrace{\int_{\Gamma_\mathrm{a}} V_1 \frac{\partial V_1}{\partial r}\, \mathrm{d}S}_{I_3} - \underbrace{\int_{\Gamma_\mathrm{a}} V_2 \frac{\partial V_2}{\partial r}\, \mathrm{d}S}_{I_4} \\ & + \underbrace{\int_{\Gamma_\mathrm{b}} V_1 \frac{\partial V_1}{\partial r}\, \mathrm{d}S}_{I_5} + \underbrace{\int_{\Gamma_\mathrm{b}} V_2 \frac{\partial V_2}{\partial r}\, \mathrm{d}S}_{I_6}. \end{aligned} \tag{17.10}$$

The boundary conditions imply

$$I_1 = 0, \qquad I_4 = 0, \qquad I_6 = 0. \tag{17.11}$$

Therefore,

$$E_\Omega[V] = I_2 - I_3 + I_5. \tag{17.12}$$

The term I_2 contains the Mehler and constant energies; $-I_3 + I_5$ contains the internal and external Legendre energies.

17.4 Normalised coefficient convention

For the remainder of this section, let

$$\widehat{P}_{km}(\theta), \qquad \widehat{K}_{pm}(\theta), \qquad \widehat{C}_m(\theta) \tag{17.13}$$

be unit-normalised in $L^2((0,\theta_0), \sin\theta\, \mathrm{d}\theta)$. The radial functions satisfy

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p R_q\, \mathrm{d}r = r_\mathrm{a}\delta_{pq}. \tag{17.14}$$

A hat on a coefficient indicates that it multiplies the corresponding unit-normalised angular function.

Define the longitude-weighted coefficient product

$$\langle \boldsymbol{a}, \boldsymbol{b}\rangle_{m,\lambda} = \ell^\mathrm{c}_m a^\mathrm{c} b^\mathrm{c} + \ell^\mathrm{s}_m a^\mathrm{s} b^\mathrm{s}, \tag{17.15}$$

where the sine term is omitted at $m = 0$. In particular,

$$\|\boldsymbol{a}\|^2_{m,\lambda} = \ell^\mathrm{c}_m (a^\mathrm{c})^2 + \ell^\mathrm{s}_m (a^\mathrm{s})^2. \tag{17.16}$$

17.5 Angular energy identity for Legendre modes

The unit-normalised Legendre mode satisfies

$$\mathscr{L}_m \widehat{P}_{km} = \lambda_{km} \widehat{P}_{km}, \qquad \lambda_{km} = \nu_{km}(\nu_{km}+1), \tag{17.17}$$

where

$$\mathscr{L}_m A = -\frac{1}{\sin\theta}(\sin\theta A')' + \frac{m^2}{\sin^2\theta} A. \tag{17.18}$$

Because $\widehat{P}_{km}(\theta_0) = 0$ and the pole term vanishes, integration by parts yields

$$I^{(1,m)}_{kk} = \lambda_{km}. \tag{17.19}$$

For $k \neq j$ at the same m,

$$I^{(0)}_{kj} = I^{(1,m)}_{kj} = 0. \tag{17.20}$$

17.6 Internal Legendre energy: direct derivation

For one internal mode, set

$$U^{\mathrm{Int}}_{km}(r) = \left(\frac{r_\mathrm{a}}{r}\right)^{\nu_{km}+1}. \tag{17.21}$$

Then

$$r^2 (U^{\mathrm{Int}\prime}_{km})^2 = (\nu_{km}+1)^2 (U^{\mathrm{Int}}_{km})^2. \tag{17.22}$$

Using $\lambda_{km} = \nu_{km}(\nu_{km}+1)$, the radial-plus-angular integrand is

$$r^2 (U^{\mathrm{Int}\prime})^2 + \lambda_{km} (U^{\mathrm{Int}})^2 = (\nu_{km}+1)(2\nu_{km}+1)(U^{\mathrm{Int}})^2. \tag{17.23}$$

Moreover,

$$\begin{aligned} \int_{r_\mathrm{a}}^{r_\mathrm{b}} (U^{\mathrm{Int}}_{km})^2 \, \mathrm{d}r &= \int_{r_\mathrm{a}}^{r_\mathrm{b}} \left(\frac{r_\mathrm{a}}{r}\right)^{2\nu_{km}+2} \mathrm{d}r \\ &= \frac{r_\mathrm{a}}{2\nu_{km}+1} \left[1 - \left(\frac{r_\mathrm{a}}{r_\mathrm{b}}\right)^{2\nu_{km}+1}\right] \\ &= \frac{r_\mathrm{a}}{2\nu_{km}+1} \left[1 - \mathrm{e}^{-(2\nu_{km}+1)s}\right]. \end{aligned} \tag{17.24}$$

Therefore the energy of that coefficient pair is

$$\varepsilon^{\mathrm{Int},\Omega}_{km} = r_\mathrm{a}^3 (\nu_{km}+1) \left[1 - \mathrm{e}^{-(2\nu_{km}+1)s}\right] \|\widehat{\boldsymbol{a}}^{\mathrm{Int}}_{km}\|^2_{m,\lambda}. \tag{17.25}$$

Summing gives

$$E^{\mathrm{Int}}_{\Omega} = r_\mathrm{a}^3 \sum_{k,m} (\nu_{km}+1) \left[1 - \mathrm{e}^{-(2\nu_{km}+1)s}\right] \|\widehat{\boldsymbol{a}}^{\mathrm{Int}}_{km}\|^2_{m,\lambda}. \tag{17.26}$$

17.7 External Legendre energy: direct derivation

For

$$U_{km}^{\mathrm{Ext}}(r) = \left(\frac{r}{r_{\mathrm{a}}}\right)^{\nu_{km}}, \tag{17.27}$$

one has

$$r^2 (U_{km}^{\mathrm{Ext}\prime})^2 = \nu_{km}^2 (U_{km}^{\mathrm{Ext}})^2, \tag{17.28}$$

and hence

$$r^2 (U^{\mathrm{Ext}\prime})^2 + \lambda_{km} (U^{\mathrm{Ext}})^2 = \nu_{km}(2\nu_{km}+1)(U^{\mathrm{Ext}})^2. \tag{17.29}$$

The radial integral is

$$\begin{aligned}\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} (U_{km}^{\mathrm{Ext}})^2 \, \mathrm{d}r &= \frac{r_{\mathrm{a}}}{2\nu_{km}+1}\left[\left(\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}\right)^{2\nu_{km}+1} - 1\right] \\ &= \frac{r_{\mathrm{a}}}{2\nu_{km}+1}\left[\mathrm{e}^{(2\nu_{km}+1)s} - 1\right].\end{aligned} \tag{17.30}$$

Thus

$$E_{\Omega}^{\mathrm{Ext}} = r_{\mathrm{a}}^3 \sum_{k,m} \nu_{km}\left[\mathrm{e}^{(2\nu_{km}+1)s} - 1\right] \|\widehat{\boldsymbol{a}}_{km}^{\mathrm{Ext}}\|_{m,\lambda}^2. \tag{17.31}$$

17.8 Pointwise cancellation of internal–external cross terms

For the same (k, m, σ),

$$r^2 U_{km}^{\mathrm{Int}\prime} U_{km}^{\mathrm{Ext}\prime} = -\nu_{km}(\nu_{km}+1) U_{km}^{\mathrm{Int}} U_{km}^{\mathrm{Ext}}. \tag{17.32}$$

The angular contribution is

$$I_{kk}^{(1,m)} = \nu_{km}(\nu_{km}+1) I_{kk}^{(0)}. \tag{17.33}$$

Therefore the integrand in Equation (17.7) vanishes pointwise in r:

$$r^2 U_{km}^{\mathrm{Int}\prime} U_{km}^{\mathrm{Ext}\prime} I_{kk}^{(0)} + U_{km}^{\mathrm{Int}} U_{km}^{\mathrm{Ext}} I_{kk}^{(1,m)} = 0. \tag{17.34}$$

Cross terms with different k, m, or parity vanish by angular or longitude orthogonality. Hence

$$\mathcal{E}_{\Omega}(V^{\mathrm{Int}}, V^{\mathrm{Ext}}) = 0. \tag{17.35}$$

17.9 Mehler angular boundary identity

The unit-normalised Mehler function satisfies

$$\mathscr{L}_m \widehat{K}_{pm} = -\mu_p \widehat{K}_{pm}. \tag{17.36}$$

Multiplying by $\widehat{K}_{pm}$ and integrating gives

$$\begin{aligned} I_{pp}^{(1,m)} &= -\mu_p \int_0^{\theta_0} \widehat{K}_{pm}^2 \sin\theta \, \mathrm{d}\theta + \sin\theta_0 \widehat{K}_{pm}(\theta_0)\widehat{K}'_{pm}(\theta_0) \\ &= -\mu_p + \Xi_{pm}^{\mathrm{Meh}}, \end{aligned} \tag{17.37}$$

where

$$\Xi_{pm}^{\mathrm{Meh}} = \sin\theta_0 \widehat{K}_{pm}(\theta_0)\widehat{K}'_{pm}(\theta_0). \tag{17.38}$$

The radial eigenvalue identity is

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} r^2 (R_p')^2 \, \mathrm{d}r = \mu_p \int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p^2 \, \mathrm{d}r = \mu_p r_\mathrm{a}. \tag{17.39}$$

Substitution into the generic factorisation gives the cancellation

$$(\mu_p r_\mathrm{a}) + r_\mathrm{a} \left(-\mu_p + \Xi_{pm}^{\mathrm{Meh}} \right) = r_\mathrm{a} \Xi_{pm}^{\mathrm{Meh}}. \tag{17.40}$$

Therefore

$$E_\Omega^{\mathrm{Meh}} = r_\mathrm{a}^3 \sum_{p \geq 1} \sum_m \Xi_{pm}^{\mathrm{Meh}} \| \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}} \|_{m,\lambda}^2. \tag{17.41}$$

17.10 Orthogonality of distinct Mehler radial indices

For $p \neq q$,

$$\int R_p R_q \, \mathrm{d}r = 0. \tag{17.42}$$

Multiplying the radial eigenvalue equation for R_p by R_q and integrating also gives

$$\int r^2 R_p' R_q' \, \mathrm{d}r = \mu_p \int R_p R_q \, \mathrm{d}r = 0. \tag{17.43}$$

Both radial factors in Equation (17.7) therefore vanish, regardless of whether K_{pm} and K_{qm} are angularly orthogonal. This is the reason the three-dimensional Mehler energy is diagonal even though the Mehler angular functions do not share a lateral self-adjoint boundary condition.

17.11 Constant-family energy

The unit-normalised constant angular function satisfies

$$\mathscr{L}_m \widehat{C}_m = 0. \tag{17.44}$$

Hence

$$I_{00}^{(1,m)} = \sin\theta_0 \widehat{C}_m(\theta_0) \widehat{C}_m'(\theta_0) =: \Xi_{0m}^{\mathrm{Cons}}. \tag{17.45}$$

For $m \geq 1$, the identity $\widehat{C}_m' = m \widehat{C}_m / \sin\theta$ gives

$$\Xi_{0m}^{\mathrm{Cons}} = m [\widehat{C}_m(\theta_0)]^2. \tag{17.46}$$

For $m = 0$, $\widehat{C}_0$ is constant and

$$\Xi_{00}^{\mathrm{Cons}} = 0. \tag{17.47}$$

Because $R_0' = 0$ and $\int R_0^2 \, \mathrm{d}r = r_\mathrm{a}$,

$$E_\Omega^{\mathrm{Cons}} = r_\mathrm{a}^3 \sum_m \Xi_{0m}^{\mathrm{Cons}} \| \widehat{\boldsymbol{d}}_{0m}^{\mathrm{Cons}} \|_{m,\lambda}^2. \tag{17.48}$$

17.12 Mehler–constant cross terms

For $p \geq 1$,

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p R_0 \, \mathrm{d}r = 0 \tag{17.49}$$

by radial orthogonality, and

$$\int r^2 R_p' R_0' \, \mathrm{d}r = 0 \tag{17.50}$$

because $R_0' = 0$. Hence

$$\mathcal{E}_\Omega(V^{\mathrm{Meh}}, V^{\mathrm{Cons}}) = 0. \tag{17.51}$$

17.13 Legendre–V_2 cross terms

The exact gradient orthogonality of the mixed boundary decomposition gives

$$\mathcal{E}_\Omega\left(V^{\mathrm{Int}}+V^{\mathrm{Ext}},V^{\mathrm{Meh}}+V^{\mathrm{Cons}}\right)=0. \tag{17.52}$$

The proof uses Green's identity:

$$\mathcal{E}_\Omega(V_1,V_2)=\int_{\partial\Omega}V_1\frac{\partial V_2}{\partial n}\,\mathrm{d}S-\int_\Omega V_1\Delta V_2\,\mathrm{d}\Omega. \tag{17.53}$$

The volume term is zero. On the lateral boundary $V_1=0$; on the two spherical boundaries $\partial_n V_2=0$. The boundary term therefore vanishes.

17.14 Complete three-dimensional Parseval-type identity

Combining all families gives

$$E_\Omega[V]=E_\Omega^{\mathrm{Int}}+E_\Omega^{\mathrm{Ext}}+E_\Omega^{\mathrm{Meh}}+E_\Omega^{\mathrm{Cons}}. \tag{17.54}$$

In fully expanded coefficient form,

$$\begin{aligned}E_\Omega[V]=\;&r_{\mathrm{a}}^3\sum_{k,m}(\nu_{km}+1)\left[1-\mathrm{e}^{-(2\nu_{km}+1)s}\right]\|\widehat{\boldsymbol{a}}_{km}^{\mathrm{Int}}\|_{m,\lambda}^2\\&+r_{\mathrm{a}}^3\sum_{k,m}\nu_{km}\left[\mathrm{e}^{(2\nu_{km}+1)s}-1\right]\|\widehat{\boldsymbol{a}}_{km}^{\mathrm{Ext}}\|_{m,\lambda}^2\\&+r_{\mathrm{a}}^3\sum_{p\geq1}\sum_m\Xi_{pm}^{\mathrm{Meh}}\|\widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}}\|_{m,\lambda}^2\\&+r_{\mathrm{a}}^3\sum_m\Xi_{0m}^{\mathrm{Cons}}\|\widehat{\boldsymbol{d}}_{0m}^{\mathrm{Cons}}\|_{m,\lambda}^2.\end{aligned} \tag{17.55}$$

This is an exact identity for every finite expansion and extends to the energy closure modulo additive constants whenever the displayed non-negative series converges.

17.15 Conversion from unnormalised angular coefficients

If a coefficient g_{km}^{Int} multiplies the unnormalised function $P_{\nu_{km}}^m$, then

$$\widehat{g}_{km}^{\mathrm{Int}}=g_{km}^{\mathrm{Int}}\left[\int_0^{\theta_0}(P_{\nu_{km}}^m)^2\sin\theta\,\mathrm{d}\theta\right]^{1/2}. \tag{17.56}$$

The same rescaling applies to the sine coefficient and to the external family. If the Mehler or constant functions are not unit-normalised, their coefficients must be multiplied by the corresponding angular norms before Equation (17.55) is used. Thus the energy is normalisation-invariant even though the numerical values of individual coefficients are not.

18 Exact full-gradient energy on a fixed spherical-cap surface

The second spectral object is the energy of the complete three-dimensional gradient restricted to one spherical-cap surface. Fix

$$r=\beta,\qquad r_{\mathrm{a}}\leq\beta\leq r_{\mathrm{b}}, \tag{18.1}$$

and let

$$\Sigma_\beta = \{(r,\theta,\lambda) : r = \beta, 0 \le \theta \le \theta_0, 0 \le \lambda < 2\pi\}. \tag{18.2}$$

Unlike the volume energy of Section 17, no radial integration is present. The radial Sturm–Liouville orthogonality that diagonalises the V_2 family in the shell is therefore unavailable. The exact surface energy contains Mehler–Mehler, Mehler–constant, and Legendre–Mehler cross terms.

18.1 Definition and component form

This section applies to finite R-SCHA expansions and, more generally, to harmonic potentials in $H^1(\Omega)$ whose full gradient has a trace in $L^2(\Sigma_\beta)^3$. This trace assumption is part of the domain of the quadratic form. Membership in $H^1(\Omega)$ alone does not imply it, in particular at $\beta = r_\mathrm{a}$ or r_b. A general weak harmonic potential has a conormal derivative in a negative-order trace space, which need not be an L^2 function. Thus the surface formulae below do not follow for arbitrary finite-energy weak solutions solely from the volume Parseval identity. The coefficient formulae are exact for finite sums. An infinite formula is understood as the limit of these finite quadratic forms only when the corresponding partial gradients converge in $L^2(\Sigma_\beta)^3$; the existence of a trace for an H^1 limit alone does not justify taking this boundary limit term by term.

The full-gradient surface energy is

$$E_\beta[V] = \int_{\Sigma_\beta} |\nabla V|^2 \, \mathrm{d}S. \tag{18.3}$$

Because

$$\mathrm{d}S = \beta^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda, \tag{18.4}$$

one has

$$E_\beta[V] = \int_0^{2\pi} \int_0^{\theta_0} \left[\beta^2 \left(\left.\frac{\partial V}{\partial r}\right|_{r=\beta} \right)^2 + \left(\left.\frac{\partial V}{\partial \theta}\right|_{r=\beta} \right)^2 + \frac{1}{\sin^2\theta} \left(\left.\frac{\partial V}{\partial \lambda}\right|_{r=\beta} \right)^2 \right] \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda. \tag{18.5}$$

It is useful to write

$$E_\beta[V] = E_\beta^r[V] + E_\beta^h[V], \tag{18.6}$$

where the two terms contain the radial derivative and the two tangential derivatives, respectively.

18.2 Generic bilinear form on the surface

For separated modes

$$\phi_i = r_\mathrm{a} c_i U_i(r) A_i(\theta) L_{m_i}^{\sigma_i}(\lambda), \qquad \phi_j = r_\mathrm{a} c_j U_j(r) A_j(\theta) L_{m_j}^{\sigma_j}(\lambda), \tag{18.7}$$

Fourier orthogonality gives

$$\begin{aligned} \mathcal{E}_\beta(\phi_i, \phi_j) &:= \int_{\Sigma_\beta} \nabla\phi_i \cdot \nabla\phi_j \, \mathrm{d}S \\ &= r_\mathrm{a}^2 c_i c_j \delta_{m_i m_j} \delta_{\sigma_i \sigma_j} \ell_{m_i}^{\sigma_i} \Big[\beta^2 U_i'(\beta) U_j'(\beta) I_{ij}^{(0)} \\ &\qquad + U_i(\beta) U_j(\beta) I_{ij}^{(1,m_i)} \Big], \end{aligned} \tag{18.8}$$

with

$$I_{ij}^{(0)} = \int_0^{\theta_0} A_i A_j \sin\theta \, \mathrm{d}\theta, \tag{18.9a}$$

$$I_{ij}^{(1,m)} = \int_0^{\theta_0} \left[A_i' A_j' + \frac{m^2}{\sin^2\theta} A_i A_j \right] \sin\theta \, \mathrm{d}\theta. \tag{18.9b}$$

Equation Equation (18.8) is the exact surface analogue of the volume factorisation. It contains no radial inner product and hence no factor δ_{pq} for distinct radial eigenfunctions.

18.3 Angular Green–Lagrange identity

Let

$$\mathscr{L}_m A = -\frac{1}{\sin\theta}(\sin\theta A')' + \frac{m^2}{\sin^2\theta} A. \tag{18.10}$$

For sufficiently regular A and B,

$$I^{(1,m)}(A,B) = \left[A(\theta) \sin\theta B'(\theta) \right]_0^{\theta_0} + \int_0^{\theta_0} A(\mathscr{L}_m B) \sin\theta \, \mathrm{d}\theta. \tag{18.11}$$

The pole term vanishes for regular functions. This identity determines all cross-family angular energy integrals.

For unit-normalised functions,

$$\mathscr{L}_m \widehat{P}_{km} = \lambda_{km} \widehat{P}_{km}, \qquad \widehat{P}_{km}(\theta_0) = 0, \tag{18.12a}$$

$$\mathscr{L}_m \widehat{K}_{pm} = -\mu_p \widehat{K}_{pm}, \tag{18.12b}$$

$$\mathscr{L}_m \widehat{C}_m = 0. \tag{18.12c}$$

18.4 Angular overlap notation

For fixed m, define

$$Q_{kp}^{(m)} = \int_0^{\theta_0} \widehat{P}_{km} \widehat{K}_{pm} \sin\theta \, \mathrm{d}\theta, \tag{18.13a}$$

$$M_{pq}^{(m)} = \int_0^{\theta_0} \widehat{K}_{pm} \widehat{K}_{qm} \sin\theta \, \mathrm{d}\theta, \tag{18.13b}$$

$$H_{pq}^{(m)} = I^{(1,m)}(\widehat{K}_{pm}, \widehat{K}_{qm}), \tag{18.13c}$$

$$H_{p0}^{(m)} = I^{(1,m)}(\widehat{K}_{pm}, \widehat{C}_m), \tag{18.13d}$$

$$H_{00}^{(m)} = I^{(1,m)}(\widehat{C}_m, \widehat{C}_m). \tag{18.13e}$$

The Green identity gives the explicit relations

$$I^{(1,m)}(\widehat{P}_{km}, \widehat{K}_{pm}) = -\mu_p Q_{kp}^{(m)}, \tag{18.14a}$$

$$I^{(1,m)}(\widehat{P}_{km}, \widehat{C}_m) = 0, \tag{18.14b}$$

$$H_{pq}^{(m)} = -\mu_q M_{pq}^{(m)} + \sin\theta_0 \widehat{K}_{pm}(\theta_0) \widehat{K}_{qm}'(\theta_0), \tag{18.14c}$$

$$H_{p0}^{(m)} = \sin\theta_0 \widehat{K}_{pm}(\theta_0) \widehat{C}_m'(\theta_0), \tag{18.14d}$$

$$H_{00}^{(m)} = \sin\theta_0 \widehat{C}_m(\theta_0) \widehat{C}_m'(\theta_0). \tag{18.14e}$$

Although Equation (18.14c) is written with μ_q, symmetry of the energy form implies the equivalent formula with p and q interchanged. Their equality is the Green–Lagrange Wronskian identity for the two Mehler parameters.

18.5 Coefficient-vector notation

Use the longitude-weighted product

$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle_{m,\lambda} = \ell_m^{\mathrm{c}} a^{\mathrm{c}} b^{\mathrm{c}} + \ell_m^{\mathrm{s}} a^{\mathrm{s}} b^{\mathrm{s}}, \tag{18.15}$$

with the sine term omitted at $m = 0$. Denote the unit-angular-normalisation coefficient vectors by

$$\widehat{\boldsymbol{a}}_{km}^{\mathrm{Int}}, \quad \widehat{\boldsymbol{a}}_{km}^{\mathrm{Ext}}, \quad \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}}, \quad \widehat{\boldsymbol{d}}_{0m}^{\mathrm{Cons}}. \tag{18.16}$$

18.6 Internal–internal block

At $r = \beta$,

$$U_{km}^{\mathrm{Int}}(\beta) = \left(\frac{r_{\mathrm{a}}}{\beta} \right)^{\nu_{km}+1}, \qquad \beta U_{km}^{\mathrm{Int}\prime}(\beta) = -(\nu_{km}+1) U_{km}^{\mathrm{Int}}(\beta). \tag{18.17}$$

Angular orthogonality removes $k \neq j$. For $k = j$,

$$\beta^2 (U^{\mathrm{Int}\prime})^2 + \lambda_{km} (U^{\mathrm{Int}})^2 = (\nu_{km}+1)(2\nu_{km}+1)(U^{\mathrm{Int}})^2. \tag{18.18}$$

Thus

$$E_\beta^{\mathrm{Int,Int}} = r_{\mathrm{a}}^2 \sum_{k,m} (\nu_{km}+1)(2\nu_{km}+1) \left(\frac{r_{\mathrm{a}}}{\beta} \right)^{2\nu_{km}+2} \| \widehat{\boldsymbol{a}}_{km}^{\mathrm{Int}} \|_{m,\lambda}^2. \tag{18.19}$$

18.7 External–external block

Similarly,

$$U_{km}^{\mathrm{Ext}}(\beta) = \left(\frac{\beta}{r_{\mathrm{a}}} \right)^{\nu_{km}}, \qquad \beta U_{km}^{\mathrm{Ext}\prime}(\beta) = \nu_{km} U_{km}^{\mathrm{Ext}}(\beta), \tag{18.20}$$

and

$$E_\beta^{\mathrm{Ext,Ext}} = r_{\mathrm{a}}^2 \sum_{k,m} \nu_{km}(2\nu_{km}+1) \left(\frac{\beta}{r_{\mathrm{a}}} \right)^{2\nu_{km}} \| \widehat{\boldsymbol{a}}_{km}^{\mathrm{Ext}} \|_{m,\lambda}^2. \tag{18.21}$$

18.8 Internal–external block

For the same (k, m),

$$\begin{aligned} &\beta^2 U_{km}^{\mathrm{Int}\prime}(\beta) U_{km}^{\mathrm{Ext}\prime}(\beta) + \lambda_{km} U_{km}^{\mathrm{Int}}(\beta) U_{km}^{\mathrm{Ext}}(\beta) \\ &\quad = -\nu_{km}(\nu_{km}+1) U_{km}^{\mathrm{Int}} U_{km}^{\mathrm{Ext}} + \nu_{km}(\nu_{km}+1) U_{km}^{\mathrm{Int}} U_{km}^{\mathrm{Ext}} = 0. \end{aligned} \tag{18.22}$$

Different k values are angularly orthogonal. Hence

$$E_\beta^{\mathrm{Int,Ext}} = 0. \tag{18.23}$$

This cancellation is valid on every fixed radius and does not require radial integration.

18.9 Legendre–constant blocks

The radial derivative of the constant radial mode is zero, and Equation (18.14b) gives zero horizontal cross energy. Therefore,

$$E_\beta^{\mathrm{Int,Cons}} = E_\beta^{\mathrm{Ext,Cons}} = 0. \tag{18.24}$$

This exact vanishing is stronger than the general statement that the complete V_1 and V_2 subspaces are non-orthogonal on one surface: the non-zero Legendre–V_2 coupling comes entirely from the Mehler family.

18.10 Internal–Mehler block

Using Equation (18.14a),

$$\begin{aligned} E_\beta^{\mathrm{Int,Meh}} = 2r_{\mathrm{a}}^2 \sum_m \sum_k \sum_{p\geq 1} &\langle \widehat{\boldsymbol{a}}_{km}^{\mathrm{Int}}, \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}} \rangle_{m,\lambda} Q_{kp}^{(m)} \\ &\times \left[\beta^2 U_{km}^{\mathrm{Int}\prime}(\beta) R_p'(\beta) - \mu_p U_{km}^{\mathrm{Int}}(\beta) R_p(\beta)\right]. \end{aligned} \tag{18.25}$$

Equivalently,

$$\begin{aligned} E_\beta^{\mathrm{Int,Meh}} = 2r_{\mathrm{a}}^2 \sum_{m,k,p} &\langle \widehat{\boldsymbol{a}}_{km}^{\mathrm{Int}}, \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}} \rangle_{m,\lambda} Q_{kp}^{(m)} U_{km}^{\mathrm{Int}}(\beta) \\ &\times \left[-\beta(\nu_{km}+1) R_p'(\beta) - \mu_p R_p(\beta)\right]. \end{aligned} \tag{18.26}$$

This term is generally non-zero.

18.11 External–Mehler block

The corresponding expression is

$$\begin{aligned} E_\beta^{\mathrm{Ext,Meh}} &= 2r_{\mathrm{a}}^2 \sum_m \sum_k \sum_{p\geq 1} \langle \widehat{\boldsymbol{a}}_{km}^{\mathrm{Ext}}, \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}} \rangle_{m,\lambda} Q_{kp}^{(m)} \\ &\quad \times \left[\beta^2 U_{km}^{\mathrm{Ext}\prime}(\beta) R_p'(\beta) - \mu_p U_{km}^{\mathrm{Ext}}(\beta) R_p(\beta)\right] \\ &= 2r_{\mathrm{a}}^2 \sum_{m,k,p} \langle \widehat{\boldsymbol{a}}_{km}^{\mathrm{Ext}}, \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}} \rangle_{m,\lambda} Q_{kp}^{(m)} U_{km}^{\mathrm{Ext}}(\beta) \\ &\quad \times \left[\beta \nu_{km} R_p'(\beta) - \mu_p R_p(\beta)\right]. \end{aligned} \tag{18.27}$$

18.12 Mehler–Mehler block

No δ_{pq} is available at fixed β. The exact block is

$$\begin{aligned} E_\beta^{\mathrm{Meh,Meh}} = r_{\mathrm{a}}^2 \sum_m \sum_{p\geq 1} \sum_{q\geq 1} &\langle \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}}, \widehat{\boldsymbol{d}}_{qm}^{\mathrm{Meh}} \rangle_{m,\lambda} \\ &\times \left[\beta^2 R_p'(\beta) R_q'(\beta) M_{pq}^{(m)} + R_p(\beta) R_q(\beta) H_{pq}^{(m)}\right]. \end{aligned} \tag{18.28}$$

Separating diagonal and off-diagonal terms gives

$$E_\beta^{\mathrm{Meh,Meh}} = r_{\mathrm{a}}^2 \sum_{m,p} \|\widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}}\|_{m,\lambda}^2 \left[\beta^2 (R_p')^2 + R_p^2 H_{pp}^{(m)}\right] + 2r_{\mathrm{a}}^2 \sum_m \sum_{1\le p<q} \langle \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}}, \widehat{\boldsymbol{d}}_{qm}^{\mathrm{Meh}} \rangle_{m,\lambda} \times \left[\beta^2 R_p' R_q' M_{pq}^{(m)} + R_p R_q H_{pq}^{(m)}\right], \tag{18.29}$$

where every radial quantity in the last two equations is evaluated at $r = \beta$.

18.13 Mehler–constant block

Because $R_0' = 0$, only the horizontal energy survives:

$$E_\beta^{\mathrm{Meh,Cons}} = 2r_{\mathrm{a}}^2 R_0 \sum_m \sum_{p\ge 1} \langle \widehat{\boldsymbol{d}}_{pm}^{\mathrm{Meh}}, \widehat{\boldsymbol{d}}_{0m}^{\mathrm{Cons}} \rangle_{m,\lambda} R_p(\beta) H_{p0}^{(m)}. \tag{18.30}$$

This block vanishes in the three-dimensional energy because $\int R_p R_0 \, \mathrm{d}r = 0$, but it generally survives on one radius.

18.14 Constant–constant block

The exact constant block is

$$E_\beta^{\mathrm{Cons,Cons}} = r_{\mathrm{a}}^2 R_0^2 \sum_m H_{00}^{(m)} \|\widehat{\boldsymbol{d}}_{0m}^{\mathrm{Cons}}\|_{m,\lambda}^2. \tag{18.31}$$

The $m = 0$ term is the additive-potential gauge and has zero energy.

18.15 Complete exact surface-energy formula

Collecting all blocks gives

$$E_\beta[V] = E_\beta^{\mathrm{Int,Int}} + E_\beta^{\mathrm{Ext,Ext}} + E_\beta^{\mathrm{Meh,Meh}} + E_\beta^{\mathrm{Cons,Cons}} + E_\beta^{\mathrm{Int,Meh}} + E_\beta^{\mathrm{Ext,Meh}} + E_\beta^{\mathrm{Meh,Cons}}. \tag{18.32}$$

The omitted pair blocks vanish exactly:

$$E_\beta^{\mathrm{Int,Ext}} = E_\beta^{\mathrm{Int,Cons}} = E_\beta^{\mathrm{Ext,Cons}} = 0. \tag{18.33}$$

18.16 Block Gram-matrix form

Order the coefficient vector by the four families. The exact surface Gram matrix has the block structure

$$\boldsymbol{G}_\beta = \begin{bmatrix} \boldsymbol{G}_{\mathrm{II}} & \boldsymbol{0} & \boldsymbol{G}_{\mathrm{IM}} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{G}_{\mathrm{EE}} & \boldsymbol{G}_{\mathrm{EM}} & \boldsymbol{0} \\ \boldsymbol{G}_{\mathrm{IM}}^{\mathsf{T}} & \boldsymbol{G}_{\mathrm{EM}}^{\mathsf{T}} & \boldsymbol{G}_{\mathrm{MM}} & \boldsymbol{G}_{\mathrm{MC}} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{G}_{\mathrm{MC}}^{\mathsf{T}} & \boldsymbol{G}_{\mathrm{CC}} \end{bmatrix}. \tag{18.34}$$

Hence

$$E_\beta[V] = \boldsymbol{c}^{\mathsf{T}} \boldsymbol{G}_\beta \boldsymbol{c}. \tag{18.35}$$

The compact quadratic form is equivalent to, not a replacement for, the fully expanded family expression Equation (18.32).

18.17 Positive semidefiniteness and null space

By definition,

$$\boldsymbol{c}^{\mathsf{T}}\boldsymbol{G}_{\beta}\boldsymbol{c} = \int_{\Sigma_{\beta}} |\nabla V|^2 \, \mathrm{d}S \geq 0. \tag{18.36}$$

For a finite coefficient set, the Gram matrix is therefore symmetric positive semidefinite. For an infinite expansion, the quadratic form is defined only on coefficients whose represented field has the required L^2 gradient trace; an infinite matrix is not automatically a bounded operator on an unweighted coefficient space.

On this domain, the functional null space of the continuous full-gradient form consists only of additive constants. Indeed, if a harmonic function with classical traces on an open surface patch satisfies

$$\nabla V = 0 \quad \text{on a non-empty open subset of } \Sigma_{\beta}, \tag{18.37}$$

then its tangential derivatives vanish, so its trace is constant on that patch, and its radial derivative vanishes there as well. After subtracting the constant, both the Dirichlet and Neumann Cauchy data vanish on an analytic non-characteristic surface patch. Unique continuation for harmonic functions gives

$$V \equiv \text{constant} \quad \text{in the connected shell.} \tag{18.38}$$

For a weak harmonic potential on a boundary patch, the same conclusion uses its constant Dirichlet trace and zero conormal trace there: after subtracting the constant, zero extension across the patch is weakly harmonic, so interior unique continuation applies. For an interior patch the potential is already smooth.

Consequently, for a finite coefficient family the matrix kernel is the preimage of constant potentials under the coefficient-to-function map. If the nonconstant modes are independent and one additive gauge mode is included, this kernel is one-dimensional; if no constant is represented, the kernel is trivial provided the family is independent. Additional kernel directions arise from redundant parametrisations. This injectivity statement supplies no positive lower bound uniform in truncation degree. Discrete sampling or observation of only selected components can introduce further null spaces.

18.18 Mean-square surface energy

The cap area is

$$|\Sigma_{\beta}| = 2\pi\beta^2(1 - \cos\theta_0). \tag{18.39}$$

A mean-square field measure is

$$\overline{E}_{\beta}[V] = \frac{E_{\beta}[V]}{2\pi\beta^2(1 - \cos\theta_0)}. \tag{18.40}$$

This normalisation changes the overall scale but not the family cross-term structure.

19 Simplified Legendre-dominant spectrum on a spherical-cap surface

The third spectral object is a one-dimensional spectrum obtained from the Legendre part of the fixed-surface energy. It is motivated by the fact that the exact surface energy in Section 18 contains non-diagonal Mehler and cross-family terms and therefore does not give a diagonal spectrum in the original R-SCHA modes. Allocating its cross terms to individual degrees requires an additional

convention. The simplified spectrum removes those terms by an explicit approximation and bins the remaining diagonal Legendre energies according to the non-integer degree.

The derivation in this section makes two points precise. First, the Legendre-only surface energy is itself exact once V_2 has been omitted. Second, the omission of V_2 is a modelling approximation, not a consequence of the three-dimensional gradient orthogonality.

19.1 The simplifying approximation

Write

$$V = V_1 + V_2, \qquad V_1 = V^{\mathrm{Int}} + V^{\mathrm{Ext}}. \tag{19.1}$$

The exact surface energy is

$$E_\beta[V] = E_\beta[V_1] + E_\beta[V_2] + 2\mathcal{E}_\beta(V_1, V_2). \tag{19.2}$$

The simplified construction assumes

$$E_\beta[V] \approx E_\beta[V_1]. \tag{19.3}$$

Equivalently, it neglects

$$E_\beta[V_2] + 2\mathcal{E}_\beta(V_1, V_2). \tag{19.4}$$

No exact R-SCHA identity sets Equation (19.4) to zero on a fixed radius. The error can be bounded by Cauchy–Schwarz:

$$|E_\beta[V] - E_\beta[V_1]| \le E_\beta[V_2] + 2\sqrt{E_\beta[V_1]E_\beta[V_2]}. \tag{19.5}$$

For $E_\beta[V_1] > 0$, a verified ratio $E_\beta[V_2]/E_\beta[V_1] \le \epsilon^2$ therefore gives a relative error bound $2\epsilon + \epsilon^2$ with respect to the retained energy. Spatial localisation near the lateral boundary, by itself, is not such a bound: large field gradients in a narrow buffer can carry appreciable surface energy.

19.2 Exact diagonal form of the retained Legendre energy

Within V_1, the internal and external families are angularly diagonal, and the internal–external cross term cancels exactly on every radius. Hence

$$E_\beta[V_1] = E_\beta^{\mathrm{Int}} + E_\beta^{\mathrm{Ext}}. \tag{19.6}$$

For unit-normalised angular functions, the individual mode energies are

$$\varepsilon_{km}^{\mathrm{Int}}(\beta) = r_{\mathrm{a}}^2(2\nu_{km}+1)(\nu_{km}+1)\left(\frac{r_{\mathrm{a}}}{\beta}\right)^{2\nu_{km}+2} \|\widehat{\boldsymbol{a}}_{km}^{\mathrm{Int}}\|_{m,\lambda}^2, \tag{19.7a}$$

$$\varepsilon_{km}^{\mathrm{Ext}}(\beta) = r_{\mathrm{a}}^2(2\nu_{km}+1)\nu_{km}\left(\frac{\beta}{r_{\mathrm{a}}}\right)^{2\nu_{km}} \|\widehat{\boldsymbol{a}}_{km}^{\mathrm{Ext}}\|_{m,\lambda}^2. \tag{19.7b}$$

Therefore,

$$E_\beta^{\mathrm{Int}} = \sum_{k,m} \varepsilon_{km}^{\mathrm{Int}}(\beta), \tag{19.8a}$$

$$E_\beta^{\mathrm{Ext}} = \sum_{k,m} \varepsilon_{km}^{\mathrm{Ext}}(\beta), \tag{19.8b}$$

$$E_\beta[V_1] = \sum_{k,m} \left[\varepsilon_{km}^{\mathrm{Int}}(\beta) + \varepsilon_{km}^{\mathrm{Ext}}(\beta)\right]. \tag{19.8c}$$

Equation Equation (19.8) is the complete, unbinned Legendre surface-energy expression.

The two mode factors in Equation (19.7) are the diagonal blocks derived in Equations (18.19) and (18.21): the radial and horizontal contributions add to $(\nu+1)^2+\nu(\nu+1)=(\nu+1)(2\nu+1)$ for an internal mode, and to $\nu^2+\nu(\nu+1)=\nu(2\nu+1)$ for an external mode.

19.3 Raw line spectrum in non-integer degree

Before binning, the exact Legendre energy may be represented as the discrete measure

$$\mathrm{d}\mathscr{S}_\beta^{\mathrm{Leg}}(\nu)=\sum_{k,m}\left[\varepsilon_{km}^{\mathrm{Int}}(\beta)+\varepsilon_{km}^{\mathrm{Ext}}(\beta)\right]\delta(\nu-\nu_{km})\,\mathrm{d}\nu, \tag{19.9}$$

where δ is the Dirac measure. Its total mass is

$$\int_0^\infty \mathrm{d}\mathscr{S}_\beta^{\mathrm{Leg}}(\nu)=E_\beta[V_1]. \tag{19.10}$$

This form preserves the exact non-integer degrees and does not require an arbitrary binning convention.

19.4 Asymptotic root spacing and cap-order bins

For fixed m and sufficiently large root index, the Dirichlet cap roots have the approximate spacing

$$\Delta\nu\sim\frac{\pi}{\theta_0}. \tag{19.11}$$

Define the intervals

$$\mathcal{B}_j=[j\Delta\nu,(j+1)\Delta\nu),\qquad j=0,1,2,\ldots. \tag{19.12}$$

The corresponding cap-order coordinate is

$$\kappa=\frac{\theta_0}{\pi}\nu, \tag{19.13}$$

so that a bin of width $\Delta\nu=\pi/\theta_0$ has unit width in κ.

19.5 Binned internal, external, and total spectra

Define

$$\mathscr{S}_j^{\mathrm{Int}}(\beta)=\frac{1}{\Delta\nu}\sum_{\substack{k,m\\ \nu_{km}\in\mathcal{B}_j}}\varepsilon_{km}^{\mathrm{Int}}(\beta), \tag{19.14a}$$

$$\mathscr{S}_j^{\mathrm{Ext}}(\beta)=\frac{1}{\Delta\nu}\sum_{\substack{k,m\\ \nu_{km}\in\mathcal{B}_j}}\varepsilon_{km}^{\mathrm{Ext}}(\beta), \tag{19.14b}$$

$$\mathscr{S}_j^{\mathrm{Leg}}(\beta)=\mathscr{S}_j^{\mathrm{Int}}(\beta)+\mathscr{S}_j^{\mathrm{Ext}}(\beta). \tag{19.14c}$$

Using $1/\Delta\nu=\theta_0/\pi$,

$$\mathscr{S}_j^{\mathrm{Leg}}(\beta)=\frac{\theta_0}{\pi}\sum_{\substack{k,m\\ \nu_{km}\in\mathcal{B}_j}}\Bigg\{r_{\mathrm{a}}^2(2\nu_{km}+1)(\nu_{km}+1)\left(\frac{r_{\mathrm{a}}}{\beta}\right)^{2\nu_{km}+2}\|\widehat{\boldsymbol{a}}_{km}^{\mathrm{Int}}\|_{m,\lambda}^2 + r_{\mathrm{a}}^2(2\nu_{km}+1)\nu_{km}\left(\frac{\beta}{r_{\mathrm{a}}}\right)^{2\nu_{km}}\|\widehat{\boldsymbol{a}}_{km}^{\mathrm{Ext}}\|_{m,\lambda}^2\Bigg\}. \tag{19.15}$$

This is the fully expanded binned simplified spectrum.

19.6 Recovery of the retained energy from the bins

If the bins cover every retained Legendre degree exactly once, then

$$\begin{aligned}\sum_j \Delta\nu\, \mathscr{S}_j^{\mathrm{Leg}}(\beta) &= \sum_{k,m} \left[\varepsilon_{km}^{\mathrm{Int}}(\beta) + \varepsilon_{km}^{\mathrm{Ext}}(\beta)\right] \\ &= E_\beta[V_1]. \end{aligned} \tag{19.16}$$

The binned spectrum therefore conserves the retained Legendre energy. It does not recover the omitted V_2 and cross-family energy.

19.7 Mean-square spectral density

Dividing by the cap area gives the mean-square spectral density

$$\overline{\mathscr{S}}_j^{\mathrm{Leg}}(\beta) = \frac{\mathscr{S}_j^{\mathrm{Leg}}(\beta)}{2\pi\beta^2(1-\cos\theta_0)}. \tag{19.17}$$

Its bin integral recovers the mean-square Legendre field on the cap.

19.8 Cumulative spectrum

Define

$$\mathscr{C}_J(\beta) = \sum_{j=0}^{J} \Delta\nu\, \mathscr{S}_j^{\mathrm{Leg}}(\beta). \tag{19.18}$$

Then

$$\lim_{J\to\infty} \mathscr{C}_J(\beta) = E_\beta[V_1] \tag{19.19}$$

whenever the Legendre energy series converges. The energy in a degree range $[\nu_1, \nu_2)$ is obtained by summing only the bins or line-spectrum masses within that interval.

19.9 Nominal horizontal scale

The angular Laplace–Beltrami eigenvalue of degree ν is

$$\lambda_\nu = \nu(\nu+1). \tag{19.20}$$

At radius β, define

$$\kappa_h(\nu;\beta) = \frac{\sqrt{\nu(\nu+1)}}{\beta} \tag{19.21}$$

and

$$L_{\mathrm{nom}}(\nu;\beta) = \frac{2\pi\beta}{\sqrt{\nu(\nu+1)}}. \tag{19.22}$$

For large degree,

$$L_{\mathrm{nom}}(\nu;\beta) \sim \frac{2\pi\beta}{\nu+1/2}. \tag{19.23}$$

These quantities label the analytical scale of a mode. They do not, by themselves, constitute a theorem about the resolving power of a coefficient estimate from finite observations.

19.10 Relation to ordinary spherical-harmonic spectra

A global spherical harmonic of integer degree n has angular eigenvalue $n(n+1)$. Matching eigenvalues gives the continuous equivalent degree

$$n_{\mathrm{eq}}(n_{\mathrm{eq}}+1) = \nu(\nu+1), \tag{19.24}$$

so that the non-negative solution is

$$n_{\mathrm{eq}} = \nu. \tag{19.25}$$

Thus the R-SCHA non-integer degree can be compared directly with a continuous extension of the global spherical-harmonic degree. The cap-order coordinate $\kappa = (\theta_0/\pi)\nu$ instead measures the number of approximate root spacings across the cap and should not be confused with n_{eq}.

19.11 Unnormalised coefficients

If $P^m_{\nu_{km}}$ is not unit-normalised, define

$$\mathcal{N}^P_{km} = \int_0^{\theta_0} [P^m_{\nu_{km}}(\cos\theta)]^2 \sin\theta \, \mathrm{d}\theta. \tag{19.26}$$

Then the squared coefficient norm in Equation (19.7) is replaced by

$$\|\widehat{\boldsymbol{a}}_{km}\|^2_{m,\lambda} = \mathcal{N}^P_{km} \left[\ell^{\mathrm{c}}_m (g_{km})^2 + \ell^{\mathrm{s}}_m (h_{km})^2\right]. \tag{19.27}$$

This gives the complete formula directly in terms of the unnormalised Legendre coefficients.

Part V
Representation and Analytical Approximation

20 Mapping between spherical harmonics and R-SCHA

A globally defined harmonic potential restricted to a spherical-cap shell may be represented by its spherical-harmonic series or by the boundary-adapted R-SCHA series. The coefficient systems are different because the two representations use different angular spectra and different radial boundary conditions. Nevertheless, once the spherical-harmonic model has been expressed in the cap-centred coordinate system, its boundary data determine the R-SCHA coefficients by orthogonal projection.

The construction in this section follows the boundary-projection principle of R-SCHA: the internal and external Legendre coefficients are determined from the normal derivatives on the lower and upper spherical boundaries, while the Mehler and constant coefficients are determined from the potential trace on the lateral boundary [9, 8, 7]. The derivation below makes the normalisation factors explicit and separates the mapping into angular, radial, and two-boundary algebraic operators.

20.1 Coefficient-space rotation of spherical harmonics

This subsection supplies the coefficient rotation that is otherwise hidden in the phrase "rotate the spherical-harmonic model to the cap frame." Let $Y_{\ell m}$ denote a declared complex spherical-harmonic

convention and let $\boldsymbol{A}_{\mathrm{c}}$ be the proper orthogonal rotation defined in Equation (15.32). Define the degree-ℓ rotation matrix by the identity

$$Y_{\ell m}\left(\boldsymbol{A}_{\mathrm{c}}^{-1}\widehat{\boldsymbol{r}}\right)=\sum_{q=-\ell}^{\ell}D_{mq}^{\ell}(\boldsymbol{A}_{\mathrm{c}})Y_{\ell q}(\widehat{\boldsymbol{r}}). \tag{20.1}$$

If

$$V_{\ell}(r,\widehat{\boldsymbol{r}})=f_{\ell}(r)\sum_{q=-\ell}^{\ell}c_{\ell q}Y_{\ell q}(\widehat{\boldsymbol{r}}), \tag{20.2}$$

then invariance of the scalar potential under a passive coordinate change and Equation (20.1) give

$$\widetilde{c}_{\ell q}=\sum_{m=-\ell}^{\ell}D_{mq}^{\ell}(\boldsymbol{A}_{\mathrm{c}})c_{\ell m}. \tag{20.3}$$

Thus, if the coefficient arrays are stored as column vectors ordered by $m=-\ell,\ldots,\ell$, the matrix form associated with the defining identity Equation (20.1) is

$$\widetilde{\boldsymbol{c}}_{\ell}=\left[\boldsymbol{D}^{\ell}(\boldsymbol{A}_{\mathrm{c}})\right]^{\mathsf{T}}\boldsymbol{c}_{\ell}. \tag{20.4}$$

The transpose is forced by the index placement in Equation (20.1); replacing it by $\boldsymbol{D}^{\ell}\boldsymbol{c}_{\ell}$ would correspond to the alternative defining convention with the two matrix indices interchanged. The rotation is block diagonal in degree: it mixes the $2\ell+1$ orders at a fixed ℓ but never mixes different degrees or the internal and external radial branches.

For orthonormal complex spherical harmonics with the Condon–Shortley phase, use the standard active Cartesian z–y–z Euler decomposition of the matrix $\boldsymbol{A}_{\mathrm{c}}$, explicitly

$$\boldsymbol{A}_{\mathrm{c}}=\boldsymbol{R}_{z}^{\mathrm{a}}(\alpha)\boldsymbol{R}_{y}^{\mathrm{a}}(\beta)\boldsymbol{R}_{z}^{\mathrm{a}}(\gamma), \tag{20.5}$$

where the superscript a denotes an active right-handed rotation (the transposes of the passive matrices in Section 15). With the index placement of Equation (20.1),

$$D_{mq}^{\ell}(\alpha,\beta,\gamma)=\mathrm{e}^{-\mathrm{i}m\gamma}d_{mq}^{\ell}(\beta)\mathrm{e}^{-\mathrm{i}q\alpha}, \tag{20.6}$$

The reversed placement of the first and third Euler phases follows from the defining identity: $\boldsymbol{D}(\boldsymbol{AB})=\boldsymbol{D}(\boldsymbol{B})\boldsymbol{D}(\boldsymbol{A})$. The coefficient action $\boldsymbol{D}^{\mathsf{T}}$ consequently has the usual composition order. The small-d element has the finite sum

$$\begin{aligned}d_{mq}^{\ell}(\beta)=&\sqrt{(\ell+m)!(\ell-m)!(\ell+q)!(\ell-q)!}\\&\times\sum_{t=t_{\min}}^{t_{\max}}\frac{(-1)^{t-m+q}\left(\cos\dfrac{\beta}{2}\right)^{2\ell+m-q-2t}\left(\sin\dfrac{\beta}{2}\right)^{q-m+2t}}{(\ell+m-t)!\,t!\,(q-m+t)!\,(\ell-q-t)!},\end{aligned} \tag{20.7}$$

with

$$t_{\min}=\max(0,m-q),\qquad t_{\max}=\min(\ell+m,\ell-q). \tag{20.8}$$

The Euler angles must be extracted from the same active/passive and axis-order convention used to define $\boldsymbol{A}_{\mathrm{c}}$. The convention-independent definition Equation (20.1) should therefore be treated as primary.

Real cosine–sine coefficients are rotated by converting each degree block to its complex conjugate-symmetric form, applying Equation (20.3), and converting back. If a Condon–Shortley phase or Schmidt normalisation is changed, the real–complex conversion matrix changes accordingly, while the scalar field remains unchanged. The identity is exact because both coefficient blocks represent the same scalar spherical-harmonic degree subspace in two rotated orthonormal frames.

20.2 Prerequisite: rotation to the cap-centred frame

Let the target cap be centred at global co-latitude θ_{c} and longitude λ_{c}. Before applying the mapping formulae, the global spherical-harmonic model must be represented in the cap-centred spherical coordinates

$$(r, \theta, \lambda),$$

where the cap centre is the local north pole $\theta = 0$. The required coordinate rotation is defined in Section 15.

The exact rotation is the degree-preserving coefficient transformation derived in the preceding subsection. The mapping formulae below assume that this transformation has already been applied. Coefficients referred to the original global polar axis cannot be inserted directly because their angular basis is not the cap-centred basis used by the regional boundary problem.

20.3 Rotated spherical-harmonic potential

Let $r_{\mathrm{ref}} > 0$ be the reference radius of the rotated spherical-harmonic model. Define the coefficient vectors

$$\boldsymbol{c}^{\mathrm{i}}_{nm} = \begin{bmatrix} g^{\mathrm{i}}_{nm} \\ h^{\mathrm{i}}_{nm} \end{bmatrix}, \tag{20.9a}$$

$$\boldsymbol{c}^{\mathrm{e}}_{nm} = \begin{bmatrix} g^{\mathrm{e}}_{nm} \\ h^{\mathrm{e}}_{nm} \end{bmatrix}, \tag{20.9b}$$

$$\boldsymbol{t}_m(\lambda) = \begin{bmatrix} \cos(m\lambda) \\ \sin(m\lambda) \end{bmatrix}. \tag{20.9c}$$

For $m = 0$, the second coefficient in each vector is zero.

The rotated spherical-harmonic potential is

$$\begin{aligned} V^{\mathrm{SH}}(r, \theta, \lambda) = r_{\mathrm{ref}} \sum_{n=0}^{L_{\mathrm{i}}} \sum_{m=0}^{n} \left(\frac{r_{\mathrm{ref}}}{r}\right)^{n+1} P^m_n(\cos\theta) \left(\boldsymbol{c}^{\mathrm{i}}_{nm}\right)^{\mathsf{T}} \boldsymbol{t}_m(\lambda) \\ + r_{\mathrm{ref}} \sum_{n=0}^{L_{\mathrm{e}}} \sum_{m=0}^{n} \left(\frac{r}{r_{\mathrm{ref}}}\right)^{n} P^m_n(\cos\theta) \left(\boldsymbol{c}^{\mathrm{e}}_{nm}\right)^{\mathsf{T}} \boldsymbol{t}_m(\lambda). \end{aligned} \tag{20.10}$$

All associated Legendre functions and coefficients in Equation (20.10) must use a declared and consistent normalisation and phase convention.

20.4 R-SCHA coefficient vectors

For the Legendre part of R-SCHA, define

$$\boldsymbol{a}^{\mathrm{Int}}_{km} = \begin{bmatrix} g^{\mathrm{Int}}_{km} \\ h^{\mathrm{Int}}_{km} \end{bmatrix}, \tag{20.11a}$$

$$\boldsymbol{a}^{\mathrm{Ext}}_{km} = \begin{bmatrix} g^{\mathrm{Ext}}_{km} \\ h^{\mathrm{Ext}}_{km} \end{bmatrix}. \tag{20.11b}$$

For the V_2 families, define

$$\boldsymbol{d}^{\mathrm{Meh}}_{pm} = \begin{bmatrix} G^m_p \\ H^m_p \end{bmatrix}, \qquad p \geq 1, \tag{20.12a}$$

$$\boldsymbol{d}^{\mathrm{Cons}}_{0m} = \begin{bmatrix} G^m_0 \\ H^m_0 \end{bmatrix}. \tag{20.12b}$$

The same scalar mapping weights act on the cosine and sine coefficients. Consequently, no cosine–sine mixing occurs after the spherical-harmonic model has been rotated to the cap-centred axis.

20.5 Angular overlap matrix

For fixed order m, define the overlap between an integer-degree spherical harmonic and a non-integer cap eigenfunction by

$$\mathcal{I}^{(m)}_{n,km} = \int_0^{\theta_0} P^m_n(\cos\theta) P^m_{\nu_{km}}(\cos\theta) \sin\theta \, \mathrm{d}\theta. \tag{20.13}$$

The cap norm is

$$\mathcal{N}^P_{km} = \int_0^{\theta_0} \left[P^m_{\nu_{km}}(\cos\theta)\right]^2 \sin\theta \, \mathrm{d}\theta, \tag{20.14}$$

and the normalised projection weight is

$$\mathcal{Q}^{(m)}_{n,km} = \frac{\mathcal{I}^{(m)}_{n,km}}{\mathcal{N}^P_{km}}. \tag{20.15}$$

When the cap eigenfunctions are unit normalised, $\mathcal{N}^P_{km} = 1$.

The overlap admits both its defining integral and a closed Sturm–Liouville boundary formula. Let

$$x_0 = \cos\theta_0.$$

If

$$\nu_{km}(\nu_{km}+1) \neq n(n+1),$$

then

$$\mathcal{I}^{(m)}_{n,km} = \frac{\left(1-x_0^2\right) P^m_n(x_0) \left.\dfrac{\partial P^m_\nu(x_0)}{\partial x}\right|_{\nu=\nu_{km}}}{\nu_{km}(\nu_{km}+1) - n(n+1)}. \tag{20.16}$$

This formula follows from the Green–Lagrange identity and

$$P^m_{\nu_{km}}(x_0) = 0.$$

If the two eigenvalues coincide, the quotient is interpreted by its continuous limiting value, equivalently by the defining integral.

20.6 Normal-derivative matching on the lower boundary

The first R-SCHA subpotential satisfies

$$V_1 = 0 \quad \text{on } \Gamma_{\theta_0},$$

and carries the radial normal-derivative data. On the lower boundary, equality of the spherical-harmonic and R-SCHA outward normal derivatives is equivalent to

$$-\left.\frac{\partial V^{\mathrm{SH}}}{\partial r}\right|_{r=r_{\mathrm{a}}} = -\left.\frac{\partial V_1}{\partial r}\right|_{r=r_{\mathrm{a}}}. \tag{20.17}$$

For every pair (k, m), define the projected lower-boundary vector

$$\boldsymbol{b}^{\mathrm{a}}_{km} = \sum_{n=m}^{L_{\mathrm{i}}} (n+1) \left(\frac{r_{\mathrm{ref}}}{r_{\mathrm{a}}}\right)^{n+2} \mathcal{Q}^{(m)}_{n,km} \boldsymbol{c}^{\mathrm{i}}_{nm} - \sum_{n=m}^{L_{\mathrm{e}}} n \left(\frac{r_{\mathrm{a}}}{r_{\mathrm{ref}}}\right)^{n-1} \mathcal{Q}^{(m)}_{n,km} \boldsymbol{c}^{\mathrm{e}}_{nm}. \tag{20.18}$$

Projection of Equation (20.17) onto $P^m_{\nu_{km}}$ gives

$$(\nu_{km}+1)\, \boldsymbol{a}^{\mathrm{Int}}_{km} - \nu_{km} \boldsymbol{a}^{\mathrm{Ext}}_{km} = \boldsymbol{b}^{\mathrm{a}}_{km}. \tag{20.19}$$

20.7 Normal-derivative matching on the upper boundary

On the upper boundary,

$$\left.\frac{\partial V^{\mathrm{SH}}}{\partial r}\right|_{r=r_{\mathrm{b}}} = \left.\frac{\partial V_1}{\partial r}\right|_{r=r_{\mathrm{b}}}. \tag{20.20}$$

Define the projected upper-boundary vector

$$\boldsymbol{b}^{\mathrm{b}}_{km} = -\sum_{n=m}^{L_{\mathrm{i}}} (n+1) \left(\frac{r_{\mathrm{ref}}}{r_{\mathrm{b}}}\right)^{n+2} \mathcal{Q}^{(m)}_{n,km} \boldsymbol{c}^{\mathrm{i}}_{nm} + \sum_{n=m}^{L_{\mathrm{e}}} n \left(\frac{r_{\mathrm{b}}}{r_{\mathrm{ref}}}\right)^{n-1} \mathcal{Q}^{(m)}_{n,km} \boldsymbol{c}^{\mathrm{e}}_{nm}. \tag{20.21}$$

Projection gives

$$-(\nu_{km}+1) \left(\frac{r_{\mathrm{a}}}{r_{\mathrm{b}}}\right)^{\nu_{km}+2} \boldsymbol{a}^{\mathrm{Int}}_{km} + \nu_{km} \left(\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}\right)^{\nu_{km}-1} \boldsymbol{a}^{\mathrm{Ext}}_{km} = \boldsymbol{b}^{\mathrm{b}}_{km}. \tag{20.22}$$

20.8 Two-boundary linear system

Introduce

$$\alpha_{km} = \left(\frac{r_{\mathrm{a}}}{r_{\mathrm{b}}}\right)^{\nu_{km}+2}, \tag{20.23a}$$

$$\beta_{km} = \left(\frac{r_{\mathrm{b}}}{r_{\mathrm{a}}}\right)^{\nu_{km}-1}, \tag{20.23b}$$

$$D_{km} = \beta_{km} - \alpha_{km}. \tag{20.23c}$$

For the positive Dirichlet spectrum $\nu_{km} > 0$ and $r_{\mathrm{b}} > r_{\mathrm{a}} > 0$,

$$D_{km} > 0. \tag{20.24}$$

Equations (20.19) and (20.22) form the 2×2 system

$$\begin{bmatrix} \nu_{km}+1 & -\nu_{km} \\ -(\nu_{km}+1)\alpha_{km} & \nu_{km}\beta_{km} \end{bmatrix} \begin{bmatrix} \boldsymbol{a}^{\mathrm{Int}}_{km} \\ \boldsymbol{a}^{\mathrm{Ext}}_{km} \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}^{\mathrm{a}}_{km} \\ \boldsymbol{b}^{\mathrm{b}}_{km} \end{bmatrix}. \tag{20.25}$$

Its solution is

$$\boldsymbol{a}_{km}^{\mathrm{Int}} = \frac{\beta_{km}\boldsymbol{b}_{km}^{\mathrm{a}} + \boldsymbol{b}_{km}^{\mathrm{b}}}{(\nu_{km}+1)\,D_{km}}, \tag{20.26a}$$

$$\boldsymbol{a}_{km}^{\mathrm{Ext}} = \frac{\alpha_{km}\boldsymbol{b}_{km}^{\mathrm{a}} + \boldsymbol{b}_{km}^{\mathrm{b}}}{\nu_{km}D_{km}}. \tag{20.26b}$$

These equations map the rotated spherical-harmonic coefficients to the internal and external R-SCHA Legendre coefficients.

20.9 Fully expanded Legendre mapping

Substitution of the lower-boundary projection Equation (20.18) and the upper-boundary projection Equation (20.21) into Equation (20.26) gives

$$\begin{aligned}\boldsymbol{a}_{km}^{\mathrm{Int}} = \frac{1}{(\nu_{km}+1)\,D_{km}}\Bigg\{ &\sum_{n=m}^{L_{\mathrm{i}}}(n+1)\Bigg[\beta_{km}\left(\frac{r_{\mathrm{ref}}}{r_{\mathrm{a}}}\right)^{n+2} \\ &\qquad - \left(\frac{r_{\mathrm{ref}}}{r_{\mathrm{b}}}\right)^{n+2}\Bigg]\mathcal{Q}_{n,km}^{(m)}\boldsymbol{c}_{nm}^{\mathrm{i}} \\ &+ \sum_{n=m}^{L_{\mathrm{e}}} n\left[\left(\frac{r_{\mathrm{b}}}{r_{\mathrm{ref}}}\right)^{n-1} - \beta_{km}\left(\frac{r_{\mathrm{a}}}{r_{\mathrm{ref}}}\right)^{n-1}\right]\mathcal{Q}_{n,km}^{(m)}\boldsymbol{c}_{nm}^{\mathrm{e}}\Bigg\},\end{aligned} \tag{20.27}$$

and

$$\begin{aligned}\boldsymbol{a}_{km}^{\mathrm{Ext}} = \frac{1}{\nu_{km}D_{km}}\Bigg\{ &\sum_{n=m}^{L_{\mathrm{i}}}(n+1)\Bigg[\alpha_{km}\left(\frac{r_{\mathrm{ref}}}{r_{\mathrm{a}}}\right)^{n+2} \\ &\qquad - \left(\frac{r_{\mathrm{ref}}}{r_{\mathrm{b}}}\right)^{n+2}\Bigg]\mathcal{Q}_{n,km}^{(m)}\boldsymbol{c}_{nm}^{\mathrm{i}} \\ &+ \sum_{n=m}^{L_{\mathrm{e}}} n\left[\left(\frac{r_{\mathrm{b}}}{r_{\mathrm{ref}}}\right)^{n-1} - \alpha_{km}\left(\frac{r_{\mathrm{a}}}{r_{\mathrm{ref}}}\right)^{n-1}\right]\mathcal{Q}_{n,km}^{(m)}\boldsymbol{c}_{nm}^{\mathrm{e}}\Bigg\}.\end{aligned} \tag{20.28}$$

Remark 20.1 (The case $\nu_{km} = 0$). The external solution Equation (20.26b) contains ν_{km} in the denominator. The non-integer Dirichlet spectrum used for the Legendre family is positive for the standard cap problem, so this degeneracy does not occur there. The zero separation eigenvalue is represented separately by the constant-function family.

20.10 Lateral trace after the Legendre mapping

At the lateral boundary,

$$\theta = \theta_0,$$

the Legendre R-SCHA modes vanish:

$$P_{\nu_{km}}^{m}(\cos\theta_0) = 0. \tag{20.29}$$

Therefore,

$$V^{\mathrm{SH}}(r,\theta_0,\lambda) = V_2(r,\theta_0,\lambda). \tag{20.30}$$

For every order m, define the vector-valued spherical-harmonic lateral trace

$$\boldsymbol{f}_m(r) = r_{\mathrm{ref}} \sum_{n=m}^{L_{\mathrm{i}}} \left(\frac{r_{\mathrm{ref}}}{r}\right)^{n+1} P_n^m(\cos\theta_0)\boldsymbol{c}_{nm}^{\mathrm{i}} + r_{\mathrm{ref}} \sum_{n=m}^{L_{\mathrm{e}}} \left(\frac{r}{r_{\mathrm{ref}}}\right)^{n} P_n^m(\cos\theta_0)\boldsymbol{c}_{nm}^{\mathrm{e}}. \tag{20.31}$$

Then

$$V^{\mathrm{SH}}(r,\theta_0,\lambda) = \sum_{m=0}^{\infty} \boldsymbol{f}_m^{\mathsf{T}}(r)\boldsymbol{t}_m(\lambda). \tag{20.32}$$

20.11 Radial projection notation

Define the spherical-harmonic radial functions

$$I_n(r) = \left(\frac{r_{\mathrm{ref}}}{r}\right)^{n+1}, \tag{20.33a}$$

$$E_n(r) = \left(\frac{r}{r_{\mathrm{ref}}}\right)^{n}. \tag{20.33b}$$

For $p \geq 1$, define

$$\mathcal{J}_{pn}^{\mathrm{i}} = \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} I_n(r)R_p(r)\ \mathrm{d}r, \tag{20.34a}$$

$$\mathcal{J}_{pn}^{\mathrm{e}} = \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} E_n(r)R_p(r)\ \mathrm{d}r. \tag{20.34b}$$

For the constant radial mode, define

$$\mathcal{J}_{0n}^{\mathrm{i}} = \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} I_n(r)R_0\ \mathrm{d}r, \tag{20.35a}$$

$$\mathcal{J}_{0n}^{\mathrm{e}} = \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} E_n(r)R_0\ \mathrm{d}r. \tag{20.35b}$$

Closed forms for all four integrals are given in Equations (11.54) to (11.56) and (11.58).

Retain general radial norms

$$\mathcal{N}_p^R = \int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R_p^2(r)\ \mathrm{d}r. \tag{20.36}$$

Under the convention adopted in Section 11,

$$\mathcal{N}_p^R = r_{\mathrm{a}}, \qquad p = 0, 1, 2, \ldots. \tag{20.37}$$

20.12 Mehler coefficient mapping

Project Equation (20.30) onto $R_p(r)$ and the cosine or sine function of order m. Radial and longitude orthogonality give

$$\boldsymbol{d}_{pm}^{\mathrm{Meh}} = \frac{r_{\mathrm{ref}}}{r_{\mathrm{a}}\mathcal{N}_p^R K_p^m(\theta_0)} \left[\sum_{n=m}^{L_{\mathrm{i}}} \mathcal{J}_{pn}^{\mathrm{i}} P_n^m(\cos\theta_0)\boldsymbol{c}_{nm}^{\mathrm{i}} + \sum_{n=m}^{L_{\mathrm{e}}} \mathcal{J}_{pn}^{\mathrm{e}} P_n^m(\cos\theta_0)\boldsymbol{c}_{nm}^{\mathrm{e}} \right], \qquad p \geq 1. \tag{20.38}$$

With $\mathcal{N}_p^R = r_\mathrm{a}$,

$$\boldsymbol{d}_{pm}^{\mathrm{Meh}} = \frac{r_\mathrm{ref}}{r_\mathrm{a}^2 K_p^m(\theta_0)} \left[\sum_{n=m}^{L_\mathrm{i}} \mathcal{J}_{pn}^\mathrm{i} P_n^m(\cos\theta_0) \boldsymbol{c}_{nm}^\mathrm{i} + \sum_{n=m}^{L_\mathrm{e}} \mathcal{J}_{pn}^\mathrm{e} P_n^m(\cos\theta_0) \boldsymbol{c}_{nm}^\mathrm{e} \right]. \tag{20.39}$$

Remark 20.2 (Normalisation dependence of the denominator). The leading factor in the potential expansion is r_a, and the radial functions in this work satisfy $\int R_p^2 \, \mathrm{d}r = r_\mathrm{a}$. The resulting denominator is $r_\mathrm{a}^2 K_p^m(\theta_0)$. If unit-normalised radial functions are used instead, the denominator becomes $r_\mathrm{a} K_p^m(\theta_0)$. A mapping formula must therefore be interpreted together with its radial normalisation convention.

20.13 Constant coefficient mapping

Projection onto the constant radial function gives

$$\boldsymbol{d}_{0m}^{\mathrm{Cons}} = \frac{r_\mathrm{ref}}{r_\mathrm{a} \mathcal{N}_0^R \mathcal{C}_m(\theta_0)} \left[\sum_{n=m}^{L_\mathrm{i}} \mathcal{J}_{0n}^\mathrm{i} P_n^m(\cos\theta_0) \boldsymbol{c}_{nm}^\mathrm{i} + \sum_{n=m}^{L_\mathrm{e}} \mathcal{J}_{0n}^\mathrm{e} P_n^m(\cos\theta_0) \boldsymbol{c}_{nm}^\mathrm{e} \right]. \tag{20.40}$$

Under the present convention, $\mathcal{N}_0^R = r_\mathrm{a}$, so

$$\boldsymbol{d}_{0m}^{\mathrm{Cons}} = \frac{r_\mathrm{ref}}{r_\mathrm{a}^2 \mathcal{C}_m(\theta_0)} \left[\sum_{n=m}^{L_\mathrm{i}} \mathcal{J}_{0n}^\mathrm{i} P_n^m(\cos\theta_0) \boldsymbol{c}_{nm}^\mathrm{i} + \sum_{n=m}^{L_\mathrm{e}} \mathcal{J}_{0n}^\mathrm{e} P_n^m(\cos\theta_0) \boldsymbol{c}_{nm}^\mathrm{e} \right]. \tag{20.41}$$

20.14 Operator form of the complete mapping

Collect the rotated spherical-harmonic coefficients into

$$\boldsymbol{c}_\mathrm{SH} = \begin{bmatrix} \boldsymbol{c}^\mathrm{i} \\ \boldsymbol{c}^\mathrm{e} \end{bmatrix}, \tag{20.42}$$

and collect the R-SCHA coefficients into

$$\boldsymbol{c}_\text{R-SCHA} = \begin{bmatrix} \boldsymbol{a}^{\mathrm{Int}} \\ \boldsymbol{a}^{\mathrm{Ext}} \\ \boldsymbol{d}^{\mathrm{Meh}} \\ \boldsymbol{d}^{\mathrm{Cons}} \end{bmatrix}. \tag{20.43}$$

For specified finite input and output coefficient sets, the mapping has the linear form

$$\boldsymbol{c}_\text{R-SCHA} = \boldsymbol{M}_{\mathrm{SH}\to\text{R-SCHA}} \boldsymbol{c}_\mathrm{SH}. \tag{20.44}$$

The matrix is assembled from:

1. angular overlap blocks $\mathcal{Q}^{(m)}_{n,km}$;
2. the two-boundary radial factors α_{km} and β_{km};
3. radial overlap blocks $\mathcal{J}^{\mathrm{i}}_{pn}$ and $\mathcal{J}^{\mathrm{e}}_{pn}$;
4. angular boundary traces $P^m_n(\cos\theta_0)$, $K^m_p(\theta_0)$, and $\mathcal{C}_m(\theta_0)$.

For fixed cap geometry, reference radius, truncation, and normalisation, this matrix is a fixed linear operator between the two coefficient spaces.

20.15 Exactness and truncation

The analytical mapping is exact at the function-space level when:

1. the spherical-harmonic series represents the target harmonic potential exactly in the shell;
2. the complete Legendre and radial R-SCHA systems are retained;
3. all admissible azimuthal orders are included;
4. all function normalisations and reference radii are treated consistently;
5. the boundary data lie in the mixed-problem trace spaces and the series converge in the energy sense of Section 13.

For finite truncations, the V_2 trace is its Fourier–radial projection. The Legendre part simultaneously reproduces both projected radial normal traces only for angular modes whose internal and external radial branches are both retained. If the two truncations are unequal, deleting one branch of a paired coefficient solution generally changes both radial traces; the resulting model is a truncation of the full expansion, not an exact two-boundary projection on those unpaired modes. For example, on the hemispherical cap $\theta_0 = \pi/2$, the target $V = r^3 P_3(\cos\theta)$ requires its external $\nu = 3$ branch. Omitting that branch while retaining the internal $\nu = 3$ mode cannot match both radial derivatives. All omitted modes contribute to the approximation error.

20.16 Conditioning of the two-boundary mapping

The determinant of the two-boundary matrix is

$$\det \boldsymbol{A}_{km} = \nu_{km}\,(\nu_{km}+1)\,D_{km}. \tag{20.45}$$

Writing

$$s = \ln(r_{\mathrm{b}}/r_{\mathrm{a}}),$$

one has

$$D_{km} = \mathrm{e}^{(\nu_{km}-1)s} - \mathrm{e}^{-(\nu_{km}+2)s}. \tag{20.46}$$

For a fixed angular degree ν_{km} and $s \to 0$, the expansion is

$$D_{km} = (2\nu_{km}+1)\,s + \mathcal{O}(s^2). \tag{20.47}$$

This asymptotic is not uniform in unbounded ν_{km}. For each fixed mode, as the two spherical boundaries approach one another, separation of the internal and external Legendre coefficients becomes increasingly sensitive to boundary-data errors. This is a conditioning issue of the two-boundary decomposition, not a failure of harmonicity.

20.17 Reverse mapping, harmonic extension, and ill-posedness

The forward restriction and re-expansion

$$\mathrm{SH} \longrightarrow \text{R-SCHA}$$

is well defined once the global harmonic field, cap geometry, target basis, and normalisation are specified. The reverse direction requires a more careful distinction between exact analytic continuation and coefficient recovery.

Suppose that two harmonic functions are defined on the same connected larger shell and agree on a non-empty open subset of the regional shell. Their difference is harmonic and vanishes on an open set; unique continuation for harmonic functions therefore forces the two functions to agree throughout the connected shell. Consequently, an exact harmonic continuation, if it exists in a prescribed connected global domain, is unique.

The actual obstructions are the following. First, a general regional R-SCHA field need not admit any harmonic continuation to a full spherical shell: singularities or incompatible analytic structure may occur outside the regional domain. Second, analytic continuation from a proper subdomain is severely ill posed; small perturbations of regional data may produce large changes in a recovered global coefficient sequence. Third, practical R-SCHA models contain only finitely many coefficients or finite-dimensional projections. The injectivity of such a finite-data map depends on the chosen global coefficient space and the rank of the regional projection matrix.

Accordingly, there is no inverse defined on the whole regional coefficient space. On a specified finite-dimensional global spherical-harmonic space, exact restriction to an open three-dimensional region is injective by unique continuation. Its composition with a finite coefficient projection is injective only if the resulting matrix has full column rank; unique continuation alone does not prove that rank condition. When an inverse on the exact range exists, it may be poorly conditioned. Regularisation can stabilise coefficient recovery, but is not a requirement for uniqueness of an exact full-rank finite system.

21 Exact coefficient projection and finite-dimensional approximation

The analytical R-SCHA series is determined by boundary data. The coefficients depend on the chosen basis and normalisation and, for the complete mixed-boundary problem, are obtained by spectral projection. This section collects the exact projection formulae for the two mixed subproblems and then formulates a finite truncation as an orthogonal or Gram projection in a declared Hilbert space.

21.1 Spectral data spaces and energy-solution data spaces

Two levels of regularity must be distinguished. For the purpose of defining Fourier–Sturm–Liouville projection coefficients, it is sufficient to assume that the radial normal-data densities of V_1 satisfy

$$G_{\mathrm{a}}, G_{\mathrm{b}} \in L^2\left((0,\theta_0)\times(0,2\pi), \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\lambda\right), \tag{21.1}$$

and that the lateral data of V_2 satisfy

$$F \in L^2\left((r_{\mathrm{a}}, r_{\mathrm{b}})\times(0,2\pi), \mathrm{d}r \, \mathrm{d}\lambda\right). \tag{21.2}$$

These assumptions define the boundary coefficient sequences and yield L^2 reconstruction of the corresponding traces.

They are not, by themselves, the full hypotheses for an $H^1(\Omega)$ weak solution. To invoke the variational existence and uniqueness theory of Equation (4.24), the Dirichlet datum must satisfy

$$F \in H^{1/2}(\Gamma_D) = \mathrm{Tr}_{\Gamma_D} H^1(\Omega), \tag{21.3}$$

and the combined Neumann datum must define an element of

$$G_N \in \left(H_{00}^{1/2}(\Gamma_N)\right)'. \tag{21.4}$$

Square-integrable densities on the smooth spherical faces provide admissible weak Neumann functionals through the natural duality pairing, but an arbitrary L^2 lateral function need not be the trace of an H^1 function. Thus square integrability is sufficient for spectral projection, whereas the trace and dual-trace conditions are required for the finite-energy mixed problem.

21.2 Longitude projection

Let

$$L_m^{\mathrm{c}} = \cos(m\lambda), \qquad L_m^{\mathrm{s}} = \sin(m\lambda), \tag{21.5}$$

with norms

$$\ell_m^{\mathrm{c}} = \begin{cases} 2\pi, & m = 0, \\ \pi, & m \geq 1, \end{cases} \qquad \ell_m^{\mathrm{s}} = \pi \quad (m \geq 1). \tag{21.6}$$

For a boundary function f, its longitude coefficients are

$$f_m^{\sigma} = \frac{1}{\ell_m^{\sigma}} \int_0^{2\pi} f(\lambda) L_m^{\sigma}(\lambda)\, \mathrm{d}\lambda. \tag{21.7}$$

All R-SCHA coefficient formulae preserve m and longitude parity because the boundary geometry is axisymmetric about the cap pole.

21.3 Legendre projection of the V_1 data

Let

$$P_{km}(\theta) = P_{\nu_{km}}^{m}(\cos\theta) \tag{21.8}$$

and

$$\mathcal{N}_{km}^{P} = \int_0^{\theta_0} P_{km}^2(\theta) \sin\theta\, \mathrm{d}\theta. \tag{21.9}$$

The exact projections of the lower and upper normal data are

$$b_{km}^{\Gamma,\sigma} = \frac{1}{\mathcal{N}_{km}^{P} \ell_m^{\sigma}} \int_0^{2\pi} \int_0^{\theta_0} G_{\Gamma}(\theta,\lambda) P_{km}(\theta) L_m^{\sigma}(\lambda) \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\lambda, \tag{21.10}$$

where $\Gamma \in \{\mathrm{a}, \mathrm{b}\}$. The internal and external coefficient vectors are then the unique solution of the two-boundary system Equation (5.20). Explicitly,

$$a_{km}^{\mathrm{Int},\sigma} = \frac{\beta_{km} b_{km}^{\mathrm{a},\sigma} + b_{km}^{\mathrm{b},\sigma}}{(\nu_{km}+1)(\beta_{km} - \alpha_{km})}, \tag{21.11a}$$

$$a_{km}^{\mathrm{Ext},\sigma} = \frac{\alpha_{km} b_{km}^{\mathrm{a},\sigma} + b_{km}^{\mathrm{b},\sigma}}{\nu_{km}(\beta_{km} - \alpha_{km})}. \tag{21.11b}$$

Thus the V_1 coefficient map is the composition of an orthogonal angular projection with a 2×2 algebraic boundary-separation map.

21.4 Fourier–radial projection of the V_2 data

Define

$$F_m^\sigma(r) = \frac{1}{\ell_m^\sigma} \int_0^{2\pi} F(r,\lambda) L_m^\sigma(\lambda)\ \mathrm{d}\lambda. \tag{21.12}$$

Let

$$\mathcal{N}_p^R = \int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p^2(r)\ \mathrm{d}r. \tag{21.13}$$

The general normalisation-dependent formula is

$$d_{pm}^\sigma = \frac{1}{r_\mathrm{a} \mathcal{N}_p^R K_p^m(\theta_0)} \int_{r_\mathrm{a}}^{r_\mathrm{b}} F_m^\sigma(r) R_p(r)\ \mathrm{d}r, \qquad p \geq 1, \tag{21.14a}$$

$$d_{0m}^\sigma = \frac{1}{r_\mathrm{a} \mathcal{N}_0^R C_m(\theta_0)} \int_{r_\mathrm{a}}^{r_\mathrm{b}} F_m^\sigma(r) R_0\ \mathrm{d}r. \tag{21.14b}$$

Under the convention $\mathcal{N}_p^R = r_\mathrm{a}$, the denominator contains r_a^2. These are exact Fourier–Sturm–Liouville coefficients of the lateral trace.

21.5 Reconstruction of the boundary data

If the component systems are complete and the data belong to the stated L^2 spaces, then

$$G_\Gamma(\theta,\lambda) = \sum_{m,\sigma} \sum_k b_{km}^{\Gamma,\sigma} P_{km}(\theta) L_m^\sigma(\lambda), \tag{21.15}$$

$$F(r,\lambda) = r_\mathrm{a} \sum_{m,\sigma} \left[d_{0m}^\sigma R_0 C_m(\theta_0) + \sum_{p\geq 1} d_{pm}^\sigma R_p(r) K_p^m(\theta_0) \right] L_m^\sigma(\lambda) \tag{21.16}$$

in the corresponding L^2 sense. At this level the equalities are boundary spectral reconstructions. If, in addition, the data satisfy Equations (21.3) and (21.4) and the resulting coefficient sequence has finite Dirichlet energy, then, after fixing the constant gauge, substitution into the interior series converges in $H^1(\Omega)$ to the unique weak harmonic solution of the mixed problem. For merely L^2 lateral data, one should not identify the projected series with an H^1 energy solution without a separate regularity argument.

21.6 Finite-dimensional subspaces

Let

$$\mathcal{V}_N = \mathrm{span}\{\phi_1, \ldots, \phi_N\} \tag{21.17}$$

be a finite R-SCHA subspace of the declared real Hilbert space. For a target f in that Hilbert space $\mathcal{H}$, the best approximation is

$$f_N = \operatorname*{arg\,min}_{v \in \mathcal{V}_N} \|f - v\|_{\mathcal{H}}^2. \tag{21.18}$$

The projection theorem gives the orthogonality conditions

$$\langle f - f_N, \phi_i \rangle_{\mathcal{H}} = 0, \qquad i = 1, \ldots, N. \tag{21.19}$$

Writing

$$f_N = \sum_{j=1}^N c_j \phi_j \tag{21.20}$$

gives the Gram system

$$\sum_{j=1}^{N} G_{ij}^{\mathcal{H}} c_j = b_i^{\mathcal{H}}, \tag{21.21}$$

where

$$G_{ij}^{\mathcal{H}} = \langle \phi_i, \phi_j \rangle_{\mathcal{H}}, \qquad b_i^{\mathcal{H}} = \langle \phi_i, f \rangle_{\mathcal{H}}. \tag{21.22}$$

21.7 Existence and uniqueness of the finite projection

The Gram matrix is symmetric positive semidefinite. It is positive definite if and only if the selected functions are linearly independent in $\mathcal{H}$. The best-approximation function in the finite-dimensional subspace is always unique; linear independence is required for a unique coefficient vector representing it. If the basis contains an additive-constant gauge mode and the chosen norm is the Dirichlet-energy seminorm, uniqueness is recovered by quotienting out constants or by fixing one gauge condition.

21.8 Orthogonal and non-orthogonal projections

When the basis is orthogonal in $\mathcal{H}$,

$$c_i = \frac{\langle f, \phi_i \rangle_{\mathcal{H}}}{\langle \phi_i, \phi_i \rangle_{\mathcal{H}}}. \tag{21.23}$$

This is the situation for the fixed-m Legendre boundary projection and for the radial Sturm–Liouville projection. On a fixed spherical-cap surface, the complete four-family system is not generally orthogonal; the exact surface Gram matrix of Section 18 must then be retained.

21.9 Galerkin formulation of the mixed Laplace problem

Let $a(u, v) = \mathcal{E}_\Omega(u, v)$ be the Dirichlet form on the affine space satisfying the prescribed essential boundary trace. A finite Galerkin solution V_N is characterised by

$$a(V_N, w_N) = \ell(w_N) \qquad \text{for all } w_N \in \mathcal{V}_N^0, \tag{21.24}$$

where $\mathcal{V}_N^0$ is the associated homogeneous test space and ℓ is the boundary functional induced by the Neumann data. The essential trace must be imposed by a lifting; a finite trace approximation replaces the boundary datum by a different one. Thus this Galerkin formulation is not, for arbitrary trial/test spaces, identical to the L^2 boundary projections above. In the harmonic energy spaces of Section 13, retaining the exact coefficients in an energy-orthogonal finite mode set does give its energy projection. That statement and the two-boundary matching condition must not be conflated: the latter requires retaining the paired radial branches for each matched angular mode.

21.10 Truncation as orthogonal projection

Let P_N denote the orthogonal projector onto a nested sequence $\mathcal{V}_N \subset \mathcal{H}$. Then

$$\|f - P_N f\|_{\mathcal{H}} = \inf_{v \in \mathcal{V}_N} \|f - v\|_{\mathcal{H}}. \tag{21.25}$$

If the union of the subspaces is dense, then

$$\|f - P_N f\|_{\mathcal{H}} \longrightarrow 0. \tag{21.26}$$

For the Legendre and radial component systems, this follows from their Sturm–Liouville completeness.

21.11 Mapping as a composition of projections

The spherical-harmonic to R-SCHA map in Section 20 may be written abstractly as

$$\mathcal{M} = \mathcal{P}_{V_1}^{\text{radial traces}} \oplus \mathcal{P}_{V_2}^{\text{lateral trace}}, \tag{21.27}$$

where the first component projects the two normal traces onto the cap Legendre system and solves the two-boundary algebraic equations, while the second projects the lateral trace onto the Fourier–radial system. The boundary inner products are those declared above; changing them changes the projection operator. The analytical coefficient map does not depend on the numerical method used to evaluate its integrals.

22 Boundary regularity and analytical convergence

The R-SCHA basis satisfies the Laplace equation and its defining boundary conditions exactly. Approximation questions arise because boundary traces may have limited regularity, may be incompatible where boundary surfaces meet, or may be represented by a finite spectral subspace. This section concerns only the analytical consequences of those facts: edge constraints, mixed-boundary regularity, alternative boundary conditions, convergence in L^2 and Dirichlet-energy norms, and convergence of derivatives on strictly interior subdomains. Only function-space regularity, spectral convergence, and boundary compatibility are considered.

22.1 The two circular edges of the shell

The three boundary surfaces meet along the lower and upper circular edges

$$\mathcal{C}_{\mathrm{a}} = \{(r,\theta,\lambda) : r = r_{\mathrm{a}}, \theta = \theta_0, 0 \le \lambda < 2\pi\}\,, \tag{22.1a}$$

$$\mathcal{C}_{\mathrm{b}} = \{(r,\theta,\lambda) : r = r_{\mathrm{b}}, \theta = \theta_0, 0 \le \lambda < 2\pi\}\,. \tag{22.1b}$$

For compactness, write

$$\mathcal{C} = \mathcal{C}_{\mathrm{a}} \cup \mathcal{C}_{\mathrm{b}}. \tag{22.2}$$

The standard R-SCHA decomposition uses

$$P_{\nu_{km}}^{m}(\cos\theta_0) = 0 \tag{22.3}$$

for every internal and external Legendre mode, and

$$R_p'(r_{\mathrm{a}}) = R_p'(r_{\mathrm{b}}) = 0, \qquad R_0' = 0 \tag{22.4}$$

for every Mehler and constant radial mode. Substitution into the field and tensor expressions of Section 14 gives the following edge identities for finite expansions, and for infinite expansions whose

differentiated traces converge at the edges:

$$V\big|_{\mathcal{C}} = V^{\mathrm{Meh}}\big|_{\mathcal{C}} + V^{\mathrm{Cons}}\big|_{\mathcal{C}}, \tag{22.5a}$$
$$B_x\big|_{\mathcal{C}} = B_x^{\mathrm{Int}}\big|_{\mathcal{C}} + B_x^{\mathrm{Ext}}\big|_{\mathcal{C}} + B_x^{\mathrm{Meh}}\big|_{\mathcal{C}} + B_x^{\mathrm{Cons}}\big|_{\mathcal{C}}, \tag{22.5b}$$
$$B_y\big|_{\mathcal{C}} = B_y^{\mathrm{Meh}}\big|_{\mathcal{C}} + B_y^{\mathrm{Cons}}\big|_{\mathcal{C}}, \tag{22.5c}$$
$$B_z\big|_{\mathcal{C}} = 0, \tag{22.5d}$$
$$T_{xx}\big|_{\mathcal{C}} = T_{xx}^{\mathrm{Int}}\big|_{\mathcal{C}} + T_{xx}^{\mathrm{Ext}}\big|_{\mathcal{C}} + T_{xx}^{\mathrm{Meh}}\big|_{\mathcal{C}} + T_{xx}^{\mathrm{Cons}}\big|_{\mathcal{C}}, \tag{22.5e}$$
$$T_{xy}\big|_{\mathcal{C}} = T_{xy}^{\mathrm{Int}}\big|_{\mathcal{C}} + T_{xy}^{\mathrm{Ext}}\big|_{\mathcal{C}} + T_{xy}^{\mathrm{Meh}}\big|_{\mathcal{C}} + T_{xy}^{\mathrm{Cons}}\big|_{\mathcal{C}}, \tag{22.5f}$$
$$T_{yy}\big|_{\mathcal{C}} = T_{yy}^{\mathrm{Int}}\big|_{\mathcal{C}} + T_{yy}^{\mathrm{Ext}}\big|_{\mathcal{C}} + T_{yy}^{\mathrm{Meh}}\big|_{\mathcal{C}} + T_{yy}^{\mathrm{Cons}}\big|_{\mathcal{C}}, \tag{22.5g}$$
$$T_{zx}\big|_{\mathcal{C}} = T_{zx}^{\mathrm{Int}}\big|_{\mathcal{C}} + T_{zx}^{\mathrm{Ext}}\big|_{\mathcal{C}} + T_{zx}^{\mathrm{Meh}}\big|_{\mathcal{C}} + T_{zx}^{\mathrm{Cons}}\big|_{\mathcal{C}}, \tag{22.5h}$$
$$T_{zy}\big|_{\mathcal{C}} = T_{zy}^{\mathrm{Meh}}\big|_{\mathcal{C}} + T_{zy}^{\mathrm{Cons}}\big|_{\mathcal{C}}, \tag{22.5i}$$
$$T_{zz}\big|_{\mathcal{C}} = T_{zz}^{\mathrm{Meh}}\big|_{\mathcal{C}}. \tag{22.5j}$$

These relations are exact structural consequences of the boundary-adapted basis. In particular, the vertical field component must vanish on both circular edges:

$$B_z\big|_{\mathcal{C}} = 0.$$

If the target field or the supplied boundary data do not satisfy the same edge behaviour, a finite R-SCHA expansion cannot reproduce all components there simultaneously. This pointwise edge obstruction does not preclude energy convergence: the edges have zero surface measure, and an H^1 limit need not have convergent edge derivatives. No localisation or rate of the finite-truncation error follows from the edge identity alone.

22.2 Corner compatibility and boundary regularity

The shell has piecewise smooth boundary surfaces that meet along $\mathcal{C}_{\mathrm{a}}$ and $\mathcal{C}_{\mathrm{b}}$. Classical smoothness up to these edges requires compatibility among the traces prescribed on the adjoining surfaces. For the standard decomposition,

$$V_1 = 0 \quad \text{on } \Gamma_{\theta_0},$$

while the radial normal derivatives of V_1 are prescribed on Γ_{a} and Γ_{b}. Likewise,

$$\frac{\partial V_2}{\partial r} = 0 \quad \text{on } \Gamma_{\mathrm{a}} \cup \Gamma_{\mathrm{b}},$$

while its value is prescribed on Γ_{θ_0}.

Weak solutions remain well defined under the assumptions in Section 4, but incompatible edge data reduce boundary regularity. The resulting harmonic solution may belong to $H^1(\Omega)$ while possessing weaker higher-order regularity near the circular edges. Since magnetic tensor components contain second derivatives of the potential, they are more sensitive to this loss of regularity than the potential or vector field.

The following analytical phenomena should therefore be distinguished:

1. **exact edge constraints**, such as Equation (22.5d), which hold for every basis mode;
2. **spectral truncation effects**, caused by projecting a boundary trace onto only finitely many modes;
3. **incompatible boundary traces**, which reduce the regularity of the exact mixed-boundary solution.

22.3 Standard Dirichlet–Neumann formulation

The standard R-SCHA boundary conditions are

$$\Delta V = 0 \quad \text{in } \Omega, \tag{22.6a}$$

$$V = F \quad \text{on } \Gamma_{\theta_0}, \tag{22.6b}$$

$$\frac{\partial V}{\partial n} = G_{\mathrm{a}} \quad \text{on } \Gamma_{\mathrm{a}}, \tag{22.6c}$$

$$\frac{\partial V}{\partial n} = G_{\mathrm{b}} \quad \text{on } \Gamma_{\mathrm{b}}. \tag{22.6d}$$

The name Dirichlet–Neumann refers to the lateral Dirichlet condition and the two radial Neumann conditions.

The properties established here are:

1. the two complementary subpotentials are gradient-orthogonal;
2. the mixed problem has no additive-constant ambiguity because the Dirichlet part of the boundary has positive area;
3. the Legendre and radial eigenproblems are self-adjoint and separable after the regular pole condition is imposed;
4. the exact three-dimensional energy has the Parseval-type decomposition in Section 17;
5. the mapping from spherical harmonics has a clear boundary-data interpretation.

Its principal practical limitation is the edge structure in Equation (22.5). Finite truncation cannot remove that structural restriction.

22.4 Neumann–Neumann formulation

An alternative imposes normal derivatives on all three boundary components:

$$\Delta V = 0 \quad \text{in } \Omega, \tag{22.7a}$$

$$\frac{\partial V}{\partial n} = F_N \quad \text{on } \Gamma_{\theta_0}, \tag{22.7b}$$

$$\frac{\partial V}{\partial n} = G_{\mathrm{a}} \quad \text{on } \Gamma_{\mathrm{a}}, \tag{22.7c}$$

$$\frac{\partial V}{\partial n} = G_{\mathrm{b}} \quad \text{on } \Gamma_{\mathrm{b}}. \tag{22.7d}$$

The lateral Legendre-type eigenfunctions then satisfy

$$\frac{\partial}{\partial \theta} P_\nu^m(\cos\theta)\bigg|_{\theta=\theta_0} = 0. \tag{22.8}$$

Because all boundary conditions are Neumann, solvability requires the compatibility condition

$$\int_{\Gamma_{\theta_0}} F_N \, \mathrm{d}S + \int_{\Gamma_{\mathrm{a}}} G_{\mathrm{a}} \, \mathrm{d}S + \int_{\Gamma_{\mathrm{b}}} G_{\mathrm{b}} \, \mathrm{d}S = 0, \tag{22.9}$$

and the potential is determined only up to an additive constant.

The degree-zero Neumann angular eigenfunction must be separated when a formula divides by its eigenvalue. This is a removable degeneracy of that formula, not a singularity of the constant

potential itself. The standard DN complementary orthogonality proof does not carry over to an arbitrary NN splitting: neither factor necessarily vanishes on each boundary face. Existence and uniqueness modulo constants follow from the weak Neumann problem when the prescribed functional annihilates constants. No general comparison of approximation rates with the DN formulation follows without specifying the data, truncations, and norm.

22.5 Dirichlet–Dirichlet formulation

A second alternative prescribes the potential on all three boundary components:

$$\Delta V = 0 \quad \text{in } \Omega, \tag{22.10a}$$

$$V = F_D \quad \text{on } \Gamma_{\theta_0}, \tag{22.10b}$$

$$V = G_{\mathrm{a}} \quad \text{on } \Gamma_{\mathrm{a}}, \tag{22.10c}$$

$$V = G_{\mathrm{b}} \quad \text{on } \Gamma_{\mathrm{b}}. \tag{22.10d}$$

The corresponding Dirichlet radial eigenfunctions may be written

$$R_p^{\mathrm{DD}}(r) = \left(\frac{2r_{\mathrm{a}}}{sr}\right)^{1/2} \sin\left[\frac{p\pi}{s}\ln\left(\frac{r}{r_{\mathrm{a}}}\right)\right], \qquad p = 1, 2, \ldots. \tag{22.11}$$

They satisfy

$$R_p^{\mathrm{DD}}(r_{\mathrm{a}}) = R_p^{\mathrm{DD}}(r_{\mathrm{b}}) = 0 \tag{22.12}$$

and

$$\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} \left[R_p^{\mathrm{DD}}(r)\right]^2 \mathrm{d}r = r_{\mathrm{a}}. \tag{22.13}$$

The all-Dirichlet weak problem is unique for a prescribed global trace in $H^{1/2}(\partial\Omega)$. Separate values on the three faces must be restrictions of such a trace; arbitrary facewise data cannot be assembled without checking their behaviour at the edges. Classical smoothness has stronger compatibility requirements. The corresponding complementary splitting is not generally energy-orthogonal. Its convergence rate cannot be ranked against another boundary formulation without additional regularity and approximation assumptions.

22.6 Neumann–Dirichlet formulation

The remaining mixed alternative uses a lateral Neumann condition and radial Dirichlet conditions:

$$\Delta V = 0 \quad \text{in } \Omega, \tag{22.14a}$$

$$\frac{\partial V}{\partial n} = F_N \quad \text{on } \Gamma_{\theta_0}, \tag{22.14b}$$

$$V = G_{\mathrm{a}} \quad \text{on } \Gamma_{\mathrm{a}}, \tag{22.14c}$$

$$V = G_{\mathrm{b}} \quad \text{on } \Gamma_{\mathrm{b}}. \tag{22.14d}$$

Its radial functions are again given by Equation (22.11), while the angular roots satisfy Equation (22.8). The complementary decomposition is again exactly gradient-orthogonal. Indeed, choose V_1 with zero lateral normal derivative and the prescribed radial values, and V_2 with zero radial values and the prescribed lateral normal derivative. Green's identity written as

$$\mathcal{E}_\Omega(V_1, V_2) = \int_{\partial\Omega} V_2 \partial_n V_1 \, \mathrm{d}S = 0 \tag{22.15}$$

has a zero integrand on each boundary face. The edge conditions and energy weights change, so the DN coefficient formulae cannot be reused unchanged.

Table 1: Qualitative comparison of the principal boundary-condition families used in three-dimensional R-SCHA. The first letter refers to the lateral boundary and the second to the two radial boundaries.

Type	Lateral condition	Radial condition	Principal mathematical features
DN	Dirichlet	Neumann	Standard formulation; unique mixed problem; exact V_1–V_2 gradient orthogonality; clear energy decomposition; structural restrictions on the circular edges.
NN	Neumann	Neumann	Requires flux compatibility and gauge fixing; DN complementary orthogonality is not automatic; isolate the degree-zero mode in eigenvalue-dependent formulae.
DD	Dirichlet	Dirichlet	Unique for admissible global $H^{1/2}$ trace; sine-type radial functions; complementary gradient orthogonality is not automatic.
ND	Neumann	Dirichlet	Unique mixed problem; exact complementary gradient orthogonality; radial sine modes; changed edge compatibility and energy weights.

22.7 Comparison of the four boundary-condition families

The standard DN assignment is the formulation developed in this handbook. Both complementary mixed assignments admit exact gradient orthogonality; changing the assignment requires deriving its own spectra, coefficient maps, and edge conditions.

22.8 Truncation and Gibbs-type behaviour

Let $V_{\boldsymbol{N}}$ denote a finite R-SCHA expansion with truncation vector

$$\boldsymbol{N} = (N_{\mathrm{Int}}, N_{\mathrm{Ext}}, P_{\max}, M_{\mathrm{Meh}}, M_{\max})\,. \tag{22.16}$$

Even when every retained mode satisfies the boundary conditions exactly, the partial sum may oscillate near a discontinuity or a region of low boundary regularity. For classical one-dimensional Fourier-type partial sums a jump can produce persistent Gibbs overshoot. An analogous quantitative assertion for this multidimensional expansion requires a specified summation procedure and regularity class; harmonicity of every partial sum alone gives no overshoot constant or localisation estimate.

The severity increases with derivative order. A mode with characteristic angular degree ν contributes approximately the factors

$$1, \qquad \frac{\nu}{r}, \qquad \frac{\nu^2}{r^2} \tag{22.17}$$

to the potential, field, and second-order tensor, respectively. Therefore, small high-degree coefficient errors can be inconspicuous in V but visible in $\boldsymbol{B}$ and dominant in $\boldsymbol{T}$.

For compatible traces, convergence can be rapid on every compact interior subdomain while remaining non-uniform near the circular edges. At an actual jump or edge singularity, the boundary layer is therefore a property of the spectral approximation and of the regularity class, not a failure of the Laplace equation satisfied by each partial sum.

22.9 Convergence of the Legendre component

For fixed m, the cap Legendre eigenfunctions form a complete orthogonal system in

$$L^2\left((0,\theta_0),\sin\theta\ \mathrm{d}\theta\right).$$

Let

$$f_m(\theta)=\sum_{k=m}^{\infty} c_{km}\widehat{P}_{km}(\theta), \tag{22.18}$$

where the angular functions are unit normalised. Then

$$\left\| f_m-\sum_{k=m}^{K} c_{km}\widehat{P}_{km}\right\|^2_{L^2_{\sin\theta}} = \sum_{k=K+1}^{\infty} |c_{km}|^2 \longrightarrow 0. \tag{22.19}$$

If

$$\sum_{k=m}^{\infty} \lambda_{km}\,|c_{km}|^2<\infty, \tag{22.20}$$

then the expansion also converges in the angular energy norm:

$$\begin{aligned}\int_0^{\theta_0}\left[\left|\frac{\mathrm{d}}{\mathrm{d}\theta}\left(f_m-f_{m,K}\right)\right|^2+\frac{m^2}{\sin^2\theta}\,|f_m-f_{m,K}|^2\right]\sin\theta\ \mathrm{d}\theta\\ =\sum_{k=K+1}^{\infty}\lambda_{km}\,|c_{km}|^2\longrightarrow 0.\end{aligned} \tag{22.21}$$

Faster coefficient decay requires regularity relative to the spectral operator and its endpoint conditions; classical smoothness alone does not ensure those conditions. For example, a smooth function with nonzero Dirichlet endpoint value need not have rapidly decaying Dirichlet spectral coefficients. Completeness alone gives no convergence rate.

22.10 Convergence of the radial component

The normalised radial functions satisfy

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_pR_q\ \mathrm{d}r=r_\mathrm{a}\delta_{pq}.$$

For

$$f(r)=\sum_{p=0}^{\infty} c_pR_p(r), \tag{22.22}$$

Parseval's identity gives

$$\left\| f-\sum_{p=0}^{P} c_pR_p\right\|^2_{L^2(r_\mathrm{a},r_\mathrm{b})} = r_\mathrm{a}\sum_{p=P+1}^{\infty}|c_p|^2\longrightarrow 0. \tag{22.23}$$

If

$$\sum_{p=1}^{\infty} \mu_p \left|c_p\right|^2 < \infty, \tag{22.24}$$

then

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} r^2 \left| \frac{\mathrm{d}}{\mathrm{d}r} \left(f - f_P\right) \right|^2 \mathrm{d}r = r_\mathrm{a} \sum_{p=P+1}^{\infty} \mu_p \left|c_p\right|^2 \longrightarrow 0. \tag{22.25}$$

22.11 Three-dimensional energy convergence

Let

$$\{\phi_j\}_{j\geq 1}$$

be the complete real R-SCHA modes after removal of the constant gauge mode, and let

$$\mathcal{E}\left(\phi_i, \phi_j\right) = e_j \delta_{ij}, \qquad e_j > 0. \tag{22.26}$$

For

$$V = \sum_{j=1}^{\infty} c_j \phi_j, \tag{22.27}$$

the condition

$$\sum_{j=1}^{\infty} e_j \left|c_j\right|^2 < \infty \tag{22.28}$$

implies convergence in the Dirichlet-energy norm:

$$\left| V - \sum_{j=1}^{J} c_j \phi_j \right|_{\mathcal{E}}^2 = \sum_{j=J+1}^{\infty} e_j \left|c_j\right|^2 \longrightarrow 0. \tag{22.29}$$

This statement uses the exact three-dimensional orthogonality established in Section 13. It does not imply that the same modes are orthogonal on a single fixed-radius cap surface.

22.12 Interior convergence of derivatives

Let

$$\Omega_\delta = \left\{ \boldsymbol{x} \in \Omega : \mathrm{dist}\left(\boldsymbol{x}, \partial\Omega\right) > \delta \right\}, \qquad \delta > 0. \tag{22.30}$$

Choose representatives with a common volume mean, so that the Poincaré inequality controls the H^1 norm by the energy norm. Suppose that the truncation error

$$e_{\boldsymbol{N}} = V - V_{\boldsymbol{N}}$$

is harmonic in Ω. Interior elliptic regularity gives, for every non-negative integer q,

$$\left\| e_{\boldsymbol{N}} \right\|_{H^{q+1}(\Omega_\delta)} \leq C_{q,\delta} \left\| e_{\boldsymbol{N}} \right\|_{H^1(\Omega)}. \tag{22.31}$$

Therefore, energy convergence implies convergence of all derivatives on every strictly interior subdomain. The estimate is not uniform as $\delta \to 0$, so derivative errors can be small in the interior and large near the boundary.

This estimate concerns subsets separated from every boundary face, including both spherical faces. Reducing only the angular cap radius does not by itself make a set strictly interior to the shell.

22.13 Trace convergence

Let V_N be a sequence of harmonic R-SCHA partial sums converging to V in $H^1(\Omega)$. Continuity of the trace operator gives

$$\operatorname{Tr} V_N \longrightarrow \operatorname{Tr} V \quad \text{in } H^{1/2}(\partial\Omega), \tag{22.32}$$

by the global trace theorem on a Lipschitz domain [4]. For harmonic H^1 functions, Green's formula also defines a continuous weak conormal trace in $H^{-1/2}(\partial\Omega)$, so their normal derivatives converge in that space already under the stated H^1 convergence. Classical-strength traces need more regularity. If $V_N \to V$ in $H^2(\Omega)$, then

$$\frac{\partial V_N}{\partial n} \longrightarrow \frac{\partial V}{\partial n} \quad \text{in} \prod_{\Gamma\in\{\Gamma_{\mathrm{a}},\Gamma_{\mathrm{b}},\Gamma_{\theta_0}\}} H^{1/2}(\Gamma) \tag{22.33}$$

with facewise Sobolev spaces. The unit normal jumps across the circular edges, so a global $H^{1/2}(\partial\Omega)$ assertion for the scalar normal component does not follow from the H^2 trace theorem.

22.14 Compatibility at the circular edges

The Dirichlet and Neumann traces prescribed on adjoining faces are not independent if a classical solution is required up to the circular edges. For example, if $V_1 = 0$ on the lateral face and V_1 is continuously differentiable up to an edge, both its longitude and radial derivatives tangent to the lateral face vanish at the edge. Consequently its radial Neumann data must vanish there. If V_2 is C^1 up to the same edge, its lateral Dirichlet datum must satisfy $\partial_r F = 0$ at the corresponding radial endpoint. Failure of these conditions does not necessarily destroy the weak solution, but it changes the local Sobolev regularity and introduces edge singular components.

A local edge analysis may involve power terms of the schematic form

$$V = V_{\mathrm{reg}} + \sum_j c_j \rho^{\gamma_j} \Psi_j(\varphi, z), \tag{22.34}$$

where ρ is distance from the edge, φ is the local angle between boundary faces, and the singular exponents γ_j are determined by the associated angular mixed-boundary pencil. The explicit exponents for the spherical-cap shell depend on the local dihedral geometry and on the selected Dirichlet–Neumann assignment. Logarithmic factors can occur at resonant exponents, so the displayed power sum is not a general expansion theorem. No complete edge expansion or sharp regularity classification is proved here.

22.15 Termwise differentiation of infinite series

Let

$$V = \sum_{j=1}^{\infty} c_j \phi_j \tag{22.35}$$

be an R-SCHA expansion. Convergence in the Dirichlet-energy norm justifies the gradient in the weak L^2 sense. Classical termwise differentiation of order q on a compact interior set $K \Subset \Omega$ follows, for example, if

$$\sum_{j=1}^{\infty} |c_j| \|\phi_j\|_{C^q(K)} < \infty. \tag{22.36}$$

More generally, harmonic interior regularity converts H^1 convergence into $H^{q+1}(K)$ convergence, as stated in Equation (22.31). Thus every derivative converges on strictly interior subdomains even when boundary convergence is slower.

22.16 Analytical meaning of finite truncation

A finite R-SCHA expansion is an exact harmonic function in the selected finite-dimensional space. Its error is an approximation error relative to the target boundary data or target harmonic function. It is not a residual of the Laplace equation. If P_N is the orthogonal projector in the energy space, then

$$|V - P_N V|_{\mathcal{E}}^2 = \sum_{j>N} e_j |c_j|^2, \tag{22.37}$$

for an energy-orthogonal ordering. On a fixed surface, where the complete basis is not diagonal, the error is instead governed by the corresponding Gram operator.

22.17 Convergence hierarchy

The analytical convergence statements form the hierarchy

$$\text{mixed-data convergence} \Longrightarrow H^1(\Omega) \text{ convergence} \Longrightarrow H^q(\Omega_\delta) \text{ convergence for every } q, \tag{22.38}$$

where the first arrow uses convergence of the lateral datum in $H^{1/2}(\Gamma_D)$ and of the radial Neumann functional in $(H_{00}^{1/2}(\Gamma_N))'$, together with continuity of the mixed solution operator. For spectral series, the weighted energy-tail condition provides the alternative criterion. In particular, L^2 convergence of a lateral boundary expansion alone does not imply the middle step; the trace-space distinction in Equations (21.3) and (21.4) must be respected. Uniform or classical convergence up to the circular edges requires stronger boundary regularity and compatibility and cannot be inferred solely from componentwise Sturm–Liouville completeness.

Part VI
Synthesis

23 Summary, scope, and questions not addressed here

This section records the scope of the results without repeating their derivations. References below identify the boundary assumptions and the spaces in which each assertion is used.

23.1 Boundary-adapted representation

For the mixed problems of Section 4, a harmonic H^1 potential admits the boundary-adapted decomposition $V = V_1 + V_2$ of Section 13. The first part has zero lateral Dirichlet trace; the second has zero spherical-boundary normal trace in the weak sense. Their Dirichlet-energy pairing vanishes. The constant potential is a gauge mode and is removed when an energy norm is required.

The separated factors are derived in Sections 5, 6 and 8 to 11. Individual retained modes satisfy the Laplace equation and their homogeneous boundary conditions. For an infinite series, these modewise identities alone do not establish convergence, the existence of an H^1 sum, or convergence

of boundary derivatives. The trace and energy hypotheses in Section 22 are needed for those conclusions.

Boundary coefficient projection and Dirichlet-energy projection are different constructions. In particular, simultaneous projection of both spherical Neumann traces by the V_1 part requires both radial branches for each retained angular mode. Independently truncating the internal and external families does not in general preserve that two-boundary projection property; see Sections 20 and 21.

23.2 Energy and differential identities

The shell Dirichlet energy is diagonal in the energy-orthogonal modes of Section 17. For infinite expansions the corresponding sum is understood through convergence in the energy space.

On a fixed cap $r = \beta$, the full-gradient energy is defined only when the gradient trace belongs to $L^2(\Sigma_\beta)$, as assumed in Section 18. This condition holds for finite separated expansions but does not follow from $H^1(\Omega)$ membership alone. The finite-basis surface Gram form retains the cross terms derived there. Discarding V_2 gives the Legendre-only spectrum of Section 19; it does not give the exact surface energy of a general four-family field. Its accuracy requires control of the omitted surface-gradient contribution.

For classical harmonic potentials in the interior, $\boldsymbol{B} = -\nabla V$ and $\boldsymbol{T} = -\operatorname{Hess} V$ satisfy symmetry, zero tensor trace, and zero field divergence. These identities and the orthogonal-frame transformations are derived in Sections 14 to 16. Boundary fluxes of weak solutions use conormal-trace pairings rather than pointwise derivatives unless the additional boundary regularity has been established.

23.3 One-surface continuation

The R-SCHA2D construction in Section 12 uses different boundary information from the closed-shell problem. Its degree-$-1/2$ radial branch is not obtained by collapsing the two-boundary radial eigenproblem. Existence of a continuation, uniqueness conditional on the specified data, and stability are separate questions. For example, an additive gauge is irrelevant to field data, and a radial derivative on one open cap does not by itself distinguish both radial branches.

The fixed-order Legendre eigenfunction completeness theorem does not by itself establish a Riesz-basis or frame result for the combined R-SCHA2D continuation family. No such result is proved here. This is a limitation of the present treatment rather than an assertion about all published results.

23.4 Further analysis outside the present scope

The following topics require assumptions or analysis beyond the identities and conditional convergence statements derived in this handbook:

1. quantitative coefficient and truncation bounds in specified boundary trace spaces, including their dependence on aperture and shell thickness;
2. a classification of boundary-edge singularities and the corresponding compatibility conditions for each assignment of mixed data;
3. a physically interpretable organisation of fixed-surface Gram eigenvectors and estimates for the Legendre-only approximation. A finite real symmetric Gram matrix already has an orthogonal eigendecomposition by the spectral theorem; its existence is not an open problem;
4. representation and continuation results for the combined one-surface family in a specified function space;

5. explicit representations on non-spherical domains. The variational mixed-boundary decomposition can still be formulated on suitable Lipschitz domains; the particular separated spherical functions used here generally cannot.

These are directions not developed in this handbook, not a list of problems certified to be unresolved in the mathematical literature.

24 Notation register

This register consolidates the notation used throughout the handbook. Symbols are grouped by mathematical role rather than by first occurrence. When a symbol has more than one conventional use, its meaning is determined by indices and context; the principal cases are stated explicitly below.

24.1 General conventions

Symbol or convention	Meaning
$\mathbb{R}$, $\mathbb{C}$	Sets of real and complex numbers.
$\mathbb{N}$	Set of natural numbers; the lower index limit is stated whenever inclusion of zero matters.
i, e	Imaginary unit and base of the natural logarithm.
d	Differential symbol, as in $\mathrm{d}r$, $\mathrm{d}\theta$, and $\mathrm{d}\Omega$.
δ_{ij}	Kronecker delta.
$\boldsymbol{I}$	Identity matrix or identity operator of the dimension implied by context.
$\boldsymbol{0}$	Zero vector of the dimension implied by context.
$(\cdot)^{\mathsf{T}}$	Matrix or vector transpose.
$\operatorname{diag}(\boldsymbol{A})$	Diagonal matrix formed from the diagonal entries of $\boldsymbol{A}$, or the diagonal vector when this is clear from context.
$\operatorname{tr}\boldsymbol{A}$, $\det\boldsymbol{A}$	Trace and determinant of a square matrix or second-rank tensor.
$\operatorname{spec}(\boldsymbol{A})$	Spectrum, or set of eigenvalues, of $\boldsymbol{A}$.
$\ker(\boldsymbol{A})$, $\operatorname{ran}(\boldsymbol{A})$	Null space and range of a linear operator or matrix.
$\langle u, v\rangle$	Inner product; the integration domain and weight are specified locally.
$\|u\|$, $\lvert u\rvert$	Norm and absolute value or modulus. Subscripts identify the relevant norm.
$H^q(\Omega)$, $L^2(\Omega)$	Sobolev and square-integrable function spaces.
$\operatorname{Tr} u$	Boundary trace of a Sobolev function.
$\mathcal{O}(\cdot)$	Asymptotic order symbol.
$\operatorname{atan2}(Y, X)$	Quadrant-safe two-argument arctangent.

24.2 Geometry and boundary sets

Symbol	Meaning
$\Omega = \Omega_{\theta_0}^{r_\mathrm{a}, r_\mathrm{b}}$	Three-dimensional spherical-cap shell: $r_\mathrm{a} < r < r_\mathrm{b}$, $0 \le \theta < \theta_0$, and $0 \le \lambda < 2\pi$.
r_a, r_b	Inner and outer radii of the shell, with $0 < r_\mathrm{a} < r_\mathrm{b}$.
θ_0	Angular radius, or half-angle, of the spherical cap.
r_ref or a	Reference radius used by a spherical-harmonic model or by a declared normalisation convention. It is conceptually distinct from r_a.
Γ_a	Inner spherical boundary $r = r_\mathrm{a}$.
Γ_b	Outer spherical boundary $r = r_\mathrm{b}$.
Γ_{θ_0}	Lateral conical boundary $\theta = \theta_0$.
$\partial\Omega$	Closed boundary $\Gamma_\mathrm{a} \cup \Gamma_\mathrm{b} \cup \Gamma_{\theta_0}$.
$\mathcal{C}_\mathrm{a}$, $\mathcal{C}_\mathrm{b}$	Lower and upper circular edges at $(r, \theta) = (r_\mathrm{a}, \theta_0)$ and (r_b, θ_0).
$\boldsymbol{n}$	Outward unit normal on $\partial\Omega$.
$\mathrm{d}\Omega$	Volume element $r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\lambda$.
$\mathrm{d}S$	Boundary-surface element; its explicit form depends on the boundary component.
s	Logarithmic shell thickness, $s = \ln(r_\mathrm{b}/r_\mathrm{a})$.
Ω_δ	Strictly interior subdomain whose distance from $\partial\Omega$ exceeds $\delta > 0$.
θ_tar	Maximum co-latitude of the region retained for interpretation.
$\Delta\theta_\mathrm{buf}$	Angular buffer width $\theta_0 - \theta_\mathrm{tar}$.

24.3 Coordinates and orthonormal frames

Symbol	Meaning
(r, θ, λ)	Spherical radius, co-latitude, and longitude in the frame currently under discussion.
(X, Y, Z)	Global geocentric Cartesian coordinates.
$(X_{\mathrm{c}}, Y_{\mathrm{c}}, Z_{\mathrm{c}})$	Cap-centred Cartesian coordinates.
$(\theta_{\mathrm{c}}, \lambda_{\mathrm{c}})$	Global co-latitude and longitude of the cap centre.
(r, θ', λ')	Cap-centred spherical coordinates. The radius is invariant under rotation.
$\boldsymbol{e}_X$, $\boldsymbol{e}_Y$, $\boldsymbol{e}_Z$	Global geocentric Cartesian unit vectors.
$\boldsymbol{e}_r$, $\boldsymbol{e}_\theta$, $\boldsymbol{e}_\lambda$	Spherical orthonormal unit vectors.
$\boldsymbol{e}_x$, $\boldsymbol{e}_y$, $\boldsymbol{e}_z$	Local north–east–down unit vectors, with $\boldsymbol{e}_x = -\boldsymbol{e}_\theta$, $\boldsymbol{e}_y = \boldsymbol{e}_\lambda$, and $\boldsymbol{e}_z = -\boldsymbol{e}_r$.
$\boldsymbol{g}_p$, $\boldsymbol{g}^p$	Covariant and contravariant coordinate bases.
g_{pq}, g^{pq}	Covariant and contravariant metric components.
h_p	Scale factor of an orthogonal coordinate system. In spherical coordinates, $h_r = 1$, $h_\theta = r$, and $h_\lambda = r \sin\theta$.
Γ^s_{pq}	Christoffel symbols of the second kind.
$\boldsymbol{Q}(\theta, \lambda)$	Matrix whose columns are $\boldsymbol{e}_r$, $\boldsymbol{e}_\theta$, and $\boldsymbol{e}_\lambda$ expressed in the geocentric Cartesian frame.
$\boldsymbol{C}_{\mathrm{NED}}(\theta, \lambda)$	Matrix whose columns are the local north, east, and down unit vectors expressed in the geocentric Cartesian frame.
$\boldsymbol{A}_{\mathrm{c}}$	Proper orthogonal rotation from the global Cartesian frame to the cap-centred Cartesian frame.
$\boldsymbol{R}_{\mathrm{new}\leftarrow\mathrm{old}}$	Direction-cosine matrix transforming physical components from an old orthonormal frame to a new one.
$\boldsymbol{M}(\boldsymbol{R})$	6×6 matrix induced by $\boldsymbol{R}$ for the six independent components of a symmetric second-rank tensor.

24.4 Differential operators, fields, and tensors

Symbol	Meaning
∇, $\nabla\cdot$, $\nabla\times$, Δ	Gradient, divergence, curl, and Laplace operators.
V	Scalar magnetic potential.
V_1, V_2	Complementary harmonic subpotentials associated with the two mixed boundary-value subproblems.
V^{Int}, V^{Ext}, V^{Meh}, V^{Cons}	Internal Legendre, external Legendre, Mehler, and constant-function contributions to the complete R-SCHA potential.
$\boldsymbol{B}$	Magnetic field, $\boldsymbol{B} = -\nabla V$.
B_x, B_y, B_z	North, east, and downward magnetic-field components.
$\operatorname{Hess} V$	Covariant Hessian of the scalar potential, represented by physical components in an orthonormal frame.
$\boldsymbol{T}$	Second-order magnetic-gradient tensor, $\boldsymbol{T} = -\operatorname{Hess} V$.
T_{xx}, T_{xy}, T_{yy}, T_{zx}, T_{zy}, T_{zz}	Six independent components of the symmetric magnetic-gradient tensor in the local north–east–down frame.
$V_{,p}$	Partial derivative of V with respect to coordinate u^p.
$V_{;pq}$	Covariant second derivative of V.
$V_{;\hat{p}\hat{q}}$	Physical Hessian component in an orthonormal basis.
$\lVert\boldsymbol{T}\rVert_{\mathrm{F}}$	Frobenius norm of the magnetic-gradient tensor.
$\operatorname{tr}\boldsymbol{T} = 0$	Trace-free condition equivalent to $\Delta V = 0$ and $\nabla\cdot\boldsymbol{B} = 0$.

24.5 Indices and separation parameters

Symbol	Meaning
m	Non-negative azimuthal order. Sine terms are omitted for $m = 0$.
k	Degree-like index of a non-integer Legendre root, with $k = m, m+1, \ldots$ at fixed azimuthal order m.
p	Radial eigenfunction index. The value $p = 0$ denotes the constant radial mode; $p \geq 1$ denotes oscillatory Mehler radial modes.
n, ℓ	Integer spherical-harmonic degree.
i	Observation index, tensor index, or generic row index, according to context.
j	Basis-function, coefficient, or generic column index, according to context.
q	Generic derivative order, data component, cap index, or Sobolev order, according to context.
ν	Continuous associated-Legendre degree.
ν_{km}	Admissible non-integer degree for order m, selected by the cap boundary condition; ν_{mm} is the first root.
$\nu = -\frac{1}{2} + \mathrm{i}\tau$	Complex-degree parameterisation used by the Mehler family.
λ_{km}	Angular Sturm–Liouville eigenvalue $\nu_{km}(\nu_{km}+1)$. The indexed symbol is distinct from the longitude λ.
μ	Radial Sturm–Liouville eigenvalue, $\mu = -\nu(\nu+1)$ for the V_2 separation.
μ_0, μ_p	Zero radial eigenvalue and positive oscillatory eigenvalues.
τ, τ_p	Mehler parameter and its discrete radial values, $\tau_p = p\pi/s$.
σ, ρ	Longitude parity labels, normally c for cosine and s for sine.

24.6 Legendre functions and related quantities

Symbol	Meaning
$P_\nu^m(x)$	First-kind associated Legendre or Ferrers function of continuous degree ν and integer order m, evaluated for $-1 < x \le 1$.
$P_\nu^m(\cos\theta)$	Angular Legendre factor written in co-latitude.
$P_{\nu_{km}}^m(\cos\theta_0) = 0$	Homogeneous lateral Dirichlet root condition defining the standard cap degrees.
$\widehat{P}_{km}(\theta)$	Cap-unit-normalised Legendre eigenfunction.
$\mathcal{N}_{km}^P$	Squared cap norm of $P_{\nu_{km}}^m(\cos\theta)$.
λ_{km}	Angular eigenvalue $\nu_{km}(\nu_{km}+1)$.
$F_m(\nu)$	Root function $P_\nu^m(\cos\theta_0)$.
$\partial_\nu P_\nu^m$	Derivative of the Legendre function with respect to its continuous degree.
$j_{m,q}$	qth positive zero of the Bessel function J_m, used in small-cap root estimates.
$\Delta\nu$	Approximate cap-degree spacing, $\Delta\nu = \pi/\theta_0$.
$\mathcal{B}_j$	Non-integer degree bin used in the simplified surface spectrum.
ℓ_{eq}	Continuous equivalent global spherical-harmonic degree obtained by matching the angular Laplace–Beltrami eigenvalue.

24.7 Mehler and constant angular functions

Symbol	Meaning
$K_\tau^m(\theta)$	Unnormalised regular Mehler angular function, represented by $P_{-1/2+\mathrm{i}\tau}^m(\cos\theta)$ under the declared convention.
$K_p^m(\theta)$	Mehler angular function evaluated at $\tau = \tau_p$.
$\widehat{K}_p^m(\theta)$	Cap-unit-normalised Mehler angular function.
$F_\tau^m(\theta)$	Real hypergeometric-series factor used in the stable normalised Mehler representation.
$\delta_k^m(\tau)$	Coefficient of the Mehler hypergeometric series.
$\widetilde{\mathcal{N}}_\tau^m(\theta)$	Explicit prefactor in the completely normalised Mehler representation.
$\mathcal{C}_m(\theta)$	Normalised zero-eigenvalue angular function used by the constant family.
$P_0^m(\theta)$	Regularised degree-zero angular branch before or after a declared normalisation; it is not the classical identically zero Ferrers polynomial for $m > 0$.
I_m	Squared cap norm of the unnormalised constant angular function.
$\Phi(z,\alpha,v)$	Lerch transcendent used in the closed constant-function norm.
Ξ_{pm}^{Meh}	Boundary-energy factor of a normalised Mehler angular mode.
Ξ_{0m}^{Cons}	Boundary-energy factor of a normalised constant angular mode.

24.8 Classical one-surface R-SCHA2D quantities

Symbol	Meaning
$\mathcal{C}_{\theta_0}(a)$	Spherical-cap observation surface of radius a used in R-SCHA2D.
n_k^m	Real non-integer degree satisfying the lateral Neumann condition $\partial_\theta P^m_{n_k^m}(\cos\theta_0) = 0$.
$R_2(r)$	Repeated-root radial continuation $A_2(a/r)^{1/2}[\ln(r/a) + 2]$ selected by $R_2'(a) = 0$.
$P^m_{-1/2}(\cos\theta)$	Degree-$-1/2$ conical angular factor of the complementary one-surface potential.
V_1^{2D}, V_2^{2D}	The derivative-root Legendre contribution and the complementary repeated-root contribution of classical R-SCHA2D.

24.9 Radial functions

Symbol	Meaning
t	Logarithmic radial coordinate $t = \ln(r/r_{\mathrm{a}})$. In the constant-angular derivation, the same letter is used locally for $\ln\tan(\theta/2)$; the context distinguishes the two uses.
$R_p(r)$	Normalised oscillatory radial eigenfunction for $p \geq 1$.
R_0	Normalised constant radial eigenfunction.
$\mathcal{A}_p$	Normalisation factor appearing in the closed form of R_p.
$\mathcal{N}_p^R$	Squared radial norm $\int_{r_{\mathrm{a}}}^{r_{\mathrm{b}}} R_p^2 \, \mathrm{d}r$.
μ_p	Radial eigenvalue $\frac{1}{4} + (p\pi/s)^2$ for $p \geq 1$.
$R_p^{\mathrm{DD}}(r)$	Sine-type radial eigenfunction associated with radial Dirichlet conditions.
$I_n(r)$	Internal-type spherical-harmonic radial power $(r_{\mathrm{ref}}/r)^{n+1}$.
$E_n(r)$	External-type spherical-harmonic radial power $(r/r_{\mathrm{ref}})^n$.
$\mathcal{J}^{\mathrm{i}}_{pn}$, $\mathcal{J}^{\mathrm{e}}_{pn}$	Radial overlaps of R_p with I_n and E_n.
D_{pn}	Denominator $(n + \frac{1}{2})^2 + (p\pi/s)^2$ used in closed radial overlaps.

24.10 R-SCHA coefficients and truncation indices

Symbol	Meaning
g_{km}^{Int}, h_{km}^{Int}	Cosine and sine coefficients of the internal Legendre family.
g_{km}^{Ext}, h_{km}^{Ext}	Cosine and sine coefficients of the external Legendre family.
G_p^m, H_p^m	Cosine and sine coefficients of the Mehler family.
G_0^m, H_0^m	Cosine and sine coefficients of the constant family.
$\boldsymbol{a}_{km}^{\mathrm{Int}}$, $\boldsymbol{a}_{km}^{\mathrm{Ext}}$	Two-component cosine–sine coefficient vectors for the Legendre families.
$\boldsymbol{d}_{pm}^{\mathrm{Meh}}$, $\boldsymbol{d}_{0m}^{\mathrm{Cons}}$	Two-component cosine–sine coefficient vectors for the Mehler and constant families.
$C_m(\lambda)$, $S_m(\lambda)$	Generic cosine and sine coefficient combinations used in differentiation formulae.
C_{km}^{Int}, S_{km}^{Int}	Internal Legendre longitude combinations.
C_{km}^{Ext}, S_{km}^{Ext}	External Legendre longitude combinations.
C_{pm}^{Meh}, S_{pm}^{Meh}	Mehler longitude combinations.
C_{0m}^{Cons}, S_{0m}^{Cons}	Constant-family longitude combinations.
N_{Int}	Maximum root index retained in the internal Legendre sequence.
N_{Ext}	Maximum root index retained in the external Legendre sequence.
$P_{\max}$	Maximum positive radial index in the Mehler truncation.
M_{Meh}	Maximum azimuthal order in the Mehler truncation.
$M_{\max}$	Maximum azimuthal order retained in the constant sequence.
$\boldsymbol{N}$	Truncation vector $(N_{\mathrm{Int}}, N_{\mathrm{Ext}}, P_{\max}, M_{\mathrm{Meh}}, M_{\max})$.
L_{i}, L_{e}	Maximum integer degrees of the internal and external source spherical-harmonic models.

24.11 Function spaces and energy quantities

Symbol	Meaning
$\mathscr{H}(\Omega)$	Space of weakly harmonic H^1 functions in Ω.
$\mathscr{H}_1$	Harmonic subspace with zero lateral Dirichlet trace.
$\mathscr{H}_2$	Harmonic subspace with zero radial normal derivative on the two spherical boundaries.
$\dot{H}^1(\Omega)$	Quotient space $H^1(\Omega)/\mathbb{R}$ removing additive constants.
$\mathcal{E}(u,v)$	Dirichlet-energy pairing $\int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}\Omega$.
$\lvert u\rvert_{\mathcal{E}}$	Dirichlet-energy seminorm.
$E_\Omega[V]$	Three-dimensional Dirichlet energy $\int_\Omega \lvert\nabla V\rvert^2 \, \mathrm{d}\Omega$.
E_Ω^{Int}, E_Ω^{Ext}, E_Ω^{Meh}, E_Ω^{Cons}	Exact three-dimensional energies of the four R-SCHA families.
Σ_β	Spherical-cap surface at the fixed radius $r = \beta$.
$E_\beta[V]$	Full-gradient energy on Σ_β, provided the gradient trace exists in $L^2(\Sigma_\beta)$.
$E_\beta^r[V]$, $E_\beta^h[V]$	Radial and horizontal contributions to the fixed-surface energy.
$\boldsymbol{G}_\beta$	Exact surface-energy Gram matrix of the selected finite R-SCHA basis.
$\mathscr{S}_j(\beta)$	Simplified Legendre-dominant surface spectral density in degree bin $\mathcal{B}_j$.
A_{km}^{Int}, A_{km}^{Ext}, A_{pm}^{Meh}, A_{0m}^{Cons}	Longitude-weighted squared coefficient amplitudes used in energy formulae.
$\kappa_{\mathrm{h}}(\nu;\beta)$	Nominal horizontal wavenumber $\sqrt{\nu(\nu+1)}/\beta$.
$L_{\mathrm{nom}}(\nu;\beta)$	Nominal horizontal wavelength $2\pi\beta/\sqrt{\nu(\nu+1)}$.

24.12 Flux quantities

Symbol	Meaning
$\Phi_{\mathrm{a}}[V]$	Outward magnetic flux through the inner spherical boundary.
$\Phi_{\mathrm{b}}[V]$	Outward magnetic flux through the outer spherical boundary.
$\Phi_{\theta_0}[V]$	Outward magnetic flux through the lateral conical boundary.
$\Phi_{\partial\Omega}[V]$	Total outward flux through the closed shell boundary.
$\mathcal{R}_{\mathrm{div}}$	Pointwise or discretised divergence residual $\nabla\cdot\boldsymbol{B} = \operatorname{tr}\boldsymbol{T}$.
$\mathcal{R}_{\Phi}$	Integrated closed-boundary flux residual $\Phi_{\mathrm{a}} + \Phi_{\mathrm{b}} + \Phi_{\theta_0}$.
$\varepsilon_{\mathrm{div}}$, ε_{Φ}	Dimensionless divergence and relative flux residuals.

24.13 Spherical-harmonic to R-SCHA mapping

Symbol	Meaning
$\boldsymbol{c}^{\mathrm{i}}_{nm}$, $\boldsymbol{c}^{\mathrm{e}}_{nm}$	Two-component cosine–sine coefficient vectors of the rotated internal and external spherical-harmonic models.
$\boldsymbol{t}_m(\lambda)$	Longitude vector $[\cos(m\lambda), \sin(m\lambda)]^{\mathsf{T}}$.
$\mathcal{I}^{(m)}_{n,km}$	Angular overlap between P^m_n and $P^m_{\nu_{km}}$ on the cap.
$\mathcal{Q}^{(m)}_{n,km}$	Normalised angular projection weight $\mathcal{I}^{(m)}_{n,km}/\mathcal{N}^P_{km}$.
$\boldsymbol{b}^{\mathrm{a}}_{km}$, $\boldsymbol{b}^{\mathrm{b}}_{km}$	Projected lower- and upper-boundary normal-derivative data.
α_{km}	Radial factor $(r_{\mathrm{a}}/r_{\mathrm{b}})^{\nu_{km}+2}$.
β_{km}	Radial factor $(r_{\mathrm{b}}/r_{\mathrm{a}})^{\nu_{km}-1}$.
D_{km}	Two-boundary denominator $\beta_{km} - \alpha_{km}$.
$\boldsymbol{f}_m(r)$	Order-m vector-valued spherical-harmonic potential trace on $\theta = \theta_0$.
$\boldsymbol{c}_{\mathrm{SH}}$	Rotated spherical-harmonic coefficient vector.
$\boldsymbol{c}_{\text{R-SCHA}}$	R-SCHA coefficient vector for the declared index set.
$\boldsymbol{M}_{\mathrm{SH}\to\text{R-SCHA}}$	Linear coefficient map; it is a finite matrix only after the input and output index sets have been specified and truncated.

24.14 Additional surface-spectrum notation

Symbol	Meaning
$\varepsilon^{\mathrm{Int}}_{km}(\beta)$, $\varepsilon^{\mathrm{Ext}}_{km}(\beta)$	Individual exact Legendre mode energies on the fixed surface.
$\mathscr{S}^{\mathrm{Leg}}_j(\beta)$	Binned simplified Legendre-dominant spectrum. Its bin integral recovers $E_\beta[V_1]$, not the complete $E_\beta[V]$.
$Q^{(m)}_{kp}$	Legendre–Mehler angular L^2 overlap on the cap.
$M^{(m)}_{pq}$, $H^{(m)}_{pq}$	Mehler angular L^2 and angular-energy Gram entries.

24.15 Normalisation and sign conventions

The coefficient values are meaningful only together with the adopted basis normalisation. This handbook uses the following principal conventions.

1. The magnetic field is the negative gradient:

$$\boldsymbol{B} = -\nabla V. \tag{24.1}$$

2. The magnetic-gradient tensor is the negative covariant Hessian:

$$\boldsymbol{T} = -\operatorname{Hess} V. \tag{24.2}$$

3. The local Cartesian frame is north–east–down.
4. The radial eigenfunctions satisfy

$$\int_{r_\mathrm{a}}^{r_\mathrm{b}} R_p(r) R_q(r)\ \mathrm{d}r = r_\mathrm{a} \delta_{pq}. \tag{24.3}$$

5. Cap-unit-normalised angular functions satisfy

$$\int_0^{\theta_0} A_m^2(\theta) \sin\theta\ \mathrm{d}\theta = 1. \tag{24.4}$$

6. Longitude functions are not divided by their $\sqrt{\pi}$ or $\sqrt{2\pi}$ norms. The corresponding factors appear explicitly in energy and orthogonality formulae.
7. Sine coefficients with $m = 0$ are zero and are normally omitted.
8. Orthogonal coordinate transformations act on column-vector physical components according to

$$\boldsymbol{v}_\mathrm{new} = \boldsymbol{R}_{\mathrm{new}\leftarrow\mathrm{old}} \boldsymbol{v}_\mathrm{old},$$

and on second-rank tensors according to

$$\boldsymbol{T}_\mathrm{new} = \boldsymbol{R}_{\mathrm{new}\leftarrow\mathrm{old}} \boldsymbol{T}_\mathrm{old} \boldsymbol{R}_{\mathrm{new}\leftarrow\mathrm{old}}^{\mathsf{T}}.$$

24.16 Potentially ambiguous symbols

The following symbol pairs should not be conflated.

1. λ is longitude, whereas $\lambda_{km} = \nu_{km}(\nu_{km} + 1)$ is an angular eigenvalue.
2. $s = \ln(r_\mathrm{b}/r_\mathrm{a})$ is the logarithmic shell thickness, whereas s_m may denote a sine coefficient in a local generic longitude combination.
3. $\mathcal{C}_m(\theta)$ is a constant-family angular function, whereas $\mathcal{C}_\mathrm{a}$ and $\mathcal{C}_\mathrm{b}$ are circular boundary edges.
4. A_{km}^{Int} and related symbols in Section 17 are squared coefficient amplitudes, whereas $A(\theta)$ in a separated mode denotes a generic angular function.
5. D_{km} is the two-boundary mapping denominator, whereas D_{pn} is the denominator in the radial overlap formula.
6. α_{km} is a radial mapping factor, whereas Euler-angle symbols used in coordinate rotations are local to that context.
7. $R_p(r)$ denotes a radial basis function, whereas $\boldsymbol{R}$ denotes an orthogonal rotation matrix.
8. $\boldsymbol{G}_\beta$ denotes a surface-energy Gram matrix, whereas G_p^m denotes a Mehler cosine coefficient.
9. $I_n(r)$ is an internal-type radial power, I_m is a constant-function angular norm, and $\boldsymbol{I}$ is the identity matrix.
10. The letter t denotes the logarithmic radial coordinate in Section 11; it is also used locally for the logarithmic angular coordinate in the derivation of the constant modes.

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